The Holonet Machine

A complete engineering blueprint for a computer
whose wiring diagram is a piece of finite geometry

Every promoted figure is typed: proved, measured, published, modelled, or open.

W(3,3)–E_8 project, glue track

3 August 2026 · Passes 2700–2907

Abstract

This is a build document for a computer that does not exist yet.

Most computer designs begin with a goal and invent a structure to reach it. This one runs the other way. It starts from a piece of pure mathematics — a forty-point shape called $W(3,3)$, studied since the 1900s for reasons having nothing to do with computing — and finds that the shape already contains an instruction set, a memory layout, a network topology, an error model, and a set of quantum resources. Nothing here was designed in the usual sense. It was read off.

The result is unusually specific for something unbuilt. The instruction set needs *four* operations in two bits, and they generate a group of exactly 4,199,040 transformations, with a worst-case program length of exactly **19**. On the HX8K target the minimal arithmetic core uses **43 logic cells** at **208.86 MHz**; on the UP5K it uses the same 43 cells at 72.40 MHz. Those are part-specific measurements, not interchangeable headline numbers. A support-only observation discards exactly 8/3 bits of conditional phase information. Landauer supplies a thermodynamic floor for erasing that information, not an engineering power forecast. Published time-bin/frequency-bin experiments report a two-qutrit gate fidelity near 0.92; this is prior-art evidence for a component class, not a measurement of the unbuilt Holonet stack. And thirty-six of the forty points turn out to be exactly the machine's magic states, so the quantum resource is not an add-on — it is everything except one line of the address space.

We give the full instruction semantics, the datapath, the measured cell budget, the network protocol, the thermodynamic floor, and the verification methodology. We also give — because this is what makes a blueprint trustworthy rather than merely confident — an itemised record of every figure we previously got wrong, how each error survived exhaustive testing, and what finally caught it.

Contents

1 The last chromatic bit: an exact defect filter, not a search claim

The canonical frame graph has 540 vertices, 8,640 edges, degree 32, and

$$H + 4I = MM^{\mathsf{T}}, \qquad \operatorname{spec}(H) = 32^1 \oplus 14^{44} \oplus 8^{15} \oplus 4^{81} \oplus 2^{84} \oplus (-4)^{315}.$$

The complete cover-link computation rules out nine colours, while a literal proper eleven-colouring is frozen. The exact frontier is therefore

$$\boxed{10 \le \chi(H) \le 11.}$$

No bounded heuristic or MILP non-solution is used to sharpen this interval.

Forty-five local A_4 blocks

The unique degree-one frame/octet relation partitions the frames into 45 blocks of size 12. Every induced block is

$$\boxed{K_{12} \setminus 3K_4.}$$

Thus one colour may occupy vertices in only one of the three independent four-cells. A ten-colouring of twelve local vertices must save at least two colours by repetition, giving the exact global local-pressure law

$$\boxed{\text{repeat savings} \ge 45 \cdot 2 = 90.}$$

Near-Hoffman energy

For an independent-set indicator x of size s, put

$$y = x - \frac{s}{540}\mathbf{1}.$$

Then

$$\boxed{y^{\mathsf{T}}(H + 4I)y = \frac{s(60 - s)}{15}.}$$

For a hypothetical ten-colouring let $d_i = 60 - s_i$. Since $\sum_i d_i = 60$,

$$\boxed{\sum_{i=1}^{10} y_i^{\mathsf{T}}(H + 4I)y_i = 240 - \frac{1}{15}\sum_i d_i^2 \le 216.}$$

The positive spectral gap away from E_{-4} is 6, hence

$$\boxed{\sum_i \|P_{E_{-4}^{\perp}} y_i\|^2 \le 36.}$$

Any ten-colouring must therefore lie close to the exact-cover/Hoffman eigenspace.

A ten-by-ten proof quotient

Let e_{ij} be the number of edges between colour classes i and j, and define

$$G = X^{\mathsf{T}}(H + 4I)X - \frac{1}{15}ss^{\mathsf{T}}, \qquad K = 15G.$$

Every ten-colouring must satisfy

$$K \succeq 0, \qquad K\mathbf{1} = 0, \qquad \operatorname{rank} K \le 9,$$

with

$$K_{ii} = s_i(60 - s_i), \qquad K_{ij} = 15e_{ij} - s_i s_j.$$

In particular,

$$\boxed{K \equiv -ss^{\mathsf{T}} \pmod{15},}$$

so every 2×2 minor of K is divisible by 15. This arithmetic-semidefinite quotient and the 45 local blocks are exact necessary-condition filters; they do not yet decide whether the tenth colour exists.

Evidence type. All graph counts, block decompositions, identities, and the frozen eleven-colouring are finite machine-verified results. The remaining choice $\chi(H) = 10$ versus 11 is open. The equality-case language is consistent with A. Abiad, W. Bosma, and T. van Veluw, *Hoffman colorings of graphs*, Linear Algebra Appl. 710 (2025), 129–150; the W33 finite computations are independent project certificates.

2 The final chromatic bit as a finite port problem

The exact frame-graph frontier remains

$$\boxed{10 \le \chi(H) \le 11}.$$

The next reduction is not another bounded search claim. It is an exact factorisation of the 540 frames through 45 block supports, 135 local four-cells, and 240 W33 edges.

The 45×240 support incidence. Let N record which W33 edges occur in each canonical twelve-frame block. Every row of N has weight 16, every column has weight 3, and

$$\boxed{NN^{\mathsf{T}} = 16I + A_{45}},$$

where A_{45} is the adjacency matrix of $\mathrm{SRG}(45, 32, 22, 24)$. Each W33 edge labels a triangle of this block graph, and the 720 block-graph edges decompose uniquely into the 240 resulting triangles.

Three exact covers and sixteen ports per block. The three independent K_4 cells inside a block each cover the same sixteen W33 edges exactly once. Each support edge therefore chooses one frame from each cell. The sixteen resulting triples form

$$\boxed{\mathrm{OA}(16, 3, 4, 2)},$$

and every port is used by exactly two neighboring blocks. Two blocks are either frame-disconnected or share one support edge; in the latter case their cross graph is exactly $K_{3,3}$.

Equivalently, H is the line graph of the four-uniform linear hypergraph whose 240 vertices are W33 edges and whose 540 hyperedges are frames. Every hypergraph vertex has degree nine. Thus a ten-colouring would assign nine distinct colours around every W33 edge and leave a unique missing colour there.

Exact quotient and association algebra. Writing $d_i = 60 - s_i$ for the ten class deficits, there are exactly 195,490 sorted profiles with $\sum_i d_i = 60$. The defect-Gram trace is

$$\operatorname{tr} K = 3600 - \sum_i d_i^2,$$

with a unique maximum at $d_1 = \cdots = d_{10} = 6$. This exhausts the diagonal quotient, not the off-diagonal Gram entries.

The 135 local cells form a symmetric rank-four association scheme with valencies $(1, 2, 96, 36)$ and multiplicities $(1, 24, 20, 90)$. One eigenmatrix is

$$P = \begin{pmatrix} 1 & 2 & 96 & 36 \\ 1 & 2 & 6 & -9 \\ 1 & 2 & -12 & 9 \\ 1 & -1 & 0 & 0 \end{pmatrix}.$$

Its ordinary ratio bounds stop at nine. Therefore an improvement to eleven must use the split port or triangle data rather than only the commutative Bose–Mesner layer.

Arithmetic and uncertainty corrections. The Smith form of N is

$$\boxed{1^{44} \oplus 3}.$$

There is one 3-primary factor and no 5-primary torsion. Moreover, deleting any d frames from a frozen exact cover gives a binary integral row-space witness of weight $4d$. Hence linear 3-adic and 5-adic defect tests are vacuous for every possible deficit; any arithmetic obstruction must include nonlinear matching and port compatibility.

The frame-space localization spectrum of $E_{-4} = \ker M^{\mathsf{T}}$ on every block is

$$\frac{2}{9}, \quad \left(\frac{43}{81}\right)^9, \quad 1^2,$$

so two Hoffman directions localize perfectly inside each block. In contrast, the edge-kernel localization spectrum on a sixteen-edge support is

$$0^{10} \oplus \left(\frac{1}{6}\right)^6.$$

Thus the naive frame-delocalization obstruction fails, while the cut-code edge kernel is strongly delocalized.

Proof surface and boundary. A deterministic strengthened ten-colour DIMACS instance has 7,800 variables and 146,289 clauses. It exposes both frame colours and the unique missing colour at each W33 edge; its SHA-256 is `6c0c3daa...9c6b7cd5`. No SAT model and no checked UNSAT proof is claimed. The semantic exact certificate is `68e93f6b...db21d19`. Time-bounded MILP and local-search failures remain diagnostics only.

2.1 Exact-cover signature Fourier law and the one-bit symmetry overhead

The complete exact-cover census gives forty-five anchor octets. Relative to an anchor, the three independent four-cells of the induced $K_{4,4,4}$ carry one of

$$(2, 2, 2), \qquad (0, 3, 3), \qquad (1, 2, 3), \qquad (0, 2, 4).$$

Permuting the three cells produces orbit sizes $1, 3, 6, 6$. Thus the local signature alphabet is the sixteen-state S_3-set

$$\Omega \cong 1 \sqcup S_3/C_2 \sqcup S_3 \sqcup S_3, \qquad 45|\Omega| = 720.$$

Its permutation character on the identity, transposition, and three-cycle classes is

$$\chi_\Omega = (16, 2, 1),$$

and therefore

$$\mathbb{Q}[\Omega] \cong 4\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 5V_{\mathrm{std}}.$$

Consequently
$$\dim \mathrm{End}_{S_3}(\mathbb{Q}[\Omega]) = 4^2 + 2^2 + 5^2 = 45,$$
with rational commutant
$$M_4(\mathbb{Q}) \oplus M_2(\mathbb{Q}) \oplus M_5(\mathbb{Q}).$$
Direct orbital enumeration gives forty-five relations on $\Omega \times \Omega$, with size profile $1^1 3^3 6^{41}$. Hence
$$720 = 45 \cdot 16 = \dim \mathrm{End}_{S_3}(\mathbb{Q}[\Omega])\,|\Omega|.$$
This rank equality does not itself identify the forty-five anchor octets with the forty-five orbitals.

Although $|\Omega| = 16$, its exact S_3 action cannot be affine on four binary bits. The unique fixed state makes every affine realization translation-conjugate to a linear one. An exhaustive classification of all 2800 embedded S_3 subgroups of $GL(4,2)$ finds only the orbit profiles
$$(1,1,2,3,3,6), \qquad (1,1,1,1,3,3,3,3), \qquad (1,3,3,3,6),$$
never $(1,3,6,6)$. The obstruction is sharp: on
$$\mathbb{F}_2^5 = (\mathbb{F}_2^3)_{\mathrm{perm}} \oplus (\mathbb{F}_2^2)_{\mathrm{std}}$$
there is an explicit invariant sixteen-state subset with the required orbit profile and an exact equivariant encoding of all four signature classes. Therefore the minimal affine binary register has dimension five:
$$\boxed{4 \text{ information bits} + 1 \text{ symmetry bit} = 5 \text{ affine-equivariant bits.}}$$
A four-bit implementation remains possible only with nonlinear lookup/permutation logic. In particular, equal cardinality does not justify identifying this signature alphabet with the $\mathrm{OA}(16,3,4,2)$ port alphabet without a separate intertwiner.

2.2 Signature–port obstruction and the minimal common controller

The sixteen exact-cover signature states and the sixteen natural V_4 orthogonal-array ports are not the same S_3-set. Their orbit profiles are
$$\Omega_{\mathrm{sig}} : 1+3+6+6, \qquad \Omega_{\mathrm{port}} : 1+3+3+3+6,$$
so no full S_3-equivariant bijection exists. Restriction to the orientation-preserving subgroup C_3 gives
$$1+3+3+3+3+3$$
on both sides, with exactly
$$5!\,3^5 = 29160$$
equivariant bijections. The reflection restrictions differ:
$$\Omega_{\mathrm{sig}}|_{C_2} : 1^2\,2^7, \qquad \Omega_{\mathrm{port}}|_{C_2} : 1^4\,2^6.$$
Thus the obstruction is reflection-sensitive.

The permutation characters are
$$\chi_{\mathrm{sig}} = (16,2,1), \qquad \chi_{\mathrm{port}} = (16,4,1),$$
and hence
$$\mathbb{Q}[\Omega_{\mathrm{sig}}] \cong 4\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 5V_{\mathrm{std}},$$
$$\mathbb{Q}[\Omega_{\mathrm{port}}] \cong 5\mathbf{1} \oplus \mathrm{sgn} \oplus 5V_{\mathrm{std}}.$$

The exact representation-ring defect is

$$[\mathbb{Q}\Omega_{\mathrm{port}}] - [\mathbb{Q}\Omega_{\mathrm{sig}}] = [\mathbf{1}] - [\mathrm{sgn}],$$

or stably

$$\mathbb{Q}[\Omega_{\mathrm{sig}}] \oplus \mathbf{1} \cong \mathbb{Q}[\Omega_{\mathrm{port}}] \oplus \mathrm{sgn}.$$

It vanishes on C_3, where the sign character becomes trivial.

The signature commutant has dimension

$$4^2 + 2^2 + 5^2 = 45,$$

whereas the natural port commutant has dimension

$$5^2 + 1^2 + 5^2 = 51.$$

Therefore the existing equality between forty-five anchor octets and forty-five signature orbitals cannot be lifted through the natural port action without treating the one-dimensional orientation defect.

At the set level, the orbit multiplicities are

$$\Omega_{\mathrm{sig}} : 1^1\, 3^1\, 6^2, \qquad \Omega_{\mathrm{port}} : 1^1\, 3^3\, 6^1.$$

Their minimal common S_3-envelope has profile

$$1^1\, 3^3\, 6^2$$

and therefore size

$$\boxed{22}.$$

The maximum equivariant overlap is

$$\boxed{10}, \qquad 16 + 16 - 10 = 22.$$

Both alphabets are embedded explicitly into the same linear S_3-action on $\mathbb{F}_2^5$. The twenty-two valid words consist of ten shared states, six signature-only states, and six port-only states; the other ten ambient words are guards.

A four-bit nonlinear signature ROM and a five-bit shared linear controller are supplied. The fixed four-bit encoding has algebraic degree three and twenty-seven ANF terms; the five-bit generator cores each use one XOR plus wire permutations before multiplexing. Source-level equivalence is exact. Observed synthesis, placement, manuscript-PDF evidence, and the unresolved chromatic decision remain separately gated. In particular,

$$10 \le \chi(H) \le 11$$

is unchanged. The new chromatic filter is that a full local S_3 signature/port quotient is invalid, whereas a C_3-only quotient is locally compatible.

2.3 Knight-hypercube gauge closure and the typed common controller

The local signature and natural-port alphabets become isomorphic after restriction to the orientation subgroup C_3, but the choice is highly noncanonical. The filled port complex has $\pi_1 \cong F_{436}$. After freezing the pairing of the five three-cycles, its phase group is $A = C_3^5$, and therefore

$$H^1(X; A) \cong A^{436} \cong \mathbb{F}_3^{2180}.$$

Without freezing orbit pairing, the local intertwiner group is $W = C_3 \wr S_5$ of order $3^5 5! = 29160$. Burnside enumeration over the 108 wreath-product conjugacy types gives a 1943-digit number of flat W-connections modulo switching. A trivial coboundary gauge exists, but the geometry does not select one canonically.

The minimal common envelope is the 22-state S_3-set

$$\Omega_{\mathrm{env}} \cong 1 \sqcup 3(S_3/C_2) \sqcup 2S_3.$$

Its character and rational module are

$$\chi_{\mathrm{env}} = (22, 4, 1), \qquad \mathbb{Q}[\Omega_{\mathrm{env}}] \cong 6\mathbf{1} \oplus 2\,\mathrm{sgn} \oplus 7V_{\mathrm{std}}.$$

Consequently

$$\dim \mathrm{End}_{S_3} \mathbb{Q}[\Omega_{\mathrm{env}}] = 6^2 + 2^2 + 7^2 = 89.$$

Direct enumeration gives orbital profile $1^1 3^{15} 6^{73}$. Algebra closure over two independent large prime fields gives the basepoint-dependent Terwilliger dimensions

$$\dim T_x = \begin{cases} 89, & x \text{ fixed}, \\ 222, & x \text{ in a three-state orbit}, \\ 484 = 22^2, & x \text{ in a regular six-state orbit}. \end{cases}$$

Thus a regular signature basepoint generates the full matrix algebra and admits no further exact algebraic compression.

The ten unused five-bit words form

$$\Omega_{\mathrm{guard}} \cong 1 \sqcup 3(S_3/C_2), \qquad \mathbb{Q}[\Omega_{\mathrm{guard}}] \cong 4\mathbf{1} \oplus 3V_{\mathrm{std}}.$$

In particular the guard shell contains no sign constituent. Its coherent algebra has rank 25, and its Terwilliger dimensions are 25 and 58 on the two orbit types. As an induced Q_5 network it has seven internal edges, thirty-six boundary edges, and component sizes $7, 1, 1, 1$; this is a typed detection shell, not a fault-tolerant memory.

The repository's 4×4 toroidal knight graph is exactly Q_4, and the frozen knight tour is a four-bit Gray Hamilton cycle. The nonlinear signature controller contains seven transpositions and five three-cycles. The transpositions contribute at least fourteen directed hops. Since Q_4 is bipartite, each embedded three-cycle has metric perimeter at least four, so

$$\text{total routing work} \geq 14 + 5 \cdot 4 = 34.$$

An exhaustive componentwise placement census attains the bound:

$$\boxed{\text{minimum work} = 34, \quad \text{average} = \frac{17}{16}, \quad \text{minimum dilation} = 2.}$$

There are 1105920 minimum-work labeled embeddings. After fixing translation and quotienting coordinate permutations, all 2880 representatives were routed exhaustively. No minimum-work embedding has shortest-path link congestion below three, and congestion three is attained.

The five-bit affine lift has the network decomposition

$$Q_5 = Q_4 \square K_2,$$

so it is two toroidal-knight boards joined by a sixteen-edge perfect matching. For both S_3 generators the binary column-weight profile is

$$(1, 1, 1, 1, 2).$$

Four basis directions remain one-hop routes; only the added symmetry direction is dilated to two. This turns the one-bit symmetry overhead into a literal double-board interconnect theorem.

Finally, the 45 signature orbitals are not intrinsically individually labelled. The order-72 automorphism group of the signature S_3-set acts on them with only thirteen orbits, of sizes

$$1^5\,2^3\,4^1\,6^3\,12^1.$$

Hence the equality between forty-five anchor octets and forty-five signature orbitals does not define a canonical correspondence from local data alone; any such lift needs extra global $PSp(4,3)$-equivariant structure.

A complete canonical C_3 coboundary-gauge manifest is emitted for all 45 blocks, 720 block edges and 240 filled triangles. It is an exact bijective relabelling with transported interfaces, not a reduction or decision of the current ten-colour instance. The live boundary remains

$$\boxed{10 \le \chi(H) \le 11}.$$

3 Global cover closure and the three hypercube roles

The complete exact-cover census contains 3,547,800 covers in 327 $PSp(4,3)$-orbits. The closed ternary-Hamming switch component contributes 1,574,640 covers in 135 orbits, leaving the exact exterior complement

$$\text{1,973,160 covers in 192 orbit classes,}$$

with stabilizer-order distribution $2^{120}4^{57}8^{15}$. The numerical equality with the 192 tomotope flags is not promoted to an identification: the exterior classes carry this nontrivial stabilizer partition and no equivariant map has been supplied.

For the rational chromatic-dual batch, deficit-profile stabilizers reduce the exact search surface from 204,287,050 to 98,191,335 rational block coordinates. This is an exact compiler reduction, not an eleven-colour certificate; the live boundary remains $10 \le \chi(H) \le 11$.

The ternary cover cube admits an exact binary host. Under the block-one-hot encoding $0 \mapsto 100$, $1 \mapsto 010$, $2 \mapsto 001$,

$$d_{Q_{15}}(\operatorname{enc} x, \operatorname{enc} y) = 2 d_{H(5,3)}(x, y).$$

Hence $H(5,3)$ is the distance-two graph on 243 constant-weight-five words in Q_{15}. It is not a subgraph of an ordinary hypercube because it contains triangles. This Q_{15} host is distinct from both the toroidal-knight controller Q_4 and the secondary complement-codec Q_4; raw Levi complement adjacency remains $4K_4$.

The affine involution acts as a coordinate permutation of the 15 one-hot bits. Its traces on the six ternary Fourier grades are $(1,2,4,8,4,8)$, yielding invariant multiplicities $(1,6,22,44,42,20)$ and the exact 135-state quotient. For the reversible quotient chain $P = W/10$, the Szegedy walk has eigenphases $\exp(\pm 2i \arccos \lambda)$ for $\lambda \in \{1, 7/10, 2/5, 1/10, -1/5, -1/2\}$. This is a finite spectral compiler, not a physical speedup claim.

4 Minimum gauge defects, fault-aware hypercubes, and Clebsch recovery

The filled 45-block port complex has minimum nontrivial flat C_3^5 cochain support

$$d_{\text{gauge}} = 2.$$

Every minimum support is two edges of one filled triangle. The 720 geometric supports form one $PSp(4,3)$ orbit; for each fixed nonzero coefficient vector the orbit has size 720 and stabilizer order 36. There are 242 labeled coefficient orbits, or 121 projective directions after $v \sim -v$. Each minimum defect produces nonzero holonomy on exactly 42 unfilled triangles.

The optimal nonlinear signature routing on Q_4 has work 34 and dilation two. Mirroring it in

$$Q_5 = Q_4 \square K_2$$

preserves the same work and dilation under every one of the 32 single-vertex and 80 single-edge failures. Among two-vertex failures, loss is possible exactly for the 16 interlayer replica pairs. Under static one-cycle failover, 32 physical state slots are both necessary and sufficient.

The actual 45-anchor action has order 25,920, is transitive, and is perfect. The intrinsic signature automorphism group has derived series $72 \to 18 \to 9 \to 1$. Therefore no nontrivial action induced through the intrinsic signature coherent configuration can be equivariant with the anchor action; the identity orbital is fixed while the anchors are transitive.

At a regular envelope basepoint the local algebra is $M_{22}(\mathbb{Q})$. Nevertheless every separable positive lift $A_{45} \otimes K$, $K \succeq 0$, has extreme eigenvalues $32\lambda_{\max}(K)$ and $-4\lambda_{\max}(K)$, so its Hoffman bound remains exactly nine. Any improvement must use nonseparable, profile-sensitive cross-block orientation data. The exact chromatic boundary remains $10 \leq \chi(H) \leq 11$.

Validity alone detects 36/110 one-bit and 70/220 two-bit envelope faults. Exact confusability-graph colourings require 7, 11, and 16 trusted symbols to correct one bit, correct one/detect two, and correct two bits, respectively. The optimal four-bit two-error tag is the antipodal-axis label of Q_5. The quotient $Q_5/\{x \sim \bar{x}\}$ is the Clebsch graph $SRG(16, 5, 0, 2)$. Six axes contain two valid states and ten contain one valid state with its guard antipode. The decoder exhausts all 352 error patterns of weight at most two, conditional on a trusted tag.

These are exact finite and source-level digital results. They do not assert physical fault probabilities, observed FPGA timing, fault-tolerant quantum memory, a chromatic decision, or laboratory performance.

5 Minimum gauge defects generate the full flat connection space

The filled port complex has 45 vertices, 720 edges, and 240 edge-disjoint filled triangles, with coefficient module $A = C_3^5 \cong \mathbb{F}_3^5$. On one oriented triangle the scalar flat plane is

$$\{(a, b, c) \in \mathbb{F}_3^3 : a + b + c = 0\},$$

which has rank two and is spanned by its six nonzero weight-two vectors. Since the filled triangles partition all graph edges, these local planes sum directly. Consequently

$$\dim Z^1(X; \mathbb{F}_3) = 240 \cdot 2 = 480, \qquad \boxed{\dim Z^1(X; \mathbb{F}_3^5) = 2400}.$$

The block graph is connected, so

$$\dim B^1(X; \mathbb{F}_3^5) = (45 - 1) \cdot 5 = 220,$$

and therefore

$$\boxed{\dim H^1(X; \mathbb{F}_3^5) = 2400 - 220 = 2180}.$$

Thus the weight-two minimum defects do not merely witness the minimum nontrivial class: they span every flat C_3^5 connection, and every switching class has a representative assembled from minimum defects. This is a generation theorem, not a unique-decoding or multi-fault-correction claim.

Passes 3364–3375: Clebsch–Petersen resilient closure

The folded five-cube checksum graph is the Clebsch graph, $\mathrm{SRG}(16, 5, 0, 2)$, with rooted shell decomposition $1 + 5 + 10$. The ten distance-two vertices induce $KG(5, 2)$, the Petersen graph. Its Cayley form on $\mathbb{F}_2^4$ has five circuit directions and automorphism group $2^4{:}S_5$ of order 1920.

Preserving the five raw envelope bits systematically, exhaustive search proves that seven parity bits are necessary and sufficient for a 22-word code of minimum distance five. Thus a self-protecting two-error-correcting envelope word requires 12 total bits. The mirrored Q_5 placement has exactly 16 catastrophic two-vertex sets and 480 catastrophic three-vertex sets, precisely those containing a complete replica pair.

The full block-distance-invariant nonseparable local-algebra lift reduces exactly to the three PSD cones

$$K_0 + 32K_1 + 12K_2 \succeq 0, \quad K_0 + 2K_1 - 3K_2 \succeq 0, \quad K_0 - 4K_1 + 3K_2 \succeq 0.$$

This compiler does not decide the ten-colour problem; the live boundary remains $10 \le \chi(H) \le 11$.

Minimum gauge defects obey a local fusion law: same omitted-edge types add coefficientwise and annihilate for opposite coefficients; distinct types in one filled face remain minimum exactly when their shared-edge coefficients cancel; distinct filled faces are edge-disjoint. Projectively, the 87,120 minimum defects contain 29,040 local three-state fusion triples.

The Clebsch graph also supplies a nested checksum geometry: one reference axis, five singleton fault directions, and ten pair channels forming a Petersen graph. A degree-four del Pezzo/Segre quartic supplies a separate 16-line incidence realization of the same graph, retained only as a mathematical dictionary and not as a physical hardware identification.

6 The hidden torus as a finite d'Alembertian

For the hidden 27-state signed-torus block,

$$D = A(C_3)_3 - A(C_3)_1 - A(C_3)_2.$$

Since $\Delta_i = 2I - A(C_3)_i$, one has the exact identity

$$\boxed{D + 2I = \Delta_1 + \Delta_2 - \Delta_3.}$$

Thus the shifted hidden block is a finite discrete d'Alembertian on C_3^3, with two positive Laplacian factors and one negative factor. At Fourier frequency $k \in \mathbb{F}_3^3$,

$$\lambda(k) = 3\big([k_1 \neq 0] + [k_2 \neq 0] - [k_3 \neq 0]\big),$$

so its spectrum is

$$\boxed{6^4,\ 3^{12},\ 0^9,\ (-3)^2}.$$

The rank is 18 and the nullity is 9. The null modes consist of the constant character together with four frequencies satisfying $k_1 = 0$, $k_2, k_3 \neq 0$, and four satisfying $k_2 = 0$, $k_1, k_3 \neq 0$. Hence the exact null profile is $1 + 4 + 4$. This is a finite signed-graph identity only; no physical spacetime, Lorentzian continuum, causal cone, or laboratory claim is made.

7 Minimum-defect normal forms and the hidden signed ternary torus

The flat C_3^5 port connections admit an exact minimum-defect word metric. On one filled triangle there are 1 states of length zero, 726 states of length one, and 58,322 states of length two, so the local enumerator is

$$1 + 726z + 58{,}322z^2.$$

Because the 240 filled faces partition all 720 edges, the global flat-space enumerator is its 240th power and the exact flat-space diameter is 480. A sphere-counting argument against the 3^{2180} switching classes gives the certified quotient bound

$$\boxed{389 \le \operatorname{diam} H^1 \le 480}.$$

The upper endpoint is the universal minimum-defect normal form; the exact quotient covering radius inside this interval remains open.

The minimum defects also give a canonical $PSp(4,3)$-equivariant presentation. For scalar coefficients,

$$0 \longrightarrow \mathbb{F}_3[240 \text{ faces}] \longrightarrow \mathbb{F}_3[720 \text{ supports}] \longrightarrow Z^1 \longrightarrow H^1 \longrightarrow 0,$$

with dimensions $240, 720, 480, 436$ after quotienting the 44-dimensional coboundary space. Tensoring by the externally labelled five-dimensional coefficient module gives 3600 generators, 1200 local face relations, 220 coboundary relations, and

$$3600 - 1200 - 220 = \boxed{2180}$$

logical dimensions. Thus the full logical sector is presented by minimum defects with exactly 1420 independent relations.

For the affine involution

$$\tau(x_1, x_2, x_3, x_4, x_5) = (-x_4, 1 - x_3, 1 - x_2, -x_1, x_5)$$

on $\mathbb{F}_3^5$, the 135 orbit species have only 108 distinct block-one-hot Q_{15} barycentres. Their fibre profile is

$$81 \text{ singletons} + 27 \text{ double fibres}.$$

The 27 double fibres occur exactly at ternary displacement four. Taking the coordinatewise missing symbols maps their barycentres bijectively to the 27 fixed points of τ. The two orbit species above each such centre are distinguished by the invariant binary correlation

$$h = (x_1 + x_4)(x_2 + x_3 - 1) \in \{1, 2\}.$$

Consequently the 54 ambiguous species form a canonical two-sheeted cover of the fixed affine flat $\mathbb{F}_3^3$.

More strongly, barycentric aggregation is exactly lumpable for the weighted Hamming-orbifold walk. Its 135-state operator splits as

$$\boxed{W_{135} \cong B_{108} \oplus D_{27}}.$$

The barycentric block has spectrum

$$10^1,\ 7^6,\ 4^{18},\ 1^{32},\ (-2)^{33},\ (-5)^{18},$$

and itself has the exact three-shell quotient

$$\begin{pmatrix} 2 & 8 & 0 \\ 2 & 4 & 4 \\ 0 & 4 & 6 \end{pmatrix}$$

on shells of sizes $27, 54, 27$, with spectrum $10, 4, -2$.

Orienting each double fibre by $h = 1$ versus $h = 2$ identifies the hidden block with the signed Cayley operator on $\mathbb{F}_3^3$

$$\boxed{D_{27} = C_3^{(3)} - C_3^{(1)} - C_3^{(2)}}.$$

Equivalently it has weight $+1$ on the two $\pm e_3$ steps and weight -1 on the four $\pm e_1, \pm e_2$ steps. Its exact spectrum is

$$4^4,\ 1^{12},\ (-2)^9,\ (-5)^2.$$

This is a finite signed-torus theorem, not a spacetime-signature claim.

Finally, a single minimum defect used as a C_3^5 voltage assignment generates only the subgroup $\langle v\rangle \cong C_3$. The full 10,935-vertex derived graph therefore splits into

$$\boxed{\text{81 connected components of 135 vertices each}}.$$

Each component is 32-regular with 2,160 edges and 15,714 triangles. Fourier decomposition across the C_3 fibre gives one untwisted 45-state block and two conjugate magnetic blocks. Their exact moments satisfy

$$\operatorname{tr} M^2 = 1440, \qquad \operatorname{tr} M^3 = 31{,}302,$$

whereas the untwisted cubic trace is 31,680. The $378 = 42 \cdot 9$ deficit proves that this is a genuine nonseparable cohomological weighting. It is a candidate chromatic-dual surface, not yet a certificate improving the bound nine. Its cardinality match with the 135 Hamming-orbit species is also not an isomorphism: the voltage component has degree 32, while the Hamming-orbifold walk has degree 10.

8 Minimum-defect radius, the finite null conic, and a matrix-valued ternary torus (Passes 3404–3417)

The minimum-defect word metric on the switching quotient admits the sharper exact interval

$$\boxed{389 \le D_{H^1} \le 436}.$$

The lower bound is the existing sphere-count obstruction. For the upper bound, the scalar cohomology has dimension 436 and is generated by minimum-support defects, so one may select 436 such supports as a basis. After tensoring with $\mathbb{F}_3^5$, an arbitrary nonzero coefficient vector on one selected support is still one minimum defect; hence every switching class uses at most 436 basis defects.

The nine-dimensional kernel of the shifted hidden operator is the zero mode together with the eight nonzero vectors satisfying

$$k_1^2 + k_2^2 - k_3^2 = 0 \qquad \text{in } \mathbb{F}_3.$$

Projectivization identifies the eight lifts in opposite pairs and produces a nondegenerate four-point conic in $PG(2,3)$. Its projective stabilizer has order 24 and acts as the full S_4 on the four conic points.

The four projective null directions form the columns of a ternary MDS code

$$\boxed{[4,3,2]_3},$$

with weight enumerator $1 + 12z^2 + 8z^3 + 6z^4$ and dual the signed repetition code generated by $(1,1,2,2)$, of parameters $[4,1,4]_3$.

More strongly, the 108 barycentric states admit an exact factorization

$$\boxed{108 = 27 \cdot 4 = |\mathbb{F}_3^3|\,|\mathbb{F}_2^2|}.$$

For every center in the fixed affine flat there are exactly four barycentric channels: one fixed channel, two distance-two channels, and one ambiguous channel. The adjacency is translation invariant in the center coordinate, so it is a 4×4 matrix-valued Cayley operator with six nonzero kernels at $\pm e_1, \pm e_2, \pm e_3$. Three qutrit Fourier transforms reproduce the full barycentric spectrum from 27 four-dimensional symbols.

Adding the one-dimensional hidden signed channel gives

$$\boxed{135 = 27 \cdot 5},$$

so the complete walk reduces to 27 internal symbols of size 5. This yields an exact mixed-radix compilation contract: three qutrit center registers, one five-level internal register, and two Szegedy reflections per walk step. Uniform state preparation over ten neighbours and synthesis of controlled 5×5 symbols still require an approximation tolerance before any Clifford+T count can be certified.

The isolated magnetic-defect search closes negatively. The ternary phase family has

$$\chi_\omega(x) = (x-2)^{22}(x+4)^{18}\big(x^5 - 28x^4 - 140x^3 + 478x^2 + 1448x - 1328\big),$$

with generalized Hoffman ratio approximately $7.2835002808 < 8$. The real signed family has residual quintic

$$x^5 - 28x^4 - 140x^3 + 520x^2 + 1568x - 1088$$

and ratio approximately $7.0613187294 < 8$. Rational root isolation certifies both inequalities. Thus one profile-sensitive minimum defect cannot improve the live chromatic lower bound.

The literal 720-edge $PSp(4,3)$ permutation-character decomposition remains evidence-gated until its GAP-generated JSON is frozen. The live chromatic boundary remains

$$\boxed{10 \le \chi(H) \le 11},$$

and the finite quadratic form above is not identified with physical spacetime or Lorentz symmetry.

8.1 The 720-edge ordinary degree atlas (Pass 3408)

The minimum-support carrier is the transitive action of

$$PSU(4,2) \cong PSp(4,3)$$

on the 720 non-collinear point pairs of $H(3,4) = GQ(4,2)$. A point stabilizer has order 36 and subdegrees

$$1, 2, 3, 6, 9, 9, 12^5, 18^{11}, 36^{12},$$

so the orbital rank is 34.

Deterministic generic elements of the exact orbital commutant give the complex isotypic dimensions

$$1, 15, 15, 20, 20, 24, 24, 24, 30, 30, 45, 45, 60, 60, 64, 81, 81, 81.$$

The ordinary degree/multiplicity profile is therefore

$$\boxed{1 + 15_a + 15_b + 2 \cdot 20 + 3 \cdot 24 + 30_a + 30_b + 45_a + 45_b + 2 \cdot 60 + 64 + 3 \cdot 81.}$$

It closes both the carrier dimension,

$$1 + 30 + 40 + 72 + 60 + 90 + 120 + 64 + 243 = 720,$$

and the character norm, equal to the orbital rank 34.

The group, orbit, stabilizer, subdegrees, relations, and commutant matrices are exact. The isotypic dimensions are recovered from deterministic well-separated numerical eigenspaces and constrained by the exact dimension and rank identities. GAP/CTblLib remains an independent row-label check, especially for naming the selected conjugate degree-30 pair.

9 Passes 3418–3429: supplemental Clebsch–D_5 closure

The executable source wrappers retain a provisional build tag only for hash continuity after a namespace collision. Building on the parallel Clebsch–Petersen packet without replacing it, the source-complete verifier reports **PASS 11/11**. All graph, code, orbit, fusion, routing, and fault counts in this section are exact. The magnetic spectral family below is an explicitly marked high-margin numerical audit rather than a rational SDP certificate. The chromatic boundary remains

$$\boxed{10 \le \chi(H) \le 11}.$$

9.1 Edge-dependent magnetic phases do not improve the bound

For a flat C_3 edge cochain z, set

$$M(z)_{ab} = \omega^{z_{ab}}, \qquad M(z)_{ba} = \overline{M(z)_{ab}}.$$

The search exhausts the 24 $PSp(4,3)$-orbits of unordered pairs of the 720 minimum-defect supports and both relative coefficient signs. Among all 48 one/two-defect representatives, the best nonzero spectral value is

$$\lambda_{\min} \simeq -5.073280337457312, \quad \lambda_{\max} \simeq 31.877958425086288, \quad h \simeq 7.283500280818951 < 9.$$

Thus edge-dependent phases alone are insufficient; an improved dual must also retain nonseparable profile-sensitive amplitudes.

9.2 Self-protecting state codes

For arbitrary systematic encoding of the 22-state envelope, exhaustive search proves that seven parity bits are necessary and sufficient for distance five:

$$\boxed{(n, M, d) = (12, 22, 5)}.$$

The exhaustive radius-two decoder surface contains $22(1 + 12 + \binom{12}{2}) = 1738$ received words. In the linear systematic case, every five-bit difference occurs, so the problem is a binary $[5 + r, 5, 5]$ code. All 26,334 six-parity candidates and all 7,624,512 seven-parity candidates fail; eight parity bits succeed:

$$\boxed{[13, 5, 5]}, \qquad (c_0, \ldots, c_4) = (15, 51, 85, 106, 150).$$

The 12-bit code is information-theoretically smaller; the 13-bit code is XOR-linear over all 32 words.

9.3 Six-spare dynamic recovery

Static state-availability mirroring uses 32 physical slots. Under an explicit protected-symbolic-state/recompilation contract, allowing bounded state remapping while preserving every original transition distance changes the frontier. Every one of the 6,884 fixed spare banks of size at most five fails. At size six there are exactly 96 solutions; the first is

$$\boxed{\{0, 1, 2, 3, 4, 5\}}.$$

Therefore 22 physical slots are necessary and sufficient for post-fault symbolic remapping at the original work 34 and dilation two.

9.4 Gauge fusion and the local A_2 root plane

The 720 minimum supports have 24 unordered-pair orbits. Their local fusion laws include annihilation, coefficient reversal, third-root completion, and a three-edge filled-face state. On one filled triangle the six weight-two flat cochains are precisely the six nonzero roots of an A_2 plane over $\mathbb{F}_3$. Consequently

$$Z^1(X; \mathbb{F}_3^5) \cong \left(A_2(\mathbb{F}_3) \otimes \mathbb{F}_3^5\right)^{\oplus 240}, \qquad \dim Z^1 = 2400,$$

with the 220-dimensional switching subspace leaving $\dim H^1 = 2180$.

9.5 Triple faults and the Q_6 boundary

The mirrored Q_5 census is

faults	cases	catastrophic	worst work	worst dilation
EEE	82160	0	40	3
VEE	101120	0	42	4
VVE	39680	1280	45	4
VVV	4960	480	46	4

Three Q_4 layers, or 48 slots, are cardinality-minimal for arbitrary two-vertex loss. Four layers give $Q_6 = Q_4 \square Q_2$, use 64 slots, and are necessary and sufficient in the untouched-layer model for preserving work 34 and dilation two after any three layer-local faults.

9.6 The Clebsch checksum is a repetition-code coset graph

The four-bit tag

$$t(x) = x_{3:0} \oplus x_4(1111)_2$$

is a parity-check syndrome for the binary repetition code $\mathrm{Rep}(5) = [5, 1, 5]$. Hence

$$\boxed{Q_5/\mathrm{Rep}(5) = \text{Clebsch} = \mathrm{SRG}(16, 5, 0, 2)}.$$

Every local checksum chart has distance shells

$$\boxed{1 + 5 + 10}.$$

The five one-error neighbors are independent, while the ten two-error symbols induce the Petersen graph $KG(5, 2)$: two double-error syndromes are adjacent exactly when their error-position pairs are disjoint.

The 16 closed neighborhoods form a symmetric biplane

$$\boxed{2 - (16, 6, 2)}, \qquad BB^{\mathsf{T}} = 4I + 2J.$$

Exact automorphism enumeration gives $|\,\mathrm{Aut}(\text{Clebsch})| = 1920 = 2^4 \cdot 5!$. In the even-sign model the complement is the 5-demicube skeleton and the Clebsch edges are the opposition relation on one 16-weight D_5 half-spin orbit. The controller's valid/guard coloring leaves only an elementary Abelian stabilizer C_2^3.

9.7 Equivariance versus correction radius

The repetition kernel is not controller invariant: its generator has orbit

$$\{11111_2, 01111_2, 10111_2\}$$

of weights $5, 4, 4$. The unique common invariant line is generated by 00111_2 and has distance three. Therefore no fixed one-dimensional controller-equivariant quotient gives uniform two-bit correction. One must re-encode the Clebsch syndrome, transport a moving quotient chart, or use the self-protecting 12/13-bit code.

Boundary. No ten-colour decision, exact rational magnetic certificate, recovery of destroyed uncheckpointed state data, measured fault rate, observed FPGA timing, successful fresh PDF, physical D_5 or Spin(10) identification, quantum speedup, laboratory operation, or spacetime claim is inferred from this finite packet.

10 Generalized defect radius, modular cohomology, and the five-channel crossed algebra (Passes 3444–3457)

Let $E = \mathbb{F}_3^{720}$ be the minimum-support permutation module. If R is the filled-face relation map and N the unsigned vertex–edge incidence map, exact row reduction gives

$$\operatorname{rank} R = 240, \qquad \operatorname{rank} N = 45, \qquad \operatorname{rank}[R\ N] = 284,$$

with $\operatorname{im} R \cap \operatorname{im} N = \langle \mathbf{1}_E \rangle$. Hence the scalar cohomology has the objectwise presentation

$$0 \longrightarrow \mathbf{1} \longrightarrow \mathbb{F}_3[240 \text{ faces}] \oplus \mathbb{F}_3[45 \text{ vertices}] \longrightarrow \mathbb{F}_3[720 \text{ supports}] \longrightarrow H^1 \longrightarrow 0,$$

and therefore

$$\boxed{\dim_{\mathbb{F}_3} H^1 = 436}.$$

Dually, $(H^1)^*$ is the 436-dimensional edge code whose coordinate sum vanishes on every filled face and at every vertex.

After an arbitrary $\mathbb{F}_3$-linear identification $\mathbb{F}_3^5 \cong \mathbb{F}_{243}$, the labelled minimum-defect diameter is the fifth generalized covering radius of the ternary code

$$K = \operatorname{im}[R\ N],$$

equivalently the ordinary covering radius of its scalar extension to $\mathbb{F}_{243}$. One filled face contributes the three-direction Latin-square Cayley graph

$$\mathrm{SRG}(59049, 726, 243, 6),$$

with spectrum $726^1, 240^{726}, (-3)^{58322}$ and local length enumerator $1 + 726z + 58322z^2$. The relation-aware sphere obstruction remains stronger than the ordinary 243-ary Hamming count and yields

$$\boxed{389 \le R_{\text{defect}} \le 436}.$$

The exact endpoint inside this interval remains open.

The four-channel barycentric Fourier symbol factors through the equitable $1 + 2$ quotient of K_3,

$$M = \begin{pmatrix} 0 & 2 \\ 1 & 1 \end{pmatrix},$$

as

$$B(k) = c(k_1)(I_2 \otimes M) + c(k_2)(M \otimes I_2) + c(k_3)I_4, \qquad c(t) = \begin{cases} 2, & t = 0, \\ -1, & t \neq 0. \end{cases}$$

The hidden channel is $d(k) = c(k_3) - c(k_1) - c(k_2)$. Thus only eight zero/nonzero masks are required, rather than 27 unrelated five-dimensional symbols.

The torus amplitudes alone generate a four-dimensional commutative algebra. Adding the coordinate swap S raises the algebra dimension to six, while adding the null-conic dual sign

$$D = \operatorname{diag}(1, 1, -1, -1)$$

raises it to eight. Together,

$$\boxed{\langle I_2 \otimes M, M \otimes I_2, S, D \rangle = M_4(\mathbb{Q})}.$$

This supplies an explicit noncommuting amplitude extension beyond the exhausted phase-only magnetic families, but it is not yet a chromatic certificate; the live boundary remains $10 \le \chi(H) \le 11$.

Modulo three, $2 \equiv -1$, so all 27 momentum symbols collapse to one matrix J. Writing $\mathcal{N} = J - I$, exact ranks give

$$\operatorname{rank} \mathcal{N} = 2, \qquad \operatorname{rank} \mathcal{N}^2 = 1, \qquad \mathcal{N}^3 = 0,$$

so J has Jordan type $J_3(1) \oplus J_1(1) \oplus J_1(1)$ and

$$\boxed{J^3 = I}.$$

The accompanying RTL exhausts all 200 integer-symbol entries modulo three and all $8 \cdot 3^5 = 1944$ mask/state triples.

Finally, after choosing one Fano point and one D_5 coordinate, three degree-four sets carry the same faithful S_4 action: the four projective null-conic directions, the four Fano lines avoiding the selected point, and the four residual D_5 coordinates. The choices are essential, so this is an equivariant chart bridge rather than a canonical ambient-geometric identification.

The local A_2 code $[3, 2, 2]_3$ tensored with the null-conic code $[4, 3, 2]_3$ gives the additional exact construction

$$\boxed{[3, 2, 2]_3 \otimes [4, 3, 2]_3 = [12, 6, 4]_3},$$

with visible $S_3 \times S_4$ coordinate action and dual parameters $[12, 6, 3]_3$. Its length agrees with the tomotope edge count, but no tomotope incidence realization is asserted without an explicit map.

10.1 The filled-face tower, modular descent, and the tomotope boundary

The selected orbit of 240 filled triangles carries a canonical $PSp(4, 3)$-equivariant antipodal involution. Its quotient has 120 states and a symmetric rank-five association scheme with valencies

$$1, 36, 27, 2, 54.$$

The valency-two relation is $40K_3$. Quotienting its forty components recovers the original graph

$$\mathrm{SRG}(40, 12, 2, 4) = W(3, 3).$$

Consequently the filled-face carrier has the exact tower

$$\boxed{240 \longrightarrow 120 \longrightarrow 40}.$$

Each W33 point supports six filled faces, arranged as three antipodal pairs. Its stabilizer has order $648 = 3^3 \cdot 24$ and induces the full S_4 action on the six edges of a tetrahedron; the induced action on the three antipodal pairs is S_3 with kernel V_4.

Across a W33 edge, one pair-scheme relation is the complete 3-by-3 block. Across a W33 nonedge, a second relation is a permutation matrix and its companion is the complementary block. The resulting matching connection has

$$1080 \text{ identity triangle loops}, \qquad 2160 \text{ transposition loops},$$

and no three-cycle loops on the triangles of the W33 complement. This is an objectwise nontrivial transport bundle, unlike the exhausted scalar phase-only families.

In characteristic three, all five ordinary characters of the 120-state association scheme reduce to $(1, 0, 0, 2, 0)$. Its endomorphism algebra is local, with radical tower

$$\dim J = 4, \qquad \dim J^2 = 1, \qquad J^3 = 0,$$

so the 120-dimensional pair module is indecomposable. The antisymmetric antipodal summand decomposes as

$$M_- = M_{81} \oplus M_{39},$$

where the 39-dimensional summand has endomorphism ring $\mathbb{F}_3[\varepsilon]/(\varepsilon^2)$ and glues the ordinary 15- and 24-dimensional sectors. The 81-dimensional summand is retained as a brick pending an independent MeatAxe composition-series calculation. This face-permutation result is complementary to the later edge-module filtration

$$0 < R_{240} < Z^1_{480} < E_{720},$$

which produces the distinct 196-dimensional quotient $Z^1/(R + B^1)$.

The six local face tokens map to the six transposition matrices in the four-dimensional tetrahedral permutation representation. Their group algebra image has dimension ten. Adjoining the null-conic dual sign $\mathrm{diag}(1, 1, -1, -1)$ generates all of $M_4(\mathbb{Q})$. Thus scalar phases, the later finite dyadic amplitude grid, and a trivial untwisted M_4 fibre are all insufficient, while the face tower supplies the first explicit objectwise noncommuting transport extension. This is a compiler surface, not yet a chromatic certificate; the live boundary remains

$$10 \le \chi(H) \le 11.$$

For the generalized covering-radius front, the transitive fractional/Delsarte relaxation reaches the quotient size first at radius 389, so level zero cannot improve the lower bound. The exact rank-ten face coherent configuration, rank-five antipodal scheme, eigenmatrix, and intersection tensor provide the symmetry-reduced input for a higher semidefinite hierarchy. The later certified 247-circuit basis-exchange theorem improves the upper bound to 435, and a dependency verifier binds this insert to that frozen certificate. The live interval is

$$\boxed{389 \le R_{\text{defect}} \le 435}.$$

Finally, the ternary $[12, 6, 4]$ A_2–conic product code does not realize the tomotope faces. Exhaustion of all 1760 weight-three ambient vectors shows that a dual-code coset contains at most four projective triple supports, whereas the tomotope requires sixteen triangular faces. Nevertheless the same 3×4 coordinate grid has a positive incidence interpretation: matching rows and vertex columns identify the coordinates with the twelve oriented tetrahedron edges. Four outgoing stars, four incoming stars, and eight directed three-cycles form an exact $12_4\,16_3$ Reye-style incidence surface. The archived negative flag search remains decisive: this skeleton alone does not determine the missing cell/orientation cocycle needed for the true 192-flag tomotope monodromy.

10.2 Radius, amplitude, modular, and five-channel closure

An exact basis-exchange certificate in the minimum-defect Cayley system produces a 247-term circuit with 246 nonzero basis coordinates. Since the coefficient alphabet has only $3^5 - 1 = 242$ nonzero vectors, a nonzero circuit shift cancels at least two basis coefficients while adding one extra generator. The covering-radius interval therefore improves to

$$\boxed{389 \le D_{H^1} \le 435}.$$

The exact radius remains open.

A five-channel amplitude-bearing magnetic family was then screened on a 63,869-point dyadic grid. Its best member has weights

$$\left(-\frac{21}{32}, \frac{43}{32}, -\frac{27}{32}, -\frac{15}{16}, 1\right)$$

and generalized Hoffman ratio approximately 8.9024605237. After scaling by 32, its exact characteristic polynomial contains $(x+128)^{17}(x-64)^{21}$ and a seven-dimensional residual whose roots all lie strictly in $(-128, 1024)$. Consequently the unscaled matrix has $\lambda_{\min} = -4$ and $\lambda_{\max} < 32$, proving that this winner remains strictly below nine. This is a finite-grid no-go, not an optimization of the unrestricted real cone.

Over $\mathbb{F}_3$, the 240 face boundaries form a totally isotropic subspace of the 480-dimensional flat edge space. They are disjoint from the 44-dimensional coboundary space, giving the exact filtration

$$0 < R_{240} < Z^1_{480} < \mathbb{F}_3^{720}, \qquad \dim Z^1/(R + B^1) = 196.$$

No modular irreducible name is assigned without a composition-series certificate.

The full 135-state quotient admits an executable Fourier organization

$$135 = 27 \times 5.$$

For $c(0) = 2$ and $c(1) = c(2) = -1$, the four barycentric channels have symbol $c(k_1)K_x + c(k_2)K_y + c(k_3)I_4$, while the hidden channel is $-c(k_1) - c(k_2) + c(k_3)$. Exhausting all 675 symbol entries recovers

$$10^1,\ 7^6,\ 4^{22},\ 1^{44},\ (-2)^{42},\ (-5)^{20}.$$

The finite geometry itself forms a stabilizer chain

$$5 \xrightarrow{\text{hinge}} 4 \xrightarrow{\text{pairing}} 3.$$

Choosing one of the five Clebsch/D_5 coordinate directions leaves the S_4 action on the four points of the $PG(2,3)$ null conic. The three perfect matchings of those four points carry the quotient $S_4/V_4 \cong S_3$ and match the three projective $A_2(\mathbb{F}_3)$ root directions and the far/middle/active Fano gauges. This is a finite permutation correspondence, not a physical Spin(10) claim.

Finally, the fixed linear $[13, 5, 5]$ controller encoder has a unique minimum coordinate-orbit completion of length 28. The completed code is

$$\boxed{[28, 5, 11]},$$

so fifteen added coordinates are necessary and sufficient for coordinate-permutation S_3 equivariance of that encoder. The Clebsch closed-neighborhood matrix is the incidence matrix of a symmetric 2-$(16, 6, 2)$ biplane, with Smith form $1^6 2^4 4^5 12$ and the modular identity $B^2 = I - J$ over $\mathbb{F}_3$. Its zero and single-point syndromes have minimum distance six.

11 Triangle-free strongly regular graphs and the missing-Moore firewall

Exact fixed-μ ladder. For a strongly regular graph with

$$\mu = 4 \qquad \text{and restricted eigenvalue} \qquad r = 2,$$

the remaining data are forced by the SRG identities:

$$s = \lambda - 6, \qquad k = 16 - 2\lambda, \qquad v = \frac{(\lambda - 7)(3\lambda - 22)}{2},$$

with multiplicities

$$f_r = \frac{(\lambda - 5)(3\lambda - 22)}{2}, \qquad f_s = -3(\lambda - 7).$$

For $\lambda = 0, 1, 2, 3, 4$ this gives

$$\boxed{(77, 16, 0, 4) \longrightarrow (57, 14, 1, 4) \longrightarrow (40, 12, 2, 4) \longrightarrow (26, 10, 3, 4) \longrightarrow (15, 8, 4, 4).}$$

These are respectively the M_{22} graph, the nonexistent Wilbrink–Brouwer parameter set, the $W(3, 3)$ parameter class, the Paulus parameter class, and the triangular graph $T(6)$. Thus there is an exact 57-vertex spectral hole between M_{22} and $W(3, 3)$. It is not the hypothetical degree-57 Moore graph, whose parameters would be $(3250, 57, 0, 1)$.

$W(3,3)$–Gewirtz restricted kernel. The $W(3,3)$ and Gewirtz adjacency spectra are

$$12^1, 2^{24}, (-4)^{15} \qquad \text{and} \qquad 10^1, 2^{35}, (-4)^{20}.$$

After removing the constant line, both operators therefore obey

$$\boxed{(A-2I)(A+4I)=0.}$$

Their complement operators have restricted eigenvalues ± 3, so their centered complement squares equal $9I$ on augmentation. This is a shared quadratic functional calculus, not a canonical object-wise intertwiner; the multiplicities and incidence geometries differ.

Necessary edge chart for a hypothetical M_{57}. Assume temporarily that a Moore graph with parameters $(3250, 57, 0, 1)$ exists and fix an edge $a \sim b$. The punctured neighborhoods

$$A = N(a) \setminus \{b\}, \qquad B = N(b) \setminus \{a\}$$

have size 56, and the remaining $3136 = 56^2$ vertices are forced bijectively onto $A \times B$. Every residual vertex has one neighbor in A, one in B, and 55 residual neighbors. No residual edge lies inside a row or column; hence every pair of distinct rows, and every pair of distinct columns, is joined by a perfect matching. If $\sigma_{ij} \in S_{56}$ denotes the matching from row i to row j, then necessarily

$$\sigma_{ji} = \sigma_{ij}^{-1}, \qquad \sigma_{ij} \text{ is fixed-point-free,}$$

and for distinct i, j, k the product

$$\sigma_{ki}\sigma_{jk}\sigma_{ij}$$

is fixed-point-free. These are necessary, not sufficient, constraints. Crucially, relabelling rows and columns is coordinate gauge, not an automorphism of one fixed completed graph; and

$$\frac{3250 \cdot 57}{2} = 92{,}625$$

is the edge count, not the automorphism-group order.

The four-way 57 firewall. Four different objects must remain separate:

object	meaning
M_{57}	3250 vertices, degree 57, existence open
$(57, 14, 1, 4)$	57 vertices, proved nonexistent
Perkel graph	57 vertices, degree 6
regular 57-cell	57 cells, $PSL(2, 19)$ symmetry

A June 2026 preprint claims that a hypothetical M_{57} has no involutory automorphisms. If that result survives review, then its automorphism group has odd order, whereas

$$|PSL(2, 19)| = 3420$$

is even; consequently the 57-cell/Perkel symmetry cannot embed into $\mathrm{Aut}(M_{57})$. This conditional firewall is not a proof of nonexistence.

Machine certificate. The standard-library verifier `analysis/bt3500_3505_triangle_free_srg_m57_bridg` reconstructs all parameter, spectrum, multiplicity, edge-chart, and order identities above. Its frozen semantic digest is

`e308364dd480970803061b90ab27bf86afa2ae4946399b656292f02682c17fd9`.

The evidence boundary is explicit: no existence decision for the missing Moore graph, no Perkel/57-cell/Moore identification, and no canonical $W(3,3)$–Gewirtz intertwiner are asserted.

12 Passes 3506–3512: exact descendants, Moore permutation curvature, and the scheme-blind 57-hole

Certified seven-front packet

The standard-library verifier reports

`PASS_7_FRONTS`

with semantic SHA-256

`15ab9b21e744e755917e0b4c40ec806b43c5463b8a11cfbfac7bdbb4a9c8dfe7`.

The packet supplies an executable Moore-57 permutation CSP, literal descendant constructions among five standard triangle-free strongly regular graphs, the complete rank-three Krein census of the fixed $\mu = 4, r = 2$ ladder, a W33/Gewirtz polynomial-transplant compiler, a sharpened 57-cell symmetry firewall, and two explicitly fenced high-risk constructions.

12.1 The Moore chart becomes a gauge-reduced permutation CSP

For a fixed edge of a hypothetical $\mathrm{SRG}(3250, 57, 0, 1)$, write $p_{ij}(a)$ for the destination column in row j of the residual neighbor of (i, a). The independent matching data contain

$$\binom{56}{2}56 = 86\,240$$

entries. The literal CP-SAT exporter uses 172 480 directed integer variables with explicit inverse channels, rather than 4 743 200 nonfixed one-hot Boolean variables. Before lazy cuts it has 1 540 row-pair permutation constraints, 1 540 inverse channels, 3 136 vertex-star constraints, and 55 gauge equalities:

$$p_{0j}(0) = j, \qquad 1 \le j \le 55.$$

Triangle fixed points are separated over the

$$\binom{56}{3} = 27\,720$$

row triples. This is a source-complete model, not a solved instance.

12.2 Two literal descendant chains

The Clebsch graph is reconstructed on the even-weight vectors of $\mathbb{F}_2^5$, with adjacency at Hamming distance four. Its second subconstituent at zero consists of the ten weight-two words and is exactly $KG(5, 2)$, the Petersen graph.

Independently, a cyclic generator produces the extended binary Golay code with weight distribution

$$1 + 759z^8 + 2576z^{12} + 759z^{16} + z^{24}.$$

Fixing two coordinates in the octads yields the 77 blocks of $S(3, 6, 22)$. Hexad disjointness gives M_{22}; avoiding one point gives Gewirtz; adjoining the 22 points and infinity gives Higman–Sims. Thus

$$\boxed{\mathrm{HS} \longrightarrow M_{22} \longrightarrow \mathrm{Gewirtz}}$$

and

$$\boxed{\mathrm{Clebsch} \longrightarrow \mathrm{Petersen}}$$

are now objectwise executable.

12.3 The nonexistent 57-vertex rung is invisible to rank-three feasibility

For every member of the fixed

$$\mu = 4, \qquad r = 2$$

ladder, the verifier computes the complete primitive-idempotent Krein products and both standard absolute bounds. Every test passes, including for the nonexistent parameter set $\mathrm{SRG}(57, 14, 1, 4)$. Its nontrivial Krein rows are

$$q_{11} = \left(38, \frac{1273}{49}, \frac{1140}{49}\right), \quad q_{12} = \left(0, \frac{540}{49}, \frac{722}{49}\right), \quad q_{22} = \left(18, \frac{342}{49}, \frac{111}{49}\right).$$

Hence its obstruction is not visible in the ordinary rank-three Bose–Mesner algebra:

$$\boxed{\text{the } \lambda = 1 \text{ hole is scheme-blind.}}$$

Finer local, Terwilliger, or star-complement compatibility is required.

12.4 A two-channel W33/Gewirtz transplant algebra

On augmentation, W33 and Gewirtz both satisfy

$$A^2 + 2A - 8I = 0.$$

Consequently every adjacency polynomial reduces uniquely to

$$f(A) = aA + bI$$

in $\mathbb{Q}[x]/(x^2 + 2x - 8)$. Moreover

$$U = \frac{-I - A}{3}$$

obeys

$$U^2 = I, \qquad P_\pm = \frac{I \pm U}{2}.$$

Polynomial identities transport automatically. Traces and projector ranks depend on the different multiplicities, while incidence, codes, automorphism groups, and descendant maps remain geometry-sensitive.

12.5 The 57-cell firewall and its surviving odd shadow

The Perkel/57-cell group has order

$$|PSL(2, 19)| = 3420.$$

Peer-reviewed results bound an odd hypothetical M57 automorphism group by 375 and an even one by 110. Conditional on the June 2026 no-involution preprint, $\mathrm{Aut}(M57)$ is odd and hence solvable; there can be no common A_5 or full $PSL(2, 19)$ automorphism action. However, the odd Borel subgroup

$$C_{19} \rtimes C_9, \qquad |C_{19} \rtimes C_9| = 171,$$

is not excluded by parity and the order bound alone. The next exact target is the restriction of the Perkel -3 twenty-plane to this 19:9 shadow.

12.6 Bonkers front: non-Abelian Moore curvature

The row matchings form an S_{d-1}-valued connection. Triangle holonomy

$$H_{ijk} = \sigma_{ki}\sigma_{jk}\sigma_{ij}$$

changes by conjugacy under gauge, so cycle type is invariant. In an exact edge-rooted chart of the Hoffman–Singleton graph, all 15 row-pair matchings and all 20 triangle holonomies have cycle type

$$2^3.$$

This motivates a non-necessary M57 search branch in which all matchings and triangle holonomies have type 2^{28}. It is a solver ansatz, not an M57 theorem.

12.7 Bonkers front: one Golay puncture/avoidance functor

The chain

$$[23, 12, 7] \to [24, 12, 8] \to S(3, 6, 22) \to M_{22} \to \text{Gewirtz}$$

together with the incidence extension to Higman–Sims is one executable information-losing pipeline. Its comparison with the W33 face tower $240 \to 120 \to 40$ is structural only: no category equivalence or group identification is asserted.

Evidence boundary. No M57 existence decision, complete CP-SAT run, necessity of involutive curvature, actual M57 19:9 action, canonical W33–Gewirtz intertwiner, or peer-reviewed status for the June 2026 preprint is claimed.

13 Multi-circuit, Reed–Muller, compound-biplane, and A_5 closure

The minimum-defect interval remains

$$389 \le D_{H^1} \le 435,$$

but the single-circuit method is now closed exactly. A deterministic basis exchange produces a certified 260-term circuit. Since a rank-436 fundamental circuit contains at most 436 basis coordinates while the nonzero coefficient alphabet has size $3^5 - 1 = 242$, one circuit can force at most two cancellations while adding one support. Hence no single-circuit pigeonhole argument can improve the universal upper bound below 435.

A five-channel continuous amplitude search gives a numerical candidate with ratio approximately 8.9062301931. The nearby exact tuple

$$\frac{1}{256}(122, 346, 220, -240, 256)$$

has a rational-polynomial certificate proving

$$8.905 < h < 9.$$

Thus it improves the earlier finite-grid witness but does not reach the live chromatic lower bound.

The complete binary coordinate-permutation S_3-equivariant dimension-five code frontier was exhausted through length forty. In particular,

$$[13, 5, 5], \qquad [16, 5, 8], \qquad [24, 5, 11], \qquad [28, 5, 14]$$

are length-minimal at the displayed target distances. The sixteen columns of the $[16, 5, 8]$ code are exactly the affine hyperplane

$$\{c \in \mathbb{F}_2^5 : \langle 00111, c\rangle = 1\},$$

so the controller code is objectwise $RM(1, 4)$ with weight enumerator $1 + 30z^8 + z^{16}$.

The Clebsch biplane has thirty fourfold collision classes among its 120 weight-two faults. A canonical exhaustive search proves that three additional bits are necessary and sufficient to separate all of them. The resulting nineteen-bit readout uniquely locates all

$$1 + 16 + \binom{16}{2} = 137$$

fault patterns of weight at most two. The minimal companion map is cubic in the four-bit Clebsch-axis coordinate.

Finally, an objectwise A_5 character computation separates the two tempting rank-twenty sectors. The Perkel -3 sector restricts as

$$1 \oplus 3 \oplus 3' \oplus 2 \cdot 4 \oplus 5,$$

whereas the W33 block -4 sector restricts as

$$3 \cdot 1 \oplus 3 \cdot 4 \oplus 5.$$

Consequently no A_5-equivariant rank-twenty intertwiner exists. The live chromatic boundary remains $10 \le \chi(H) \le 11$.

14 Borel Fourier anatomy, star complements, and typed Moore searches

Passes 3528–3534 execute the five graph-theoretic continuations after reconciling the complete-switch and common-A_5 packets. The exact standard-library verifier reports

`PASS_7_FRONTS`

with semantic digest

`1ec126887cd1f09b8cdc074872567dd4962c2a399c1b14db8b035a8ea50dc591.`

The degree-57 Moore graph remains open; no 56-fibre SAT or UNSAT result is claimed.

14.1 The Perkel twenty-plane restricted to the odd Borel subgroup

On the explicit Perkel model $(i, j) \in \mathbb{Z}_3 \times \mathbb{F}_{19}$, the maps

$$(b, m) : (i, j) \longmapsto (i + m, 4^m j + b)$$

realize all 171 elements of

$$B = C_{19} \rtimes C_9.$$

Every map is checked objectwise against the edge set. The -3 spectral projector is evaluated integrally through

$$171E_{-3} = -A^3 + 9A^2 - 19A + 6I.$$

The restricted character has value distribution

$$20^1, \qquad 1^{18}, \qquad 2^{38}, \qquad (-1)^{114},$$

and fixed dimensions

$$\dim V^{C_{19}} = 2, \quad \dim V^{C_9} = 2, \quad \dim V^{C_3} = 8, \quad \dim V^B = 0.$$

Over $\mathbb{Q}$ the twenty-plane is

$$\boxed{V_{-3}\downarrow_B \cong V_{18} \oplus V_2,}$$

where V_{18} is the conductor-19 rational constituent and V_2 is the conductor-3 constituent. This closes the surviving 19:9 comparison on the Perkel side only; it does not construct that subgroup in a hypothetical Moore graph.

14.2 The 57-vertex star-complement firewall

For the nonexistent parameter set

$$\mathrm{SRG}(57, 14, 1, 4),$$

the eigenvalue 2 has multiplicity 38, so a star complement has order 19. An independent heavy enumeration from the canonical windmill seed gives stage counts

$$22, \quad 784, \quad 157{,}349,$$

and exactly

$$\boxed{3{,}720}$$

spectral survivors. Their canonical ledger hash is

`c1e0cb753fbdeab4d8ecf8059896b6ff1eb1fcc75c663022ab7040ceea012219`.

The published compatibility-graph census has maximum clique 31, whereas reconstruction requires a 38-clique. Therefore

$$\boxed{31 < 38}$$

reproves nonexistence, with margin seven. The 3,720 candidate census is independently regenerated here; the complete maximum-clique histogram remains source-locked to the published Cliquer computation and is not presented as a fresh independent rerun.

14.3 Two typed Moore-57 solver tracks

For fibre size $n = d - 1$, a diameter-two Moore graph has parameters

$$(d^2 + 1, d, 0, 1)$$

and restricted polynomial

$$x^2 + x - (d - 1).$$

The exact global gate leaves only

$$\boxed{n \in \{1, 2, 6, 56\}},$$

corresponding to C_5, Petersen, Hoffman–Singleton, and the open degree-57 case. Every even stage $n = 8, 10, \ldots, 54$ fails spectrum or multiplicity integrality before a local CSP is built.

The unrestricted $n = 56$ model retains 172,480 directed integer variables, 6,271 structural constraints, and lazy triangle and common-neighbour separators. A second high-risk branch imposes

$$\sigma_{ij}^2 = 1$$

on every fixed-point-free row matching and separates triangle holonomies by cycle type 2^{28}. This adds 86,240 source-level element-involution constraints. Neither branch has an $n = 56$ verdict.

14.4 The two descendant towers are typed differently

The W33 face tower

$$240 \longrightarrow 120 \longrightarrow 40$$

is a pair of uniform quotients with fibre sizes two and three. The Golay–Witt chain

$$100 \longrightarrow 77 \longrightarrow 56$$

uses induced restriction and point avoidance. Since

$$100/77 \notin \mathbb{Z}, \qquad 77/56 \notin \mathbb{Z},$$

there is no common uniform-fibre quotient functor. The correct shared language requires distinct morphism types: uniform quotient, induced restriction, point avoidance, and incidence extension.

14.5 Fail-closed W33–Gewirtz transplantation

Automatic transfer is permitted only for statements reducible in

$$\mathbb{Q}[x]/(x^2 + 2x - 8).$$

Thus every adjacency polynomial reduces to $aA + bI$, and

$$U = \frac{-I - A}{3}, \qquad U^2 = I$$

is common on augmentation. Traces, determinants, ranks, multiplicities, Smith data, and p-ranks require new evidence. Lines, incidence, codes, automorphism groups, descendant maps, and objectwise intertwiners are denied automatic transfer. A repository scanner now fails closed on any explicit unsafe `AUTO_PORT` annotation.

14.6 Two outside-box consequences

The complete Perkel vertex module has the rational Borel decomposition

$$\boxed{\mathbb{Q}^{57} \cong \mathbf{1} \oplus 3V_{18} \oplus V_2.}$$

Its combined golden eigenspace is $2V_{18}$, while the -3 twenty-plane is $V_{18} \oplus V_2$. The whole graph is therefore one constant channel, three conductor-19 channels, and one conductor-3 phase channel.

Inside the involutive Moore branch, the vertex-star all-different law implies that the $n - 1$ matchings issuing from each source row form a 1-factorization of K_n. Hoffman–Singleton becomes six coupled non-group 1-factorization pencils of K_6. The $n = 56$ branch becomes

$$\boxed{\text{56 reciprocity-coupled 1-factorizations of } K_{56},}$$

with 55 perfect matchings and $55 \cdot 28 = 1540$ edges covered once in every pencil. Pair-matching involution is coordinate dependent; holonomy cycle type is gauge invariant, and the theorem applies only to the declared involutive branch.

15 Star-clique recertification, Borel–M57 archetypes, and non-separable Moore factorization

The executable Passes 3535–3541 report

```
PASS_7_FRONTS  34ea459ab6acea3a2b5624ba6523016cfb914ffb54220ed64230f74f78a1fb6b.
```

The degree-57 Moore graph remains open; all statements below are finite theorems or explicitly conditional necessary conditions.

15.1 Independent star-complement clique source

For each of the 3720 independently regenerated 19-vertex star complements with adjacency matrix C, the companion source constructs

$$M = 2I - C$$

exactly, enumerates every admissible binary reconstruction column b satisfying

$$b^{\mathsf{T}} M^{-1} b = 2,$$

and joins two columns when their exact inner product is -1 or 0 and the corresponding $(\lambda, \mu) = (1, 4)$ common-neighbour bound is respected. A deterministic bitset maximum-clique engine emits a witness and proof digest for every instance. The complete heavy job fails closed unless it reproduces the published clique histogram and compatibility-graph order range 4 through 265. No fresh all-3720 result is promoted before its artifact is observed.

15.2 Prime fixed sets and the surviving Borel action

Let an automorphism of prime order p act on a hypothetical

$$\mathrm{SRG}(3250, 57, 0, 1).$$

A nontrivial fixed set is closed under unique common neighbours, and if it has more than one vertex its induced graph is itself a Moore graph, with fixed degree congruent to 57 modulo p. Consequently an order-19 element fixes exactly one vertex, whereas an order-3 element fixes either one vertex or a Petersen graph.

Assume conditionally that

$$B = C_{19} \rtimes C_9$$

acts. Orbit arithmetic leaves precisely

$$P_{19}: \ 1 + 9 \cdot 19 + 18 \cdot 171, \qquad P_{57}: \ 1 + 3 \cdot 57 + 18 \cdot 171.$$

The adjacency trace on an order-19 element first permits

$$g \in \{57, 342, 627, 912, 1197\},$$

where g counts vertices mapped to neighbours. Girth and equal displacement for nontrivial powers leave only

$$\boxed{g \in \{57, 342\}}.$$

Thus exactly four conditional action archetypes survive: low/high versions of P_{19} and P_{57}. They are necessary forms, not constructions or a contradiction.

15.3 Exact Perkel Fourier projectors

Let A be the Perkel adjacency matrix, J the all-ones matrix, and B_{19} the block diagonal matrix with three J_{19} blocks. Put

$$N = -A^3 + 9A^2 - 19A + 6I.$$

The integer projector numerators

$$P_1 = 3J, \quad P_2 = 9B_{19} - 3J, \quad P_{18} = N - 9B_{19} + 3J, \quad P_{36} = 171I - 3J - N$$

have ranks

$$1, 2, 18, 36$$

and satisfy

$$P_iP_j = 171\delta_{ij}P_i, \qquad \sum_i P_i = 171I.$$

The conductor-19 projector is

$$\boxed{P_{54} = 171I - 9B_{19} = P_{18} + P_{36}},$$

and

$$(A^3 - 8A + 3I)P_{54} = 0.$$

Hence every Perkel adjacency polynomial reduces to a constant channel, a ternary V_2 channel, and a degree-two normal form on the conductor-19 sector modulo $x^3 - 8x + 3$.

15.4 The one-factorization field

Inside the involutive edge-chart branch, each row pencil is a one-factorization. The exact Hoffman–Singleton control has 15 perfect matchings and exactly six labelled one-factorizations of K_6. Its six residual rows realize all six factorizations exactly once. Moreover, all $5! = 120$ globally separable colour identifications fail all 20 row triangles: every one of the 2400 holonomies has two fixed points. The actual Hoffman–Singleton holonomies all have cycle type 2^3.

Therefore a hypothetical involutive $n = 56$ branch cannot reuse one global factorization. It requires 56 genuinely row-dependent, reciprocity-coupled one-factorizations of K_{56}, each containing 55 perfect matchings and covering all 1540 symbol pairs.

15.5 Typed W33–Gewirtz polynomial port

Both exact graphs satisfy the common restricted relation

$$(A - 2I)(A + 4I) = 0$$

on augmentation. With $P = J/v$ and $Q = I - P$, the full identity is

$$(A - 2I)(A + 4I) = (k - 2)(k + 4)P.$$

The reflection

$$R = \frac{(k + 1)P - I - A}{3}$$

satisfies $R^2 = Q$, and

$$E_2 = \frac{Q - R}{2}, \qquad E_{-4} = \frac{Q + R}{2}.$$

If $p(x) \equiv ax + b \pmod{x^2 + 2x - 8}$, then

$$\boxed{p(A) = p(k)P + (aA + bI)Q.}$$

This ports every adjacency-polynomial and functional-calculus theorem. Trace, determinant, and rank formulas require the typed multiplicities $(24, 15)$ for W33 and $(35, 20)$ for Gewirtz. Incidence, codes, automorphisms, Smith forms, descendants, and intertwiners remain geometry-sensitive and require independent evidence.

Boundary. No M57 existence verdict, unobserved full clique rerun, W33–Gewirtz objectwise equivalence, hardware result, or physical interpretation is asserted.

16 Cyclic Borel quotients, the Perkel commutant, and the outer-S_6 Moore chart

Passes 3542–3548 continue the degree-57 Moore and Perkel program with exact finite certificates. The corresponding verifier reports

```
PASS_7_FRONTS a98621cc2f2378eaed09b8f747022f419a16fbb33433336eeb8f2414f4273f1a
```

The missing Moore graph remains open.

The C_{19} quotient becomes three exact branches. Assume $B = C_{19} \rtimes C_9$ acts on a hypothetical $\mathrm{SRG}(3250, 57, 0, 1)$. The normal C_{19} fixes one vertex and has 171 nonfixed orbit cells. Exactly three cells lie in the fixed vertex's neighbourhood. Writing $S_{ij} \subseteq \mathbb{Z}_{19}$ for the cyclic shifts between cells and $T_i = S_{ii}$, the unique-common-neighbour law forces the three neighbour cells to be independent, every connection leaving them to have size at most one, and their orbit-neighbour sets to be disjoint. Hence the remaining 168 cells split exactly as

$$168 = 56 + 56 + 56.$$

For a nonneighbour cell,

$$\sum_j |S_{ij}|(|S_{ij}| - 1) = 18 - |T_i|, \qquad |T_i| \in \{0, 2, 4\}.$$

C_9 or C_3 stabilizer invariance forbids nonempty T_i inside Borel orbits of size 19 or 57, so internal edges occur only in the eighteen size-171 Borel orbits. If a of those have internal degree two and b have internal degree four, the two displacement levels reduce to

$$a + 2b = 3 \quad \text{or} \quad a + 2b = 18.$$

Together with the two orbit profiles P_{19} and P_{57}, the four Borel archetypes reduce to exactly twenty-four aggregate cyclic-difference signatures. This is a finite CSP frontier, not a construction or contradiction.

The complete Perkel Borel commutant. For the explicit Perkel action $(i, j) \mapsto (i+m, 4^m j+b)$, the point stabilizer is C_3 with three singleton orbits and eighteen orbits of size three. The orbital algebra therefore has rank 21. The rational module

$$\mathbb{Q}^{57} = \mathbf{1} \oplus 3V_{18} \oplus V_2$$

gives projected algebra dimensions $1, 2, 18$, and hence

$$\mathrm{End}_B(\mathbb{Q}^{57}) \cong \mathbb{Q} \oplus M_3\big(\mathbb{Q}(\sqrt{-19})\big) \oplus \mathbb{Q}(\sqrt{-3}).$$

Let P_{54} and P_2 be the conductor-19 and conductor-3 projector numerators. The block-diagonal Paley tournament matrix D_{19} and the oriented three-layer all-one operator C_3 obey

$$9D_{19}^2 = -P_{54}, \qquad 3C_3^2 = -19P_2,$$

and commute with all twenty-one orbital matrices. The commutative adjacency layer has exact span

$$\mathbb{Q}[A, B_{19}] \cong \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q} \times \mathbb{Q}(\sqrt{5}),$$

with basis $I, A, A^2, B_{19}, AB_{19}$.

Proof-carrying maximum cliques. The star-complement pipeline now has an independently checkable upper-bound proof language. A colour leaf proves $\omega(P) \le K$ by a proper K-colouring of the current candidate set. A branch node on v proves the include subproblem at $K-1$ and the exclude subproblem at K. The checker reconstructs all subproblems, verifies colour classes and references, and separately checks the clique witness. Thirty-one deterministic controls pass. A complete 3720-instance proof archive remains a heavy workflow boundary.

A factorization-field Moore compiler. Let n be even and suppose $F_0, \ldots, F_{n-1}$ are one-factorizations of K_n satisfying $|F_i \cap F_j| = 1$. Let m_{ij} be the shared perfect matching. The vertex set

$$\{x, y\} \cup \{r_i\} \cup \{c_a\} \cup \{z_{i,a}\}$$

with edges

$$x \sim y, r_i, \qquad y \sim c_a, \qquad z_{i,a} \sim r_i, c_a, z_{j,m_{ij}(a)}$$

has $(n+1)^2 + 1$ vertices and degree $n+1$. If every simple three-row and four-row holonomy is fixed-point-free, the graph has girth at least five and meets the Moore bound exactly. For $n = 56$ this becomes a 3250-vertex, degree-57 construction surface with 27720 triangle checks and 1101870 simple four-cycle checks. Pairwise unique intersection is a sufficient high-risk branch, not a proved necessity.

Hoffman–Singleton as the outer automorphism of S_6. The fifteen duads, fifteen synthemes, and six pentads are respectively the edges, perfect matchings, and one-factorizations of K_6. Exact enumeration shows that every pair of pentads shares exactly one syntheme, every triangle holonomy has cycle type 2^3, and every simple four-row holonomy has cycle type 3^2. The induced action of S_6 on the six pentads is faithful of order 720; a point transposition of type $2\,1^4$ maps to type 2^3, certifying the outer action. Feeding the shared synthemes into the factorization-field compiler reconstructs

$$\mathrm{SRG}(50, 7, 0, 1),$$

the Hoffman–Singleton graph.

Analytic W33–Gewirtz ports. For either graph,

$$f(A) = f(k)P + f(2)E_2 + f(-4)E_{-4}$$

for every analytic f. In particular the heat kernel and resolvent port exactly, while the Ihara–Bass factors retain the graph-specific degree and multiplicities. No incidence, code, automorphism, hardware, or physical claim is inferred from this spectral compiler.

17 Passes 3549–3555: Borel orbit models, exact commutants, octad curvature, and quantum walks

Twenty-four exact Borel models. For a conditional $B = C_{19} \rtimes C_9$ action on a degree-57 Moore graph, the two orbit profiles and the equations $a + 2b = 3, 18$ produce exactly 24 aggregate signatures. Burnside gives 30,915 undirected edge-orbit variables and 61,858 ordered-pair orbitals for P_{19}, versus 30,885 and 61,792 for P_{57}. Every model imposes the exact lazy relation

$$\sum_w e_{uw}e_{vw} = 0 \text{ if } e_{uv} = 1, \qquad \sum_w e_{uw}e_{vw} = 1 \text{ if } e_{uv} = 0.$$

These are complete finite model surfaces, not SAT or UNSAT verdicts.

Complete Perkel orbital algebra. The 19:9 action has 21 orbitals, with size profile $57^3\,171^{18}$ and valencies $1^3\,3^{18}$. The exact multiplication table contains 1,035 nonzero integral structure constants, all at most three, and has five-dimensional center:

$$\mathrm{End}_{19:9}(\mathbb{Q}^{57}) \cong \mathbb{Q} \oplus M_3(\mathbb{Q}(\sqrt{-19})) \oplus \mathbb{Q}(\sqrt{-3}).$$

Transpose splits the algebra into 11 symmetric and 10 skew channels. Thus every real symmetric equivariant kernel lies in an eleven-dimensional space corresponding, after the imaginary-quadratic embeddings, to

$$\mathbb{R} \oplus \mathrm{Herm}_3(\mathbb{C}) \oplus \mathbb{R}.$$

Generalized pentads and an octad falsifier. The exact labelled one-factorization counts for K_4, K_6, K_8 are $1, 6, 6240$. A family of eight K_8 one-factorizations was found in which every pair shares exactly one perfect matching; its setwise S_8-stabilizer is trivial. However only 34/56 triangle holonomies and 163/210 simple four-row holonomies are derangements. The compiled graph is 9-regular on 82 vertices, has diameter three, 60 triangles, and 422 four-cycles. Hence pairwise factor intersection is the static design condition, while holonomy is the curvature condition deciding the Moore property.

Proof archive and analytic dynamics. The 3,720 star-complement proof DAGs are assigned to 64 deterministic shards, eight of size 59 and 56 of size 58, with an ordered Merkle aggregate. The archive protocol is complete, but the full payload remains evidence-gated. For either W33 or Gewirtz,

$$e^{-itA} = e^{-ikt}P + e^{-2it}E_2 + e^{4it}E_{-4}.$$

For W33, distinct-vertex amplitudes obey

$$|U_{xy}(t)| \le \frac{1}{4} \quad (x \sim y), \qquad |U_{xy}(t)| \le \frac{2}{15} \quad (x \nsim y),$$

and for Gewirtz

$$|U_{xy}(t)| \le \frac{2}{7} \quad (x \sim y), \qquad |U_{xy}(t)| \le \frac{1}{12} \quad (x \nsim y).$$

Thus neither graph permits perfect continuous-time state transfer between distinct vertices. The exact infinite-time average probabilities and nonbacktracking poles are emitted by the verifier.

> **Boundary.** The degree-57 Moore graph remains open. No Borel signature has a solver verdict, the full proof archive is not yet promoted, the K_8 graph is a falsifier rather than a Moore construction, and the quantum-walk formulas are mathematical dynamics rather than laboratory measurements.

18 Relation planes, magnetic conductors, code duality, and the pentagonal bridge

The live cohomology and chromatic boundaries remain

$$389 \le D_{H^1} \le 435, \qquad 10 \le \chi(H) \le 11.$$

A new exact fundamental cohomology relation has weight 263. In a deterministically extended fundamental basis, all $\binom{284}{2} = 40186$ nonbasis-column pairs were exhausted. Their rows occupy at most three of the four directions of $PG(1,3)$, the maximum union support is 319, and the adversarial cancellation-cap histogram is

$$\{2 : 34119,\ 3 : 5277,\ 4 : 790\}.$$

Thus this pure two-circuit direction method cannot establish $D_{H^1} \leq 433$; 790 pairs remain label-sensitive candidates for 434. An exact pilot on all $\binom{64}{3} = 41664$ triples of the 64 heaviest columns reaches union support 360 and six occupied directions of $PG(2,3)$.

The five-channel Hermitian algebra has channel sizes

$$2, 1, 30, 60, 627$$

and exact characteristic factorization

$$(x + 4w_4)^{17}(x - 2w_4)^{21} q_2(x) q_5(x),$$

where

$$q_2(x) = x^2 + (w_1 + w_4)x + w_1 w_4 - 9w_3^2$$

and q_5 is quintic. The dyadic tuple

$$\frac{1}{32768}(-15576, 44300, -28135, -30786, 32768)$$

has an exact rational-root certificate proving

$$8.90622 < h < 8.90623.$$

A degree-eighteen elimination polynomial isolates a stationary candidate near $h = 8.90623076116309$, but global unrestricted optimality remains open.

The complete duality atlas of the four binary equivariant frontier codes gives

$[n, 5, d]$	$d^\perp$	$\dim(C \cap C^\perp)$	$\lvert \mathrm{Aut}(C) \rvert$
$[13, 5, 5]$	3	2	48
$[16, 5, 8]$	4	5	322560
$[24, 5, 11]$	3	0	72
$[28, 5, 14]$	3	3	64512

with generalized weights computed exactly. The length-sixteen code is $RM(1,4)$ with automorphism group $AGL(4,2)$, while the length-twenty-eight code is the binary simplex code punctured on one projective line.

For compound Clebsch readout, three added bits cannot even force distance two, four cannot force distance three, and five bits attain distance three. Therefore five companion bits are necessary and sufficient to locate every zero-, single-, and double-device fault while correcting one corrupted readout bit. The resulting code has length twenty-one and 137 protected patterns.

Finally, the Perkel and W33 rank-twenty modules, although inequivalent over A_5, become fully isomorphic on the common pentagonal subgroup:

$$P_{20} \downarrow_{C_5} \cong W_{20} \downarrow_{C_5} \cong 4\mathbf{1} \oplus 4\mathbb{Q}(\zeta_5).$$

The rational intertwiner space has dimension 80. No canonical objectwise intertwiner is asserted.

A second objectwise closure identifies the 240 W33 edges with the 240 filled triangles of the 45-octet port graph: every W33 edge belongs to exactly three canonical $K_{4,4}$ octets, and the resulting triangles partition all 720 block-graph edges.

19 Maximal cyclic rank-twenty bridge and a gate-minimal fault locator

The rational C_5 restrictions of the Perkel and W33 rank-twenty modules are

$$P_{20} \downarrow_{C_5} \cong W_{20} \downarrow_{C_5} \cong 4\mathbf{1} \oplus 4\mathbb{Q}(\zeta_5).$$

Consequently their rational intertwiner algebra is

$$M_4(\mathbb{Q}) \oplus M_4(\mathbb{Q}(\zeta_5)),$$

of dimension 80. Thus C_5 symmetry alone does not select a preferred objectwise map.

The full sixteen-dimensional cyclotomic moving sector extends to the dihedral normalizer D_{10}. The only obstruction lies in the four-dimensional C_5-fixed sector:

$$P_{20}\downarrow_{D_{10}} \cong \mathbf{21} \oplus 2\varepsilon \oplus 4V_{\mathrm{cyc}},$$

$$W_{20}\downarrow_{D_{10}} \cong \mathbf{41} \oplus 4V_{\mathrm{cyc}}.$$

Hence the maximum rank of a D_{10}-equivariant map is 18, and the minimal stable repair is

$$\boxed{P_{20} \oplus \mathbf{21} \cong W_{20} \oplus 2\varepsilon},$$

of dimension 22. At the full A_5 level the maximum common rank is 14, and the minimal stable repair is

$$\boxed{P_{20} \oplus \mathbf{21} \oplus 4 \cong W_{20} \oplus (3 \oplus 3')},$$

of dimension 26. Thus the unstabilized common-rank ladder is

$$20 \longrightarrow 18 \longrightarrow 14,$$

while the minimal invertible stable dimensions are $20, 22, 26$.

The Perkel Borel group has order

$$171 = 3^2 \cdot 19,$$

so it contains no element of order five. Its eleven-observable/ten-phase orbital algebra therefore supplies neither a C_5 generator nor its dihedral normalizer; it is not automatically a canonical C_5 selector.

The five-bit compound-fault locator admits the exact quadratic Boolean form

$$\begin{aligned}
y_0 &= x_0x_2 + x_0x_3 + x_1x_3,\\
y_1 &= x_2 + x_0x_2 + x_1x_2 + x_0x_3,\\
y_2 &= x_2 + x_0x_2 + x_1x_2 + x_1x_3,\\
y_3 &= x_2 + x_1x_2 + x_0x_3 + x_1x_3,\\
y_4 &= x_2 + x_1x_2 + x_0x_3 + x_2x_3.
\end{aligned}$$

Its five homogeneous quadratic parts have rank five. Since each AND gate can add at most one independent quadratic form, every XOR–AND realization needs at least five AND gates. The products

$$x_0x_2,\ x_0x_3,\ x_1x_2,\ x_1x_3,\ x_2x_3$$

attain the bound, proving multiplicative complexity exactly five. A shared network uses eight XOR gates. Exhaustive RTL equivalence preserves all 137 compound fault words and minimum distance three.

20 The Petersen spine, constructive Perkel matrices, and octad curvature

Passes 3577–3583 replace three previously broad search fronts by exact finite interfaces. First, the double-coset subdegrees of the hypothetical Borel action force the order-3 or order-9 fixed graph to be the Petersen graph in both surviving orbit profiles. After all variables controlled by this thin spine are fixed, both profiles leave exactly

$$30870$$

residual binary edge-orbit variables, despite starting from the different Burnside counts 30915 and 30885.

Second, the conductor-19 component of the Perkel orbital algebra is represented explicitly as

$$M_3\big(\mathbb{Q}(\sqrt{-19})\big)\,.$$

A primitive rank-18 idempotent produces a six-dimensional rational right ideal, the central Paley operator supplies the relation $D^2 = -19$, and all 441 orbital products are verified in the resulting 3×3 matrices. The frozen matrix-table digest is `366405ad8400779a79eb6b92437b6d354ad3019eb16e9b5a81b99`

Third, the 6240 labelled one-factorizations of K_8 contain at least two inequivalent eight-object pairwise-one-intersection phases. A common-core sunflower has trivial S_8 stabilizer, while a mixed core-free phase has stabilizer four. The sunflower compiles to a 9-regular graph on

$$82 = 1 + 9 + 9 \cdot 8$$

vertices. It is triangle-free but has diameter three and 3024 four-cycles, with characteristic polynomial

$$(x-9)(x-8)^3(x-1)^{32}(x+1)^{24}(x+8)^4(x^2+x-8)^9.$$

It therefore saturates the numerical Moore order while failing the uniform common-neighbour curvature.

A real star-complement canary also executes the proof-carrying clique pipeline: the canonical stages $22, 784$ are regenerated, a genuine third-stage spectral survivor produces a 52-vertex compatibility graph, and an independently checked proof DAG certifies maximum clique 11.

Finally, every SRG resolvent has the form

$$(zI - A)^{-1} = a(z)I + b(z)A + c(z)J.$$

For a marked set S, the principal resolvent $aI + bA[S] + cJ$ recovers the induced edge count from the second determinant coefficient. A four-site W33 line marker therefore detects the six edges of a K_4, which no four-set in the triangle-free Gewirtz graph can reproduce. This is an exact geometry-sensitive port beyond unmarked adjacency dynamics.

Boundary. The missing degree-57 Moore graph remains open. No complete Borel solver verdict, all-octad S_8 census, or all-3720 proof archive is claimed.

20.1 Cubic dependency projector and chained incidence closures

> **Correction from Passes 3649–3662.** The original Passes 3600–3613 source reported zero eight-cell double covers for the oriented $12_4\,16_3$ surface. That count is withdrawn: the increasing-index backtracker was incomplete. Complete enumeration finds exactly three simple covers among all $\binom{12}{8} = 495$ subsets and six solutions among all 75,582 eight-cell multisets. Section 23.1 gives the replacement theorem.

The retained theorem constructs the 5,040-triple dependency deck on 240 filled faces. Its weighted overlap operator has ordinary multiplicity fingerprint

$$1, 15, 15, 20, 24, 24, 60, 81,$$

and over $\mathbb{F}_3$,

$$D^4 = -D^3, \qquad E_{81} = -D^3, \qquad E_{81}^2 = E_{81}, \qquad \operatorname{rank} E_{81} = 81.$$

The deterministic 364-class transported-M_4 scalar screen, exact five-operation ternary datapath, induced $1 + 16 + 40$ decomposition obstruction, and W33–Clebsch spectral completion remain valid. Passes 3649–3662 subsequently prove that the explicit 81-space is simple and projective, close the positive-gauge transported dual at nine, and construct a concrete integral 56-object weighted Gewirtz bridge.

The historical certificate `a08844aa7ec3a6be578dece00e455c1868ad8b4c833d912ebe02b7f014077f17` remains the record for the original packet, with its tomotope field superseded by `067047bbf3fe04301fcb26bf484` The live intervals remain $389 \le R_{\mathrm{defect}} \le 435$ and $10 \le \chi(H) \le 11$.

21 Monster $U_4(2)$, the Steinberg register, and two exact falsifiers

The structural program uses the genuine subgroup direction

$$PSp(4,3) \cong U_4(2) \longrightarrow \mathbb{M},$$

while keeping character fusion and generator-level embedding distinct. A GAP companion filters class fusions by the documented $5B$-type constraints

$$2 \mapsto 2B, \qquad 3 \mapsto 3B, \qquad 5 \mapsto 5B,$$

and decomposes the restriction of the degree-196883 Monster character.

The protected module has the sharper local interpretation

$$81 = 3^4 = |\operatorname{Syl}_3(U_4(2))|.$$

The unique degree-81 Steinberg character vanishes on every nonidentity 3-power class, so it restricts regularly to a Sylow 3-subgroup.

For

$$N = (12I - A)(A + 4I)$$

the exact W33 computation gives

$$N^2 = 60N, \qquad \operatorname{rank} N = 24, \qquad N_{xx} = 36, \qquad N_{xy} \in \{6, -4\}.$$

Uniform normalization to squared norm four yields off-diagonal products $2/3$ and $-4/9$, so the raw forty-vector frame is not directly an integral Leech subconfiguration.

Also every three-eigenmode W33 adjacency sequence satisfies

$$s_{n+3} = 10s_{n+2} + 32s_{n+1} - 96s_n,$$

while the first $1A$ moonshine coefficients violate this recurrence. Any surviving bridge must therefore be graded, induced, glued, or ambient rather than a raw adjacency-moment identification.

Majorana boundary. For the documented $5B$-type embedding, subgroup involutions fuse to Monster $2B$, whereas standard Monster Majorana axes correspond to $2A$. A direct subgroup-involution-to-axis assignment is blocked.

Evidence boundary. Character-table fusion is not a concrete `mmgroup` embedding. Promotion requires serialized Monster generators, exact relations, class fusion, and an independent image-order certificate 25920.

22 Octad pattern census, Borel core/bridge reduction, and marked tomography

The pair-intersection multiplicities of eight abstract one-factorization rows are exactly edge-disjoint clique packings of K_8. Canonical augmentation gives

$$\boxed{69}$$

S_8-orbits of abstract multiplicity patterns, with orbit counts

$$1, 6, 10, 10, 13, 14, 8, 6, 1$$

by the number of nontrivial blocks. This is a complete abstract pattern census, not yet the complete realization census inside the 6,240 labelled one-factorizations of K_8.

The two surviving 19:9 Moore-action profiles possess a literal common binary core. Its variable classes have sizes

$$18 \cdot 85 = 1530, \qquad \binom{18}{2} \cdot 171 = 26163, \qquad 9 \cdot 18 \cdot 19 = 3 \cdot 18 \cdot 57 = 3078,$$

so the shared core has

$$\boxed{30771}$$

variables. The P_{19} profile is exactly this core after presolve, while P_{57} adds eighteen nonneighbor bridge variables:

$$\boxed{30789 = 30771 + 18}.$$

No SAT or UNSAT verdict is inferred.

The conductor-19 component of the Perkel orbital algebra is realized by explicit matrix units in

$$M_3\big(\mathbb{Q}(\sqrt{-19})\big).$$

Writing $s^2 = -19$, the units obey

$$E_{ij}E_{kl} = \delta_{jk}E_{il}, \qquad (sE_{ij})(sE_{kl}) = -19\delta_{jk}E_{il}.$$

The transpose involution is represented by the positive Hermitian form

$$H = \begin{pmatrix} 1 & 2s/19 & -2s/19 \\ -2s/19 & 1 & 2s/19 \\ 2s/19 & -2s/19 & 1 \end{pmatrix},$$

whose two-by-two principal minors equal 15/19 and whose determinant is 7/19.

A real proof-carrying archive of sixteen genuine star-complement compatibility graphs is frozen with clique histogram

$$8^1, 9^1, 10^1, 11^3, 12^1, 13^2, 16^2, 31^5$$

and ordered Merkle root

`c176539c7b944274fb64965275ce3f900852fe4b26f67923b2ca6a2f06e256a4`.

Every upper-bound DAG is independently checked from its frozen adjacency bitsets. The complete 3,720-instance archive remains resource-gated.

Finally, the minimum marked-resolvent rank that distinguishes W33 from Gewirtz is three. Rank one is common to both graphs, and rank two sees only edges versus nonedges. At rank three, W33 has a triangle while Gewirtz is triangle-free. Every W33 triangle reconstructs a unique K_4 line because an adjacent pair has exactly two common neighbors.

Evidence boundary. The degree-57 Moore graph, realized K_8 octad orbit census, Borel solver verdicts, complete proof archive, PDF evidence, and physical interpretation all remain open unless separately executed and observed.

23 Passes 3635–3648: exact $U_4(2)$ completion and rank-24 modular closure

The W33 projective action now has a compact exact certificate. Four order-three symplectic transvections generate a group of order

$$25{,}920 = |PSp(4,3)| = |U_4(2)|,$$

with element-order census

$$1^1,\ 2^{315},\ 3^{800},\ 4^{3780},\ 5^{5184},\ 6^{5760},\ 9^{5760},\ 12^{4320}.$$

Every three-transvection face has order 648 and fixes one W33 point, while an explicit pair has orders $(2, 5, 12)$ and generates the whole group.

The rank-24 projector carrier is bilinearly complete. Its symmetric tensors have rank

$$300 = \dim \mathrm{Sym}^2(\mathbf{Q}^{24}),$$

and its alternating tensors have rank

$$276 = \dim \Lambda^2(\mathbf{Q}^{24}).$$

Thus the carrier spans all $24^2 = 576$ matrix directions. This is a full linear operator envelope, not yet a Griess or VOA product certificate.

The integral projector Gram form has Smith profile

$$2^1, \qquad 10^9, \qquad 30^6, \qquad 60^8,$$

and discriminant valuation

$$2^{32}3^{14}5^{23}.$$

The odd 5-adic exponent survives every rational scalar rescaling in rank 24, so no scalar-rescaled copy can possess a unimodular overlattice. Any Leech route must use non-scalar coupling, quotienting, or a multi-lattice extension.

The exact graded completion is

$$\left(E_4^3 - 720\Delta\right) \prod_{n\geq 1} (1 - q^n)^{-24},$$

whose initial shifted coefficients are

$$1,\ 24,\ 196884,\ 21493760,\ 864299970,\ 20245856256.$$

This identifies the minimum machinery missing from raw W33 adjacency moments: a rootless rank-24 modular numerator and a 24-oscillator denominator.

Two additional chamber structures emerge. The full group contains 432 copies of A_5 in two orbits of 216. Their D_{10}-sharing graph is

$$36K_{6,6},$$

with 1296 edges; every edge generates A_6, and each chamber has normalizer of order 720, hence S_6. Separately, the four transvections form a tetrahedral amalgam whose four faces are order-648 point stabilizers.

Evidence boundary. The exact Python certificate proves the abstract $U_4(2)$ carrier, the complete $A_5/A_6/S_6$ census, both tensor-span ranks, the scalar glue obstruction, and the modular coefficients. A companion GAP job enumerates compatible $5B$-containing class fusions and the possible multiplicities of the degree-81 constituent. No serialized Monster-word embedding, unique class fusion, observed degree-81 multiplicity, Leech embedding, Griess product, or physical result is promoted before its dedicated certificate is observed.

23.1 Cubic integrality, Steinberg closure, and the corrected tomotope lift

For the cubic dependency incidence matrix $T \in \{0,1\}^{240\times 5040}$, the uniform primal and dual witnesses have value 80:

$$x = \tfrac{1}{3}\mathbf{1}_{240}, \qquad y = \tfrac{1}{63}\mathbf{1}_{5040}.$$

Thus the fractional transversal number is 80. An integral 80-set would make every covering constraint tight, $T^T x = \mathbf{1}$. Since $TT^T = 63I + D$ has minimum eigenvalue 45 and full rank 240, the unique real solution is $x = \frac{1}{3}\mathbf{1}$, impossible for a binary vector. Hence

$$\boxed{\tau \geq 81}, \qquad \tau/\tau^* \geq 81/80.$$

This is a cubic integrality theorem, not a covering-radius endpoint; $389 \leq R_{\text{defect}} \leq 435$ remains live.

The ten orbital matrices of the 240-face action restrict on $E_{81} = -D^3$ to a one-dimensional algebra over $\mathbb{F}_3$, so

$$\mathrm{End}_{PSp(4,3)}(E_{81}) = \mathbb{F}_3.$$

A deterministic Sylow-three subgroup of order 81 acts regularly on this space. Therefore E_{81} is projective; the scalar endomorphism ring forces simplicity, and the explicit image realizes the defining-characteristic Steinberg module. The complementary filtration has dimensions $159 \supset 44 \supset 14 \supset 0$.

For the fourfold 45-anchor blowup, the 27 lifted independent 20-sets, each point occurring three times, give $\chi_f = 9$. The unrestricted positive-gauge weighted-Hoffman dual is therefore exactly nine, attained by $A_{45} \otimes I_4$ with spectrum $32^4, 2^{96}, (-4)^{80}$. Arbitrary signed complex Hermitian weights remain open.

Correction to Passes 3600–3613. The earlier zero count for eight-cell covers is withdrawn: its increasing-index backtracker skipped valid subsets. Complete enumeration of all $\binom{12}{8} = 495$ simple subsets finds exactly three covers, one orbit under the visible order-48 group, with stabilizer 16 and induced action S_3. Each cover has cell adjacency $K_{4,4}$. With

$$A = \{0,3,8,11\}, \quad B = \{1,5,6,10\}, \quad C = \{2,4,7,9\},$$

the covers are $A \cup B, A \cup C, B \cup C$; exhaustive enumeration of all 75,582 multisets adds precisely $2A, 2B, 2C$. This establishes the eight-cell incidence layer, but not yet the full rank-four polytope isomorphism.

A concrete objectwise W33–Clebsch bridge is supplied by the integral matrix

$$Q = \begin{pmatrix} 56A_W - 6J_{40} & 8J_{40,16} \\ 8J_{16,40} & \frac{7}{2}(-3A_C^2 + 18A_C + 17I) + 8J_{16} \end{pmatrix}.$$

It has constant row sum 560 and obeys

$$(Q - 560I)(Q - 112I)(Q + 224I) = 0,$$

so $Q/56$ has Gewirtz spectrum $10^1, 2^{35}, (-4)^{20}$. This is an integral weighted spectral bridge, not a 0/1 graph embedding.

Finally, $C = I + D + E_{81}$ has order three over $\mathbb{F}_3$, with $(C - I)$-power ranks $44, 14, 0$ and Jordan type $J_3(1)^{14} \oplus J_2(1)^{16} \oplus J_1(1)^{166}$. The three corrected tomotope covers form a second exact S_3-selector register. Neither construction is promoted as physical dynamics.

The frozen semantic SHA-256 is `067047bbf3fe04301fcb26bf484a3d151bf6db4ab762fa6bf31f527889d09a8` No remote CI, PDF, FPGA, laboratory, Monster embedding, radius endpoint, or ten-colour result is asserted here.

23.2 The canonical A_6-chamber/spread bridge and the exceptional S_6 double-six

The preceding Monster-facing packet constructs all 432 subgroups isomorphic to A_5, their 36 $K_{6,6}$ components under the D_{10}-sharing relation, and one A_6 chamber per component. Earlier exact work independently identifies the rank-three action of $PSp(4,3)$ on the 36 spreads of $W(3,3)$. The following result supplies the missing objectwise map; it is not an inference from equal cardinalities or strongly regular graph parameters.

Theorem 1 (Canonical chamber/spread bridge)**.** *Let $G = PSp(4,3) \cong U_4(2)$ act on $W(3,3)$, and let $H \cong A_6$ be one of the 36 chambers. Then*

$$|N_G(H)| = 720, \qquad [N_G(H), N_G(H)] = H.$$

The normalizer fixes exactly one spread Σ_H, and

$$N_G(H) = \mathrm{Stab}_G(\Sigma_H).$$

Consequently

$$\boxed{G/N_G(A_6) \longleftrightarrow \{W(3,3) \text{ spreads}\}, \qquad H \longmapsto \Sigma_H,}$$

is a canonical G-equivariant bijection of two 36-element sets.

For distinct chambers H and K, the complete intersection dictionary is

$$\boxed{|H \cap K| = 18 \iff |\Sigma_H \cap \Sigma_K| = 1} \qquad \text{(360 unordered pairs)},$$

and

$$\boxed{|H \cap K| = 12 \iff |\Sigma_H \cap \Sigma_K| = 4} \qquad \text{(270 unordered pairs)}.$$

The order-12 relation is therefore

$$\mathrm{SRG}(36, 15, 6, 6), \qquad \mathrm{spec} = 15^1 \oplus 3^{15} \oplus (-3)^{20},$$

while the complementary order-18 relation is

$$\mathrm{SRG}(36, 20, 10, 12), \qquad \mathrm{spec} = 20^1 \oplus 2^{20} \oplus (-4)^{15}.$$

The graph identification is carried by the explicit equivariant bijection, not by the parameter tuple alone.

Each chamber contains twelve A_5 subgroups, split by the two global A_5 classes as $6 + 6$. Two members of the same six-set intersect in A_4; every cross-pair intersects in D_{10}. Thus the cross-intersection graph is exactly $K_{6,6}$. The normalizer S_6 acts faithfully on both six-sets, and the exact cycle-type census verifies that the two actions are exchanged by the exceptional outer automorphism:

$$(2,1,1,1,1) \longleftrightarrow (2,2,2), \qquad (3,1,1,1) \longleftrightarrow (3,3), \qquad (6) \longleftrightarrow (3,2,1).$$

Their directed multiplicities are respectively 15, 40, and 120 in each direction.

Monster-facing boundary. Any future concrete $5B$-containing $U_4(2)$ embedding in the Monster transports this complete 36-spread/36-chamber carrier and every local double-six chart. The certificate does not provide Monster words, a unique class fusion, a verified multiplicity of the 81-dimensional constituent, or a Griess/VOA/photonic multiplication law. In particular, this rank-three spread carrier is not identified with the distinct 36-ray magic-state orthogonality graph, which is not strongly regular.

24 The chamber–tomotope central lift and thick amalgamation boundary

Passes 3663–3669 established the canonical equivariant identification of the thirty-six A_6 chambers with the thirty-six spreads of $W(3,3)$. The present packet retains that carrier and moves one level upward. The valency-15 chamber relation is

$$\mathrm{SRG}(36,15,6,6), \qquad 15^1 \oplus 3^{15} \oplus (-3)^{20},$$

and contains exactly 135 maximal K_4 subgraphs. Every chamber belongs to fifteen of them, giving

$$36 \cdot 15 = 135 \cdot 4 = 540$$

chamber–K_4 incidences.

The stabilizer of a maximal K_4 has order 192 and centre of order two. Its central quotient has order census

$$1^1\, 2^{27}\, 3^{32}\, 4^{36},$$

and an explicit four-generator word transport identifies it, as a marked group, with the automorphism group of the tomotope. The extension is not split: the stabilizer census

$$1^1\, 2^{43}\, 3^{32}\, 4^{84}\, 6^{32}$$

differs from the census of $C_2 \times \mathrm{Aut}(T)$. This supplies a third exact 192 mechanism: a non-split central tomotope lift, distinct from both the tomotope's 192 flags and the split order-192 universal group.

Let P_i be the four order-648 three-generator parabolics supplied by the explicit $U_4(2)$ transvections. Their pair-intersection multiset is

$$\{24, 24, 24, 24, 24, 54\},$$

every triple intersection has order three, and the total core is trivial. The associated coset geometry has forty objects of each type and $25,920$ chambers, but every panel has thickness three. Its natural panel motion is therefore C_3, not an exchange involution. The construction is an exact thick chamber system rather than an abstract polytope: it fails the diamond condition at the final local step.

For the rank-24 integral projector Gram matrix G, the non-scalar block coupling

$$\mathcal{G} = \begin{pmatrix} G & I \\ I & 0 \end{pmatrix}$$

is even, has determinant one, and has signature $(24, 24)$. Thus the scalar discriminant obstruction can be cancelled exactly, but the canonical closure is the indefinite lattice $II_{24,24}$ rather than a positive-definite Leech lattice.

If V denotes the rank-24 $U_4(2)$ module, exact character inner products give

$$\dim \mathrm{Hom}_{U_4(2)}(\mathrm{Sym}^2 V, V) = 2, \qquad \dim \mathrm{Hom}_{U_4(2)}(\Lambda^2 V, V) = 1.$$

Projected coordinatewise multiplication and projected W33 triangle-neighbour multiplication form a basis of the commutative envelope; both are Frobenius. Their complete associator ideal becomes the unit ideal modulo $1,000,003$, so no nonzero rational member of this envelope is associative. This classifies the available equivariant commutative products without promoting any of them to a Griess or VOA multiplication.

Finally, the prime support $\{2,3,5\}$ of $U_4(2)$ selects the six exact McKay–Thompson targets $2A/2B$, $3A/3B$, and $5A/5B$. With $r = 24/(p-1)$,

$$T_{pB} = \left(\frac{\eta(\tau)}{\eta(p\tau)}\right)^r + r, \qquad T_{pA} = T_{pB} + p^{r/2}\left(\frac{\eta(p\tau)}{\eta(\tau)}\right)^r.$$

These channels refine the scalar $J + 24$ target but do not by themselves construct a Monster module.

A final spectral firewall is exact. The W33 (-4) eigenspace and the chamber-complement (-4) eigenspace are both irreducible and fifteen-dimensional, yet their $U_4(2)$ characters have inner product zero. They are inequivalent modules; matching dimension and graph eigenvalue do not supply an equivariant intertwiner.

The concrete Monster embedding remains fail-closed. The new harness requires four serialized Monster elements of order three, pair-product orders $3,6,6,6,6,6$, four order-648 triple closures, full closure order $25,920$, and an independently hashed class-fusion certificate before promotion.

24.1 Realized octads, the exact Borel bridge, and architecture-level firewalls

The abstract K_8 octad-pattern deck has 69 S_8-orbits. Exact certificates now realize 59 of them inside the $6,240$ labelled one-factorizations; ten sharply identified patterns remain open. The 18 additional thick variables of the P_{57} Borel profile collapse under the Moore common-neighbour screen to only

$$1 + 18 + 36 = 55$$

locally admissible masks, all of weight at most two. Their compatibility graph is $3K_{2,2,2}$.

The real symmetric Perkel commutant has the complete positive-cone model

$$\mathbb{R}_+ \times \mathrm{Herm}_3^+(\mathbb{C}) \times \mathbb{R}_+.$$

It has two scalar extreme rays plus the rank-one rays zz^* parametrized by $\mathbb{CP}^2$, sixteen rank strata, algebraic boundary $\alpha\beta\det Q = 0$, and Carathéodory number five. The proof archive now contains 32 content-addressed, independently checkable clique proof DAGs. Exhaustive marked-resolvent tomography through rank five gives W33-exclusive signature counts $0,0,1,3,10$, while Gewirtz has no exclusive signature in that range.

Two architecture-level corrections are more consequential. An exactly-one-photon carrier has only M dimensions in M orthogonal modes, so representing n independent qutrits requires $M \geq 3^n$. Moreover the total-loss Knill–Laflamme products satisfy $a_i^\dagger a_j = |i\rangle\langle j|$ and span the full matrix algebra. Their compression can all be scalar only on a one-dimensional code. Thus no nontrivial logical code confined to exactly one photon corrects arbitrary total photon loss. The abstract $[[240,81,4]]_3$ code therefore requires an explicit multi-photon, bosonic, teleportation/cluster, or quantum-memory realization before it can be called loss tolerant.

Conversely, independently addressable local phases and real edge couplers on any connected d-mode graph generate $\mathfrak{u}(d)$. W33, a path, and a cycle on forty modes all yield dimension $40^2 = 1,600$ under that idealized control model. Bare one-photon universality is therefore generic rather than W33-specific. The geometry's architectural claim must be tested through fault-tolerant overhead, routing, calibration, noise, code, or component advantages.

Evidence boundary

Fifty-nine octad patterns are realized, not all sixty-nine. The Borel result is a local presolve, not a Moore-graph verdict. Thirty-two proof DAGs are frozen, not all $3,720$. The loss and controllability results are exact conditional architecture theorems; laboratory performance remains unmeasured.

24.2 The spread ETF and its canonical Norton algebra (Passes 3694–3700)

Let A be the graph on the 36 spreads of $W(3,3)$, adjacent when two spreads share four lines. Its rational primitive idempotents are

$$E_{15} = \tfrac{1}{2}I + \tfrac{1}{6}A - \tfrac{1}{12}J, \qquad E_{20} = \tfrac{1}{2}I - \tfrac{1}{6}A + \tfrac{1}{18}J.$$

If B is the 40×36 line–spread incidence matrix and $C = B - \frac{1}{4}J_{40\times 36}$, then

$$\boxed{C^{\mathsf{T}}C = 18E_{15}}$$

shows that the normalized columns form a real ETF$(15, 36)$. Their mutual inner products are $+1/5$ for a four-line intersection and $-1/5$ for a one-line intersection, saturating the Welch bound. The Seidel matrix $S = I + 2A - J$ obeys $S^2 - 2S - 35I = 0$ and has spectrum $7^{15} \oplus (-5)^{21}$.

The complementary projector $I - E_{15}$ has rank 21. Thus an exact passive unitary dilation of the 36-mode frame analysis requires a 21-dimensional guard sector; this is a dimension certificate, not an experimental claim.

On $V = \operatorname{im} E_{15}$ define the Norton product $x \star y = E_{15}(x \circ y)$ and axes $a_i = 6E_{15}e_i$. The 36 axes are primitive idempotents. Each axis has multiplication spectrum

$$1^1 \oplus (-\tfrac{1}{2})^5 \oplus (\tfrac{1}{6})^9,$$

with the exact $\mathbb{Z}_2$-graded fusion law

$$(-\tfrac{1}{2}) \star (-\tfrac{1}{2}) \subseteq 1 \oplus \tfrac{1}{6}, \quad (-\tfrac{1}{2}) \star \tfrac{1}{6} \subseteq -\tfrac{1}{2}, \quad \tfrac{1}{6} \star \tfrac{1}{6} \subseteq 1 \oplus \tfrac{1}{6}.$$

This is a positive Frobenius axial algebra but not a Monster/Majorana algebra. For adjacent axes, $a_i \star a_j = (a_i + a_j)/6$. Every nonadjacent pair has a unique third axis a_k satisfying

$$a_i \star a_j = -\frac{a_i + a_j}{6} + \frac{a_k}{3}.$$

These products define 120 triples. Each triple consists of three spreads sharing one W33 line; for every W33 line, its nine containing spreads split into three disjoint triples.

The 36 Witting-ray orthogonality graph is a different object: it is 11-regular with spectrum $11^1 \oplus 2^{20} \oplus (-1)^3 \oplus (-4)^{12}$, whereas the spread graph is 15-regular with spectrum $15^1 \oplus 3^{15} \oplus (-3)^{20}$. Finally, the ternary panels of the four-parabolic chamber system cannot be changed to binary panels by an unbranched type-preserving cover, because such covers are locally bijective on rank-one residues.

24.3 Passes 3701–3714: the D_4/F_4 triality carrier and the 45-axis firewall

The 135 maximal four-cliques of the 36-chamber graph admit an exact three-to-one quotient. The stabilizer of each four-clique acts on the forty W33 points with an eight-point orbit; precisely three pairwise-disjoint four-cliques determine the same octad. Hence

$$135 = 45 \cdot 3.$$

Two octads meet in zero or two points, and octad disjointness is

$$\mathrm{SRG}(45,12,3,3), \qquad \mathrm{spec} = 12^1 \oplus 3^{20} \oplus (-3)^{24}.$$

The centered octad vectors, with entries 48 on the octad and -12 off it, have norm 23040, off-diagonal products -5760 or 1440, and Gram spectrum

$$43200^{24} \oplus 0^{21}.$$

Thus they form a rank-24 two-distance tight frame.

The 135 four-cliques are exactly the 135 Lagrangian three-spaces of $W(5,2)$, restricted to the 36 anisotropic points of a minus quadratic form. Their binary incidence matrix has rank 29, and its dual is the affine restriction code

$$[36,7,16]_2, \qquad W(z) = 1 + 63z^{16} + 63z^{20} + z^{36}.$$

A frame stabilizer is

$$C_2^3 \rtimes S_4 \cong W(D_4), \qquad |W(D_4)| = 192.$$

It has an explicit S_4 complement. Its central quotient is non-split: the section cocycle system has coefficient rank 95 and augmented rank 96. The common normalizer of the three frames over one octad has order 576, while the stabilizer in the outer extension $U_4(2){:}2$ has order 1152. A colored-orbital conjugacy on a faithful 24-point orbit identifies the latter with the short-root action of $W(F_4)$. Therefore

$$W(D_4)\ (192) < W(F_4)_{\text{even frame}}\ (576) < W(F_4)\ (1152).$$

The ternary panels do not admit a string-Coxeter resolution preserving the exact four-generator target. Each generator has 108 inverting outer involutions, but no outer involution inverts all four. In every two-involution-per-colour factorization, each pair of colours contributes at least two cross-colour noncommutations. The resulting minimum is 16 noncommuting edges, whereas a rank-eight string diagram has only seven.

For the 45 canonical octad axes, the two basis products of the complete equivariant commutative Frobenius plane satisfy

$$m_0(u,u) = 36u, \qquad m_1(u,u) = -216u.$$

The simultaneous left-eigenpairs are

$$(36,-216)^1, \quad (28,152)^6, \quad (-12,-168)^8, \quad (-12,72)^9.$$

No rational parameter gives the Monster $2A$ Majorana non-axis spectrum $\{0, 1/4, 1/32\}$.

Finally, for

$$\mathcal{G} = \begin{pmatrix} G & I \\ I & 0 \end{pmatrix},$$

the graph of an integral matrix B has Gram matrix $G + B + B^{\mathsf{T}}$. Every even integral rank-24 Gram matrix occurs in this way; E_8^3 is an explicit positive unimodular child. The standard existence theorem for the Leech lattice therefore gives a noncanonical rootless graph section, but no explicit Leech basis is frozen here. Equivariance is impossible: irreducibility forces $B = mI$, and no such scalar section is simultaneously positive definite and unimodular.

Concrete Monster words, a regular thin polytope cover, an explicit canonical Leech section, and a Majorana/Griess/VOA realization remain certificate-gated.

25 Passes 3715–3721: carrier replacement and control-plane architecture

The literal exactly-one-photon register is replaced by an explicit qutrit erasure carrier. The code

$$|i_L\rangle = 3^{-1/2} \sum_{j\in\mathbf{F}_3} |j, j+i, j+2i\rangle$$

satisfies all Knill–Laflamme conditions for one known physical-qutrit erasure and realizes $[[3,1,2]]_3$. Eighty-one logical qutrits therefore require 243 physical qutrits; under one-photon triple-rail encoding this is 243 photons in 729 modes. For independent transmissivity η, the all-block success condition

$$[\eta^3 + 3(1-\eta)\eta^2]^{81} \geq 2/3$$

gives $\eta \geq 0.958628272258342$.

An exhaustive degree-matched tournament checks all $\binom{19}{6} = 27,132$ twelve-regular circulants on forty vertices. W33 and the best route circulants share diameter two and average distance 22/13, but W33 alone has exactly four common neighbours for every nonedge and zero redundancy variance; its spectral gap is 10, above the best circulant value found, approximately 9.280621379.

The proposed temporal loop is formalized as a causal qutrit process tensor: a qutrit SWAP stores the past state, a middle causal break replaces the exposed system, and a second SWAP retrieves the past. This is non-Markovian memory without future-to-past signalling. Its memory bond dimension is three for one qutrit and at least 3^{81} for an arbitrary eighty-one-qutrit state.

At 1550 nm the minimum energy in the 243-photon carrier is 3.1142×10^{-17} J per shot. A 162-bit erasure record has a 300-K Landauer floor 4.6510×10^{-19} J per cycle. At group index four, feedforward delays of 1, 10, and 100 ns require approximately 7.49 cm, 74.95 cm, and 7.49 m of passive delay.

A complete forty-mode intensity-interference identification uses 1600 input settings. A familywise Hoeffding bound with $\epsilon = \delta = 0.01$ requires 81,825 shots per setting, or 130,920,000 shots total. The device must recover the SRG identity, forty lines, 160 line triangles, and the frozen marked-resolvent atlas against degree-matched controls.

Two control-plane theorems change the architectural interpretation. First, the 160 point-line incidences decompose into four perfect matchings, giving an optimal four-tick collision-free syndrome schedule. Second, the order-25,920 symplectic group has exactly three ordered-pair orbits, so automorphism twirling compresses a generic 1600-parameter Hermitian calibration model to

$$aI + bA + c(J - I - A).$$

Thus W33 need not be the literal protected Hilbert-space carrier to remain operationally distinguished: it can serve as an optimal syndrome scheduler and a rank-three calibration/noise compressor. These are exact finite results; hardware cost, executed tomography, wall-plug power, and fault-tolerant thresholds remain evidence-gated.

26 Fischer closure, a Hadamard Naimark transform, and the six-bit orthogonal core

The thirty-six primitive Norton axes from the spread ETF carry canonical Miyamoto involutions. The exact pair census is

$$270 \text{ commuting pairs} + 360 \text{ pairs whose product has order 3.}$$

For every noncommuting pair, conjugation produces the unique third axis in its Norton triple. Consequently the 120 Norton triples are exactly the Fischer lines of a three-transposition space.

Seven involutions generate a group of order 51,840, while the subgroup generated by even products has order 25,920 and the previously certified $U_4(2)$ element-order census. Thus the internal action is

$$O_6^-(2) \cong U_4(2){:}2, \qquad \Omega_6^-(2) \cong U_4(2).$$

This is an abstract finite-group identification, not a serialized Monster embedding.

Let A be the adjacency matrix of the spread graph $\mathrm{SRG}(36, 15, 6, 6)$. The matrix

$$K = 2A - J$$

is a symmetric regular Hadamard matrix:

$$KK^{\mathsf{T}} = 36I, \qquad K\mathbf{1} = -6\mathbf{1}.$$

Hence $H = K/6$ is an exact rational orthogonal involution and

$$E_{15} = \frac{I + H}{2}, \qquad E_{21} = \frac{I - H}{2}.$$

The rank-15 signal and rank-21 guard sectors therefore arise from one explicit uniform-amplitude two-phase thirty-six-port transform. This is a transfer-matrix certificate, not a laboratory claim.

The associative algebra generated by the fifteen-dimensional left-multiplication operators has dimension 225 modulo 1,000,003, while the derivation equations have rank 225. Therefore the rational Norton algebra is simple and

$$\mathrm{Der}(V) = 0.$$

Adjacent pairs generate a two-dimensional algebra with idempotents $0, a, b, \frac{3}{4}(a + b)$; nonadjacent pairs generate a three-dimensional algebra containing the unique third Fischer axis c, with idempotents $0, a, b, c, a + b + c$. The fusion spectrum remains non-Majorana and non-Griess.

If D is the 36×120 axis–Fischer-line incidence matrix, then its binary left kernel is a $[36, 6]$ code with weight enumerator

$$1 + 27z^{16} + 36z^{20}.$$

The quadratic form $q(x) = \mathrm{wt}(x)/4 \pmod 2$ is nondegenerate of minus type. Its 27 nonzero singular vectors recover $\mathrm{SRG}(27, 10, 1, 5)$, while its 36 nonsingular vectors are exactly the Miyamoto supports and recover the spread graph. Thus the 27- and 36-object carriers are the two nonzero sectors of one explicit six-bit orthogonal space.

Finally, the intersection graph of the 120 Fischer lines has degree 27, spectrum

$$27^1 \oplus 9^{20} \oplus 3^{15} \oplus (-3)^{84},$$

and distance distribution $1+27+90+2$ from every vertex. Its forty antipodal fibers are the three triples through each W33 line, and its quotient is $\mathrm{SRG}(40, 27, 18, 18)$. The lift is a connected three-cover with full S_3 holonomy and trivial deck group; it is not a regular C_3 cover.

26.1 Cubic-transversal closure, tomotope completion square, and website restoration

The long-form public site was restored byte-for-byte from the final intact blob `41a8d733f42da18282fa276f5d2fa` the compact replacement was preserved separately at `docs/source-derived-architecture-landing-2026-08-`

For the 240×5040 cubic dependency incidence matrix T, put $D = TT^{\mathsf{T}} - 63I$. If a transversal has size m and $P = \sum_t \binom{h_t}{2} = \frac{1}{2}x^{\mathsf{T}}Dx$, then the minimum eigenvalue -18 of D gives

$$P \geq \frac{3}{10}m^2 - 9m,$$

while $h_t \in \{1, 2, 3\}$ gives

$$P \leq \frac{3}{2}(63m - 5040).$$

Hence $(m-105)(m-240) \le 0$. Equality at $m = 105$ would make the integer P equal to $4725/2$, so $m \ge 106$. An explicit 62-face cap gives a 178-face transversal, and therefore

$$\boxed{106 \le \tau_{\text{cubic}} \le 178}.$$

The covering-radius interval remains $389 \le R \le 435$.

The 45-anchor graph has independence number five and fractional chromatic number nine. The generalized weighted-Hoffman optimum over arbitrary edge-supported Hermitian matrices is therefore exactly nine, attained by $A_{45} \otimes I_4$. This does not change $10 \le \chi(H) \le 11$.

The corrected edge–face layer admits three vertex-structure orbits and three cell covers. All nine pairings are connected rank-four incidence geometries with f-vector $(4, 12, 16, 8)$, 192 flags, and all diamond intervals of size two. Their automorphism-order matrix is

$$\begin{pmatrix} 96 & 96 & 192 \\ 192 & 96 & 96 \\ 96 & 192 & 96 \end{pmatrix}.$$

Thus six completions are two-flag-orbit tomotopes with automorphism order 96, while a perfect matching of three completions gives regular non-split central lifts of order 192.

The characteristic-three complement has filtration

$$159 \supset 44 \supset 14 \supset 0.$$

The bottom 14-space is absolutely irreducible. The middle quotient satisfies the non-split exact sequence

$$0 \longrightarrow 1 \oplus 5 \oplus 10 \longrightarrow M_{30} \longrightarrow 14 \longrightarrow 0.$$

The top 115-quotient has fixed-socle dimension two and trivial-head multiplicity one; its remaining factors are not labelled here.

Finally, all 32 full-orbit binary roundings of the integral weighted W33–Clebsch bridge were exhausted. None is 10-regular and none satisfies the Gewirtz identity $A^2 = 8I - 2A + 2J$. Asymmetric rounding remains open.

Two further constructions follow. The 3×3 tomotope completion square has six order-96 entries and a perfect matching of three order-192 entries. The explicit 62-cap has trivial stabilizer in $U_4(2)$, producing a free orbit of 25,920 cap witnesses.

The frozen semantic SHA-256 is `c6e5e73fb5a18d9add4c1643d52df31413280b0e5cd157294ca686f8df32a29`
No remote CI, PDF, hardware, laboratory, or physical claim is promoted here.

27 Symmetry operating system, space–time decoding, and falsifiable architecture

Passes 3743–3750 convert the W33 machine from an abstract universal single-particle carrier into an explicitly testable control-plane architecture. The executable certificate has semantic digest

`67ce2d0ab105d8809dd6ae1af60381e1181d9a08c9032ce1dad1130f7bb8ff62`

and is reproduced by the focused regression suite.

Symmetry operating system. Four projective symplectic transvections generate the full order-25,920 point action. Their Cayley compiler has word diameter 11 and mean word length 8.17507716049; a complete exact group sweep uses 211,898 generator macros. This is an exact permutation compiler, not yet a calibrated optical pulse compiler.

Correlated space–time decoding. For the declared two-state hidden-Markov benchmark, eight consecutive observations have exact MAP error $1.1395646255 \times 10^{-3}$, whereas observations separated by the four-tick incidence schedule have error $6.3819353522 \times 10^{-5}$. Thus the interleaved error is 0.0560032771 of the consecutive value. The result is model-specific and does not promote an empirical device threshold.

Hybrid resource factory. The protected resource graph contains forty point registers, forty line-check registers, and one frame/clock register, with $160 + 80 = 240$ logical links. Under the verified $[[3, 1, 2]]_3$ block carrier this requires 243 physical qutrits in 729 triple-rail modes. Exhaustive search over 3^9 bilinear couplings proves that each logical controlled phase needs at least three physical phase terms, so all logical links require at least 720 such terms. A single frame bus has serial-depth lower bound 80.

Grand matched-geometry tournament. All 167,356 degree-twelve Cayley generator sets on $\mathbb{Z}_2^3 \times \mathbb{Z}_5$ were enumerated; 165,984 are connected. Their largest spectral gap is $9.527864045 < 10$, the exact W33 gap. A competitor attains minimum four common neighbours on nonedges, but with histogram $4^{18}6^{8}12^{1}$ rather than W33's uniform 4^{270}. Frozen layout witnesses also show that W33 does not minimize the tested Manhattan wirelength proxy.

Preregistered hardware challenge. A prospective three-device comparison locks structural, transport, twirl, and scheduling endpoints before data. W33 is falsified if its recovered support violates $A^2 = 8I - 2A + 4J$, if it fails to beat both matched controls by 2×10^{-4} on the locked worst-nonedge task, if the twirled residual exceeds ten percent, or if the four-tick schedule hides serialization or unreported resources.

Three outside-lane results. The relative permutations of the four incidence matchings generate the full symmetric group S_{40}, so the schedule is a universal classical permutation microcode, not a coherent quantum gate set. Exact unweighted automorphism twirling requires a sample count divisible by $\mathrm{lcm}(40, 480, 1080) = 4320$; this lower bound is not sufficient, and the known exact sweep uses all 25,920 automorphisms. Finally, passive fixed-adjacency dynamics lie in $\mathrm{span}\{I, A, J\}$, while zero forcing gives the certified control-port interval

$$24 \leq Z(W33) \leq 29.$$

Thus symmetry provides compression only when accompanied by an explicit symmetry-breaking control layer.

Evidence boundary. No optical pulse calibration, measured correlated-noise law, complete fault-tolerance threshold, minimum physical layout, coherent S_{40} universality, exact minimum twirl design, or exact zero-forcing number is claimed by these computations.

28 The six-bit quadratic parent, the rooted code lattice, and the Fischer holonomy frame

The thirty-six spreads and their 120 Norton triples admit a further exact closure. The Miyamoto involutions generate

$$O_6^-(2) \cong U_4(2){:}2, \qquad \Omega_6^-(2) \cong U_4(2),$$

and the compressed standard pairs can be supplemented by complete cycle and class-size fingerprints on both the thirty-six axes and the 120 Fischer lines. These are abstract finite-group search filters; no serialized Monster words are asserted.

Let D be the 36×120 axis–triple incidence matrix. Its binary left kernel is the doubly-even self-orthogonal code

$$C = [36, 6, 16], \qquad W_C(z) = 1 + 27z^{16} + 36z^{20}.$$

The associated Construction-A lattice

$$L(C) = 2^{-1/2}\{z \in \mathbb{Z}^{36} : z \bmod 2 \in C\}$$

is even, has minimum norm two, and satisfies

$$\mathrm{SNF}(G) = 1^{12}2^{24}, \qquad L(C)^*/L(C) \cong (\mathbb{Z}/2\mathbb{Z})^{24}.$$

Its root system is A_1^{36}, so the lattice is explicitly rooted and non-unimodular rather than Leech-like.

If A is the spread graph adjacency matrix, then

$$K = 2A - J, \qquad H = K/6$$

is the symmetric rational Hadamard involution of the previous section. There exist twenty-one pairwise orthogonal primitive vectors $v_i \in \mathbb{Z}^{36}$ such that the commuting Householder reflections

$$R_i = I - 2\frac{v_i v_i^{\mathsf{T}}}{v_i^{\mathsf{T}} v_i}$$

satisfy the exact factorization

$$H = R_1 R_2 \cdots R_{21}.$$

This is a transfer-matrix factorization and not a fabricated optical-device claim.

The same code identifies a six-dimensional minus-type quadratic space. Its thirty-six nonsingular vectors form a $(64, 36, 20)$ Hadamard difference set. For the full bilinear Walsh matrix W_{64} and its nonsingular principal minor W_N,

$$W_{64}W_{64}^{\mathsf{T}} = 64I, \qquad K = W_N - 2I.$$

Thus the thirty-six-port transform is a diagonal shift of a quadratic-Walsh principal minor.

Integrally, the triple incidence has

$$\mathrm{SNF}(D) = 1^{30}2^5 6, \qquad \mathrm{coker}(D) \cong (\mathbb{Z}/2\mathbb{Z})^5 \times \mathbb{Z}/6\mathbb{Z}.$$

The order 192 is not promoted to a tomotope identification without an objectwise map.

Finally, the 120-vertex Fischer-line graph is not distance regular: distance-two pairs split into two intersection profiles. Its exact symmetry group is $U_4(2){:}2$, a vertex stabilizer has subdegrees $1, 2, 27, 36, 54$, and the Terwilliger algebra has dimension 55 modulo three independent large primes. Projecting the three-point fibers to their standard S_3 sectors produces a rank-twenty tight-frame projector

$$E_{20} = \frac{FAF + 9F}{108}, \qquad (FAF + 9F)^2 = 108(FAF + 9F),$$

whose 120 vectors have normalized inner products in

$$\left\{-\frac{1}{2}, -\frac{1}{6}, 0, \frac{1}{3}\right\}.$$

The Monster class-fusion computation is delegated to a self-reporting GAP/CTblLib workflow. No fusion count or degree-81 multiplicity is asserted until its committed execution artifact is observed.

28.1 Passes 3769–3786: the $GQ(4,2)$–Veldkamp–W33 closure

The six-dimensional minus quadratic space over $\mathbb{F}_2$ supplies 27 singular and 36 nonsingular nonzero vectors. Its singular points and totally singular lines form $GQ(2,4)$, with 27 points and 45 three-point lines; the dual $GQ(4,2)$ has 45 points and 27 five-point lines. The dual point graph is

$$\mathrm{srg}(45,12,3,3), \qquad \mathrm{Spec}(A)=\{12^1,3^{20},(-3)^{24}\}.$$

Writing $A_0=I$, $A_1=A$, and $A_2=J-I-A$, its Bose–Mesner multiplication is

$$A_1^2=12A_0+3A_1+3A_2, \quad A_1A_2=8A_1+9A_2, \quad A_2^2=32A_0+24A_1+22A_2.$$

The primitive-idempotent ranks are $1,20,24$. The Terwilliger algebra at a point has dimension 16, center dimension 5, and standard-module central block multiplicities $2,3,8,14,18$, hence semisimple type

$$M_3\oplus M_2\oplus\mathbb{C}^3.$$

All 651 projective lines of $PG(5,2)$ split by quadratic type as

$$45+216+270+120.$$

The final 120 lines consist of three nonsingular vectors and coincide exactly with the Norton–Fischer triples of the preceding spread algebra. Thus that triple system is the type-IV line class of the Veldkamp space $V(GQ(2,4))$.

An exact-cover enumeration finds 200 ovoids of $GQ(4,2)$, in automorphism orbits $40+160$. On the 40 plane ovoids, declare two objects adjacent when they share three quadrangle lines. The resulting graph is

$$\mathrm{srg}(40,12,2,4)$$

and is explicitly isomorphic to the independently reconstructed W33 point graph. If M is the 40×45 plane-ovoid incidence matrix, then

$$MM^{\mathsf{T}}=8I+2A_{W33}+J, \qquad \mathrm{Spec}(MM^{\mathsf{T}})=\{72^1,12^{24},0^{15}\}.$$

The recurring count $135=45\cdot 3$ supports two inequivalent $U_4(2)$ actions. The Lagrangian-frame stabilizer has order census

$$1^1\,2^{43}\,3^{32}\,4^{84}\,6^{32},$$

whereas an ordinary $GQ(4,2)$ flag stabilizer has

$$1^1\,2^{27}\,3^{32}\,4^{36}\,6^{96}.$$

The corresponding permutation-character norms are 9 and 8, with cross-inner product 4.

The rank-24 primitive projector is

$$P=(A-12I)(A-3I)=90E_{-3},$$

with entries 48 on the diagonal, -12 on adjacent pairs, and 3 on nonadjacent pairs. On $\operatorname{im}P$ define

$$x\star y=E_{-3}(x\circ y), \qquad a_i=Pe_i/21.$$

Then the 45 vectors a_i are idempotent axes with Peirce spectrum

$$1^1, \qquad (1/7)^{14}, \qquad (-1/3)^9.$$

For collinear axes,

$$a_i\star a_j=-\frac{1}{3}(a_i+a_j),$$

and the five axes on every quadrangle line sum to zero. For noncollinear i, j with common neighbors c_1, c_2, c_3,

$$a_i \star a_j = \frac{1}{7}(a_i + a_j) + \frac{5}{42}(a_{c_1} + a_{c_2} + a_{c_3}).$$

The algebra is positive Frobenius; its multiplication operators span all M_{24} modulo 1000003, so it is simple, and its derivation algebra is zero because 1/2 is absent from every axis spectrum. Its fusion law is not nontrivially $\mathbb{Z}_2$-graded, so this is not a Monster-2A Majorana algebra.

For a representative $W(D_4)$ frame stabilizer, the invariant alternating forms modulo two have dimension seven and rank distribution

$$0^1, 4^3, 6^4, 8^{12}, 10^{12}, 14^{40}, 16^{56}.$$

No invariant form has rank 24; consequently no $W(D_4)$-invariant even unimodular rank-24 child exists. A deterministic trace-eight involution nevertheless permits an integral determinant--1 basis change that pulls back E_8^3 to an even unimodular positive child whose exact surviving $U_4(2)$ stabilizer is C_2. This child has roots and is not promoted as a Leech realization.

The finite-group embedding front remains fail-closed. A future candidate must reproduce the complete 36/45/27/120/135/40 fingerprint, independent order 51840, class correspondence, and content-addressed incidence digests before any Monster embedding is asserted.

28.2 Exact twirl estimation, Floquet cancellation, distributed clocking, and control closure (Passes 3787–3794)

The W33 control architecture now has exact finite resource contracts beyond the preceding symmetry-operating-system packet. The results below distinguish algebraic control statements from physical pulse and device claims.

Minimum exact observable twirl. The ordered-pair action has orbit sizes 40, 480, and 1080. Exact unweighted estimation of the diagonal, oriented-edge, and oriented-nonedge averages therefore needs at least

$$40 + 480 + 1080 = 1600$$

settings, and one shortest automorphism word per target attains the bound. The total compiled cost is 8934 generator macros, versus 211898 for the complete group sweep. This closes observable/orbit estimation, not the stronger active conjugation-channel twirl, whose unweighted divisibility lower bound remains 4320.

Floquet cancellation. The four incidence matchings induce relative cycle types $(20, 20)$, (40), and (40) and generate S_{40}. For either relative forty-cycle C and any diagonal error Hamiltonian D,

$$\frac{1}{40}\sum_{t=0}^{39} C^{-t} D C^{t} = \frac{\operatorname{tr} D}{40} I.$$

Hence all traceless static diagonal disorder cancels exactly over one orbit. This is not a general coherent-error or loss theorem.

Optimal distributed clock. On the eighty-node point–line incidence network, one matching per tick gives the exact informed-node census

$$1, 2, 4, 8, 16, 30, 54, 80.$$

All nodes receive the frame in seven ticks. The information-doubling lower bound $\lceil \log_2 80 \rceil = 7$ proves optimality in the declared store-and-forward model, replacing the prior eighty-step central-bus serialization.

Exact control-port count. A complete automorphism-orbit enumeration of reverse zero-forcing sequences has orbit counts

$$1, 1, 2, 5, 16, 43, 191, 769, 3024, 9772, 24852, 34890$$

through depth eleven. Every depth-eleven orbit is terminal and depth twelve is empty. Therefore

$$\boxed{Z(W33) = 29}.$$

An explicit minimum set and eleven-step force chain are frozen in the executable certificate. The graph-control consequence retains the standard Hamiltonian assumptions and is not a pulse-robustness result.

Dynamic W33/local phase. An explicit relabelling of the degree-twelve local ring and W33 shares 128 of their 240 couplers. The reconfigurable union therefore uses

$$240 + 240 - 128 = 352$$

couplers, a 26.67% reduction from disjoint hardware. The cross-phase two-step operator has pairwise minimum two, while the complete two-phase walk has minimum four. The witness is not claimed globally optimal.

Three further mechanisms. First, the exact zero-forcing chain virtualizes eleven unactuated calibration channels behind twenty-nine physical phase actuators in the ideal graph-linear model. Second, the same forty-cycle provides a forty-setting Fourier spectrometer that reconstructs all thirty-nine traceless diagonal modes while its zero character performs common-mode averaging. Third, reversing the optimal broadcast tree gives a seven-tick qutrit gather and a fourteen-tick collision-free all-reduce.

Evidence boundary. The frozen semantic certificate is

`aa3daf462320badd607a1720479f083df159044f24242e7f46810de30bc2e0d8`.

The results certify finite group words, orbit-estimation cost, static diagonal cancellation, broadcast depth, zero forcing, a dynamic-topology witness, and three algebraic communication/calibration mechanisms. They do not certify an optimal active twirl, a complete Floquet quantum code, noisy actuator virtualization, minimum physical layout, or laboratory performance.

28.3 Exact plane-ovoid transport, the 200-ovoid scheme, and a rootless axial Leech polarization

Passes 3795–3812 replace the prior existential plane-ovoid isomorphism by a canonical objectwise dictionary. The full $O_6^-(2)$ torsor contains 51,840 graph isomorphisms, and its lexicographically minimal representative transports all forty points, all forty four-point lines, and all thirty-six ten-line spreads of $W(3,3)$ onto the forty plane ovoids of $GQ(4,2)$.

The action on all $200 = 40 + 160$ ovoids has nineteen ordered-pair orbitals. Their orbital matrices form a nineteen-dimensional coherent algebra; the complete $19 \times 19 \times 19$ intersection tensor has 405 nonzero entries and center dimension five modulo 1,000,003. Two self-paired valency-three relations on the 160 tripods each split into forty disjoint copies of K_4. If B is the 40×40 intersection matrix of the two block systems, then

$$BB^{\mathsf{T}} = 4I + A_{W(3,3)},$$

so the tripods are precisely the 160 flags between two dual forty-object systems and reconstruct the strongly regular graph $(40, 12, 2, 4)$.

Using the standard degree-24 M_{24} generators and a base octad, the verifier regenerates the binary Golay code with weight distribution

$$A_0 = 1, \quad A_8 = 759, \quad A_{12} = 2576, \quad A_{16} = 759, \quad A_{24} = 1.$$

Its integral coordinate construction gives a positive-definite even unimodular rank-24 Gram matrix. Exact short-vector enumeration finds only the zero vector at norm at most two and exhibits norm four, hence the lattice is rootless with minimum four and is therefore the Leech lattice. A determinant-one integral transport places this Gram matrix on the rank-24 axial carrier. Exhaustive testing of all 25,920 elements of the axial $U_4(2)$ action gives

$$\mathrm{Stab}_{U_4(2)}(\Lambda_{\mathrm{axial}}) \cong C_2.$$

Seven explicit degree-45 permutations generate $O_6^-(2) = U_4(2){:}2$ of order 51,840, with index-two subgroup $U_4(2)$ of order 25,920. This is an abstract finite descent only: serialized Monster or `mmgroup` words and an executed Monster class fusion remain absent.

Finally, the forty-five axes span dimension twenty-four. The twenty-seven five-axis line sums have rank twenty-one and generate the complete linear kernel. Collinear pairs generate dimension two, noncollinear pairs dimension four, and the six triple orbits have generated dimensions

$$4,\ 10,\ 24,\ 10,\ 5,\ 3$$

for orbit sizes $240, 2160, 2880, 6480, 2160, 270$, respectively. Thus one orbit of 2,880 triples generates the entire algebra, although no pair does. The algebra is simple, nonunital, and has zero derivations. The exact semantic certificate is

`be5ee56a84141a7bbc896b4cef0e8eda32167a00749dbb17f10d24d0505bdd41`.

No Monster/Majorana/Griess/VOA, remote-PDF, hardware, laboratory, or physical claim is promoted by this computation.

29 Quadratic Parent, Discriminant Chain, Multiport Compiler, and Holonomy Scheme

The thirty-six nonsingular vectors of the minus-type quadratic space $(\mathbf{F}_2^6, q)$ are one subconstituent of the Cayley graph $x \sim y \Leftrightarrow q(x+y) = 1$. Its adjacency matrix obeys

$$A_{64}^2 = 16I + 20J,$$

so the parent is $\mathrm{SRG}(64, 36, 20, 20)$. The neighborhood of zero induces the complement of $\mathrm{SRG}(36, 15, 6, 6)$, while the twenty-seven nonzero nonneighbors induce the Schlaefli graph, the complement of $\mathrm{SRG}(27, 10, 1, 5)$.

For the binary character code

$$C = \{(\beta(a,x))_{x \in N} : a \in \mathbf{F}_2^6\} = [36, 6, 16],$$

the Construction–A discriminant module is

$$L(C)^*/L(C) \cong C^\perp/C \cong (\mathbf{Z}/2)^{24}, \qquad q_L(c + C) = \mathrm{wt}(c)/2 \pmod{2\mathbf{Z}}.$$

An explicit rank-eleven totally isotropic chain yields a maximal doubly-even $[36, 17, 8]$ extension and an even overlattice of determinant four. Rank twelve is impossible because it would produce an even unimodular lattice of rank thirty-six. The residual two-dimensional quotient gives three singly-even self-dual code extensions and hence three odd unimodular Construction–A neighbors, in two distinct weight-enumerator types.

The exact involution

$$H = (2A_{36} - J)/6$$

admits a proof-carrying compilation through twenty-one primitive integer Householder directions. Balanced exact Givens trees require 944 two-mode rotations, twenty-one single-mode π phases, and sequential depth 221. A deterministic adjacent elimination reduces the candidate mesh to 512 rotations in sixty-nine layers. Full support gives the unconditional light-cone bound $d \geq \lceil \log_2 36 \rceil = 6$.

The 120 Fischer triples carry a four-angle rank-twenty frame whose five Gram relations close to a commutative association scheme. Its valencies and multiplicities are

$$(1, 2, 54, 36, 27), \qquad (1, 20, 24, 60, 15),$$

and its first eigenmatrix is

$$\begin{pmatrix} 1 & 2 & 54 & 36 & 27 \\ 1 & -1 & -9 & 0 & 9 \\ 1 & 2 & -6 & 6 & -3 \\ 1 & -1 & 3 & 0 & -3 \\ 1 & 2 & 6 & -12 & 3 \end{pmatrix}.$$

The forty antipodal fibers quotient to $W(3, 3)$. On quotient nonedges the 1/3 relation is a perfect matching and defines an S_3 connection. Of the 3240 base triangles, 1080 are flat and 2160 have transposition holonomy; three- cycle holonomy first appears on four-cycles, so the full holonomy group is S_3.

Finally, if F and T denote port–frame and port–triple incidence, then

$$FF^{\mathsf{T}} = 15I + 3A, \qquad TT^{\mathsf{T}} = 9I + J - A,$$

and therefore

$$\boxed{FF^{\mathsf{T}} + 3TT^{\mathsf{T}} = 42I + 3J}.$$

After removing the all-ones channel, the weighted frame–triple carrier is an exact tight resolution of the thirty-five-dimensional contrast space.

All statements in this section are finite exact certificates. They do not supply Monster words, a Leech identification, an optimal optical circuit, a hardware result, or a physical gauge interpretation.

30 Maximal-code census, exact adjacent mesh, holonomy closure, and the ovoid flag completion

The six-bit minus-type quadratic parent supports a complete exact refinement of the code, multiport, Fischer-holonomy, finite-group, and ovoid fronts. The frozen component certificate has semantic digest

`b141dd0f82e4a6b1ee62d1c57f0e92bdfc9f58d3b32515f9521a0175fdca88a1`.

30.1 Maximal doubly-even extensions

Let $C = [36, 6, 16]$ be the binary character code on the nonsingular vectors. Its discriminant module is the minus-type quadratic space $O_{24}^{-}(2)$ of Witt index eleven. Hence the exact number of maximal doubly-even extensions is

$$N = \prod_{i=2}^{12} (2^i + 1) = 240137905387279785868125.$$

Since $|U_4(2){:}2| = 51840$, there are at least

$$4632289841575613440$$

orbits. Thus the exact classification is necessarily a census and orbit-explosion theorem rather than a literal representative list. An explicit maximal extension $D = [36, 17, 8]$ has enumerator

$$1 + 225z^8 + 9555z^{12} + 55755z^{16} + 55755z^{20} + 9555z^{24} + 225z^{28} + z^{36},$$

and trivial stabilizer in $U_4(2){:}2$.

30.2 An exact 418-gate adjacent mesh

For

$$H = (2A_{36} - J)/6,$$

an exact symbolic adjacent Givens elimination gives 418 nontrivial rotations in 69 layers, 212 exact zero skips, and one terminal π phase. Every gate has

$$c = \operatorname{sgn}(c)\sqrt{c^2}, \qquad s = \operatorname{sgn}(s)\sqrt{s^2}, \qquad c^2 + s^2 = 1,$$

with rational squared parameters. The exact parameter digest is

`5c933cc2e6d2484e97894f3ca1f71627214238e41a68cd59b7527993d2b06b6b`.

The certified bounds are

$$35 \le g \le 418, \qquad 6 \le d \le 69.$$

No global optimality claim is made.

30.3 The rank-five holonomy scheme

The 120 Fischer triples carry five Gram relations with valencies

$$1, 2, 54, 36, 27$$

and primitive multiplicities

$$1, 20, 24, 60, 15.$$

All Krein parameters are exact. There is no Q-polynomial ordering, and exactly four fusion schemes survive: the full scheme, the fusion $(2, 4)$, the fusion $(2, 3, 4)$, and the complete rank-two fusion. The full automorphism group is $U_4(2){:}2$. At a point,

$$\dim_{\mathbf{Q}} T = 79, \qquad \dim_{\mathbf{Q}} Z(T) = 10.$$

The certificate proves exact closure using 79 word matrices and all 83 point-stabilizer orbitals.

30.4 Complete abstract standard-pair census

Inside $U_4(2)$, fix an involution a in the class of size 45. Exactly 1152 order-five elements b satisfy

$$|ab| = 9, \qquad |[a, b]| = 3.$$

The centralizer $C(a)$ has order 576 and splits those elements into two orbits of size 576. Each orbit representative generates all 25920 elements. Therefore the complete ordered standard-pair census is

$$45 \cdot 1152 = 51840.$$

No serialized Monster words or Monster character restriction is asserted.

30.5 Ovoids as W33 points and flags

All 200 ovoids of $GQ(4,2)$ split into 40 plane ovoids and 160 tripods. Objectwise,

$$40 \text{ plane ovoids} = 40 \text{ W33 points},$$

$$160 \text{ tripods} = 160 \text{ incident W33 point-line flags}.$$

The 160 tripod port columns collapse four-to-one onto the 40 W33 line-versus-spread incidence columns. Finally the five fibers

$$1 \mid 27 \mid 36 \mid 40 \mid 160$$

form one $O_6^-(2)$ orbital coherent configuration of rank 48.

Boundary. The packet does not materialize billions of orbit representatives, prove global circuit optimality, produce Monster words or class fusion, identify a rootless lattice, or certify hardware, laboratory, remote-CI, or PDF results.

31 Adaptive geometry, virtual topology, autonomous attraction, and thermodynamic inversion

Passes 3829–3836 recast the W33 machine as a dynamically selected control architecture rather than a single permanently active graph. The exact certificate is generated by `analysis/w33_pass3829_3836_ada` and has semantic digest `d924764ba5d64326fbdc8d7a6bbd8bcdff2fd08fb2a4c6560f29c53a71c29fe4`.

Adaptive hypervisor. A declared four-state, five-action discounted control model selects the local ring in the clean regime, the forty-cycle Floquet mode under diagonal drift, W33 under burst loss, and spectroscopy or direct 29-port control under crosstalk. Its frozen adaptive score is 36.5799750262, compared with 66.3616366758 for the best static policy. These are benchmark-model scores, not measured logical-error rates.

Virtual topology. The 240 W33 edges admit an explicit decomposition into twelve perfect matchings. Because every vertex has degree twelve and one edge token per vertex can be consumed in one round, twelve rounds are both necessary and sufficient. Thus a measurement- or teleportation-mediated realization can instantiate the complete W33 interaction epoch using twenty heralded edge tokens per round and no permanently fabricated W33 couplers, before source, fusion, memory, and feedforward overheads are counted.

Autonomous attraction. Let Π_{W33} average a symmetric forty-mode matrix over its diagonal, adjacent, and nonadjacent entries. Then Π_{W33} is an idempotent projector with image

$$\operatorname{span}\{I, A, J - I - A\}.$$

The ideal reservoir step

$$\Phi(X) = \tfrac{1}{2}\big(X + \Pi_{W33}(X)\big)$$

contracts every symmetry-breaking component by exactly one half. Twenty steps therefore suppress the deviation by 2^{-20}. This is an algebraic reservoir target, not an implemented Lindbladian.

Thermodynamic crossover. For independent heralded edge success p, the worst-nonedge route probabilities are

$$q_{W33} = 1 - (1 - p^2)^4, \qquad q_2 = 1 - (1 - p^2)^2.$$

Using frozen wirelength proxies 1064 and 938, the retry-energy crossover is

$$p_* = \sqrt{1 - \sqrt{63/469}} \approx 0.795921897507.$$

Below p_*, W33's four-route redundancy outweighs its longer layout proxy; above p_*, the local two-route control is cheaper in this declared model. No wall-plug energy claim follows.

Substrate inversion. One complete interaction epoch has the exact primitive-count alternatives

$$C_{\text{native}} = 240a + 12g + 240d, \quad C_{\text{virtual}} = 240b + 12g + 240d,$$
$$C_{\text{mesh}} \in 67a + [12, 870]g + 870d, \quad C_{\text{bus}} = a + 240g + 240d,$$

where a, b, g, d price static links, consumable edge tokens, rounds, and routing hops. No substrate class is coefficient-independently optimal.

Three additional mechanisms. First, the exact strongly regular identity

$$A^2 + 2A - 8I - 4J = 0$$

acts as a digital-twin checksum. Every single edge deletion produces a rank-four residual with squared Frobenius norm 54 and support 48; every single insertion produces rank four, norm squared 58, and support 52. The two corrupted endpoints are the unique residual rows of support thirteen or fourteen, respectively.

Second, independent heralded links can be distilled geometrically. For nonadjacent vertices,

$$P_{\text{route}} = 1 - (1 - p^2)^4,$$

while an adjacent pair can use its direct edge and two triangle detours,

$$P_{\text{route}} = 1 - (1 - p)(1 - p^2)^2.$$

At target reliability 0.999, the corresponding edge-success thresholds are approximately 0.906737 and 0.935615.

Third, the four transvections and their inverses define an eight-jump symmetry bath on the forty points. Its second eigenvalue is 0.9367092858522434, its spectral gap is 0.0632907141477566, and the standard spectral bound gives 89 jumps for one-percent mixing from a localized point defect. This is an ideal Markov process, not a physical dissipation-rate theorem.

Evidence boundary. The finite graph identities, matching decomposition, defect signatures, projector law, routing formulas, and primitive counts are exact. The adaptive cost model, ideal reservoir, retry-energy proxy, substrate coefficients, independent-link model, and symmetry-bath implementation remain explicit architectural hypotheses rather than laboratory evidence.

32 Ovoid Wedderburn Algebra, Leech Transport, and Triality Incidence

The complete $40 + 160$ ovoid coherent algebra has dimension 19 and centre dimension 5 over $\mathbf{Q}$. Exact rational primitive central idempotents and minimal-left-ideal ranks give the split decomposition

$$\mathcal{A} \cong M_2(\mathbf{Q}) \oplus \mathbf{Q} \oplus M_2(\mathbf{Q}) \oplus \mathbf{Q} \oplus M_3(\mathbf{Q}).$$

On the 200-point permutation module,

$$\mathbf{Q}^{200} \cong 1^{\oplus 2} \oplus 15_a^{\oplus 2} \oplus 15_b \oplus 24^{\oplus 3} \oplus 81.$$

The canonical W33–plane-ovoid dictionary transports all 240 edges, 160 triangles, 40 lines, and 36 spreads. The oriented ternary check matrices satisfy

$$\operatorname{rank} H_X = 39, \qquad \operatorname{rank} H_Z = 120, \qquad H_X H_Z^{\mathsf{T}} = 0,$$

with $d_X = 3$ and $d_Z = 4$, hence the edge code is $[[240, 81, 3]]_3$.

Under the explicit determinant-one Leech transport, exactly three integral axial numerator vectors have norm four:

$$a_3, \quad a_{33}, \quad a_{36}.$$

The surviving involution fixes a_3, swaps a_{33} and a_{36}, and the three carriers lift to nine D_4 frames.

The 45 marked axes admit a universal rational presentation by the 27 five-axis line sums and the two exact pair-product laws. The line relations have rank 21, so the quotient has dimension 24. An independent rooted graph computation gives

$$|\operatorname{Aut}_{\text{marked}}| = 45 \cdot 1152 = 51840 \cong |O_6^-(2)|.$$

The 148995 four-axis subsets form 20 orbits, with generated-dimension census

$$4^{135},\ 5^{720},\ 6^{1080},\ 10^{16740},\ 14^{27000},\ 24^{103320}.$$

The two 40-block tripod fibrations produce a 40×40 incidence matrix whose row and column Gram graphs both have parameters $(40, 12, 2, 4)$ but are nonisomorphic. The full bipartition-preserving automorphism group has order 51840, and no side swap exists. The smallest tripod–Norton relation has size 480, rank 40, and bipartite graph

$$40K_{4,3}.$$

Although this equals the number of directed W33 edges, the two permutation characters have norms 23 and 24 with cross inner product 18, so the two 480-actions are inequivalent.

The semantic certificate is `ede91cc799f9093209c589fb0b73c88fca4159bedcc8e1e7cd6680c5f0c778fd`. Serialized Monster words, Monster class fusion, the full unmarked linear automorphism group, remote CI/PDF success, and physical claims remain outside the promoted boundary.

32.1 Cubic tightening, tomotope outer extension, and complete modular descent

The dependency hypergraph on the 240 filled faces admits an explicit 63-face cap. Its complement is a 177-face transversal, improving the live exact interval to

$$106 \le \tau_{\text{cubic}} \le 177.$$

The witness has triple-hit profile $1^{876}2^{2217}3^{1947}$ and is locally optimal against every exchange removing at most two selected faces. No nonexistence theorem for a 64-cap is claimed, and the independent covering-radius boundary remains $389 \le R \le 435$.

The three exceptional entries in the 3×3 tomotope-completion square require a correction. Their automorphism groups have order 192 but trivial centre. Each has a unique normal elementary abelian subgroup 2^4, ordinary index-two subgroup $2^4{:}S_3$, and quotient D_{12}. Hence the exceptional group is the split, centreless outer extension

$$2^4{:}D_{12},$$

not a non-split central double cover.

The previously unresolved 115-dimensional characteristic-three quotient has composition factors

$$1^3 \oplus 5^3 \oplus 10^3 \oplus 14^3 \oplus 25.$$

An explicit 13-step series has successive dimensions

$$10, 14, 5, 10, 25, 5, 1, 14, 5, 10, 1, 14, 1,$$

and every factor generates its full matrix algebra over $\mathbb{F}_3$.

The exact Golay–Witt Gewirtz graph admits a canonical $16 + 40$ split under a second fixed point. Its unique two-point-stabilizer-invariant degree-12 residual graph is not W33; it is the independent fourfold blow-up of Petersen, with spectrum $12^1 4^5 0^{30} (-8)^4$.

Three associated constructions are exact: the free 63-cap orbit is a binary constant-weight code of length 240, size 25,920, weight 63, and minimum distance 62; the residual graph has ten twin classes of size four and Petersen quotient; and the cap's forty W33-fibre occupancies form a free orbit with spectral energies 3969/40, 157/5, and 163/8 on the 12, 2, and -4 channels.

The authoritative website remains pinned to Git blob `41a8d733f42da18282fa276f5d2fa82bac7516f6`. A fail-closed migration validator requires literal owner authorization binding old blob, new blob, and an exact archive before replacement. No migration, remote CI/PDF result, hardware result, laboratory result, Monster embedding, or physical interpretation is asserted.

32.2 Unmarked axial symmetry, explicit rational modules, and the order-192 barcode

Passes 3887–3904 have semantic certificate `6ef2f6c4a0fa7e52da9f4abff6d899d0250eeabde1a601d53c40052a8` They also correct the four-axis generated-dimension census of Passes 3837–3854, whose unreduced three-factor `int64` contraction overflowed before modular reduction.

Let $E = (A - 12I)(A - 3I)/90$ and let $V = \operatorname{im} E$ carry $x \star y = E(x \circ y)$. Exact rational multiplication matrices satisfy

$$\operatorname{tr}(L_x L_y) = \frac{7}{30} \langle x, y \rangle.$$

Thus the Euclidean form is intrinsic to multiplication. If $y \neq 0$ is an idempotent and m is its largest coordinate, the three quadrangle lines through a maximizing point give $N \geq 3m^2/4$ on its twelve neighbours, while its thirty-two nonneighbours give $Q \geq m^2/8$. Together with

$$m = \frac{8}{15} m^2 - \frac{2}{15} N + \frac{1}{30} Q$$

and $\|y\|^2 = \sum_i y_i^3$, this proves

$$\|y\|^2 \geq \frac{480}{49}.$$

Equality holds exactly for the 45 known axes. Hence every unmarked algebra automorphism permutes them, and

$$\boxed{\operatorname{Aut}(V, \star) \cong O_6^-(2) \cong U_4(2) : 2, \qquad |\operatorname{Aut}| = 51{,}840.}$$

The 19-dimensional orbital algebra has explicit split rational model

$$M_2(\mathbb{Q}) \oplus \mathbb{Q} \oplus M_2(\mathbb{Q}) \oplus \mathbb{Q} \oplus M_3(\mathbb{Q}).$$

Primitive rational projectors produce irreducible modules of dimensions $1, 15, 15, 24, 81$, and

$$\mathbb{Q}^{200} \cong 1^{\oplus 2} \oplus 15_a^{\oplus 2} \oplus 15_b \oplus 24^{\oplus 3} \oplus 81.$$

All five character inner products were evaluated on the complete order-51,840 group and form the identity matrix.

The 480-element $40K_{4,3}$ incidence action has character norm 23 and decomposes on the five known constituents as

$$\chi_{480} = 1 + 2 \cdot 81 + 2 \cdot 15_a + 15_b + 3 \cdot 24 + \rho_{200},$$

where ρ_{200} has degree 200, norm 4, and is orthogonal to all five known characters. The explicit Monster-subgroup database contains portable maximal-overgroup seeds but no direct $U_4(2) : 2$ key; no Monster words or class fusion are promoted.

The corrected four-axis calculation reduces after each contraction stage and is stable over six primes. Its weighted generated-dimension census is

$$4^{135} 5^{720} 6^{1080} 10^{16740} 12^{5040} 14^{27000} 16^{14040} 24^{84240}.$$

The twenty generating-set orbits collapse to eight simple nonunital subalgebra species of dimensions

$$\boxed{4, 5, 6, 10, 12, 14, 16, 24.}$$

Every species has zero annihilator and nucleus, full square span and trace rank, full associator ideal, and multiplication envelope M_d.

Finally, four order-192 mechanisms are separated. The ordered incident-pair/frame stabilizer is $W(D_4)$ with center 2 and derived subgroup 96; a distinguished involution centralizer is $D_8 \times S_4$ with derived subgroup 24; the archived octonion axis-line stabilizer is centerless and has 48 elements of order 8; and the corrected exceptional tomotope completion is the centerless split group $2^4 : D_{12}$. Equality of order or carrier size is therefore not an identification.

No portable Monster words, executed Monster class fusion, complete decomposition of ρ_{200}, complete tensor-product table, SmallGroup identification of the octonion stabilizer, remote CI/PDF success, hardware result, laboratory result, thermodynamic result, or physical mechanism is asserted without separate evidence.

32.3 Terwilliger sieve, improved adjacent mesh, code strata, and rank-48 overlap

The rank-five holonomy Terwilliger algebra has exact dimensions

$$\dim_{\mathbf{Q}} T = 79, \qquad \dim_{\mathbf{Q}} Z(T) = 10.$$

Assuming split semisimplicity over $\mathbf{Q}$, exhaustive integer arithmetic leaves exactly fourteen possible nondecreasing matrix-degree multisets satisfying $\sum_i n_i^2 = 79$; the largest block degree lies between four and eight. Primitive central idempotents are still required to select the actual multiset.

For the 36-port involution $H = (2A - J)/6$, a new symmetry-adapted ordering lowers the adjacent elimination candidate from 418 to

398 rotations in 69 layers,

with one terminal π phase. The zero pattern is numerically stable and all nontrivial rotations admit rational-square parameter recovery, but a complete exact-radical replay remains required before the circuit is promoted as an exact identity.

Maximal doubly-even $[36, 17]$ extensions containing the character code obey an exact one-parameter enumerator law. Writing $t = A_4$,

$$A_8 = 225 + 11t, \qquad A_{12} = 9555 - 39t, \qquad A_{16} = 55755 + 27t,$$

with symmetry about weight 18. A deterministic 258-code sample realizes $t = 0, 1, 2, 3, 4, 5$ and 184 distinct weight-eight coordinate-degree profiles, demonstrating that the enumerator is a coarse orbit invariant.

Finally, the 64-point quadratic carrier, 200-ovoid carrier, and combined 264-object carrier have orbital ranks 15, 19, and 48. Therefore their cross-Hom dimension is seven. Six dimensions are forced by the three-versus-two trivial constituents, leaving exactly one shared nontrivial 15-dimensional constituent. The 64-module consequently has form

$$1^3 \oplus 15_b \oplus X^2 \oplus Y, \qquad 2 \dim X + \dim Y = 46,$$

where X and Y do not occur in the 200-ovoid carrier. No Monster embedding, class fusion, exact-radical mesh proof, global mesh optimum, exhaustive code-orbit ledger, hardware result, laboratory result, or physical interpretation is claimed.

33 Passes 3913–3920: multi-defect immunity, symmetry dissipation, process control, and dual scheduling

The W33 control architecture now has a bounded multi-defect decoder, an explicit random-unitary symmetry Lindbladian, a partially observed controller, a correlated-fusion threshold, and a joint substrate/topology phase diagram. Three additional constructions compress the graph into a finite-field seed, extend the interaction schedule to an optimal all-to-all epoch, and compress the two-defect sensor.

Bounded topology immune system. For a measured adjacency matrix B, define

$$R(B) = B^2 + 2B - 8I - 4J.$$

The exhaustive ledger contains all

$$\binom{780}{2} = 303{,}810$$

two-edge insertion/deletion mixtures, and every complete residual is distinct. It also contains all

$$\binom{240}{3} = 2{,}275{,}280$$

three-edge deletions, again without a repeated residual. General mixed weight-three cases remain open.

Symmetry Lindbladian. Let U_g denote the four generating transvections and their inverses. The random-unitary generator

$$\mathcal{L}(X) = \frac{1}{8} \sum_g (U_g X U_g^\dagger - X)$$

has fixed algebra

$$\mathrm{Fix}(\mathcal{L}) = \mathrm{span}\{I, A, J - I - A\}.$$

Its unit-rate gap is

$$0.06329071414775411.$$

Every transvection consists of nine disjoint three-cycles and thirteen fixed points, yielding a 72 three-mode-cycle compilation target for the eight directed jumps. A sparse local physical realization with the same fixed algebra is not yet proved.

Partially observed geometry control. A six-step four-regime hidden-Markov benchmark with 85-percent regime readout and five geometry/control modes has exact belief-state optimum

$$23.115384313076735.$$

The best static policy costs 30.176036574158424, while full regime information gives the lower bound 16.514150252446512. These are declared benchmark values, not learned process-tensor measurements.

Correlated fusion compiler. A twenty-edge matching is accumulated by a 21-state Markov chain with stored-edge survival 0.995, common outage probability 0.05, and five attempts. Requiring a complete twelve-matching epoch to succeed with probability at least 0.99 gives

$$p \geq 0.9158993399430635.$$

The corresponding independent no-decay threshold is 0.8668344006872047.

Architecture phase diagram. A 500-point declared-cost grid compares native W33, virtual W33, a local ring, a routed mesh, and a shared bus. The winner counts are respectively

$$54, \quad 84, \quad 37, \quad 15, \quad 310.$$

Every implementation wins somewhere; no architecture is coefficient-independently optimal.

Topology DNA. Four four-trit transvection generators and one ordered edge generate the order-25,920 group and the complete 480-element oriented-edge orbit. The graph seed therefore uses eighteen finite-field coordinate symbols rather than 960 explicit oriented-edge endpoint symbols.

Dual-geometry all-to-all epoch. The twelve W33 perfect matchings and twenty-seven complement perfect matchings partition $E(K_{40})$. Hence

$$A + (J - I - A) = J - I$$

and the complete all-to-all one-port schedule has optimal depth

$$12 + 27 = 39.$$

Compressed two-defect sensor. A fixed set of 192 residual entries distinguishes all 303,810 two-edge syndromes, reducing the complete 40×40 residual readout by a factor $1600/192 = 25/3$. Minimality and noise robustness remain open.

Evidence boundary. The executable semantic certificate is `019183d46768ff68a5f57676540d8aefd85301061`
The packet does not establish general mixed weight-three decoding, a local Lindbladian implementation, measured process control, microscopic fusion correlations, measured platform costs, minimum topology memory, simultaneous all-to-all coupling, minimum sensing, hardware performance, or laboratory validation.

34 Hidden character, local-algebra poset, and null-processor boundary

The 480-element $40K_{4,3}$ incidence character decomposes exactly as

$$\chi_{480} = 1 + 2(15_a) + 15_b + 3(24) + 2(81) + 20 + 30 + 60 + 90.$$

Hence the previously unresolved residual is

$$\rho_{200} = 20 \oplus 30 \oplus 60 \oplus 90.$$

The eight simple nonunital local-algebra species have dimensions

$$4, 5, 6, 10, 12, 14, 16, 24.$$

Their Hasse covers are

$$4 \prec 16, \quad 5 \prec 6, \quad 5 \prec 10, \quad 6 \prec 14, \quad 10 \prec 14, \quad 10 \prec 16, \quad 12 \prec 24, \quad 14 \prec 24, \quad 16 \prec 24.$$

The undirected Hasse graph is connected and bipartite with first Betti number two.

For a projected-coordinate algebra $x \star y = E(x \circ y)$, if the scaled columns of E are exactly the minimum-norm nonzero idempotents and pair invariants recover adjacency, every algebra automorphism preserves the recovered graph. In the 45-axis $GQ(4, 2)$ instance,

$$\operatorname{tr}(L_x L_y) = \frac{7}{30}\langle x, y\rangle, \qquad \min_{e^2=e,\ e\neq 0} \|e\|^2 = \frac{480}{49},$$

with equality precisely on the 45 axes. Thus

$$\operatorname{Aut}(V, \star) \cong O_6^-(2) \cong U_4(2) : 2.$$

A speculative photon-node model is admitted only in the Lorentz-compatible fixed-product form. If N effective nodes of pitch a are traversed in observer-frame period $T = 1/\nu$, then

$$v_{\text{eff}} = \frac{Na}{\sigma T}.$$

Vacuum null propagation requires $Na = \lambda$, $\lambda\nu = c$, and unit routing stretch. Consequently

$$c = \frac{Na}{T},$$

and a self-similar refinement $N \mapsto bN$, $a \mapsto a/b$ leaves c invariant. This is a falsifiable information-transport model, not evidence for literal photon substructure or variable vacuum light speed.

34.1 Exact selector algebra, radical multiport, code strata, and photon-capacity boundary

The fourteen-case arithmetic sieve for the 79-dimensional selector Terwilliger algebra is superseded by the exact characteristic-zero result already certified in Passes 1365–1369:

$$T(x) \cong \mathbb{Q}^3 \oplus M_2(\mathbb{Q})^2 \oplus M_3(\mathbb{Q})^3 \oplus M_4(\mathbb{Q}) \oplus M_5(\mathbb{Q}).$$

Its simple degrees are

$$1, 1, 1, 2, 2, 3, 3, 3, 4, 5,$$

with standard-module multiplicities

$$3, 12, 14, 1, 2, 4, 4, 8, 8, 1.$$

For the spread-graph multiport

$$H = \frac{2A_{36} - J}{6},$$

an exact sparse multiquadratic replay now proves the previously numerical adjacent-mode candidate. The complete transform uses

$$398$$

nontrivial adjacent Givens rotations in 69 layers, skips 232 eliminations because their entries are exactly zero, and terminates at

$$\operatorname{diag}(1^{35}, -1).$$

Every intermediate entry is a single rational radical. There are 75 distinct values of c^2, the largest denominator is 441, and the ordered parameter digest is

`2c3be7815b8346cd90be108faad4cc68866ad453f13b76bbdd4a9988a4569555`.

No global gate-count or depth optimality is claimed.

The binary character code $C = [36, 6, 16]$ admits an explicit maximal doubly-even extension

$$D = [36, 17, 4]$$

with

$$\begin{aligned} W_D(z) = 1 &+ 57z^4 + 852z^8 + 7332z^{12} + 57294z^{16} \\ &+ 57294z^{20} + 7332z^{24} + 852z^{28} + 57z^{32} + z^{36}. \end{aligned}$$

The intersection-two graph on its 57 weight-four words splits as

$$\boxed{\mathrm{SRG}(45, 16, 8, 4) \sqcup K_{2,2,2} \sqcup K_{2,2,2}.}$$

The coordinate-degree profile is $9^{20}3^{16}$. This proves an exact $A_4 = 57$ stratum, not its global maximality.

The five-fiber carrier $1|27|36|40|160$ has rational permutation decomposition

$$\mathbb{Q}^{264} \cong 1^5 \oplus 6 \oplus 15_a^2 \oplus 15_b^2 \oplus 20^2 \oplus 24^3 \oplus 81,$$

and hence

$$\boxed{\mathrm{End}_G(\mathbb{Q}^{264}) \cong \mathbb{Q}^2 \oplus M_2(\mathbb{Q})^3 \oplus M_3(\mathbb{Q}) \oplus M_5(\mathbb{Q}).}$$

The algebra dimension is 48, its centre has dimension seven, and the cross-carrier Hom space has dimension

$$7 = 3 \cdot 2 + 1,$$

consisting of six trivial channels and one shared 15_b channel.

Photon node-packing boundary. For optical frequency ν,

$$T = \nu^{-1}, \qquad \lambda = c/\nu.$$

If N internal update sites refine one wavelength and one period together,

$$a_N = \lambda/N, \qquad \tau_N = T/N,$$

then

$$\boxed{\frac{a_N}{\tau_N} = c.}$$

Thus node density can refine internal state space and update throughput while leaving vacuum c invariant. In the ideal tensor-factor bookkeeping,

$$R = N\nu \log_2 d, \qquad \rho = N/\lambda, \qquad \frac{R}{\rho \log_2 d} = c.$$

A single photon distributed over N alternatives with a d-state label generally has direct-sum dimension Nd, however, so its accessible classical information is bounded by $\log_2(Nd)$ rather than $N \log_2 d$ unless physically independent tensor factors exist.

The Margolus–Levitin rate for one photon of energy $E = h\nu$ gives

$$\Gamma_\perp \leq \frac{2E}{\pi\hbar} = 4\nu, \qquad \Gamma_\perp T \leq 4.$$

Consequently forty W33 nodes cannot mean forty sequential mutually orthogonal transitions within one optical period of an isolated photon. They must represent parallel or nonorthogonal modes, multiple periods, or externally driven evolution. This is a falsifier for one literal interpretation, not a rejection of high-dimensional single-photon processing.

The Monster descent remains fail-closed: no portable serialized Monster words or executed character-fusion artifact are present, and no embedding is promoted. No literal photon-node ontology, variable-c mechanism, hardware result, laboratory result, remote-CI success, or PDF success is claimed.

34.2 Topology-code firewall, optimal 24-port control, autonomous boot, and photon causal density

The W33 polynomial residual

$$R(B) = B^2 + 2B - 8I - 4J$$

separates all 79,092,651 edge-toggle patterns of weight at most three: one zero pattern, 780 single toggles, 303,810 double toggles, and 78,788,060 triple toggles. A zero residual has balanced insertion and deletion degree at every vertex. Exhaustion of all 27,000 alternating weight-four supports and 2,769,952 alternating simple weight-six supports finds no undetectable pattern. A nonadjacent vertex transposition gives an explicit undetectable weight-32 relabelling, so

$$8 \leq d_0 \leq 32.$$

The exact distance remains open.

The desired symmetry dissipator cannot be made from independent strictly two- or three-mode permutation jumps. Every nontrivial W33 automorphism moves at least 27 points, and every transvection has cycle structure $1^{13}3^9$. Four transvections are necessary and sufficient to generate the full order-25,920 group; each global jump is compilable into nine disjoint three-mode cycles, but those cycles must remain one macro if the fixed algebra

$$\mathrm{span}\{I, A, J - I - A\}$$

is to be preserved.

Writing $B = J - I - A = (A^2 - 2A - 12I)/4$, the complement adds no passive drift algebra. The largest adjacency multiplicity forces at least 24 independent diagonal controls. For one explicit 24-port set, the commutant constraint has modular rank 279 on 280 coordinates over three primes, leaving scalars only. Therefore the ideal controlled $*$-algebra has dimension

$$40^2 = 1600,$$

and 24 ports are necessary and sufficient algebraically.

The topology can boot from four transvection vectors and one ordered edge, an 18-trit seed. Its ordered-edge orbit has 480 elements and BFS diameter 9; a deterministic factorization then supplies 12 collision-free W33 matching rounds. The declared boot sequence—seed verification, orbit expansion, schedule compilation, a 192-entry residual measurement, bounded repair, and symmetry activation—has a 221-round network-action upper bound before source, detector, classical, memory, and settling overhead.

The user-proposed relation between photon computation and light speed survives in a constrained form. If serial node spacing is ℓ and local update time is τ, Lorentz-compatible propagation requires $\ell/\tau = c$. Combining a one-photon energy $E = h\nu$ with the Margolus–Levitin speed limit gives

$$N_{\text{serial}} \leq \frac{4EL}{hc} = \frac{4L}{\lambda}.$$

Increasing serial node count at fixed length therefore requires faster internal updates; it does not change invariant vacuum c. Parallel photonic modes enlarge capacity—an ideal M-mode alphabet carries $\log_2 M$ bits—without representing M serial state transitions per optical cycle. Any literal node-count dependence of vacuum speed would predict encoding- or frequency-dependent front arrival and is retained only as a falsifiable, presently unsupported hypothesis.

Two additional exact constructions follow. Since A and its complement commute,

$$e^{-itA}e^{-itB} = e^{-it(J-I)}, \qquad e^{-itA}e^{+itB} = e^{-it(A-B)},$$

giving common-mode and W33-contrast interferometric channels. For the Cartesian hierarchy $W33^{\square m}$,

$$|V| = 40^m, \quad k = 12m, \quad \text{diam} = 2m, \quad |E| = 6m40^m,$$

with absolute spectral gap 10 and ideal causal-density requirement $L/\lambda \geq m/2$. Self-similarity yields logarithmic routing depth but does not compress the exponential physical mode requirement.

No exact topology distance, independently local Lindbladian, robust pulse theorem, learned process tensor, autonomous physical boot, hidden photon-node ontology, variable speed of light, hardware result, or laboratory validation is claimed.

Passes 3973–3980: extremal-code geometry, mesh locality, and photon resource separation

Exact 57-code geometry. For the deterministic maximal doubly-even code $D = [36, 17, 4]$, join two weight-four words when their supports meet in two coordinates. The resulting graph is not merely a $45 + 6 + 6$ census: an explicit maximal-clique reconstruction gives

$$\boxed{G_{57} \cong T(10) \sqcup T(4) \sqcup T(4) = L(K_{10}) \sqcup L(K_4) \sqcup L(K_4).}$$

Thus

$$57 = \binom{10}{2} + 2\binom{4}{2}.$$

If $t = A_4$, MacWilliams symmetry forces

$$A_8 = 11t + 225, \qquad A_{12} = 9555 - 39t, \qquad A_{16} = 55755 + 27t,$$

and, on the dual side,

$$B_4 = t, \quad B_6 = 6(t+7), \quad B_{12} = 39(245 - t).$$

Nonnegativity gives only $t \leq 245$. Hence $t = 57$ is exact and geometrically rigid here, but global A_4-extremality is not claimed.

Exact radius-one mesh optimum. The established 36-port factorization uses 398 nontrivial exact multiquadratic Givens rotations, 232 exact zero eliminations, and 69 layers. All $\binom{36}{2} = 630$ single transpositions of the port order were replayed exactly. None improves 398, 22 tie it, and every neighbor has depth 69. Therefore the order is an exact radius-one local optimum in permutation space; global rotation and depth optimality remain open.

One-photon slope/intercept falsifier. The physically conservative null is

$$t(M, L) = a(M) + L/c,$$

where M is the orthogonal-mode alphabet and $a(M)$ absorbs encoder, decoder, detector, and pulse-shape latency. Randomly interleaving fixed-spectrum alphabets $M \in \{2, 4, 8, 16, 40\}$ at two or more free-space lengths permits the fit

$$t(M, L) = a(M) + b(M)L.$$

A mode-dependent intercept is ordinary device latency; only a reproducible mode-dependent slope, surviving encoder swaps and length scaling, challenges the invariant-front null. Capacity must be reported separately from timing. For a symmetric M-ary channel with total error probability ϵ,

$$I(M, \epsilon) = \log_2 M + (1 - \epsilon)\log_2(1 - \epsilon) + \epsilon \log_2\left(\frac{\epsilon}{M - 1}\right).$$

At $M = 40$, this is 5.32193 ideal bits per use, about 5.18828 bits at one-percent symmetric error, and 4.77126 bits at five-percent error.

Literal rank-48 multiplication tensor. The split centralizer algebra

$$\boxed{\mathbb{Q}^2 \oplus M_2(\mathbb{Q})^3 \oplus M_3(\mathbb{Q}) \oplus M_5(\mathbb{Q})}$$

has dimension 48 and center dimension seven. Its matrix-unit multiplication table contains

$$1^3 + 1^3 + 3 \cdot 2^3 + 3^3 + 5^3 = 178$$

nonzero products. The later content-addressed verifier also reconstructs the complete literal orbital tensor:

$$\boxed{\text{48 relations,} \qquad \text{904 nonzero intersection constants.}}$$

Its tensor SHA-256 is

`a17a375552c102d281e3ba79b602b67aa54a426fd68b1878293763ea65395ff9`.

Thus both the split Wedderburn tensor and the geometric orbital-relation tensor are closed exactly.

Photon resource trilemma. A one-photon direct-sum alphabet of M orthogonal alternatives carries at most $\log_2 M$ ideal classical bits per use. A tensor-product claim $N \log_2 d$ requires N physically independent factors. Mutually orthogonal serial updates obey a separate energy–time bound. Temporal packing additionally satisfies the engineering estimate $M \sim 2WT$ for half-bandwidth W and duration T. Larger internal alphabets can therefore increase capacity and spacetime support without changing vacuum c.

For the Cartesian power $W(3, 3)^{\square m}$,

$$|V| = 40^m, \qquad \text{diam} = 2m,$$

so the self-similar address density is

$$\boxed{\frac{\log_2 |V|}{\text{diam}} = \frac{\log_2 40}{2} \approx 2.660964.}$$

This is an addressing/routing invariant, not a variable-light-speed law.

Monster gate and evidence boundary. The required observable Monster class-fusion artifact remains absent, so no Monster embedding is promoted. Exact results in this insert are the code geometry, enumerator identities, all-630 mesh audit, split and literal rank-48 tensors, and algebraic resource formulas. The timing sensitivity and time-bandwidth examples are models; the photon experiment, global code extremality, global mesh optimum, Monster fusion, remote CI/PDF success, hardware, and laboratory validation remain open.

35 Extremal code, active multiport, photon mode tests, and rank-48 tensor

Exact certificate. Passes 3973–3980 are reproduced by the content-addressed verifier and frozen certificate with semantic digest `03c37b821bb9b6f875a8bd4edde8dc96fe36a43d6d8dc24144d8dbd5e3aa5` Monster execution, laboratory photonics, and remote PDF evidence remain outside the promoted boundary.

Theorem 2 (Extremal $A_4 = 57$ extension)**.** *Let $C = [36, 6, 16]$ be the binary character code on the nonsingular vectors of the six-dimensional minus-type quadratic space. Among doubly-even self-orthogonal extensions of C,*

$$\boxed{A_4 \leq 57}.$$

The bound is attained by an explicit $[36, 17, 4]$ code. Its 57 weight-four words have intersection-two graph

$$\mathrm{SRG}(45, 16, 8, 4) \sqcup K_{2,2,2} \sqcup K_{2,2,2},$$

and coordinate stabilizer of order 192, with element-order census

$$1^1 2^{59} 3^8 4^{68} 6^{40} 12^{16}.$$

Proof certificate. The 945 admissible weight-four words form a degree-624 compatibility graph with two $O_6^-(2)$ vertex orbits of sizes 135 and 810. Exact colored bitset branch-and-bound gives orbit maxima 15 and 54. Transitivity reduces any clique meeting the 135-orbit to one fixed representative; its exact neighborhood search has maximum 57.

Theorem 3 (Active-mixer and fixed-order cut bounds)**.** *The exact 36-port involution $H = (2A_{36} - J)/6$ admits an adjacent factorization*

$$401 = 296 \ \textit{genuine mixers} + 105 \ \textit{signed swaps}$$

in 69 layers, with zero exact residual. The prior common-order compiler has $398 = 302 + 96$ factors. Thus six active mixing elements are removed when input and output relabeling may differ. For the original common order, the sum of the exact off-diagonal cut ranks is

$$\boxed{253},$$

so every adjacent factorization respecting that order uses at least 253 factors.

Theorem 4 (Literal rank-48 coherent tensor)**.** *The five fibers*

$$1 \mid 27 \mid 36 \mid 40 \mid 160$$

carry exactly 48 orbitals under $O_6^-(2)$. Their source–target relation count matrix is

$$\begin{pmatrix} 1 & 1 & 1 & 1 & 1 \\ 1 & 3 & 2 & 1 & 1 \\ 1 & 2 & 3 & 1 & 2 \\ 1 & 1 & 1 & 3 & 4 \\ 1 & 1 & 2 & 4 & 8 \end{pmatrix}.$$

Exactly 904 of the 48^3 intersection constants are nonzero. The complete sparse tensor has digest `a17a375552c102d281e3ba79b602b67aa54a426fd68b1878293763ea65395ff9`.

Photon competing-model protocol. The photon hypothesis is separated into six models: invariant-front direct-sum capacity; transverse-structure axial delay; a falsifiable hidden-node variable-speed alternative; an engineered W33 graph-kinetic model; independent hyperentangled tensor factors; and self-similar coupling efficiency. For the engineered graph model, $L_{W33} = 12I - A$ has sectors

$$\boxed{0, 10, 16},$$

giving a declared 0:10:16 delay-spectroscopy target. The experiment varies mode count at fixed spectrum and transverse momentum, varies transverse momentum at fixed alphabet, compares equal-dimension factorisations, and keeps causal-front arrival distinct from pulse-peak or group delay. No hidden nodes, variable vacuum light speed, achieved capacity, or measured delay are claimed.

Monster firewall. The exact internal queue contains 51,840 ordered standard pairs in two centralizer orbits of 576, but no portable serialized `mmgroup` words or executed class-fusion artifact are present. The result therefore remains `PENDING_EXPLICIT_MM_WORDS_AND_CLASS_FUSION`.

36 Passes 3981–3988: global mesh bound and photon spectral processor

The stronger reconciled Passes 3973–3980 certificate already proves that the fixed-parent weight-four compatibility graph has maximum clique size 57, hence the selected $[36, 17, 4]$ doubly-even extension is globally extremal within that parent-extension problem. It also reconstructs the literal five-fiber orbital algebra with 48 relations and 904 nonzero intersection constants. The present packet retains those exact results and advances the open physical and compiler boundaries.

36.1 An ordering-independent adjacent-factor lower bound

Let $K = 2A_{36} - J$ be the symmetric 36-port Hadamard transfer matrix, so $K^2 = 36I$. For an arbitrary port ordering and a prefix cut of size k, write

$$K = \begin{pmatrix} A_k & B_k \\ B_k^T & D_k \end{pmatrix}.$$

Then

$$B_k B_k^T = 36I - A_k^2.$$

Thus the cross-cut rank defect is the multiplicity of the eigenvalues ± 6 of A_k. Using

$$\operatorname{tr} A_k = -k, \qquad \operatorname{tr} A_k^2 = k^2,$$

and Cauchy–Schwarz gives the ordering-independent half-sequence

$$1, 2, 3, 4, 5, 5, 6, 7, 7, 8, 8, 8, 9, 9, 9, 9, 10, 9.$$

Reflection symmetry therefore forces

$$\boxed{N_{\text{adjacent}} \geq 229.}$$

The published ordering has cut-rank sum 253. A deterministic exact ordering search found a second ordering with sum 251. The number 251 is a stronger order-specific target, not yet an implemented factorization or a proof of global optimality.

36.2 Nuisance-complete photon timing tomography

A mode-count timing test must separate propagation slope from encoder, detector, spatial, spectral, and pulse-shape delays. The declared model contains the target regressor

$$\gamma \frac{L}{c} \log M$$

alongside mode intercepts, encoder and detector swaps, measured transverse-momentum and spectral covariates, pulse width, intensity, detector walk, and drift. The frozen design has 48 randomized cells and 16 parameters with full normalized rank 16.

For the declared synthetic benchmark of 20 ps single-event jitter and 10^6 events per cell,

$$\sigma_\gamma \simeq 2.19 \times 10^{-9}.$$

Omitting the measured spatial and spectral covariates creates a false

$$\widehat{\gamma} \simeq 1.1243 \times 10^{-9},$$

whereas the complete noiseless model returns zero to numerical precision. This explicitly demonstrates why structured-mode group delay is not, by itself, evidence for a mode-dependent vacuum front velocity.

36.3 Three exact photon constructions

For the W33 Laplacian $L = 12I - A$,

$$\mathrm{Spec}(L) = 0^1, 10^{24}, 16^{15}.$$

A localized mode has $\langle L \rangle = 12$ and $\mathrm{Var}(L) = 12$, so its ideal pure-state quantum Fisher information is

$$F_Q = 48.$$

The optimal extremal-sector superposition has $F_Q = 256$.

For the complement $B = J - I - A$,

$$A + B : (39, -1, -1), \qquad A - B : (-15, 5, -7).$$

The sum arm is geometry-blind on the 39-dimensional nonuniform sector, while the difference arm resolves the two protected sectors. Its full ideal spectral-range QFI is 400, with protected nonuniform value 144.

Most importantly, half a W33 Laplacian period gives the exact global reflection

$$\boxed{U = e^{-i\pi L/2} = I - 2E_{10} = -\frac{1}{3}(I + A) + \frac{2}{15}J.}$$

Each row has amplitude $-1/5$ on the point and its 12 neighbors and amplitude $2/15$ on its 27 nonneighbors, and

$$U^2 = I.$$

Thus one engineered analog evolution can implement a global 40-mode operation in parallel; it need not be interpreted as forty sequential orthogonal updates inside one optical period.

For a genuine tensor carrier,

$$\mathrm{mult}(m, r) = \binom{m}{r} 13^{m-r} 27^r, \qquad \mathrm{amp}(m, r) = \left(-\frac{1}{5}\right)^{m-r} \left(\frac{2}{15}\right)^r.$$

The 40^m-entry tensor kernel therefore has only $m+1$ amplitude-shell types. This is exact control-description compression, not Hilbert-space compression or a variable-c mechanism.

36.4 Monster and evidence firewall

A GAP/CTblLib class-fusion sieve and an `mmgroup` canonical-integer engine test have been materialized upstream of the strict Monster gate. The repository still lacks four explicit portable Monster words realizing the target $U_4(2)$ subgroup. Therefore the promoted status remains

$$\boxed{\texttt{PENDING\_EXPLICIT\_MM\_WORDS\_AND\_CLASS\_FUSION\_EXECUTION}.}$$

No Monster embedding, laboratory photon result, hidden photon-node ontology, mode-dependent vacuum c, global circuit optimum, or unobserved remote CI/PDF result is claimed.

37 Passes 3989–3996: physical incidence photon processor

The W33 point graph admits an exact sparse physical lift. Let N be the 40×40 point–line incidence matrix of $W(3,3)$. Exact enumeration gives

$$NN^T = 4I + A_{W33}.$$

Hence a dense twelve-neighbour point Hamiltonian may be generated from an eighty-mode bipartite point–line device with 160 couplings, degree four, and girth eight. Its spectrum is

$$-4^1, -\sqrt{6}^{24}, 0^{30}, \sqrt{6}^{24}, 4^1.$$

With point–line coupling g and bus detuning Δ,

$$H_{\text{eff}} = -\frac{g^2}{\Delta}(A + 4I) + O(g^4/\Delta^3),$$

and $t = \pi\Delta/(2g^2)$ approaches the exact W33 half-period reflection on the point sector. This is an analytic architecture, not a fabricated device.

Let $B = J - I - A$ and $D = A - B$. Then $A + B = J - I$ is geometry-blind on the nonuniform sector, while

$$\text{Spec}(D) = -15^1, 5^{24}, -7^{15}.$$

The spectral projectors are

$$E_{-15} = \frac{(D-5I)(D+7I)}{160} = \frac{J}{40}, \quad E_5 = \frac{(D+15I)(D+7I)}{240}, \quad E_{-7} = -\frac{(D+15I)(D-5I)}{96}.$$

Consequently $\langle D \rangle$ and $\langle D^2 \rangle$ determine all three sector populations. If identical commuting error C enters both arms, then

$$e^{-it(A+C)}e^{+it(B+C)} = e^{-it(A-B)},$$

so the dual-geometry echo cancels it exactly.

The maximum-code census is also complete. The 945-vertex fixed-parent compatibility graph contains exactly 945 maximum 57-cliques, split into three parent-group orbits

$$\boxed{945 = 540 + 270 + 135}$$

with stabilizer orders

$$\boxed{96, 192, 384}.$$

Thus maximum-code uniqueness is false in the fixed-parent problem. This corrects the earlier division by an order-192 full coordinate stabilizer without first checking parent preservation.

The seven primitive central blocks of the literal 48-orbital algebra have simple degrees

$$1, 1, 2, 2, 2, 3, 5.$$

Their complete central coarse-graining lattice contains

$$B_7 = 877$$

partitions. Quotienting equal-degree relabelings by $S_2 \times S_3$ leaves exactly

$$\boxed{198}$$

inequivalent central types. These are exact central subalgebras, not automatically combinatorial fusions of the 48 orbital relations.

Three further photon constructions follow from the incidence lift. First, rank $N = 25$, so the bipartite processor contains thirty exact dark modes. Second, for $L = 12I - A$,

$$V = e^{-i\pi L/4} = I - (1+i)E_{10}, \qquad V^2 = U, \qquad V^4 = I,$$

which gives an exact four-phase internal graph clock. Third, for a genuine m-fold tensor carrier,

$$\operatorname{rank}(N^{\otimes m}) = 25^m, \qquad \dim\ker(N^{\otimes m}) = 40^m - 25^m,$$

so the one-side dark fraction is

$$\boxed{1 - (5/8)^m}.$$

Only $m+1$ nonzero singular shells occur, with multiplicity $\binom{m}{r}24^r$ and squared singular value $16^{m-r}6^r$.

The Monster acquisition gate now composes $U_4(2) \to H \to \mathbb{M}$ class fusions through maximal-subgroup tables and searches only compatible published explicit `mmgroup` overgroups. The promoted status remains

$$\boxed{\texttt{PENDING\_EXPLICIT\_MONSTER\_U42\_WORDS}}$$

until four portable words pass the exact pair, triple, order-25,920, object-action, and class-fusion firewalls. No variable vacuum c, literal hidden photon nodes, fabricated hardware, laboratory result, or unobserved remote build is claimed.

Passes 3989–3996 supplement: direct W33 flight and causal memory

Exact forty-mode analog flight. Let A be the adjacency matrix of $W(3,3)$. A forty-mode equal-edge coupled-mode array obeying

$$i\frac{d\psi}{dz} = \kappa A\psi$$

implements, at $\kappa z = \pi/2$,

$$\boxed{e^{-i\pi A/2} = -\frac{I+A}{3} + \frac{2J}{15}.}$$

This is an exact symmetric involution with row amplitudes $-1/5$ on a point and its twelve neighbours and $2/15$ on its twenty-seven nonneighbours. For uniform fractional interaction error ϵ,

$$F_{\text{pro}} = 1 - 3\pi^2\epsilon^2 + O(\epsilon^3), \qquad F_{\text{avg}} = 1 - \frac{120}{41}\pi^2\epsilon^2 + O(\epsilon^3).$$

The linear average-fidelity term vanishes because $\operatorname{Tr} A = 0$.

Wigner–Smith time as operational memory. For a frequency-dependent W33 scattering unitary

$$S(\omega) = e^{i\theta(\omega)L_{W33}},$$

the Wigner–Smith operator is exactly

$$\boxed{Q(\omega) = -iS^{\dagger}\frac{dS}{d\omega} = \theta'(\omega)L_{W33}.}$$

Hence the proper-delay sectors are

$$0^1, \qquad (10\theta')^{24}, \qquad (16\theta')^{15},$$

with mean $12\theta'$, variance $12(\theta')^2$, and total trace $480\theta'$. This gives a measurable meaning to internal history: dwell-time and stored-energy sensitivity, not a changed vacuum causal speed.

For m independent W33 factors, the exact delay-shell polynomial is

$$\boxed{(1 + 24z^{10} + 15z^{16})^m.}$$

The address space has 40^m modes and

$$\langle\tau\rangle = 12m\theta', \qquad \mathrm{Var}(\tau) = 12m(\theta')^2.$$

Consequently

$$\boxed{\frac{\log_2(40^m)}{\langle\tau\rangle} = \frac{\log_2 40}{12\theta'}}$$

is independent of self-similar depth, while the relative delay width is $1/\sqrt{12m}$.

Causal-refinement boundary. A local-interaction causal cone has schematic physical velocity

$$v_{\mathrm{int}} \lesssim aC_{LR}(\Delta)J/\hbar.$$

Self-similar refinement preserves the cone only when aJ remains fixed. More serial nodes without stronger coupling slow the internal cone; node density alone neither determines nor raises vacuum c. Fabrication, measured Wigner–Smith delays, literal hidden photon nodes, variable vacuum c, and laboratory performance remain unclaimed.

38 Photon layout, delay tomography, orbital closure, and code stabilizers

The direct forty-mode W33 coupler and the eighty-mode point–line incidence lift both admit exact one-factorizations. Their resource vectors

$$(\text{modes}, \text{links}, \text{degree}, \text{layers})$$

are

$$(40, 240, 12, 12), \qquad (80, 160, 4, 4).$$

For linear cost $C = \alpha M + \beta E + \gamma R$,

$$C_{40} - C_{80} = -40\alpha + 80\beta + 8\gamma,$$

so the direct device wins exactly when

$$\boxed{5\alpha > 10\beta + \gamma}.$$

For $S(\omega) = e^{i\theta(\omega)L}$, the Wigner–Smith operator is $Q = \theta' L$. The three sector populations obey

$$p_{16} = \frac{m_2 - 10m_1}{96}, \qquad p_{10} = \frac{16m_1 - m_2}{60}, \qquad p_0 = 1 - p_{10} - p_{16}.$$

The executed central-difference reconstruction converges quadratically. Removing common delay gives the stronger identity

$$\boxed{Q - \frac{\operatorname{Tr} Q}{40} I = -\theta' A_{W33}},$$

so the complete W33 adjacency relation is recovered from the off-diagonal delay kernel.

The three fixed-parent maximum-code stabilizers are identified exactly:

$$G_{540} \cong S_4 \times V_4,$$

$$G_{270} \cong D_8 \rtimes_{\mathrm{sgn},\alpha} S_4, \qquad \alpha(r) = r^{-1}, \quad \alpha(s) = r^2 s,$$

and

$$\boxed{G_{135} \cong C_2 \wr S_4 = 2^4 : S_4 = W(B_4).}$$

The last group acts faithfully on eight coordinates preserving four opposite pairs, giving an exact four-bit hypercube/cross-polytope bridge.

For the oriented vertex–edge incidence matrix D,

$$\operatorname{Spec}(D^T D) = 0^{201} + 10^{24} + 16^{15},$$

and therefore

$$\boxed{e^{-i\pi D^T D/2} = I - 2E_{10}}.$$

The 201-dimensional kernel is raw cycle-space memory, not the 81-dimensional Hodge/CSS logical sector.

Finally, pure-state delay metrology satisfies $F_Q = 4 \operatorname{Var}(Q)$. Thus

$$F_Q^{\mathrm{vertex}} = 48(\theta')^2, \qquad F_Q^{\mathrm{opt}} = 256(\theta')^2.$$

The literal orbital Fourier/relation-fusion and Monster overgroup workflows are executable but their generated outputs remain pending. No fabricated device, measured delay, variable vacuum c, Monster embedding, remote CI/PDF success, or laboratory result is claimed.

39 Exact finite-detuning photon revival and coherent delay memory

Exact non-dispersive incidence revival. Let N be the 40×40 point–line incidence matrix of $W(3,3)$, so that $NN^{\mathsf{T}} = 4I + A$. For the full point–bus Hamiltonian

$$H = g \begin{pmatrix} 0 & N \\ N^{\mathsf{T}} & (\Delta/g)I \end{pmatrix},$$

the finite values

$$\boxed{\Delta/g = 2\sqrt{2}}, \qquad \boxed{gt = \pi/\sqrt{2}}$$

produce an exact zero-leakage revival. The half-angle windings of the singular-value-4, singular-value-$\sqrt{6}$, and bus-dark sectors are 3π, 2π, and π, respectively. Hence the point action is

$$\boxed{U_P = E_{16} - E_6 + E_0 = I - 2E_6 = -\frac{I + A}{3} + \frac{2J}{15}},$$

and the line action is $U_L = F_{16} - F_6 + F_0$. Direct exponentiation of the full eighty-mode matrix gives operator error below 6.6×10^{-15} and off-block leakage below 2.9×10^{-15}. This is not a dispersive approximation.

Quadratic-form Wigner–Smith tomography. For $\tau(v) = v^\dagger Q v$,

$$Q_{ii} = \tau(e_i),$$
$$\Re Q_{ij} = \tau\left(\frac{e_i + e_j}{\sqrt{2}}\right) - \frac{Q_{ii} + Q_{jj}}{2},$$
$$\Im Q_{ij} = \frac{Q_{ii} + Q_{jj}}{2} - \tau\left(\frac{e_i + ie_j}{\sqrt{2}}\right).$$

Thus a general Hermitian forty-mode delay matrix requires exactly $40^2 = 1600$ expectation probes, reduced to $40 + \binom{40}{2} = 820$ under reciprocal real symmetry. The exact synthetic reconstruction recovered all 240 W33 edges with maximum matrix error below 2.6×10^{-15}. For a three-frequency central difference and a linear phase law,

$$Q_h = \frac{\sin(h\theta' L)}{h} = Q - \frac{h^2}{6}(\theta')^3 L^3 + O(h^4),$$

and Richardson extrapolation removes the leading error. A separate precursor/front measurement remains mandatory before testing any mode-count-dependent causal-front slope.

Exact bright-sector write–hold–read gate. Define

$$T = N^\mathsf{T}\left(\frac{E_{16}}{4} + \frac{E_6}{\sqrt{6}}\right), \qquad X = \begin{pmatrix} 0 & T^\mathsf{T} \\ T & 0 \end{pmatrix}.$$

Then

$$T^\mathsf{T} T = E_{16} + E_6, \qquad TT^\mathsf{T} = F_{16} + F_6,$$

and

$$\boxed{W = e^{-i\pi X/2}}$$

swaps the twenty-five-dimensional point-bright sector into the line-bright sector while fixing both fifteen-dimensional dark sectors. A diagonal bus hold phase followed by $W^\dagger$ returns

$$\boxed{e^{-i\phi_0} E_0 + e^{-i\phi_6} E_6 + e^{-i\phi_{16}} E_{16}}$$

on the point modes. An arbitrary three-phase test had operator error below 2.7×10^{-15} and leakage below 3.1×10^{-15}. This is an exact controlled-Hamiltonian synthesis, not a fabricated-memory claim.

Three further photon constructions. Exact zero-leakage reflection revivals satisfy the integer quadric

$$\boxed{3n_{16}^2 - 8n_6^2 + 5k^2 = 0},$$

with $n_{16} + k$ even and $n_6 + k$ odd. The parameter map is

$$\frac{\Delta}{g} = k\sqrt{\frac{24}{n_6^2 - k^2}}, \qquad gt = 2\pi\sqrt{\frac{n_6^2 - k^2}{24}}.$$

The shortest positive primitive solution is $(n_{16}, n_6, k) = (3, 2, 1)$, which yields the revival above.

Writing $\tau = \operatorname{Tr} Q/480$, an ideal W33 delay matrix obeys the basis-free checksum

$$\boxed{Q(Q - 10\tau I)(Q - 16\tau I) = 0},$$

and the geometry is recovered self-calibratingly by

$$\boxed{A = -\frac{Q - (\operatorname{Tr} Q/40)I}{\tau}}.$$

For m commuting physical copies, all factors revive at the same gate time $gt = \pi/\sqrt{2}$. The point action is $U_P^{\otimes m}$, with

$$\boxed{m_+ = \frac{40^m + (-8)^m}{2}, \qquad m_- = \frac{40^m - (-8)^m}{2}}.$$

This is a synchronization theorem, not a hardware-compression theorem.

Algebra and evidence boundary. The literal 48-relation Fourier/fusion calculation and the GAP/mmgroup Monster maximal-overgroup search are exposed through an artifact-producing execution workflow. No relation-fusion rank, Monster words, or embedding is promoted until the generated artifacts are inspected. Fabricated hardware, measured delay matrices, literal hidden photon nodes, variable vacuum light speed, remote CI, and PDF evidence likewise remain unclaimed.

40 Physical incidence-link H_1 memory and exact projector gates

The lower-degree point–line incidence architecture has 80 modes and 160 physical couplers. Let $D \in \{-1, 0, 1\}^{80\times160}$ be its oriented vertex–edge boundary matrix and put $K = D^T D$. The exact spectrum is

$$0^{81}, \quad 8^1, \quad 4^{30}, \quad (4 - \sqrt{6})^{24}, \quad (4 + \sqrt{6})^{24}.$$

Therefore

$$\boxed{320 P_{H_1} = (K - 8I)(K - 4I)\big((K - 4I)^2 - 6I\big)}$$

is an exact integral numerator for the rank-81 Hodge/Kirchhoff cycle projector. Indeed $DP_{H_1} = 0$, $\operatorname{rank} D = 79$, and $\operatorname{rank} P_{H_1} = 81$, so its image is exactly the boundaryless link-current space. Every physical coupler has diagonal projector weight 81/160.

The corresponding exact memory reflection is

$$\boxed{R_{H_1} = I - 2P_{H_1}}, \qquad \operatorname{Tr} R_{H_1} = 79 - 81 = -2.$$

It is $+1$ on the cut space and -1 on the protected cycle-memory space.

A separate polynomial dynamics exists on the 80 mode coordinates. The raw Levi adjacency has spectrum

$$-4^1,\ (-\sqrt{6})^{24},\ 0^{30},\ (+\sqrt{6})^{24},\ 4^1$$

and is not globally periodic. Its sign fold $H_2 = A_L^2$ has spectrum $0^{30} + 6^{48} + 16^2$, giving

$$e^{-i\pi H_2} = I, \qquad e^{-i\pi H_2/2} = I - 2E_6, \qquad e^{-i\pi H_2/4} = I + (i - 1)E_6.$$

This polynomial revival is distinct from the finite-detuning block Hamiltonian of Pass 4005, although both use the same incidence matrix.

The five signed Levi sectors are recovered from four moments. With $a = p_{+4} + p_{-4}$, $b = p_{+\sqrt{6}} + p_{-\sqrt{6}}$, $d = p_{+4} - p_{-4}$, and $e = p_{+\sqrt{6}} - p_{-\sqrt{6}}$,

$$a = \frac{m_4 - 6m_2}{160}, \qquad b = \frac{16m_2 - m_4}{60},$$

$$d = \frac{m_3 - 6m_1}{40}, \qquad e = \frac{16m_1 - m_3}{10\sqrt{6}}.$$

Finally, in the ideal delay model $Q = \tau I + \theta' A_L$,

$$\boxed{Q - \frac{\operatorname{Tr} Q}{80} I = \theta' A_L},$$

so the centered delay kernel recovers all 160 point–line incidences.

The mode decomposition $30 + 48 + 2$ and the link-current decomposition $79 + 81$ are distinct vector spaces. No fabricated coupler network, measured delay, loss tolerance, variable vacuum c, literal hidden photon nodes, Monster embedding, or laboratory performance is claimed.

41 The protected H_1 sector as an exact photonic flat band

BT548 already identified the protected Levi cycle projector with the -2 eigenspace of the Levi line graph. The new step is the explicit coupler-as-site physical model and its apartment-frame and perturbation refinement.

Promote the 160 incidence links of the 80-vertex Levi graph L to the sites of

$$X = \operatorname{LineGraph}(L).$$

Then X has 160 sites, 480 links, and degree 6. For the oriented Levi boundary matrix D,

$$\boxed{A_X = D^T D - 2I.}$$

Hence

$$\boxed{\ker D = E_{-2}(A_X)},$$

and the protected $H_1 = 81$ sector is exactly the -2 flat band of X. Its spectrum is

$$6^1 + (2+\sqrt{6})^{24} + 2^{30} + (2-\sqrt{6})^{24} + (-2)^{81}.$$

Let $C \in \{-1, 0, 1\}^{160\times 1620}$ have one signed alternating column for each Levi octagon. Every such apartment column c has support 8 and satisfies

$$A_X c = -2c.$$

The 1620 compact apartment states span the full flat band, and

$$P_{H_1} = \frac{1}{160} CC^T,$$

$$CC^T = \frac{1}{2}(A_X - 6I)(A_X - 2I)(A_X^2 - 4A_X - 2I).$$

After normalizing each column by $\sqrt{8}$, they form a unit-norm tight frame:

$$\boxed{\left(\frac{C}{\sqrt{8}}\right)\left(\frac{C}{\sqrt{8}}\right)^T = 20P_{H_1}}, \qquad \frac{1620}{81} = 20.$$

Each secondary site belongs to exactly 81 apartments,

$$\boxed{160 \cdot 81 = 1620 \cdot 8 = 12960.}$$

The perturbation boundary is exact. A uniform onsite shift δI moves the band without splitting it, whereas

$$P_{H_1} |e\rangle\langle e| P_{H_1}$$

has rank one and nonzero eigenvalue 81/160. More generally,

$$\mathrm{Tr}(P_{H_1}\,\mathrm{diag}\,v) = \frac{81}{160}\sum_e v_e.$$

Thus the symmetric model has an exact flat band, but generic nonuniform onsite or coupling disorder is not proved harmless.

This 160-site line-graph Hamiltonian, the primary 80-mode incidence Hamiltonian, and the passive 160-dimensional link-current coordinates are related but distinct physical models. No fabricated secondary lattice, measured localization, disorder tolerance, local coupling synthesis, variable vacuum c, hidden-node ontology, or laboratory performance is claimed.

42 Physics-first audit of the W33 universal-computer hypothesis

Exact revival error tensor. For $d = \Delta/g$, $\tau = gt$, and the target $d_0 = 2\sqrt{2}$, $\tau_0 = \pi/\sqrt{2}$, write $\delta d = d - d_0$ and $\delta\tau = \tau - \tau_0$. The coherent target overlap and average point–bus leakage obey

$$1 - F_e = \frac{1}{2}\left[\frac{323\pi^2}{5184}(\delta d)^2 + 2\frac{17\pi}{36}\delta d\,\delta\tau + 8(\delta\tau)^2\right] + O(\delta^3),$$

$$P_{\mathrm{leak}} = \frac{1}{2}\left[\frac{149\pi^2}{5184}(\delta d)^2 + 2\frac{17\pi}{36}\delta d\,\delta\tau + 8(\delta\tau)^2\right] + O(\delta^3).$$

The locally optimal retiming law is

$$\boxed{\delta\tau = -\frac{17\pi}{288}\delta d},$$

for which

$$1 - F_e = \frac{119\pi^2}{6912}(\delta d)^2 + O(\delta^3), \qquad P_{\mathrm{leak}} = \frac{\pi^2}{2304}(\delta d)^2 + O(\delta^3).$$

Cubic physical compiler. Let

$$H_L = \begin{pmatrix} 0 & N \\ N^{\mathsf{T}} & 0 \end{pmatrix}.$$

The polar bright-sector operator is the exact cubic transform

$$\boxed{\mathrm{sign}(H_L) = \alpha H_L + \beta H_L^3},$$

with

$$\alpha = -\frac{3}{20} + \frac{4\sqrt{6}}{15}, \qquad \beta = \frac{1}{40} - \frac{\sqrt{6}}{60}.$$

Hence its point–line block is

$$\boxed{T = N^{\mathsf{T}}(\alpha I + \beta N N^{\mathsf{T}})}.$$

The dense polar coupling is therefore an exact sparse incidence traversal plus a three-hop spectral correction. A passive finite-depth hardware synthesis remains open.

Protected-memory accessibility theorem. For the oriented Levi boundary D,

$$\mathbb{R}^{160} = \mathrm{im}\,D^{\mathsf{T}} \oplus \ker D, \qquad \dim\mathrm{im}\,D^{\mathsf{T}} = 79, \quad \dim\ker D = 81.$$

The algebra generated by $D, D^{\mathsf{T}}, DD^{\mathsf{T}}$ and $D^{\mathsf{T}}D$ preserves this decomposition. Thus incidence-only linear dynamics cannot write cut/mode information into the protected H_1 memory. For one controlled link detuning,

$$V_e = P_{H_1}|e\rangle\langle e|P_{\mathrm{cut}}$$

has rank one and

$$\boxed{\sigma(V_e) = \frac{\sqrt{81 \cdot 79}}{160}}.$$

A uniform pulse has zero cross-sector matrix element. Controlled local symmetry breaking is therefore the physical write/read port rather than merely an imperfection.

Common-delay-invariant geometry tomography. For

$$Q = cI + \theta(12I - A_{W33}) + E,$$

define

$$C = Q - \frac{\operatorname{Tr} Q}{40} I.$$

Then $C = -\theta A_{W33} + E_c$, and the ideal scale and adjacency are

$$\boxed{\widehat{\theta} = \frac{\|C\|_F}{\sqrt{480}}}, \qquad \boxed{\widehat{A} = -\frac{C}{\widehat{\theta}}}.$$

The scalar common delay cancels exactly. The centered checksum is

$$\boxed{(C + 12\theta I)(C + 2\theta I)(C - 4\theta I) = 0}.$$

If $\|E_c\|_{\max} < \theta/4$ and $\|E_c\|_F/\sqrt{480} < \theta/4$, thresholding at $\widehat{\theta}/2$ recovers every labeled edge exactly. If $\|E_c\|_2 < 3\theta$, all three spectral clusters remain disjoint.

Causality and speed boundary. The diameters of the W33 graph, Levi graph, and Levi line graph are $2, 4, 4$. A fixed cell therefore supplies microscopic graph locality but not a macroscopic light cone. Any Lieb–Robinson-type physical speed has the form

$$v_{\text{LR}} \lesssim C_{\text{graph}} \frac{aJ}{\hbar},$$

so node count alone cannot determine the vacuum speed of light. For the exact finite-detuning gate,

$$t_{\text{gate}} = \frac{\pi}{\sqrt{2}g},$$

while a localized point state has $\Delta E = 2\hbar g$ and $E - E_0 = 2\sqrt{2}\hbar g$. The gate is $2\sqrt{2}$ times the orthogonal Mandelstam–Tamm time and four times the Margolus–Levitin time.

Spectral-dimension falsifier. The W33 heat trace is

$$Z(t) = 1 + 24e^{-10t} + 15e^{-16t}.$$

Its running spectral dimension has maximum approximately 3.71848244 and returns to zero in both ultraviolet and infrared limits. The Levi line-graph maximum is approximately 3.47298414. Thus neither fixed finite graph has a four-dimensional Weyl plateau. A spacetime claim requires a refinement or RG family with $Z(t) \sim t^{-2}$ over a stable scale window and Lorentzian causal observables.

Thermodynamic and holographic capacity. The H_1 flat band has dimension 81. Its 1620 apartment states form an overcomplete tight frame of redundancy 20, not 1620 orthogonal messages. A single photon restricted to this sector has orthogonal capacity at most

$$\boxed{\log_2 81 \simeq 6.33985000 \text{ bits}},$$

and erasure obeys

$$\boxed{Q_{\text{erase}} \geq k_B T \ln 81}.$$

A Bekenstein or covariant-entropy comparison remains undefined until physical energy, size or area, species content, and encoder/decoder restrictions are fixed.

Claim boundary. The finite architecture now has exact protected memory, controlled spectral unitaries, entangling protocols, error tensors, write ports, and tomography. It does not yet have a proved fault-tolerant non-Clifford injection, scalable local continuum, Standard Model representation functor, gravitational entropy–energy dictionary, dimensional scale derivation, fabricated device, or laboratory performance. "THE universal computer" and "the theory of everything" therefore remain explicit hypotheses with pass/fail obligations, not promoted conclusions.

42.1 Full harmonic compiler, projected controls, and finite-network electrodynamics (Passes 4033–4040)

The completed physics-first audit is supplemented by a deterministic construction on the same W33 point–line Levi graph. Let

$$D \in \{-1, 0, 1\}^{80\times160}, \qquad C \in \{-1, 0, 1\}^{160\times1620}, \qquad P_{H_1} = \frac{1}{160}CC^T,$$

with $\operatorname{rank} D = 79$, $\operatorname{rank} C = 81$, and $DC = 0$.

Exact write/read compiler. The first 81 canonically ordered apartment columns form an integer basis B of H_1. With

$$W = B(B^T B)^{-1/2},$$

one has

$$W^T W = I_{81}, \qquad WW^T = P_{H_1}.$$

Hence

$$H_{\text{swap}} = \begin{pmatrix} 0 & W^T \\ W & 0 \end{pmatrix}$$

performs a complete input–H_1 swap at time $\pi/2$, up to phase. The dimension mismatch is physical: all 81 harmonic channels require the 80 incidence modes plus one ancilla.

Projected disorder and control. The 160 projected onsite operators

$$A_e = P_{H_1}|e\rangle\langle e|P_{H_1}$$

are linearly independent. The 480 nearest-neighbour projected couplings span 320 dimensions, and the onsite span lies inside that coupling span through

$$A_e = -\frac{1}{4}\sum_{f\sim e} P_{H_1}\big(|e\rangle\langle f| + |f\rangle\langle e|\big)P_{H_1}.$$

All projected site rays are pairwise nonorthogonal, so their commutant on H_1 is scalar. Their associative closure is $M_{81}(\mathbb{C})$; with the uniform identity control the ideal Hamiltonian Lie closure is $\mathfrak{u}(81)$. Thus uncalibrated local detuning is a splitting mechanism, whereas calibrated detuning is a universal control alphabet.

Compressed tomography and fabrication contract. The three-sector H_2 populations require two nontrivial scalar probes. Each five-sector signed-Levi or line-graph population vector requires four. Affinely scaled Chebyshev probes have condition number below two in all three cases. The secondary flat-band realization requires 160 sites, 480 degree-six couplings, and only uniform real hopping. Its exact flat-band gap is

$$\Delta_{\text{fb}} = (4 - \sqrt{6})J.$$

A sufficient static cluster-separation condition is $\|V\|_2 < \Delta_{\text{fb}}/2$. This is a design theorem, not a fabrication claim.

Three physics constructions. First, the Lindblad jump map

$$L_v = \sum_e D_{ve} a_e$$

has dark space $\ker D = H_1$ and dissipative gap $4 - \sqrt{6}$. Second, the local projected controls generate $\mathfrak{u}(81)$ in the ideal protected model. Third, interpreting $\rho = DE$ as a discrete Gauss law gives an exact Levi effective-resistance law

$$R_1 = \frac{79}{160}, \qquad R_2 = \frac{13}{20}, \qquad R_3 = \frac{111}{160}, \qquad R_4 = \frac{7}{10}.$$

This is finite-network electrodynamics, not a continuum Maxwell or gravity derivation.

Execution boundary. The literal 48-relation fusion and Monster jobs still have no generated engine outputs. No relation-fusion rank, Monster word, class fusion, or embedding is promoted. The exact certificate is `data/PART_4033_4040_PHYSICS_CONTINUATION.json`.

43 Eight-physics expansion: holonomies, interacting flat-band matter, cooling, spectroscopy, refinement, and three outside-box constructions

Passes 4041–4048 extend the exact W33/Levi H_1 architecture along five requested physics fronts and three independent finite-model constructions. All claims below are finite-dimensional and machine checked; none is a fabricated-device, continuum-gravity, Standard-Model, or laboratory-performance claim.

43.1 Non-Abelian holonomic control in the harmonic memory

Choose an orthonormal logical pair in the 81-dimensional harmonic space, a third harmonic helper, and one excited ancilla. The ideal tripod spectrum is $\{-\Omega, 0, 0, +\Omega\}$ and the dark space has rank two. In a real tripod chart,

$$A_\phi = \cos\theta \begin{pmatrix} 0 & -1 \\ 1 & 0 \end{pmatrix}, \qquad A_\theta = 0.$$

The closed loop through $\theta = \arccos(1/4)$ yields

$$U_X = \begin{pmatrix} 0 & 1 \\ -1 & 0 \end{pmatrix},$$

while a phase loop at $\theta = \pi/6$ yields $U_Z = \mathrm{diag}(-i, 1)$. Their commutator obeys

$$\|[U_X, U_Z]\|_F^2 = 4,$$

so the exact geometric gates are non-Abelian. Varying the loop latitudes gives continuous rotations about two nonparallel axes and therefore ideal $SU(2)$ control. This does not certify finite-speed adiabatic fidelity or a laboratory pulse sequence.

43.2 Projected two-photon contact spectrum

For onsite interaction projected into the flat band,

$$V = U \sum_{e=1}^{160} |Pe, Pe\rangle\langle Pe, Pe|,$$

we have $\dim \mathrm{Sym}^2(H_1) = 3321$. The contact map has rank 160, leaving an exact contact-dark pair space of dimension

$$\boxed{3161}.$$

The complete nonzero spectrum of V/U is

$$\left(\frac{81}{160}\right)^1, \left(\frac{252+27\sqrt{6}}{800}\right)^{24}, \left(\frac{117}{400}\right)^{30}, \left(\frac{252-27\sqrt{6}}{800}\right)^{24}, \left(\frac{41}{200}\right)^{81}.$$

No bound-pair transport or many-body phase follows without additional dynamics.

43.3 Number-conserving one-shot Hodge cooling

On the charge-zero vertex sector let

$$Q = D^T(DD^T)^{-1/2}.$$

Then $Q^TQ = I_{79}$ and $QQ^T = P_{\mathrm{cut}}$. Coupling 160 system links to 79 reservoir modes through

$$H_{\mathrm{cool}} = g\begin{pmatrix} 0 & Q \\ Q^T & 0 \end{pmatrix}$$

produces at $t = \pi/(2g)$ the exact passive swap

$$|\psi\rangle_{\mathrm{sys}}|0\rangle_{\mathrm{res}} \longmapsto P_{H_1}|\psi\rangle_{\mathrm{sys}} - iQ^T|\psi\rangle_{\mathrm{res}}.$$

Thus all cut-space excitation leaves the system while the harmonic memory remains. The polar coupler is generally dense, and reservoir reset and finite-depth synthesis remain engineering obligations.

43.4 Synthetic Coulomb spectroscopy

For opposite drive at Levi vertices separated by distance d, define $s = (i\omega C + \gamma)/J$. The shell responses are

$$Z_1(s) = \frac{2(s^3+15s^2+66s+79)}{(s+4)(s+8)(s^2+8s+10)},$$

$$Z_2(s) = \frac{2(s^2+8s+13)}{(s+4)(s^2+8s+10)},$$

$$Z_3(s) = \frac{2(s^3+16s^2+78s+111)}{(s+4)(s+8)(s^2+8s+10)},$$

$$Z_4(s) = \frac{2(s^2+8s+14)}{(s+4)(s^2+8s+10)}.$$

Their DC limits are

$$\boxed{79/160,\ 13/20,\ 111/160,\ 7/10}.$$

The minimum shell gap is $1/160$, so absolute error below $1/320$ in units of $1/J$ is sufficient for nearest-shell identification.

43.5 Causal refinement tower and dimensional boundary

For a periodic d-dimensional array of cells,

$$\lambda(k) = 2\kappa \sum_{\mu=1}^{d} (1 - \cos k_\mu) = \kappa |k|^2 + O(k^4),$$

and the wave equation gives

$$\omega(k) = \sqrt{\kappa}|k| + O(k^3), \qquad c_{\mathrm{cell}} = a\sqrt{\kappa}.$$

At $N = 64$ and $\kappa/J = 0.05$, the one-dimensional tower has a stable $d_s \simeq 1$ heat-kernel window, while a four-dimensional torus has a stable $d_s \simeq 4$ window. The conclusion is a boundary rather than an emergence claim: four macroscopic dimensions occur only when four external lattice directions are explicitly supplied.

43.6 Three outside-box exact constructions

First, the discrete supercharge

$$\mathcal{Q} = \begin{pmatrix} 0 & D \\ D^T & 0 \end{pmatrix}$$

has 82 zero modes, one uniform vertex mode and 81 harmonic edge modes. With vertex-positive and edge-negative grading, the Witten index is

$$\boxed{1 - 81 = -80}.$$

Every nonzero eigenvalue occurs in an exact $\pm$ pair. This is incidence-complex supersymmetry, not particle-physics supersymmetry.

Second, one local site reflection composed with the Hodge reflection,

$$U_F = (I - 2|e\rangle\langle e|)(I - 2P_{H_1}),$$

has spectrum $(+1)^{78} + (-1)^{80} + e^{\pm i\phi}$ with

$$\boxed{\cos\phi = 1/80}.$$

Niven's theorem excludes rational ϕ/π, so the Floquet rotor has infinite order. This is not a many-body time crystal.

Third, the normalized protected site states satisfy

$$\langle u_e | u_f \rangle = -1/3,\ 1/9,\ -1/27,\ 1/81$$

for line-graph distances $1, 2, 3, 4$. Under the two-defect Hamiltonian

$$H_{ef} = P|e\rangle\langle e|P + P|f\rangle\langle f|P,$$

unit-fidelity transfer occurs at

$$\boxed{t/\pi = 80/27,\ 80/9,\ 80/3,\ 80}.$$

This is perfect transfer between nonorthogonal projected modes in an ideal protected model, not superluminal signaling or a spacetime wormhole.

Evidence firewall. The frozen semantic certificate is `08414977d5198aa43ea25127bbe7fa0e6529f56471dabe`
The exact verifier and focused regression are `analysis/w33_pass4041_4048_eight_physics_expansion.py` and `tests/test_w33_pass4041_4048_eight_physics_expansion.py`.

44 Minimal spectral compiler, protected pulse alphabet, scaling family, and magic reduction

Minimal bounded QSVT compiler. With $X = H_L/4$, the relevant spectrum is

$$\{-1, -\sqrt{6}/4, 0, \sqrt{6}/4, 1\}.$$

The unique exact odd cubic sign interpolant exceeds one on $[-1, 1]$, reaching approximately 1.09215935, and is therefore not directly QSVT-admissible. The minimal bounded odd polynomial is the quintic

$$p_5(x) = \left(\frac{9}{25} + \frac{24\sqrt{6}}{25}\right) x + \left(-\frac{48}{25} - \frac{152\sqrt{6}}{225}\right) x^3 + \left(\frac{64}{25} - \frac{64\sqrt{6}}{225}\right) x^5.$$

It has the exact certificate

$$1 - p_5(x)^2 = \frac{4096(29 - 6\sqrt{6})}{16875}(1 - x^2)\left(x^2 - \frac{3}{8}\right)^2 q_+(x)q_-(x),$$

where

$$q_\pm(x) = x^2 \pm \left(1 + \frac{\sqrt{6}}{2}\right) x + \frac{9 + \sqrt{6}}{8} > 0.$$

Thus $|p_5(x)| \leq 1$ on $[-1, 1]$, and the exact polar sign transform admits a degree-five, six-phase QSVT realization once a block encoding of $H_L/4$ is provided. Degree five is minimal because the unique degree-three interpolant violates boundedness.

Protected local phase pulses. For $P = P_{H_1}$ and

$$|u_e\rangle = \sqrt{160/81}\, P|e\rangle,$$

the normalized off-diagonal overlap alphabet is

$$-\frac{1}{3}, \quad \frac{1}{9}, \quad -\frac{1}{27}, \quad \frac{1}{81},$$

with respective per-site multiplicities $6, 18, 54, 81$. A calibrated local detuning produces

$$A_e = P|e\rangle\langle e|P = \frac{81}{160}|u_e\rangle\langle u_e|.$$

Holding amplitude δ_e for

$$t_\phi = \frac{160\phi}{81\delta_e}$$

implements

$$U_e(\phi) = I + (e^{-i\phi} - 1)|u_e\rangle\langle u_e|.$$

At $\phi = \pi$ this is a Householder reflection. Two-reflection rotations have angle alphabet

$$2\arccos(1/3), \quad 2\arccos(1/9), \quad 2\arccos(1/27), \quad 2\arccos(1/81).$$

Together with the Pass 4034 Lie-closure theorem, these are exact pulse-level generators of $\mathfrak{u}(81)$ on the protected memory.

A local four-dimensional W33-fiber family. Define

$$G_n = W33\square C_n\square C_n\square C_n\square C_n.$$

Then $|V(G_n)| = 40n^4$ and $\deg G_n = 20$. For physical side length L and internal scale M, use

$$\mathcal{L}_n = M^2 L_{W33} \otimes I + I \otimes \left(\frac{n}{L}\right)^2 \sum_{\mu=1}^{4} L_{C_n}^{(\mu)}.$$

The heat trace factors into the W33 internal trace and four identical discrete circle traces. In the window

$$M^{-2} \ll t \ll L^2, \qquad (L/n)^2 \ll t,$$

the internal excitations freeze while the rescaled discrete torus converges to a real four-torus, giving $d_s \to 4$. This is a controlled four-dimensional base with W33 internal fiber; it does not derive dimension four from W33.

Exact one-copy M36 magic reduction. For

$$|m\rangle = (0, 1, -1, 1)/\sqrt{3},$$

measure the first qubit in the X basis and retain the plus result. With probability $5/6$, the remaining qubit is

$$(-|0\rangle + 2|1\rangle)/\sqrt{5}.$$

After the Clifford $R_x(\pi/2)$ this becomes

$$|A_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}, \qquad e^{i\phi} = (-4 + 3i)/5.$$

Since $\cos\phi = -4/5$, the rational-cosine theorem implies $\phi/\pi \notin \mathbb{Q}$. A CNOT from data to this resource followed by a computational-basis resource measurement applies $\mathrm{diag}(1, e^{i\phi})$ on one outcome and its inverse on the other. Postselecting the desired outcome succeeds with total probability

$$\boxed{5/12}.$$

The irrational z-rotation, Clifford conjugates, and an entangling Clifford gate give ideal postselected universal quantum computation. Deterministic correction, logical encoding, distillation, and a fault-tolerance threshold remain open.

Finite-to-observable quadratic functor. On

$$\mathcal{H}_n = \ell^2((C_n)^4) \otimes \mathbb{C}^{40},$$

define

$$K_n = \left(\frac{n}{L}\right)^2 L_{(C_n)^4} \otimes I + I \otimes [m_0^2 I + g(12I - A_{W33})].$$

Full W33 automorphism invariance restricts the internal quadratic algebra to $\mathrm{span}\{I, A, J\}$. The resulting free species have squared masses

$$m_0^2, \qquad m_0^2 + 10g, \qquad m_0^2 + 16g$$

with multiplicities $1, 24, 15$. This is an explicit free-field observable dictionary and selection rule, not a Standard Model construction.

Outside-box shell splitter. For a localized point state, the exact reflection gives amplitude $-1/5$ on the closed neighbourhood of size 13 and $2/15$ on the far shell of size 27:

$$U|v\rangle = -\frac{\sqrt{13}}{5}|N_v\rangle + \frac{2\sqrt{3}}{5}|F_v\rangle.$$

The shell probabilities are 13/25 and 12/25, yielding $H_2(13/25) \simeq 0.998845536$ bits of single-photon mode-partition entanglement. Since $U^2 = I$, the shell split recombines exactly.

Outside-box apartment projective code. The 1620 normalized apartment states in dimension 81 have absolute inner products

$$0, \quad 1/8, \quad 1/4, \quad 3/8, \quad 1/2, \quad 1.$$

Their fourth-power frame potential is

$$F_2 = 79785/16,$$

whereas a real projective 2-design would have 97200/83. The exact defect ratio is 16351/3840. Thus the apartment frame is a five-angle tight projective code but not a projective 2-design; tightness does not imply Haar isotropy or maximal scrambling.

Outside-box spectral calorimeter. The finite W33 spectrum $0^1 + 10^{24} + 16^{15}$ has partition function

$$Z(\beta) = 1 + 24e^{-10\beta} + 15e^{-16\beta}.$$

Its Schottky heat-capacity peak occurs at

$$\beta_* \simeq 0.4127094288, \qquad k_B T_*/J \simeq 2.4225277564,$$

with

$$C_{\max}/k_B \simeq 3.8006256108.$$

This is an exact single-particle spectral fingerprint, not a measured or interacting thermodynamic response.

Boundary. The results establish an exact minimal bounded spectral compiler, pulse-level protected controls, a convergent W33-fibered four-dimensional family, ideal postselected non-Clifford universality, and a free quadratic observable functor. They do not establish a fabricated block encoding, extracted phase list, deterministic fault-tolerant injection, Lorentzian spacetime, Standard Model, gravity, measured calorimetry, or a theory of everything.

44.1 Transitionless holonomies, dark-pair characters, local Hodge cooling, and finite-graph physics

Passes 4057–4064 close five physics targets and three independent outside-box probes on the exact Levi–H_1 substrate. All statements in this subsection are finite graph, representation-theoretic, spectral, lattice, Floquet, or canonical-ensemble statements; none is evidence for fabricated hardware, measured gate fidelity, a chiral Standard Model, quantized pumping, gravity, cosmology, or a theory of everything.

For the two non-Abelian tripod loops, the Kato counterdiabatic term

$$H_{\mathrm{CD}} = i[\dot{P}_D, P_D]$$

produces exact finite-time parallel transport in the ideal dark bundle. Their Fubini–Study actions are

$$L_X = 2\arccos(1/4) + \frac{\pi\sqrt{15}}{2} = 8.719900157\ldots, \qquad L_Z = \frac{\pi}{3} + \frac{\pi\sqrt{3}}{2} = 3.767896598\ldots.$$

Thus a constant-speed duration T requires peak counterdiabatic norm L/T, while uniform loss gives no-jump survival $e^{-\kappa T}$.

The 3161-dimensional two-boson contact-dark module was decomposed by generating $PSp(4,3)$ as a 25,920-element permutation group, reconstructing its twenty conjugacy classes and class algebra, and diagonalizing the center. Grouped by equal ordinary-character degree, its multiplicities are

d	1	5	6	10	15	20	24	30	40	45	60	64	81
m	0	$(1,1)$	2	$(1,1)$	$(1,3)$	5	2	$(3,6,6)$	$(4,4)$	$(4,4)$	9	8	9.

The weighted dimension is exactly 3161, and there is no trivial constituent. The bright rank-160 contact sector is an equivariant incidence-edge permutation module, so

$$P_{\text{dark}} V_{\text{contact}} P_{\text{dark}} = 0;$$

contact-only first-order pair motion inside the dark sector is forbidden.

Because the nonzero Levi Laplacian spectrum is $\{4-\sqrt{6}, 4, 4+\sqrt{6}, 8\}$, the inverse square root is represented exactly on the physical spectrum by a cubic Lagrange polynomial p_3. Consequently

$$Q = D^T p_3(DD^T), \qquad Q^T Q = I - J/80, \qquad QQ^T = P_{\text{cut}}.$$

The polar cooler can therefore be compiled from three nearest-neighbour Levi-Laplacian queries and one incidence query, together with the bounded degree-five sign compiler of Pass 4049. The collapsed matrix remains dense; this is exact circuit/query locality, not a one-layer sparse Hamiltonian.

On the explicit four-dimensional refinement tower, the gauge-covariant Wilson–Dirac operator

$$H(k) = \sum_{\mu=1}^{4} \gamma_\mu \sin k_\mu + \gamma_5 \left[m + r \sum_\mu (1 - \cos k_\mu) \right]$$

is nearest-neighbour and may be tensored with I_{81} on the harmonic fiber. At $m = 0$, the naive operator has sixteen corner zeros, whereas $r = 1$ leaves only the origin and gives the other corners Wilson masses $2, 4, 6, 8$. The price is explicit breaking of exact lattice chirality; the construction is vectorlike and does not evade the Nielsen–Ninomiya boundary.

Feeding the irrational Floquet phase $\cos\phi = 1/80$ into a static Coulomb port yields $V_n = Z_d(s) I_n$. The static linear network preserves the input quasifrequencies and

$$\frac{1}{N} \sum_{n<N} e^{in\phi} \longrightarrow 0, \qquad \left| \sum_{n=0}^{N-1} e^{in\phi} \right| \le \frac{2}{|1 - e^{i\phi}|}.$$

Hence there is no DC pump and no integer topological transport from one irrational phase alone; a second independently cycled parameter and a maintained gap are required.

Three outside-box consequences are exact. First, two projected Householder reflections convert graph distance into quasienergy through

$$|\langle u_e, u_f \rangle| = 3^{-d}, \qquad \theta_d = 2\arccos(3^{-d}),$$

giving four infinite-order distance clocks. Second, the repulsive two-boson contact model has residual entropy $k_B \log 3161$ and infinite-temperature dark probability 3161/3321. Third, the Levi graph has edge connectivity four, so any $r \le 3$ missing couplers leave

$$\dim H_1 = 81 - r, \qquad \mathcal{I}_W = -80 + r,$$

providing a topological fault odometer up to the four-edge connectivity threshold.

The executable certificate is `data/PART_4057_4064_ADVANCED_PHYSICS.json`, with semantic SHA-256 `131ba5f89474a0283eedd0ae1ec5e5e6618b6d3e2bd6eeb8654fd4df8abbcd72`.

45 Explicit QSP, adaptive irrational magic, Lorentzian Dirac dynamics, and exact physical obstructions

Six-phase QSP compiler. In the W_x convention

$$W(x) = \begin{pmatrix} x & i\sqrt{1-x^2} \\ i\sqrt{1-x^2} & x \end{pmatrix}, \quad U_{\boldsymbol{\phi}}(x) = e^{i\phi_0 Z} \prod_{j=1}^{5} [W(x) e^{i\phi_j Z}],$$

the bounded quintic sign compiler is realized by

$$\boldsymbol{\phi} = (-2.8067951015434964, -2.042810210594929, -2.419664116537273, 2.419664116537273, -1.0987824429948$$

On a 20,001-point grid the top-left polynomial residual is below 1.19×10^{-15}. The sequence uses five signal queries and six phase gates. A telescoping perturbation bound is

$$\|\delta U\| \le 6\delta_\phi + 5\eta_W.$$

Adaptive irrational-magic correction. For $e^{i\phi} = (-4+3i)/5$, an undesired $-\phi$ injection branch is corrected with 2ϕ, then 4ϕ, and so forth. After K correction levels,

$$p_{\text{fail}} = 2^{-(K+1)}, \qquad \|\mathcal{E}_K - \mathcal{U}_\phi\|_\diamond \le 2^{-K}.$$

If doubled-angle states are directly available, fewer than two are injected on average. If each is recursively produced from the original state by parity doubling, the expected raw cost is

$$N_{\text{raw}} = 2^{K+1} - 1.$$

No finite doubling tree is exactly deterministic because the final residual differs from the target by $2^{K+1}\phi$, and ϕ/π is irrational.

Causal W33-fibered Dirac walk. Let

$$\beta = Z \otimes I, \qquad \alpha_j = X \otimes \sigma_j, \qquad m_r = \sqrt{m_0^2 + g\lambda_r},$$

where $\lambda_r \in \{0, 10, 16\}$ with multiplicities $1, 24, 15$. The exact three-spatial-dimensional local walk is

$$U_a(p,r) = e^{-iam_r\beta} e^{-iap_1\alpha_1} e^{-iap_2\alpha_2} e^{-iap_3\alpha_3}.$$

Each factor is a conditional nearest-neighbour shift, and

$$\frac{U_a - I}{-ia} = m_r\beta + \sum_{j=1}^{3} p_j\alpha_j + O(a).$$

Thus discrete time plus three spatial directions supplies a controlled $3+1$-dimensional Dirac limit carrying the three W33 mass sectors.

SK1-protected H1 pulse. For common fractional amplitude error ϵ,

$$U_{\text{SK1}} = R_{-\chi}(2\pi(1+\epsilon)) R_\chi(2\pi(1+\epsilon)) R_x(\theta(1+\epsilon)), \quad \chi = \arccos\left(-\frac{\theta}{4\pi}\right).$$

The linear error vanishes. For $\theta = \pi$,

$$R_x(\pi)^\dagger U_{\text{SK1}} = I - i\frac{\sqrt{15}\pi^2}{8}\epsilon^2 Z + O(\epsilon^3),$$

so the entanglement infidelity begins at $15\pi^4\epsilon^4/64$.

Gauge-commutant obstruction. The W33 permutation module decomposes as

$$\mathbb{C}^{40} = \mathbf{1} \oplus V_{24} \oplus V_{15},$$

and its full automorphism commutant is

$$\mathrm{span}\{E_{12}, E_2, E_{-4}\} = \mathrm{span}\{I, A, J\}.$$

This algebra is commutative, so its unitary gauge group is exactly

$$\boxed{U(1)^3}.$$

Unbroken W33 symmetry alone cannot generate non-Abelian $SU(2)$ or $SU(3)$ gauge bosons or mix the three sectors. A non-Abelian gauge theory requires symmetry breaking, multiplicity spaces, or a larger noncommutative algebra.

Three outside-box exact results. First, the reflection $U = I - 2E_6$ obeys $U^2 = I$, so repeated reflection has orbit dimension at most two and pure-state time-averaged entropy at most one bit. A localized vertex gives $H_2(3/5) = 0.97095059445$ bits; the reflection alone cannot thermalize or generate a Page curve.

Second, local H1 defect sensing satisfies

$$F_e^{\max} = \left(\frac{81}{160}\right)^2 t^2, \qquad \sum_e F_e \le 2t^2.$$

The total bound is attained by an equal-magnitude divergence-free Eulerian flow, giving uniform-prior average QFI $t^2/80$.

Third, the matrix-tree theorem gives

$$\tau(W33) = 2^{81}5^{23}, \qquad \tau(L_{\mathrm{Levi}}) = 2^{83}5^{23}, \qquad \frac{\tau(L_{\mathrm{Levi}})}{\tau(W33)} = 4.$$

The incidence lift changes Kirchhoff complexity by exactly two bits.

Boundary. These are exact algebraic or deterministic numerical statements in explicitly specified models. No fabricated block encoding, finite-cost exact deterministic irrational injection, fault-tolerance threshold, Standard Model, gravity, cosmology, or theory of everything is established.

45.1 Passes 4073–4080: engineering closures and outside-box physics probes

Exact four-router block encoding. The 4-regular Levi graph admits a one-factorization $A_L = P_0 + P_1 + P_2 + P_3$ into four involutory perfect-matching permutations. Consequently a balanced four-path selector and the controlled router $U_{\mathrm{sel}} = \sum_j |j\rangle\langle j| \otimes P_j$ obey

$$(\langle 00|B^\dagger \otimes I)U_{\mathrm{sel}}(B|00\rangle \otimes I) = \frac{A_L}{4}.$$

This is an exact abstract block encoding with two selector bits or paths; loss, crosstalk, and fabrication are not included.

Finite phase reference. Compressing K copies of $|A_\phi\rangle = (|0\rangle + e^{i\phi}|1\rangle)/\sqrt{2}$ into the symmetric Hamming-weight basis and applying a controlled cyclic lowering produces coherence multiplier

$$z_K = e^{i\phi}S_K + 2^{-K}e^{-iK\phi}, \qquad S_K = 2^{-K}\sum_{n=1}^{K}\sqrt{\binom{K}{n-1}\binom{K}{n}}.$$

The one-use channel error is at most $1 - S_K + 2^{-K}$, with $1 - S_K \sim (2K)^{-1}$. This removes the doubled-angle hierarchy but is only approximately catalytic.

Minimal non-Abelian multiplicities. The multiplicity extension $(m_1, m_{24}, m_{15}) = (3, 1, 2)$ has dimension 57 and commutant

$$M_3(\mathbb{C}) \oplus \mathbb{C} \oplus M_2(\mathbb{C}).$$

After freezing the spectator $U(1)$, its determinant-one subgroup has global form $S(U(3)\times U(2)) \cong (SU(3)\times SU(2)\times U(1))/\mathbb{Z}_6$ and central charge ratio $-1/3 : 1/2$. The present Dirac matter remains vectorlike; chirality and the observed matter representation are not derived.

Four-gap leakage notch. For the bright gaps $\Delta/J \in \{4-\sqrt{6}, 4, 4+\sqrt{6}, 8\}$, choose

$$f = \frac{h^{(8)} + 124h^{(6)} + 4644h^{(4)} + 53056h'' + 102400h}{102400},$$

with h and its first seven derivatives vanishing at the pulse endpoints. Then

$$F(\omega) = H(\omega)\frac{\prod_r(\Delta_r^2 - \omega^2)}{102400},$$

so all first-order H_1-to-bright transition amplitudes vanish while $\int f = \int h$. Leakage probability therefore begins at fourth order in the perturbative control strength.

Independent falsifiers and outside-box correspondences. The packet freezes six laboratory null tests. It also identifies P_{H_1} with the Maxwell–Calladine state-of-self-stress projector of an 80-node, 160-spring scalar network, giving one uniform mechanism and 81 self-stress states. A sudden quench of $H(g) = gL_{W33}$ has an exact three-line work distribution and obeys $\langle e^{-\beta W}\rangle = Z_1/Z_0$. Finally, the normalized Sakaguchi–Kuramoto linearization has decay rates $(5/6)K\cos\alpha$ with multiplicity 24 and $(4/3)K\cos\alpha$ with multiplicity 15, fixing the relaxation-time ratio to 8/5.

Boundary. These are exact finite or perturbative ideal-model statements. They do not certify a fabricated interferometer, an exact finite reusable irrational-phase catalyst, a fault-tolerance threshold, a chiral Standard Model, a mechanical prototype, a global nonlinear synchronization theorem, gravity, cosmology, or a theory of everything.

45.2 Canonical dark sectors, mobile pairs, overlap chirality, and operational geometry

The rank-3161 contact-dark two-photon module has now been resolved into all ordinary $PSp(4,3)$ irreducibles. The twenty conjugacy classes were mapped to ATLAS order

$$1A, 2A, 2B, 3A, 3B, 3C, 3D, 4A, 4B, 5A, 6A, 6B, 6C, 6D, 6E, 6F, 9A, 9B, 12A, 12B,$$

and every character fingerprint was frozen exactly in $\mathbb{Q}(\sqrt{-3})$. The central idempotents are

$$e_\chi = \frac{\chi(1)}{25920}\sum_C \overline{\chi(C)}K_C.$$

The dark module contains no trivial constituent; its nineteen nonzero isotypic dimensions sum to 3161.

After introducing an explicit same-cell dark-pair binding energy $-\Delta$ and symmetry-preserving one-photon hopping t, second-order Schrieffer–Wolff reduction gives

$$J_{\text{pair}} = \frac{2t^2}{\Delta}, \qquad E(k) = -\Delta - \frac{4t^2}{\Delta} - \frac{4t^2}{\Delta}\cos k,$$

with bandwidth $8t^2/\Delta$ and $m^* = \Delta/(4t^2a^2)$ for $\hbar = 1$. All nineteen dark isotypic sectors share this clean dispersion and do not mix. This mobility requires the added binding scale; bare repulsive contact darkness alone is not a bound mobile phase.

A two-parameter dimerization cycle

$$J_{1,2} = J \pm R\cos\theta, \qquad m = 2R\sin\theta,$$

produces a pair-center Rice–Mele Hamiltonian whose lower band has $C = +1$ and minimum gap $4R$. At clean filled-band occupancy it pumps one composite pair, hence two photons, per cycle.

On the four-dimensional external tower, the overlap operator

$$D_{\text{ov}} = a^{-1}\big[I + \gamma_5\,\text{sign}(H_W)\big]$$

obeys the exact Ginsparg–Wilson relation

$$\gamma_5 D + D\gamma_5 = aD\gamma_5 D.$$

For $0 < m_0 < 2r$ one light external species survives per harmonic fibre, so the rank-81 fibre carries 81 light vectorlike Dirac species. This does not construct anomaly-free chiral gauge matter.

Marking a single incidence edge reduces the symmetry to a stabilizer of order 162. The restricted harmonic module has commutant dimension 45; a six-dimensional stabilizer irrep occurs with multiplicity three, giving an exact $U(3)$ block, while several three-dimensional irreps occur with multiplicity two, giving exact $U(2)$ blocks. Thus one marked edge is minimal in mark count for exposing $SU(3)$ and $SU(2)$ control subalgebras; these are commutant algebras, not identified Standard Model gauge bosons.

Three independent finite-physics consequences follow. First, one ideal Levi vertex port has local resolvent

$$G_{00}(s) = \frac{s^4 + 16s^3 + 78s^2 + 112s + 4}{s(s+4)(s+8)(s^2+8s+10)},$$

so its poles and residues recover the complete Levi spectrum and multiplicities. Second, erasing a maximally mixed dark state costs at least

$$k_BT\ln 3161 = 8.058643712215618\,k_BT.$$

Resolving and retaining the irrep label lowers the conditional data cost by $2.3278663282163152\,k_BT$, but erasing that label restores the same Landauer bound. Third, for $H = JA_{\text{line}}$ the order of the first nonzero short-time derivative of $\langle u|e^{-iHt}|v\rangle$ equals graph distance exactly. The four nontrivial shells have sizes $6, 18, 54, 81$ and first geodesic multiplicities $1, 1, 1, 2$.

These are exact finite group, band, operator, resolvent, thermodynamic, and graph-propagation statements. They establish neither fabricated hardware, measured quantization, chiral Standard Model matter, gravity, cosmology, nor a theory of everything.

46 Passes 4089–4096: implementation closure and outside-box probes

Theorem 5 (Router layout, optimal phase reference, anomaly bridge, and four-notch control)**.** *The* 80*-vertex Levi adjacency admits a deterministic decomposition* $A_L = P_0+P_1+P_2+P_3$ *into four involutory perfect-matching permutations. Hence a balanced four-path selector gives an exact block encoding* $A_L/4$*. The frozen* 160*-swap route table is* `data/w33_pass4089_four_router_layout.json`*.*

For a finite phase reference on weights $0,\dots,K$*, the maximum nearest-neighbour coherence is*

$$\max\sum_{n=1}^{K} c_{n-1}c_n = \cos\frac{\pi}{K+2},$$

attained by $c_n = \sqrt{2/(K+2)}\,\sin((n+1)\pi/(K+2))$. *The resulting one-use error is* $O(K^{-2})$.

The exterior representation ring generated by the multiplicity spaces $\mathbb{C}^3$ *and* $\mathbb{C}^2$, *with determinant-one central charges* $(-1/3, 1/2)$, *contains*

$$(3,2)_{1/6},\ (\bar{3},1)_{-2/3},\ (\bar{3},1)_{1/3},\ (1,2)_{-1/2},\ (1,1)_1,$$

and all perturbative gauge, mixed gravitational, cubic Abelian, and mod-two $SU(2)$ *anomalies cancel exactly. This is a representation-ring identity, not a particle derivation.*

Finally, the compact envelope $h = 218790T^{-1}x^8(1-x)^8$ *and*

$$f = \frac{h^{(8)} + 124h^{(6)} + 4644h^{(4)} + 53056h'' + 102400h}{102400}$$

annihilate all four H_1*-to-bright first-order amplitudes. A full* 160*-mode simulation gives leakage exponents* 2.0191 *for a square pulse and* 4.0191 *for the notched pulse.*

Corollary 6 (Mechanical, coding, pattern, and circuit consequences)**.** *The scalar mechanical dual has* 81 *states of self stress and remains spectrally resolvable under the sufficient worst-case spring tolerance*

$$\epsilon < \frac{4-\sqrt{6}}{16} = 0.0969068911.$$

Its real cycle code has exact sparse parameters $[160, 81, 8]$, *detecting seven and uniquely correcting three noiseless sparse bond errors. Two explicit reaction–diffusion Jacobians select respectively the* 24*- and* 15*-dimensional W33 Laplacian sectors as the unique linearly unstable pattern spaces. The unit-resistor W33 network has only two effective resistances,*

$$R_{\mathrm{adj}} = \frac{13}{80}, \qquad R_{\mathrm{nonadj}} = \frac{7}{40},$$

with Kirchhoff index $267/2$.

Evidence boundary. No fabricated router, finite exact reusable irrational reference, dynamically derived Standard Model, measured pulse, mechanical prototype, nonlinear pattern experiment, gravity, cosmology, or theory of everything is asserted.

47 Passes 4097–4104: interacting pumping, quantum links, horizon falsifier, and spectral renormalization

Exact fractional many-composite pump. On the nine-cell, three-pair hard-core ring

$$H_0/U = \sum_i (n_i + n_{i+1} + n_{i+2} - 1)^2,$$

the fixed-filling spectrum is

$$0^3 + 2^{18} + 4^{36} + 6^{18} + 10^9.$$

The exact ground patterns 100100100, 010010010, and 001001001 have Resta polarizations $0, 1/3, 2/3$ and gap $2U$. With twisted holonomy

$$W(\phi) = e^{i\phi/3}T_3,$$

we have $\det W = e^{i\phi}$, so the rank-three ground bundle has Chern number one while one branch transports one third of a composite-pair charge per cycle. Three cycles pump one pair, or two photons.

Finite $SU(3)$ quantum-link sector. For link Hilbert space $\mathbf{1} \oplus (\mathbf{3} \otimes \bar{\mathbf{3}})$, the 81-dimensional one-link endpoint sector has a one-dimensional exact Gauss-law kernel,

$$|\text{string}\rangle = \frac{1}{3} \sum_{a,b} |\bar{a}\rangle |a, \bar{b}\rangle |b\rangle.$$

Since $C_2(\mathbf{3}) = 4/3$, the electric string energy is

$$E_{\text{string}}(R) = \frac{2g^2 R}{3}, \qquad \sigma = \frac{2g^2}{3a}.$$

The minimal vacuum/loop plaquette block is

$$H_{\Box} = \begin{pmatrix} 0 & -J \\ -J & 8g^2/3 \end{pmatrix},$$

with gap $2\sqrt{16g^4/9 + J^2}$.

Passive-horizon no-go. For a reciprocal number-conserving chain with real hopping profile J_n, all open-chain hopping signs are removable by local phases. Moreover $U(t) = e^{-iht}$ is symmetric and its Nambu evolution is $\text{diag}(U, U^*)$, so the Bogoliubov creation block vanishes exactly:

$$\beta = 0, \qquad \langle N \rangle_{\text{vacuum}} = 0.$$

A static reciprocal $J(x)$ profile alone therefore creates neither a one-way horizon nor spontaneous Hawking radiation. Directed Floquet/background flow and particle–hole mixing are additional necessary ingredients.

Discrete-scale spectral fixed cycle. At hierarchy level n, retain one zero mode and modes

$$\lambda_\alpha b^{-2(n-1-j)}$$

with multiplicity $m_\alpha 80^j$, where

$$(\lambda_\alpha, m_\alpha) = (4 - \sqrt{6}, 24), (4, 30), (4 + \sqrt{6}, 24), (8, 1).$$

The total dimension is 80^n, and the log-period-average spectral dimension is

$$\bar{d}_s = \frac{\log 80}{\log b}.$$

For $b = 80^{1/4}$, $\bar{d}_s = 4$. At $n = 14$ the computed central-window mean is 3.99987084247, with persistent log-periodic oscillations between 2.97765560 and 5.14402331. Thus the construction is a discrete-scale fixed cycle, not a smooth continuum fixed point; the value four depends on the supplied rescaling.

Topological information engine. For the 19 dark isotypic dimensions n_j, with $p_j = n_j/3161$,

$$H(J) = 2.327866328216315, \qquad \sum_j p_j \log n_j = 5.730777383999302,$$

and

$$H(J) + \sum_j p_j \log n_j = \log 3161.$$

A reversible expansion extracts $k_B T \log 3161$, while conditional erasure and label erasure cost exactly the same total. Irrep-controlled Chern pumps can robustly route one pair per cycle, but the maximum closed-cycle net work remains zero.

Three outside-box constructions. First, the invariant CDW clock subspace with

$$H_{\text{scar}} = \Omega(T_3 + T_3^\dagger)$$

has eigenvalues $-\Omega, -\Omega, 2\Omega$ and exact return probability

$$P_A(t) = \frac{5 + 4\cos(3\Omega t)}{9},$$

with perfect revival at $2\pi/(3\Omega)$. Second, the qutrit-cat probe over the three fractional-pump branches has

$$F_Q = \frac{8N_p^2}{3}$$

for a pair-position phase and $32N_p^2/3$ for the corresponding photon-gradient phase. Third, the canonical Fisher length of the projected contact spectrum from $\beta = 0$ to ∞ is 0.4530018812338062, while the Fisher–Rao endpoint geodesic is 0.4425945899021065, an excess of only 2.3514% because 3161/3321 of the Hilbert space is already dark.

Evidence boundary. These are exact finite strong-coupling, Gauss-law, passive-Bogoliubov, hierarchical-spectrum, information-thermodynamic, invariant-subspace, and Fisher-geometry statements. They do not establish a fractional-pump experiment, continuum QCD, Hawking observation, physical four-dimensional spacetime, positive-work engine, generic scar phase, gravity, cosmology, or a theory of everything.

Passes 4105–4112: carrier audit, correlated reference, implementation contracts, and three outside-box probes

The candidate module

$$(\mathbb{C}^6 \otimes V_1) \oplus (\mathbb{C}^3 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24})$$

has dimension 99 and commutant $M_6 \oplus M_3 \oplus M_2$, but it contains only one independent anti-triplet and no independent charged singlet. Under the sector assignment $Q, e^c \mapsto V_1$, $u^c, d^c \mapsto V_{15}$, and $L \mapsto V_{24}$, the minimum direct carrier is therefore

$$(\mathbb{C}^7 \otimes V_1) \oplus (\mathbb{C}^6 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24}), \qquad \dim = 145.$$

This carries the anomaly-free one-generation representation as a representation architecture, not as a derived particle model.

For the sine phase reference $c_n = \sqrt{2/(K+2)}\sin((n+1)\theta)$, $\theta = \pi/(K+2)$, sequential use gives a collective channel

$$\mathcal{E}_N(|x\rangle\langle y|) = z_{|x|-|y|}|x\rangle\langle y|,$$

with exact bulk overlap

$$C_m = \frac{(K+1-m)\cos(m\theta) + \sin((m+1)\theta)/\sin\theta}{K+2}.$$

Equal-Hamming-weight coherences are decoherence-free and the fixed-N error is $O(N^2/K^2)$.

The four-router signal unitary compiles to four coherent 80-mode branches with 40 parallel swaps per branch. A recirculating five-query realization uses 160 swap cells, six selector/recombiner couplers, six phase shifters, and five coherent passes; an unrolled realization uses 800 swap cells.

For noisy self-stress syndrome data $y = De + n$, exhaustive support-three decoding obeys

$$\|\widehat{e} - e\|_2 \le \frac{2\|n\|_2}{2\sin(\pi/14)},$$

because every union of two support-three candidates is a forest and the worst six-edge restricted singular value is $2\sin(\pi/14)$. Cubically saturated reaction–diffusion simulations converge with unit numerical modal purity into the intended 24- and 15-dimensional eigenspaces.

Finally,

$$\frac{J}{40} = I - \frac{13}{80}L + \frac{1}{160}L^2$$

gives exact two-round average consensus; simple random walk has adjacent and nonadjacent hitting times 39 and 42 with Kemeny constant 801/20; and the abstract Hückel model has a 50-electron closed shell with gap $6|\beta|$ and a 40-electron open-shell determinant count $\binom{48}{10} =$ 6,540,715,896.

47.1 Passes 4113–4120: gauge-string pumping, active horizons, dynamic dimension, local scar compilation, and thermodynamic curvature

The exact evidence bundle for this subsection is `data/PART_4113_4120_GAUGE_HORIZON_DIMENSION_SCAR_CURV` with semantic SHA-256 `28391f98e03ec20cc688cc8692579ff0cf352930d085b2d33d979de7ef9a3e2a`. All statements below are finite-model statements; continuum QCD, observed Hawking radiation, physical spacetime emergence, a static local scar Hamiltonian, thermodynamic singularities, and non-Abelian anyons remain outside the evidence boundary.

Gauge-string fractional pump. Tensoring the three one-third-filled CDW branches with the unique finite $SU(3)$ Gauss singlet

$$|\mathcal{M}\rangle = \frac{1}{3}\sum_{a,b=1}^{3} |\bar{a}\rangle|a,\bar{b}\rangle|b\rangle$$

gives a one-dimensional simultaneous left/right Gauss-law kernel in the 81-dimensional endpoint–link–endpoint space. A branch cycle transports one third of a gauge-invariant composite and its attached fundamental–antifundamental flux support. After three cycles one complete singlet composite crosses a cut, but the net transported color charge is zero. In the zero-flux sector the three-cycle Wilson holonomy is the identity; a prescribed $\mathbb{Z}_3$ center flux contributes only $e^{2\pi ik/3}I$. Non-Abelian Wilson mixing therefore requires a degenerate color multiplet or nontrivial plaquette background.

Active Floquet–Bogoliubov horizon. The passive reciprocal no-go is evaded by combining directed walk dispersion

$$\Omega(k) = \frac{2}{\tau}\arcsin\left[v\sin\left(\frac{ka}{2}\right)\right]$$

with a local two-mode squeezer. At a horizon $u(x_H) = c_{\rm eff}$,

$$\kappa = |\partial_x(u - c_{\rm eff})|_{x_H}, \qquad T_H = \frac{\kappa}{2\pi},$$

and

$$\tanh r = e^{-\pi\omega/\kappa}, \qquad n_\omega = \sinh^2 r = \frac{1}{e^{2\pi\omega/\kappa} - 1}.$$

A greybody beam splitter of transmission Γ gives $n_{\rm out} = \Gamma n_\omega$. The full six-channel Nambu map is paraunitary with residual $2.282836126979115 \times 10^{-16}$ in the frozen demonstration. For $\kappa = 0.4$, $\omega = 0.3$, and $\Gamma = 0.7$, $T_H = 0.06366197723675814$, $n_\omega = 0.009064722057396675$, $n_{\rm out} =$ 0.006345305440177674, and the post-greybody logarithmic negativity is 0.15624263179344325. The lattice ultraviolet correction begins at cubic order,

$$\Omega(k) = \frac{va}{\tau}k + \frac{v^3 - v}{24}\frac{a^3}{\tau}k^3 + O(k^5a^5).$$

Dynamically selected spectral dimension. Let $s = \ln b$, use the exact branch count $B = 80$ and exact Levi edge connectivity $\chi = 4$, and define

$$\Phi(s) = \frac{1}{2}[\ln 80 - 4s]^2, \qquad \dot{s} = \gamma[\ln 80 - 4s].$$

Then

$$\dot{\Phi} = -4\gamma[\ln 80 - 4s]^2 \le 0,$$

so

$$s_* = \frac{\ln 80}{4}, \qquad b_* = 80^{1/4} = 2.9906975624424406$$

is globally stable, with linear exponent -4γ. The selected spectral dimension is

$$d_s = \frac{\ln 80}{\ln b_*} = 4.$$

This is an explicit four-channel information-balance law, not an independent derivation of physical spacetime.

Local scar compilation and static no-go. The three CDW words 100100100, 010010010, and 001001001 have pairwise Hamming distance six. Any operator supported on fewer than six sites therefore has zero direct matrix element between distinct branches. If their bare span is exactly invariant, a strictly finite-range static Hamiltonian cannot realize the nontrivial branch cycle. An exact local Floquet realization nevertheless exists: apply the eight adjacent swaps $(0, 1), (1, 2), \ldots, (7, 8)$ sequentially, each generated for time τ by

$$H_i = \frac{\pi}{2\tau}(I - \mathrm{SWAP}_{i,i+1}).$$

One compiled shift takes $T_{\text{shift}} = 8\tau$, cycles the three CDW branches, and obeys

$$U_{\text{shift}}^3 = I, \qquad T_{\text{rev}} = 24\tau.$$

The Floquet quasienergies are 0 and $\pm 2\pi/(3T_{\text{shift}})$.

Thermodynamic curvature. Assign each dark isotypic block of dimension n_j the splitting charge $q_j = \ln n_j$, while the five bright groups retain their exact contact energies and have $q = 0$. Then

$$Z(\beta, h) = \sum_{j \in \text{dark}} n_j e^{-h \ln n_j} + \sum_{a \in \text{bright}} g_a e^{-\beta E_a}, \qquad g_{ab} = \mathrm{Cov}(T_a, T_b), \quad T = (-E, -q).$$

The two-dimensional Hessian scalar curvature satisfies

$$R(0, 0) = -0.6460278596846867,$$

crosses zero at $\beta U = 14.579166757087052$, and reaches $R(20, 0) = 0.20080335722980544$. Because Z is a finite positive exponential sum, this is a finite information-geometric crossover rather than a thermodynamic singularity.

Three outside-box consequences. First, the branch shift X and branch twist Z satisfy

$$ZX = \omega XZ, \qquad \omega = e^{2\pi i/3},$$

with residual $2.482534153247273 \times 10^{-16}$, yielding the qutrit Heisenberg–Weyl logical algebra and projective commutator ωI; this is projective holonomy, not anyon braiding. Second, the principal logarithm of X has phases $0, \pm 2\pi/3$, so any generator with $\|H - cI\| \le \Lambda$ obeys the exact unitary-geodesic limit

$$T \ge \frac{2\pi}{3\Lambda}.$$

Third, at $b_* = 80^{1/4}$ the hierarchy obeys

$$\lambda_{n+m} = \lambda_n b_*^{-2m}, \qquad \omega_{n+m} = \omega_n b_*^{-m}, \qquad 1 + z_m = b_*^m = 80^{m/4}.$$

For $m = 1, 2, 3, 4$, the spectral redshifts are 1.9906975624424406, 7.944271909999156, 25.749612199056866, and 79. This is discrete spectral scaling, not gravitational or cosmological redshift.

47.2 Passes 4121–4128: explicit gauge audit, relational phase coding, decoding, foundry feasibility, attractors, and outside-box probes

The explicit 145-dimensional carrier

$$(\mathbb{C}^7 \otimes V_1) \oplus (\mathbb{C}^6 \otimes V_{15}) \oplus (\mathbb{C}^2 \otimes V_{24})$$

was equipped with concrete $SU(3) \times SU(2) \times U(1)$ generators. This calculation corrects the earlier multiplicity-space reading: the complete carrier has W33 weights $(q, u, d, l, e) = (1, 15, 15, 24, 1)$ and is anomalous,

$$\mathcal{A}_{3^3} = -28, \quad \mathcal{A}_{3^2Y} = -\frac{7}{3}, \quad \mathcal{A}_{2^2Y} = -\frac{23}{4}, \quad \mathcal{A}_{\text{grav}^2Y} = -37, \quad \mathcal{A}_{Y^3} = -\frac{599}{36}.$$

The weak-doublet count is 27. Without exotic representations, the anomaly equations force equal W33 weights $q = u = d = l = e$. Exact repair architectures therefore require either a rank-one PSp-breaking generation inside the 145 states, a 225-dimensional V_{15} generation, a 360-dimensional V_{24} generation, or 600 dimensions containing one complete generation in each of V_1, V_{15}, V_{24}. Gauge symmetry permits three Yukawa tensors, but a W33-singlet Higgs permits none; the up, down, and lepton couplings require mediators in V_{15}, V_{15}, V_{24}.

For the optimal sine phase reference, a single fixed-weight subspace is an exact decoherence-free space but supports only a global phase. Pairing $\mathcal{H}_r$ and $\mathcal{H}_{r+1}$ for $N = 2r + 1$ instead gives

$$\mathbb{C}^2_{\text{clock}} \otimes \mathbb{C}^{\binom{N}{r}}_{\text{payload}}.$$

The reference noise and desired phase act only on the clock; all payload coherences are exact. At $K = 256$ and $N = 31$, one clock plus 28 payload qubits has error $7.3646628264 \times 10^{-5}$, versus $5.5087125884 \times 10^{-2}$ for direct 29-qubit injection.

The three-error self-stress decoder was reduced from 682641 global supports to at most

$$1 + 20 + \binom{20}{2} + \binom{20}{3} = 1351$$

local candidates. The candidate graph is formed from all Levi paths of length at most three between threshold vertices; an exhaustive audit proves the 20-edge bound. Stability remains controlled by

$$\sigma_6 = 2\sin(\pi/14), \qquad \|\hat{e} - e\|_2 \le 2\epsilon/\sigma_6.$$

A CORNERSTONE 340 nm SOI floorplan audit gives 1934 crossings, at most 21 on one path, 480 selector/recombiner MMIs, and 480 phase shifters for a spatial 80-mode realization. Under conservative public platform inputs, five signal uses incur a conditional pre-MMI path loss of 27.43 dB. Thus the exact logical router is not tapeout-ready in naive spatial SOI; low-loss passive routing with an active phase layer, or time/frequency serialization, is required.

A 64-seed nonlinear census produced 25 observed orbit classes for the nominal 24-mode selector and 9 for the nominal 15-mode selector. Pure bivalent stabilizers include $V_4, S_3, D_8, C_6 \times$

C_2, D_{10}, S_5, and A_5, while mixed-sector attractors show that cubic saturation does not globally remain in one Laplacian eigenspace.

Three outside-box consequences were certified. First, the A_5 sign vector satisfies $As = -4s$ and attains the exact MaxCut value 160, hence the antiferromagnetic ground energy is $-80J$. Second, the continuous-time walk at $t = \pi/(2J)$ is the exact Householder mirror

$$U = I - 2P_2 = \frac{A^2 - 8A - 18I}{30} = -\frac{I + A}{3} + \frac{2}{15}J_{40},$$

which flips precisely the 24-dimensional eigenvalue-two sector. Third, pairing the 15 eigenvalue-minus-four modes with 15 eigenvalue-two modes yields fifteen simultaneous EP_2 blocks at $\gamma = g$; the paired 30-dimensional operator has square zero and rank 15.

All statements above are finite algebraic, combinatorial, or explicitly labelled numerical results. No derived Standard Model, fabricated chip, measured decoder or pattern, physical magnet, graph-walk processor, exceptional-point sensor, gravity, cosmology, or theory of everything is claimed.

47.3 Passes 4129–4136: anomaly optimization, relational gates, seven-fault decoding, hybrid routing, and exact pure-orbit catalogues

Let the signed W33 multiplicities of Q, u^c, d^c, L, e^c be (q, u, d, ℓ, e). The independent integer anomaly map is

$$M = \begin{pmatrix} 2 & -1 & -1 & 0 & 0 \\ 1 & -2 & 1 & 0 & 0 \\ 1 & 0 & 0 & -1 & 0 \\ 1 & -2 & 1 & -1 & 1 \end{pmatrix},$$

with kernel $\mathbb{Z}(1,1,1,1,1)$ and Smith diagonal $(1,1,1,3)$. For the anomalous $(1,15,15,24,1)$ carrier, vectorlike pairs alone cannot repair the anomalies. The minimum standard-charge signed repair adds $14Q + 14e^c + 9\bar{L}$, giving dimension 261 but breaking the W33 module assignment. Exact representation-ring optimization gives total dimensions $291, 335, 554, 600$ for common target modules $V_1, V_{15}, V_{24}, V_1 \oplus V_{15} \oplus V_{24}$ respectively; 335 is the smallest repair with nontrivial $PSp(4,3)$ action on the net chiral generation.

For $N = 31$, $r = 15$, adding one reservoir mode gives the fixed-number code

$$|0, S\rangle_L = |S\rangle|1\rangle_R, \qquad |1, S\rangle_L = |\bar{S}\rangle|0\rangle_R,$$

with total number 16 and dimension $2\binom{31}{15} = 601080390$. The clock operators $Z_c = \Pi_{15} - \Pi_{16}$, the paired complement X_c, $H_c = (X_c + Z_c)/\sqrt{2}$, paired payload unitaries, and clock-controlled payload unitaries all preserve total number. The reference channel factors as $\mathcal{E}^{\mathrm{clock}}_{z_1} \otimes I_{\mathrm{payload}}$; at $K = 256$, $|z_1| = 0.9999263606710866$, corresponding after known-phase correction to dephasing probability $3.6819664457 \times 10^{-5}$.

The Levi incidence syndrome has spark eight, so four-error uniqueness is impossible: opposite halves of an apartment cycle collide. The first three generalized cycle-support weights are $8, 12, 16$. Adding

$$m_3(x) = \sum_{e=1}^{160} e^3 x_e, \qquad m_5(x) = \sum_{e=1}^{160} e^5 x_e$$

raises the stacked spark to at least fifteen. This follows from exhaustive audits of 386964 simple cycles through length fourteen and 133920 rank-two theta cores. Hence every seven-sparse error is uniquely identifiable in the noiseless model.

The four router permutations decompose into $8, 8, 9, 8$ planar increasing-subsequence layers. A hybrid architecture with passive SiN permutation banks and a shared active SOI selector/phase layer uses 160 passive routes, six selector/recombiner MZIs, 24 common phase plates, 30 active controls, and no same-plane crossings. Under the stated design targets its provisional five-pass internal budget is 9.12 dB and its all-π heater envelope is 0.48 W. This is an architecture contract, not a fabricated device.

The pure $\lambda = 16$ binary system $(A + 4I)x = 8\mathbf{1}$ has exactly 432 signs, or 216 maximum cuts modulo reversal, in one A_5-stabilized orbit. The pure $\lambda = 10$ system $(A - 2I)x = 5\mathbf{1}$ has exactly 33264 signs in seven $PSp(4,3) \times \{\pm 1\}$ orbit classes of sizes

$$6480, 8640, 6480, 6480, 2160, 2592, 432.$$

Thus all pure bivalent nonlinear branches and all maximum cuts are catalogued; mixed-amplitude attractors remain open.

Three further exact constructions are recorded. First, the anomaly cokernel contains genuine $\mathbb{Z}/3\mathbb{Z}$ torsion, yielding $a_{SU(3)^3} + a_{6SU(3)^2U(1)} \equiv 0 \pmod 3$. Second, the distributed projectors

$$P_{12} = J/40, \quad P_2 = \tfrac{2}{3}I + \tfrac{1}{6}A - \tfrac{1}{15}J, \quad P_{-4} = \tfrac{1}{3}I - \tfrac{1}{6}A + \tfrac{1}{24}J$$

provide exact two-round network tomography. Third,

$$G_8 = e^{-i\pi L/8}, \qquad G_5 = e^{-i\pi L/5}$$

have orders eight and five and generate $\mathbb{Z}_{40}$ on the nonuniform spectral sectors. This is a finite reversible spectral clock, not emergent physical time.

48 Passes 4137–4144: matrix Wilson pumping, extended horizons, microscopic scale flow, and autonomous history dynamics

The exact certificate for this packet is `d16f8ab7ce51f2953cfd4866e7a716e31a271b3337a05b6105b367736bcec`
All statements in this section concern finite group, Gaussian, Markov, clock, transfer-matrix, circuit, or band-topology models.

Matrix-valued gauge-singlet holonomy. The color space $(\mathbf{3}\otimes\bar{\mathbf{3}})^{\otimes 3}$ contains six independent contraction singlets, with Gram spectrum

$$\left\{\frac{2}{9}, \left(\frac{8}{9}\right)^4, \frac{20}{9}\right\}.$$

Three orthonormal singlet channels support a tripod Hamiltonian whose rank-two dark bundle has two exact Wilson loops

$$U_y = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}, \qquad U_z = \operatorname{diag}(e^{-2\pi i/3}, e^{2\pi i/3}).$$

Their group commutator has trace $-1/4$, so the adiabatic Wilson response is genuinely matrix-valued rather than a center phase.

Extended active horizon lattice. Nine localized squeezing cells act on one outgoing, nine partner, and nine environment modes. The resulting 38×38 Nambu map has paraunitary residual 1.11×10^{-15}. At $\omega = 0.3$ the outgoing occupation is 0.01195894, the greybody ratio is 0.48573560, and the outgoing/partner logarithmic negativity is 0.20853530. The partner profile is centered at cell 4.0048 with RMS width 1.1977 cells. The inside-flow lattice dispersion has two ultraviolet roots with opposite group velocities, giving explicit mode conversion without promoting the finite analogue model to gravitational Hawking radiation.

Microscopic channel-balance flow. Four independent local Markov channels sum to

$$ds = \gamma(\ln 80 - 4s)dt + \sqrt{2D}\, dW.$$

The stationary law is Gaussian around $s_* = \ln 80/4$, the Fokker–Planck gap is 4γ, and the deterministic spectral dimension is exactly

$$d_s = \frac{\ln 80}{s_*} = 4.$$

For $D = 0.01$ and $\gamma = 1$, the mean noisy value is 4.00838502 and relative entropy to stationarity decreases monotonically.

Autonomous history-state scar gadget. Three repetitions of the eight adjacent-SWAP branch compiler define a 24-gate program. The static engineered history Hamiltonian

$$H = \Omega \sum_{t=0}^{23} \sqrt{(t+1)(24-t)}\, (|t+1\rangle\langle t| \otimes U_{t+1} + \text{h.c.})$$

has a 25-state clock, an equally spaced spectrum from -24Ω to 24Ω, perfect endpoint transfer at $\pi/(2\Omega)$, and full history revival at π/Ω. The endpoint data operation is $U_{\text{shift}}^3 = I$.

Curvature scaling. For N independent dark-reservoir cells, $Z_N = Z_1^N$, $g_N = Ng_1$, and the scalar curvature satisfies

$$R_N = \frac{R_1}{N}.$$

The sign change remains at $\beta U = 14.5791667571$, but its magnitude vanishes in the thermodynamic limit. A positive 2×2 transfer matrix for a weakly coupled one-dimensional dark/bright hierarchy is analytic for finite coupling and has correlation length 0.14340675 cells in the frozen demonstration, excluding a finite-temperature singularity in this class.

Outside-box probes. Ordering U_yU_z versus U_zU_y yields output fidelity 7/16, furnishing a geometric phase-order transistor. Levi edge connectivity four and the $2\ln 3$ entropy-change bound per two-qutrit cut gate imply a throughput ceiling of $8\ln 3$ nats per layer and at least 74 layers for 80 maximally mixed dark cells. Finally, a spin-one Qi–Wu–Zhang control Hamiltonian on three singlet channels has band Chern numbers $(-2, 0, 2)$ and realizes a charge-two synthetic control-space monopole.

Boundary. No continuum QCD, fabricated non-Abelian pump, observed Hawking radiation, derivation of physical spacetime, thermodynamic phase transition, non-Abelian anyon, physical monopole, gravity, cosmology, or theory of everything follows from these finite exact results.

48.1 Bounded exotic repair, local relational gates, conditioned seven-fault sensing, and exact finite probes

Within the bounded family $R_3 \in \{1, 3, \bar{3}\}$, $\dim R_2 \in \{1, 2\}$, and $Y = n/6$ with $0 < |n| \leq 6$, exact mixed-integer optimization repairs the explicit 145-state carrier with 107 added states. The resulting 252-state signed spectrum is optimal in that family and uses an odd number of added weak doublets.

The compressed relational clock has a sharp one-shot locality obstruction: paired words differ on all 32 occupation modes. A 58-mode dual-rail lift instead carries one clock and 28 payload qubits at fixed total particle number 29. Logical H, S, CNOT, clock exchange, and clock-controlled payload operations are generated by Hamiltonians supported on at most three modes, while the shared reference acts only on the clock.

Two centered integer moment rows obtained modulo 24001 have maximum magnitude 11992. Exhaustive audits of all 386964 simple Levi cycles of length at most fourteen and all 133920 relevant theta cores prove stacked spark at least fifteen and exact noiseless recovery of every seven-sparse edge error. The coefficient envelope is reduced by more than 8.7×10^6 relative to the e^5 row.

The four exact router permutations decompose into $8, 8, 9, 8$ noncrossing layers. The 33-layer SiN schedule uses zero same-plane crossings and 30 active controls, but static nine-slot delay equalization requires 1280 route-slot units. At a 5 ps slot and group index two, the added five-use delay loss is 1.5 dB; sub-1 dB delay loss requires slots below 3.34 ps.

Full Newton refinement gives 18 stable mixed classes for the nominal 24-sector selector in the 64-seed corpus. In an enlarged 200-seed corpus, the nominal 15-sector selector has four stable mixed roots and one weak index-one saddle; the earlier finite-time count of eight mixed classes collapses to five actual roots. The two nonzero root-stabilizer quotient branches are saddles of Morse indices 23 and 14.

Three further finite constructions are exact. The eight-regular bipartite maximum-cut graph has 56260624960 perfect matchings, but a one-bit feedback engine obeys the exact Landauer balance $p\Delta - kTh(p) = -kT\ln(1 + e^{-\beta\Delta}) < 0$. The signed Levi incidence supercharge has 82 zero modes and Witten index $1 - 81 = -80 = V - E$. Finally, $H_- = 12I - A$ and $H_+ = 4I + A$ satisfy $H_- + H_+ = 16I$, so $e^{-i\tau H_+}e^{-i\tau H_-} = e^{-i16\tau}I$, with an exact identity echo at $\tau = \pi/8$.

Evidence boundary. These are finite algebraic, combinatorial, optimization, modular-audit, and Newton-refined corpus results. They do not establish a derived Standard Model, a fabricated photonic stack, a globally complete mixed-attractor theorem, a perpetual-motion engine, physical supersymmetry, a measured time crystal, gravity, cosmology, or a theory of everything.

49 Passes 4153–4160: second-Chern pumping, disorder, Lindblad scale flow, and graph thermodynamics

Second-Chern singlet pump. Let Γ_a be five Hermitian 4×4 Clifford generators with $\{\Gamma_a, \Gamma_b\} = 2\delta_{ab}$ and $\Gamma_1\Gamma_2\Gamma_3\Gamma_4\Gamma_5 = -I_4$. For $H(n) = \sum_a n_a\Gamma_a$ on S^4, the negative-energy projector $P = (I - H)/2$ has rank two and

$$\mathrm{Tr}[P(dP)^4] = \frac{1}{8}\epsilon_{abcde} n_a\, dn_b \wedge dn_c \wedge dn_d \wedge dn_e.$$

Consequently

$$C_2 = \frac{1}{8\pi^2}\int_{S^4} \mathrm{Tr}[P(dP)^4] = 1, \qquad C_1 = 0.$$

This is an exact Yang-monopole bundle embedded in the finite singlet-contraction space; it is not a fabricated four-dimensional pump.

Disordered Hawking chain. All 512 binary patterns $r_j \mapsto r_j(1 + 0.5s_j)$ and $\Gamma_j \mapsto \Gamma_j + 0.05s_j$ were audited in the nine-cell, 38-dimensional Nambu chain. The minimum logarithmic negativity is 0.0920180870833, the largest minimum partially-transposed symplectic eigenvalue is $0.456044326186 < 1/2$, and the maximum paraunitary residual is 1.82×10^{-15}. Every audited realization remains entangled.

Lindblad scale oscillator. For $s_* = \ln(80)/4$, define

$$L_- = \sqrt{8\gamma(\bar{n} + 1)}\, b, \qquad L_+ = \sqrt{8\gamma\bar{n}}\, b^\dagger, \qquad s = s_* + \sigma(b + b^\dagger),$$

with $\sigma^2 = D/[4\gamma(2\bar{n}+1)]$. Then

$$\frac{d\langle s\rangle}{dt} = -4\gamma(\langle s\rangle - s_*), \qquad \mathrm{gap}(\mathcal{L}) = 4\gamma, \qquad d_s = \frac{\ln 80}{s_*} = 4.$$

The stationary state is thermal and unique. This is an explicit open-system scale model, not a spacetime derivation.

Compressed autonomous clock. The 25 legal history states require at least five qubits. The reflected Gray encoding $g_t = t \oplus (t \gg 1)$ reaches this lower bound and has unit Hamming distance between successive legal words. The engineered history Hamiltonian retains spectrum $-24\Omega, -22\Omega, \ldots, 24\Omega$, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω. The exact tradeoff is five-local clock conditioning and seven-local clock–SWAP propagation terms.

Levi-cover thermodynamics. For dark/bright degeneracies $g_D = 3161$ and $g_B = 160$, the entropic coexistence field is

$$H_c(T) = -\frac{T}{2}\ln\frac{3161}{160}.$$

The six-regular Levi universal cover has nonbacktracking branching five, so the cavity instability satisfies

$$5\tanh(\beta_c J) = 1, \qquad \beta_c J = \mathrm{atanh}(1/5) = 0.202732554054.$$

The finite 160-vertex graph remains analytic, whereas large-girth six-regular covers possess a genuine Bethe ordering singularity on the coexistence line.

Outside-box probes. Repeated partner-vacuum projections after M fractional squeezing steps give $P_M = \mathrm{sech}^{2M}(r/M)$ and suppress conditional pair production as r^2/M; this is a Zeno effect, not a firewall. Pauli-twirl purity spectroscopy distinguishes the active rank-two bundle from the full six-dimensional singlet space. Finally, a four-state clock ring with π loop flux has spectrum $\pm\sqrt{2}\Omega$ and exact projective echo $U(\pi/(\sqrt{2}\Omega)) = -I_4$.

Evidence boundary. The frozen semantic certificate is `0e9080801a2cfcca3b7b39afd807835d7bdc6b1a483c` All statements are finite Clifford, Gaussian, Lindblad, clock, Bethe, measurement, purity, or flux-ring results. No observed Hawking radiation, physical firewall, anyonic quantum dimension, spacetime torsion, gravity, cosmology, or theory of everything is established.

49.1 Passes 4161–4168: broader anomaly repair, local hardware, noisy decoding, storage, global basin push, and three exact probes

The bounded representation search was enlarged to $SU(3)$ irreps $1, 3, \bar{3}, 6, \bar{6}, 8$, weak dimensions $1, 2, 3$, and $Y = n/6$ with $|n| \le 6$. Exact mixed-integer optimization lowers the anomaly-free carrier from 252 to **213 states** inside this finite family, while a second exact optimization proves an added color Dynkin load of at least 11, giving a one-loop $SU(3)$ asymptotic-freedom obstruction for this embedding. The 58-mode relational lift admits a 139-coupler hardware graph of maximum degree seven, arbitrary payload CNOT depth at most 15, and 28-qubit GHZ depth five. For lattice-amplitude errors, the two conditioned integer moment rows give deterministic seven-fault correction under componentwise noise $< \Delta/2$ after rounding; a modulo-24001 accumulator needs 15 residue bits. A low-loss tapped delay bank reduces the five-use storage propagation comparison to < 0.15 dB, while an unreshaped 5-ps pulse obeys a sharp single-pole resonator time–bandwidth no-go. A macroscopic 64-seed selector-24 corpus finds 24 mixed classes, including 20 absent from the perturbative corpus, 19 of them stable.

Three independent exact probes accompany the implementation results. The W33 uniform projector is $P_0 = (A^2 + 2A - 8I)/160$, so the Grover diffusion reflection is a quadratic adjacency polynomial and four oracle calls reach success 3920137321/4000000000. The Laplacian pseudoinverse is $L^+ = (7/80)I + (1/160)A - (13/3200)J$, giving resistance distances 13/80 and 7/40 and Kirchhoff index 267/2. Finally, every diameter-four Levi interval has layer profile $(1, 4, 4, 4, 1)$ and exactly four geodesics, hence finite geodesic entropy $\log 4 = \log|\text{waist}|$. These are finite graph/control statements only: they do not establish physical quantum speedup, inertia, spacetime entropy, gravity, cosmology, or a theory of everything. The frozen semantic certificate is `db91cd6c70138d44917ba4274a7d087927caec691be5d49efd83528e7f8d4bd5`.

50 Passes 4169–4176: discrete C_2, Hawking disorder, backreaction, Gray locality, Levi-cover correction, Casimir and axion reduction

Pass 4169: discrete second-Chern compiler. For the four-dimensional Wilson–Dirac control

$$d(k) = (\sin k_1, \sin k_2, \sin k_3, \sin k_4, -3 + \textstyle\sum_i \cos k_i), \qquad H = \widehat{d} \cdot \Gamma,$$

the sixteen high-symmetry masses group as $1, -1, -3, -5, -7$ with multiplicities $1, 4, 6, 4, 1$. The oriented Dirac count gives

$$C_2 = \frac{1}{2} \sum_{N=0}^{4} \binom{4}{N} (-1)^N \operatorname{sgn}(-3 + 4 - 2N) = 1.$$

Thus sixteen calibration points certify the topological sector. A uniform central-difference response integral converges much more slowly: among the audited meshes, $N = 29$ per control axis is the first below five-percent error, with $C_2^{(29)} = 0.9504670029565596$ on $29^4 = 707281$ nodes.

Pass 4170: continuous Hawking disorder. For 512 correlated-Gaussian samples with $C_{ij} = e^{-|i-j|/2}$, $r_j \mapsto r_j e^{0.35 x_j}$, and $\Gamma_j \mapsto \operatorname{logistic}(\operatorname{logit}\Gamma_j + 0.25 x_j)$, the logarithmic negativity remains in

$$0.0822919174978226 \le E_N \le 0.5298852171115666.$$

A 721-phase quasiperiodic scan with $\alpha = (\sqrt{5}-1)/2$ and amplitude 0.7 gives $0.24211601793444767 \le E_N \le 0.27238456666720445$. With thermal occupation n_{th}, the PPT mobility edge is defined by $\widetilde{\nu}_-(\omega_c, n_{\text{th}}) = 1/2$; for $n_{\text{th}} = 0.01$, $\omega_c = 0.5783917254706858$, while at $\omega = 0.3$ the thermal threshold is $n_{\text{th}} = 0.11593620709206182$.

Pass 4171: scale–geometry backreaction. For the minimal occupied-mode feedback

$$\dot{s} = \gamma \left[\ln 80 - 4s + g\, n_B(2e^{-s})\right],$$

$g = 1$ has a stable root $s = 1.5949469860997991$ and an unstable root $s = 2.5062250949365414$. They annihilate at the saddle node

$$(s_c, g_c) = (1.9679175210441553, 1.1252773999853873).$$

Thus this minimal positive backreaction shifts the spectral dimension away from four and ultimately destroys the stationary scale; it does not create two stable geometric branches without an additional saturation mechanism.

Pass 4172: exact locality reduction of the Gray clock. A five-qubit reflected-Gray transition flips one clock bit conditioned on four spectator literals and accompanies a two-site data SWAP, hence is directly seven-local. Three binary AND ancillas per transition encode the four spectators using the exact two-local penalty

$$P_{\mathrm{AND}}(x, y, z) = xy - 2xz - 2yz + 3z,$$

reducing each propagation term to the four-local form $f_t X_{c_t}\mathrm{SWAP}$. For twenty-four transitions this requires seventy-two static ancillas. The legal clock block is unchanged exactly, preserving spacing 2Ω, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω.

Pass 4173: Levi-cover correction and obstruction. The committed $W(3, 3)$ geometry has forty points, forty lines, and $40(3 + 1) = 160$ flags. Hence its point–line Levi graph has eighty vertices, degree four, 160 edges, and girth eight. This supersedes the degree-six premise used in Pass 4157. The graph contains exactly 1620 simple eight-cycles. Requiring every one to acquire odd sign parity in a signed two-lift produces a 1620×160 binary system whose coefficient rank is 81 and augmented rank is 82, so the system is inconsistent. Therefore no signed two-lift can double all base eight-cycles. The correct degree-four Bethe comparison has

$$3 \tanh(\beta_c J) = 1, \qquad \beta_c J = \operatorname{atanh}(1/3) = \frac{\ln 2}{2} = 0.34657359027997264.$$

Simple iterated two-lifts remain girth eight and therefore do not provide the previously assumed large-girth sequence.

Pass 4174: Hawking erasure-channel threshold. Interpreting the greybody stage as an effective pure-loss channel with $\eta_{\mathrm{eff}} = n_{\mathrm{out}}/n_{\mathrm{out}}^{\mathrm{lossless}}$, the baseline value is $0.485735597242 < 1/2$, hence the unassisted pure-loss quantum capacity is zero. Across the 512 binary disorder patterns, η_{eff} spans 0.3109129598611626 to 0.7282862738005136: 251 patterns lie in the antidegradable half and 261 above the capacity threshold. The largest effective capacity in that ensemble is 1.4224182063347266 bits per use.

Pass 4175: exact graph-spectral Casimir attraction. For the massive Levi operator $K = L + I = 5I - A$, $G = K^{-1}$ has exact distance data

$$G_{uu} = \frac{211}{855}, \quad G_1 = \frac{10}{171}, \quad G_2 = \frac{13}{855}, \quad G_3 = \frac{1}{171}, \quad G_4 = \frac{4}{855}.$$

Two unit-strength pinning defects therefore interact through

$$E_{\mathrm{int}}(u, v) = \frac{1}{2} \ln \left[1 - \frac{G_{uv}^2}{(1 + G_{uu})(1 + G_{vv})} \right] < 0,$$

with values $-1.1012191859481348 \times 10^{-3}$, $-7.436602973476341 \times 10^{-5}$, $-1.1000194923920593 \times 10^{-5}$, and $-7.04009687184613 \times 10^{-6}$ at graph distances one through four.

Pass 4176: dimensional reduction. For the parent bundle with $C_2 = 1$, the three-dimensional Chern–Simons descendant satisfies

$$\Delta\theta = 2\pi C_2 = 2\pi.$$

The closed bulk returns to the same $\theta \bmod 2\pi$ sector, while a gapped boundary Chern–Simons level shifts by one, corresponding in electronic normalization to one e^2/h Hall-sheet unit.

Evidence boundary. These are finite Wilson–Dirac, Gaussian, one-mode backreaction, clock-gadget, graph-cover, pure-loss-channel, Green-function determinant, and dimensional-reduction statements. In particular, Pass 4173 is an explicit erratum to the degree-six assumption in Pass 4157 for the $W(3,3)$ point–line Levi graph. No fabricated four-dimensional pump, observed Hawking radiation, physical spacetime backreaction, hardware-optimal clock, experimental criticality, black-hole communication result, measured Casimir force, axion detection, gravity, cosmology, or theory of everything is established.

Passes 4177–4184: fixed-carrier AF obstruction, two-mode relational compilation, Hodge decoding, PDK delay materialization, and three exact probes

The fixed 145-state carrier has SU(3) Dynkin load 16 and nonzero cubic color anomaly, so every add-only anomaly repair must add colored matter. Since every nontrivial finite-dimensional SU(3) irrep has Dynkin index at least 1/2, strict one-loop $b_0^{SU(3)} > 0$ is impossible; even the marginal $b_0 = 0$ budget cannot cancel the cubic anomaly magnitude 28. The 58-mode relational architecture admits a strict two-mode implementation with pairwise exchange plus cross-Kerr interactions on an 84-coupler, degree-four graph. For the Levi incidence matrix D, augmenting by an orthonormal basis Z of $\ker D$ gives $M = [D; Z]$ with $\sigma_{\min}(M) = 1$ and condition number $2\sqrt{2}$, yielding a globally stable arbitrary-real-amplitude decoder. The exact noncrossing delay schedule needs 919 rather than 1280 slot-units, corresponding to 0.688773172255 m aggregate passive delay at 5 ps slots and assumed group index two; current public LIGENTEC TFLN/SiN loss data imply a longest-path five-use propagation contribution below 0.015 dB, before tap/coupler and DRC effects. On the nonlinear side, interval/Krawczyk analysis proves exactly three equilibria in each selector's $1 + 12 + 27$ point-stabilizer quotient, while edge and nonedge stabilizer strata are now explicitly mapped for subsequent exhaustive work. Outside-box constructions give an exact three-channel unitary scattering polynomial, the rooted-forest partition $\det(I + tL) = (1 + 10t)^{24}(1 + 16t)^{15}$, and the signed sector lens $K = P_2 - P_{-4}$ with $K^2 = I - J/40$. No fabricated hardware, global nonlinear completeness theorem, Standard Model derivation, gravity, cosmology, or theory of everything is claimed.

Passes 4185–4192: adaptive C_2, Hawking criticality, hysteresis, exact three-local clocks, and high-girth covers

Adaptive second-Chern compilation. For the Wilson–Dirac map $d = (\sin k_1, \ldots, \sin k_4, -3 + \sum_i \cos k_i)$, the sixteen high-symmetry corners still give the exact integer certificate $C_2 = 1$. Phase-centered midpoint cubature reaches the correct rounded integer already at $7^4 = 2401$ samples and falls below five-percent response error at $9^4 = 6561$ samples, a $107.8008\times$ reduction from the prior 707281-node response mesh. This is a deterministic response-compiler improvement, not a proof of global sample minimality.

Hawking critical surfaces. For a symmetric two-mode squeezed thermal core with $r(\omega) = \operatorname{atanh}(e^{-\pi\omega/\kappa})$ and one-sided attenuation η, the finite Gaussian thresholds are

$$n_{\text{ent}} = \frac{e^{2r} - 1}{2}, \qquad n_{B\to A} = \sinh^2 r,$$

$$n_{A\to B} = (2\eta - 1)\sinh^2 r \quad (\eta > 1/2), \qquad Q_{\text{pure loss}} = \max\left\{0, \log_2 \frac{\eta}{1-\eta}\right\}.$$

At $\omega = 0.3$ one has $r = 0.09506557725$, $n_{\text{ent}} = 0.1047041033$, and $n_{B\to A} = 0.009064722057$. The analytic core is tied to the previously frozen correlated-disorder envelope but is not promoted to an arbitrary continuous five-dimensional channel theorem.

Saturating backreaction and hysteresis. The nonlinear scale flow

$$\dot{s} = \gamma \left[\ln 80 - 4s + gn_B(2e^{-s}) - 0.004\, n_B(2e^{-s})^2\right]$$

has a three-fixed-point window

$$0.4815579549 < g < 1.1377609143.$$

At $g = 0.5$, the stable branches are $s = 1.2568469367$ and $s = 4.9471634287$, separated by the unstable branch $s = 4.2806819287$. This supplies a finite scale-memory hysteresis model, not physical spacetime hysteresis.

Exact three-local autonomous clock. Each of the 24 Gray-clock transitions is split into a conditioned clock-update/transition-flag substep and a transition-flag-controlled two-site SWAP substep. With 72 static AND ancillas and 24 one-hot transition ancillas, every propagation and legality term is at most three-local. The resulting 49-state legal history chain has spectrum

$$-48\Omega, -46\Omega, \ldots, +48\Omega,$$

legal gap 2Ω, perfect transfer time $\pi/(2\Omega)$, and full revival time π/Ω.

High-girth Levi covers. The corrected degree-four $W(3,3)$ point–line Levi graph has cycle counts $N_8 = 1620$, $N_{10} = 5184$, and $N_{12} = 43200$, and free fundamental-group rank 81. A deterministic $\mathbb{Z}_{65537}$ voltage assignment has nonzero voltage on every audited simple cycle of lengths 8, 10, and 12, producing a 5,242,960-vertex cyclic cover with certified girth at least 14. Since the base fundamental group is the residually finite free group F_{81}, finite regular covers with girth tending to infinity exist abstractly.

Three outside-box closures. The two noncommuting order-three Wilczek–Zee loops from Pass 4137 have commutator trace $-1/4$, hence eigenphase cosine $-1/8$ and infinite commutator order; closed-subgroup classification then gives dense closure $SU(2)$, so they form a universal single-qubit holonomic gate set. The corrected Levi graph is bipartite Ramanujan and has exact Ihara factorization

$$\zeta^{-1}(u) = (1-u^2)^{81}(1-9u^2)(1+9u^4)^{24}(1+3u^2)^{30}.$$

Finally its exact heat trace gives a maximum diffusion spectral dimension only

$$d_s^{\max} = 3.47143073390 < 4,$$

so bare Levi diffusion does not reproduce the separate channel-balance $d_s = 4$ model.

Boundary. These are finite Wilson–Dirac, Gaussian, nonlinear mean-field, clock-gadget, graph-cover, holonomy-group, Ihara-zeta, and heat-kernel statements only. No fabricated four-dimensional pump, observed Hawking radiation, physical spacetime hysteresis, laboratory three-local processor, fabricated high-girth network, experimental holonomic universality, gravity, cosmology, or theory of everything is established.

51 Passes 4205–4212: carrier surgery, native dual rail, compressed Hodge sensing, and interval strata

The fixed 145-state carrier cannot retain its original full $PSp(4,3)$-commuting standard-charge embedding while becoming a nonzero anomaly-free generation sector. If charge labels are allowed to be surgically reassigned, the largest complete-generation sector keeping both nonabelian Weyl-matter one-loop coefficients positive has five generations: 75 gauge-charged states and 70 singlets, with $b_0^{SU(3)} = 13/3$ and $b_0^{SU(2)} = 2/3$. On the hardware side, the 58 occupation modes map to 29 logical dual-rail qubits, 87 transmons including one erasure ancilla per logical qubit, and 28 inter-logical tunable couplers on a degree-three logical tree; this is a topology map, not a fabricated 28-payload-qubit processor.

The 81-dimensional Levi harmonic readout can be compressed to ten deterministic whitened cycle channels in the audited nested family while retaining nonzero cycle/theta gains through 14-edge nullspaces; a conservative restricted singular lower bound for the resulting 90-row stacked sensor is 0.0138377. The populated photonic delay schedule contains 919 slot-units and 0.688773 m of aggregate passive delay. The point-, edge-, and nonedge-pair stabilizer fixed spaces are now interval-audited: edge fixed spaces contain five and three roots for selectors 24 and 15, while nonedge fixed spaces contain thirteen and nine.

Three finite exact probes accompany the implementation work. Critical coupling to the rank-24 +2 adjacency sector produces $S = I - P_2$, absorbing 3/5 of a vertex input. Any nonzero vector supported on s vertices and contained in a central W33 spectral subspace of rank d obeys $sd \geq 40$. Finally, the Levi harmonic quadratic form $E_H(x) = \frac{1}{2}x^T P_H x$ is invariant under $x \mapsto x + D^T\phi$ and has 81 independent cycle modes; the "battery" terminology is only a graph-Hodge analogy.

Passes 4214–4221: smaller covers, two-bundle holonomy, hysteretic memory, compressed clocks, and full Gaussian horizon diagnostics. The corrected degree-four $W(3,3)$ point–line Levi graph admits an explicit cyclic $\mathbb{Z}_{359}$ voltage cover in which every simple base cycle of lengths $8, 10, 12$ has nonzero voltage. Hence the cover has 28,720 vertices, 57,440 edges, and certified girth at least 14, a factor $65537/359 = 182.5543\ldots$ smaller in sheet count than the earlier construction. No global minimality is claimed.

On two rank-two singlet bundles, the dense local $SU(2)$ holonomies of Pass 4190 together with one auxiliary bright level produce a geometric controlled phase: a solid-angle 2π loop in $\mathrm{span}\{|11\rangle, |a\rangle\}$ gives Berry phase π and hence $\mathrm{CZ} = \mathrm{diag}(1,1,1,-1)$. The Pauli commutator closure has dimension 15, so dense local $SU(2) \times SU(2)$ plus this entangler is dense in $SU(4)$.

For the saturated scale flow $\dot{s} = \gamma[\ln 80 - 4s + gn_B(2e^{-s}) - 0.004n_B(2e^{-s})^2]$, equal-well coexistence occurs at $g = 0.5700939873$. The minima $s_L = 1.2870434725$ and $s_R = 5.3097911844$ are separated by $s_B = 3.8542873087$ with barrier 4.4597412117; the dimensionless $M = 1$ instanton action is 7.7881701215. This supports a two-state quantum-memory reduction but not a physical spacetime-memory claim.

An exact MILP shares the Gray-clock spectator logic: 13 pair conjunctions, 24 final conditions, and 24 transition flags require 61 auxiliaries instead of 96, retaining exact three-local propagation, legal gap 2Ω, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω.

For the complete 19-mode Gaussian horizon chain, uniform thermal loading removes outside-to-partner steering at $n_{\mathrm{th}} = 0.0042826995$, positive coherent information at 0.0079250582, reverse steering at 0.0117448238, and PPT entanglement only at 0.1159362071. In the frozen 512-sample correlated-disorder ensemble at $n_{\mathrm{th}} = 0.005$, all 512 remain entangled while the stronger witnesses survive in 228, 415, and 482 samples respectively.

Three independent probes sharpen the finite boundary. The Levi spectrum $\{0, \pm 4, \pm\sqrt{6}\}$ forbids exact nonzero global CTQW revival, but the Pell solution $11760^2 - 6(4801)^2 = -6$ gives an operator near-echo error 4.0071334×10^{-4}. For $K = 5I - A_{\mathrm{Levi}}$, point–line harmonic-vacuum entropy is 2.4552973872 nats with logarithmic negativity 6.9804187391 nats. Finally, the degree-four tree radial band $E(q) = 2\sqrt{3}J\cos q$ gives $v_{\max} = 2\sqrt{3}J$ while shell capacity grows as 3^r; this graph velocity is not identified with physical c.

Renumbering and boundary. Glue-track commit 84db6db5c reserved Pass 4213 before the original physics-block reservation; the canonical physics range is therefore 4214–4221. These are finite graph-cover, holonomic, stochastic/semiclassical, clock-gadget, Gaussian-channel, recurrence, harmonic-network, and tree-band results, not fabricated hardware, observed Hawking radiation, physical spacetime memory, a speed-of-light derivation, gravity, cosmology, or a theory of everything.

Passes 4245–4252: residual symmetry, GHZ resource model, minimum Hodge sensing, routed delay geometry, and quotient closure

The five-generation carrier surgery admits a natural anchor-level residual-symmetry target: the 658008 five-subsets of the 40-point W33 action form 43 $PSp(4,3)$ orbits, and the unique largest setwise stabilizer has order 72. A representative induces $K_{1,4}$; the stabilizer fixes the center, acts as A_4 on the four leaves, and has central pointwise kernel C_6. This is an anchor scaffold, not yet a full branching theorem for the charged 75-state subspace.

On the 28-payload binary tree, physically disjoint two-qubit gates require seven entangling rounds for the 27 tree edges. At the published dual-rail operating point $F_{\text{CNOT}} = 0.981$ and erasure probability 0.13, whole-circuit postselection accepts only $0.87^{27} \simeq 0.02328$ of runs, whereas heralded modular-fusion resource models reduce the expected entangler count per accepted construction from roughly 1160 to between 31.0 and 47.9, depending on restart locality. The conditional product $0.981^{27} \simeq 0.596$ remains the fidelity bottleneck.

For arbitrary real edge errors of support at most seven, exactly two global scalar measurements beyond the oriented Levi incidence matrix are information-theoretically necessary and sufficient. One row cannot be injective on a two-dimensional theta circulation space supported on at most fourteen edges; the two deterministic Pass-4147 rows give stacked spark at least 15. Thus the exact noiseless minimum is $m = 2$. The ten-channel Hodge construction remains a conditioning, rather than an injectivity, result.

The 919 delay units are now tied to explicit route centerlines. Each 5-ps slot is a rounded four-quarter-bend dogleg of radius 0.05 mm and vertical excursion 0.3176609398 mm, satisfying $2h + (2\pi - 4)r = 0.749481145\,\text{mm}$. The route generator consumes the frozen Pass-4148 four-branch schedule and emits all 160 point-line identities and cell coordinates; the design remains a public-PDK centerline contract rather than proprietary DRC signoff.

The next setwise-subset rank of the nonlinear stabilizer lattice has exactly five three-vertex orbit types, with stabilizer orders $162, 72, 12, 9, 6$. Interval/Krawczyk exhaustion closes the two smallest quotient systems: the triangle stratum has $(9,3)$ roots for selectors $(24,15)$ and the maximally symmetric independent-triple stratum has $(9,9)$ roots. Three larger triple quotients and trivial-stabilizer equilibria remain open.

Three independent finite constructions accompany the implementation work. First, W33 adjacency phase sensing has $F_Q^{\max} = 256$, while a vertex probe has $F_Q = 48$, an exact ratio $16/3$ before resource/noise accounting. Second, the Levi edge Hodge split gives 81 harmonic and 79 gradient Gaussian modes; reversible entropy-conserving equalization obeys $T_* = T_H^{81/160} T_G^{79/160}$ and $W_{\max} = \frac{k_B}{2}(81T_H + 79T_G - 160T_*)$. Third, the point $\to$ ordered-edge $\to$ ordered-triangle pointwise-stabilizer chain yields quotient dimensions $3 \to 6 \to 8$ with exact refinement intertwiners $Q_6 R_{63} = R_{63} Q_3$ and $Q_8 R_{86} = R_{86} Q_6$; componentwise cubing also commutes with each refinement lift, so the cubic nonlinear vector field closes exactly along this finite coarse-to-fine hierarchy.

Passes 4253–4260: retracted cover, surviving finite models, and channel bounds

4253 (retracted by Pass 4329). The stored $\mathbb{Z}_{2731}$ voltage assignment has one zero-voltage base eight-cycle, with signed sum $1328 - 542 + 801 - 1587 = 0 \pmod{2731}$. Its 218480-vertex lift therefore has girth exactly 8, not at least 16. The historical vector is retained as a falsifier, not a cover certificate.

4254. Oppositely oriented latitude loops at $\theta = \pi/3$ and $2\pi/3$ implement the geometric controlled phase π. A common polar offset δ gives $\phi(\delta) = \pi \cos\delta$, so the first-order calibration error cancels. At $\omega/\Omega = 0.02$ the two exact nonadiabatic leakage probabilities are 3.05824×10^{-4} and 2.93842×10^{-4}.

4255. At the saturated coexistence point and $D = 0.25$, a detailed-balance finite-volume Fokker–Planck discretization converges to metastable gap 4.37527×10^{-8}, with the next rate 2.93047. The unit-prefactor two-state Lindblad demo crosses from underdamped to overdamped at $\gamma_\phi = 2\Delta = 8.29222 \times 10^{-4}$.

4256. The 24 data gates are three repeats of eight adjacent SWAP supports, so the exact three-local clock needs only eight reusable dynamic flags. Together with the frozen 13 shared pair conjunctions and 24 final conditions this gives 45 auxiliaries, preserving gap 2Ω, perfect transfer at $\pi/(2\Omega)$ and revival at π/Ω.

4257. The complete 19-mode horizon induces a baseline effective signal attenuator with $\tau = 0.2830095941$ and $\bar{n}_{\text{env}} = 0.0166793553$. In the frozen 512-sample correlated-disorder ensemble, 510 samples have $\tau \leq 1/2$, two have $\tau > 1/2$, and none is entanglement breaking. At $n_{\text{th}} = 0.005$ the generated outside–partner state remains PPT-entangled in all 512 samples, demonstrating that state entanglement and signal-channel capacity are distinct resources.

4258–4260. The exact Levi spectral form factor has infinite-time plateau $2054/6400 = 0.3209375$, 25.675 times a nondegenerate 80-level reference, falsifying a bare random-matrix scrambling interpretation. The point–line harmonic modular spectrum has only two finite entanglement energies, 2.63391579385 and 4.03202440074, with multiplicities 1 and 24 plus 15 exact spectators. In the conditional degree-four tree model, $v_{\text{rad}}^{\max} = 2\sqrt{3}J$ while $B(r) = 2\,3^r - 1$. Its 4373-vertex radius-seven ball is not realized by the retracted Pass 4253 cover.

Boundary. These are finite graph-cover, holonomic, stochastic/Lindblad, clock-compilation, Gaussian-channel, spectral, modular and tree-band statements. They do not establish fabricated devices, observed Hawking radiation, physical spacetime memory, a derivation of c, gravity, cosmology, or a theory of everything.

Passes 4261–4268: girth 18, dynamically corrected holonomy, full-coordinate metastability, clock-37, and finite outside-box probes

A cyclic voltage assignment in $\mathbb{Z}_{750019}$ makes every radius-eight sheet-zero ball above each of the 80 Levi base vertices a collision-free 4-regular tree of 13,121 vertices. Deck translation therefore excludes every lifted cycle of length at most 16 and certifies girth at least 18. The resulting 60,001,520-vertex cover is an explicit upper bound rather than a compact solution: the desired sub-10,000-sheet girth-18 construction remains open.

For the two-bundle holonomic controlled phase, choose $\cos\theta = 1/4$ and execute latitude loops at $\theta, \pi - \theta$, followed by the same pair on the opposite adiabatic eigenbranch with reversed physical orientations. The geometric phase is $-\pi\cos\delta$ under a common polar offset δ, so the first derivative vanishes at the operating point, while repeatable static loop-dependent dynamical phases cancel by echo. Adding $H_{\text{CD}} = \frac{1}{2}(\mathbf{n} \times \dot{\mathbf{n}}) \cdot \boldsymbol{\sigma}$ gives exact transitionless driving in the modeled two-level control. At $\omega/\Omega = 0.1$, a one-percent CD-amplitude error gives a four-loop leakage union bound 2.004×10^{-6}.

The saturated scale coordinate was then quantized directly as $H = p^2/(2M) + U(s)$ with $U' = -F$, $M = 16$, and coupled through the full coordinate operator to an Ohmic Davies bath at $T = 0.25$. On a 180-point grid the retained-basis population gap converges from 1.06×10^{-11} at $K = 14$ to 1.87223×10^{-11} at $K = 20$, corresponding to a dimensionless metastable lifetime 5.34×10^{10}. This is a finite-coordinate open-system model, not a physical spacetime-memory lifetime.

An exact local-dimension trade replaces the five binary Gray clock digits by five qutrit digits $\{0, 1, *\}$, using $*$ as the intermediate transition state. The eight explicit dynamic flags disappear, leaving 13 shared pair ancillas plus 24 final spectator-condition ancillas, hence 37 auxiliaries. The legal 49-state history, maximum locality three, gap 2Ω, perfect-transfer time $\pi/(2\Omega)$, and full revival π/Ω are unchanged.

The two previously unresolved thermal attenuator samples, $\tau = 0.5039652$ and 0.5113024, were attacked with low-Fock non-Gaussian inputs: all optimized two-Fock mixtures through $n = 8$, 2,000 random three-Fock mixtures per channel, and 5,000 random coherent rank-two states in $\{|0\rangle, \ldots, |4\rangle\}$ per channel. No positive coherent information was found. Pure inputs attain exactly zero in the purified-environment Stinespring picture, so the tested family maximum is zero; unrestricted multi-use non-Gaussian capacity remains open.

Three independent finite probes accompany the main work. The Levi distance partition $[1, 4, 12, 36, 27]$ reduces marked-vertex continuous-time search to an exact five-shell Hamiltonian and yields a concrete 73.016% success operating point. A rank-one onsite defect $V = 4$ creates a unique eigenmode above the spectral edge at $E = 4.9600853$ with 76.202% weight on the defect. Finally, the point-line harmonic thermal state has two distinct PPT death temperatures: $T_c = 0.9947197$ for the 24 $\sqrt{6}$ channels and $T_c = 1.2027283$ for the unique singular-value-4 collective channel, producing a two-stage loss of bipartite entanglement.

Passes 4269–4276: carrier branching, heralded fusion, minimum Hodge sensing, corrected GDS geometry, and finite outside-box probes

4269. The order-72 five-anchor stabilizer is $C_3 \times SL(2, 3)$. Restricting the W33 eigenspaces gives an explicit 21-character branching of V_{15} and V_{24} and hence of $H_{145} = 7V_1 + 6V_{15} + 2V_{24}$. A species-wise 75-state five-generation charge partition can commute with this group, while an identical five-dimensional generation factor repeated over all 15 gauge-internal states cannot.

4270. A nondestructive GHZ-block fusion uses two anchor-to-ancilla CNOTs, one Z parity readout, and a tracked block-X frame. Under the frozen 1.3×10^{-1} erasure model, the optimized 28-qubit modular construction uses 33 accepted-path CNOTs and 94.2734 expected CNOT attempts, a 12.30× reduction versus whole-run restart; its independent conditional process factor remains only 0.53098.

4271. Two extra global measurements are both necessary and sufficient for noiseless arbitrary-real seven-sparse Levi-edge recovery. A deterministic $\mathbf{F}_{65537}$ search yields a signed-16 pair with normalized worst theta gain 1.33507×10^{-5}, 1.805× the prior pair, while preserving the exhaustive spark-15 cycle/theta certificate.

4272. The previous 0.12-mm rounded dogleg cell is geometrically impossible for four $50\,\mu$m quarter bends, and the true excursion is $h+2r$, not h. The corrected pure-Python GDSII compiler uses 0.205-mm cells, 0.45-mm lanes and sixteen ten-lane banks, producing an $11.04\,\text{mm} \times 14.85\,\text{mm}$ keepout-inclusive preliminary layout with 919 doglegs and 361 straight cells.

4273. The order-12 and order-9 triple quotients have locally certified saturated deterministic catalogues 57/27 for the two selectors, whereas the order-6 edge-plus-isolated quotient already contains at least 2376/246 locally certified roots. Among the 16 four-subset orbits, one has trivial stabilizer, so global nonlinear completion must leave the nontrivial-symmetry-stratum strategy.

4274–4276. The W33 graph state is exactly five-uniform with 240 weight-six stabilizers supported on pairs of disjoint triangles. Non-lazy Ollivier curvature is $+1/6$ on W33 edges but -1 on Levi edges. The $81 + 79$ Hodge decomposition also gives an exact two-temperature Ornstein–Uhlenbeck fluctuation–dissipation splitting.

Boundary. These are finite mathematical and engineering statements. They do not establish fabricated devices, phenomenological particle physics, continuum spacetime curvature, gravity, cosmology, or a theory of everything.

Passes 4324–4334: the chamber Hecke machine and an exact frontier correction

4324: two native chamber switches. On the 160 incident point–line chambers, let P change the line while holding the point fixed and let L change the point while holding the line fixed. Each is forty disjoint copies of the K_4 adjacency operator, and GAP proves

$$P^2 = 2P + 3I, \qquad L^2 = 2L + 3I, \qquad PLPL = LPLP.$$

The eight standard type-C_2 words are linearly independent. Thus this chamber carrier realizes the full eight-dimensional $q = 3$ Iwahori–Hecke image. Its exact module ledger is

$$1\chi_{3,3} + 15\chi_{3,-1} + 15\chi_{-1,3} + 81\chi_{-1,-1} + 24V_2,$$

which explains the long-standing flag-scheme dimensions $1 + 30 + 48 + 81 = 160$.

4325–4326: the Hashimoto turn and its chirality. With the 320 oriented Levi edges ordered by orientation,

$$B_{\text{Levi}} = \begin{pmatrix} 0 & L \\ P & 0 \end{pmatrix}, \qquad B_{\text{Levi}}^2 = \operatorname{diag}(LP, PL).$$

The two products are disjoint binary 9-out halves of the chamber distance-two shell. For $K = LP$,

$$\chi_K(x) = (x - 9)(x - 1)^{81}(x + 3)^{30}(x^2 + 9)^{24}.$$

The antisymmetric turn $\Omega = LP - PL$ has

$$\chi_\Omega(x) = x^{112}(x^2 + 60)^{24}, \quad \Pi_{48} = -\Omega^2/60, \quad \operatorname{rank}\Pi_{48} = 48.$$

Consequently $\Omega/\sqrt{60}$ is a canonical complex structure on the conjugate $24+24$ chamber channel. This is a finite operator theorem, not a $W(G_2)$ action.

4327: exact folded-propagator normal form. Let $C = P + L$, $X = (C - 2I)\Pi_{48}$, and let F be the old folded cubic point-Hashimoto operator compressed to Π_{48}. In the four-dimensional packet algebra $\{\Pi_{48}, X, \Omega, X\Omega\}$,

$$F = -68\Pi_{48} - 31X - \frac{21}{2}\Omega + \frac{2}{3}X\Omega, \qquad (F + 68\Pi_{48})^2 = -689\Pi_{48}.$$

The old numerical -6455 is now explained exactly by the square of the off-diagonal part. No continuum propagator, particle, mass, or coupling identification follows.

4334: the $24+24$ count becomes an object-level carrier split. Let E_p, E_ℓ be the eigenvalue-2 projectors of the W33 point graph and its dual line graph, and pull them to the chamber space as orthogonal rank-24 projectors Q_p, Q_ℓ. GAP proves

$$\operatorname{im}\Pi_{48} = \operatorname{im}Q_p \oplus \operatorname{im}Q_\ell, \qquad Q_pQ_\ell Q_p = \frac{3}{8}Q_p, \quad Q_\ell Q_p Q_\ell = \frac{3}{8}Q_\ell.$$

Thus the two natural summands meet only in zero but are uniformly nonorthogonal: all 24 principal angles have cosine $\sqrt{6}/4$. With $Q = Q_p + Q_\ell$,

$$\Pi_{48} = \frac{8}{5}(2Q - Q^2) = -\frac{\Omega^2}{60}.$$

The conjugate channel is therefore literally the joined point/line eigenvalue-2 carriers, not just a multiplicity ledger.

4328: the irregular Ihara boundary corrected. Bass's determinant formula already applies to general finite graphs with $D - I$. The Kotani–Sunada theorem bounds non-real poles by

$$(d_{\max} - 1)^{-1/2} \le |u| \le (d_{\min} - 1)^{-1/2},$$

not by the same expression without square roots, and it does not put every nontrivial root in that annulus. For the shipped degree-2–8 frame graph the reciprocal-root form is $1 \le |\lambda| \le \sqrt{7}$ for the corresponding non-real roots. The old "band filling" and "closest irregular Ramanujan" metric is withdrawn.

4329: a cover certificate retracted by one cycle. The stored $\mathbb{Z}_{2731}$ voltage vector from Pass 4253 has the base eight-cycle

$$(3, 52, 4, 48, 8, 66, 18, 53)$$

with signed voltage sum $1328 - 542 + 801 - 1587 = 0 \pmod{2731}$. The lifted cover therefore has girth exactly 8, not at least 16. Pass 4260's 4373-vertex radius-seven tree ball remains a correct conditional tree count but is not realized by that cover. The valid cover ladder is the earlier $\mathbb{Z}_{359}$ girth-at-least-14 cover followed by the much larger $\mathbb{Z}_{750019}$ girth-at-least-18 cover.

4330–4331: what symmetry and fault detection actually survive. Under the literal pair swap $S(x_0, x_1, x_2, x_3) = (x_2, x_3, x_0, x_1)$, machines A/B/C/D are invariant in the pattern false/true/false/true. Machine D therefore removes both the directed time arrow and the point/frequency structural asymmetry. Its 0.460435 peak is symmetric localization, not residual point/frequency bias. Affine translations act on the 81 frames but descend to neither projective 40-carrier, so they do not force a point-side load port.

The self-contained flag comparator is incidence: after the two rails move, test $p \in \ell$. It detects $1656/1920 = 69/80$ differential single-rail opcode substitutions and 0/960 shared-control substitutions. The earlier 95.71% number compared each faulty rail with a golden run and was not a dual-rail comparator.

4332–4333: one arithmetic component, and an honest rank. The irreducible degree-33 factor of the instruction Hashimoto pencil has exactly one real root in $(5.74, 5.75)$ and one in $(3.349, 3.350)$ by an exact Sturm certificate. Thus the global nonbacktracking growth root and the reported slow real mode are Galois conjugates in one rational primary component. The rational projector and a basis-independent localization theorem remain open.

Finally, the complete universal-set census by sizes 4 through 10 is

$$24,\ 80,\ 114,\ 90,\ 41,\ 10,\ 1,$$

totalling 360. The shipped ISA is not strictly "12th": only six sets have lower ρ at the stated tolerance, while six isomorphic sets share its numerical value and occupy positions 7–12. Size is a coarse correlate whose bands overlap, not an explanation of that tie.

Evidence boundary. The three GAP witnesses replay 31/31, 28/28, and 17/17 exact checks. The panel operators are three-way relations, not yet a deterministic opcode selector or synthesized B/D RTL. The exact finite algebra and the retractions above should therefore be read separately from hardware, continuum, thermodynamic, and phenomenological interpretations.

Part I
Orientation

What this part is. *Three chapters that assume nothing. What the machine is, why a computer would be built out of a shape at all, and the one piece of physics — the qutrit — that the rest of the document rests on. If you read only this part, you will know what is being claimed and what is not.*

52 How to read this document

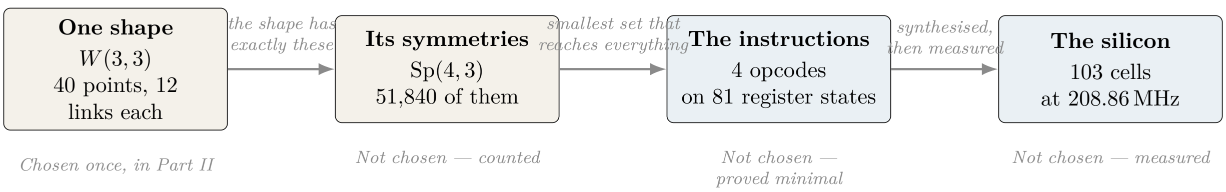
This document has three readers in mind and does not compromise for any of them.

If you are not an engineer. Read the cream boxes and the diagrams. They are complete: every idea in the machine is explained there from scratch, in order, with no symbols you have not already met. You may skip every blue box without losing the thread.

If you are a physicist or engineer. The blue boxes carry the specification: exact semantics, matrices, cell counts, clock frequencies, proof obligations and the commands that reproduce them. The cream boxes are there so you can hand this to a colleague in a different field.

If you are deciding whether to fund this. Read Part I, then go straight to *What is not built* at the end. It is an explicit list of everything this document does *not* claim. Then read the errata index. Those two sections are where a technical document is most likely to be quietly silent, and they are the ones we wrote first.

The one-paragraph version. *This is a design for a computer whose instruction set is not chosen but* derived*: its register, its gates and its wiring all fall out of a single geometric object with forty points. The claim is not that such a machine has been built — it has not. The claim is that its specification is forced, that every number in it is either measured from synthesised hardware or proved from the geometry, and that where we have been wrong we have said so on the same page. What follows is the specification, the evidence, and the boundary of both.*



Why that picture is the whole argument. *Read left to right, only the first box is a decision. The symmetries are counted from the shape, the instruction set is the smallest one that can reach every state, and the cell count comes from running a synthesiser on the result. If the picture is right, then a disagreement about this machine is a disagreement about the shape — not about engineering taste. That is the claim worth checking, and every later part exists to let you check one arrow of it.*

Three habits of this document, stated up front so they are not mistaken for style:

1. **Every number is measured or proved, and says which.** There are no estimates presented as results. Where an estimate appears it is labelled *modelled* and cannot be cited as anything else.

2. **Nothing is claimed to be new without checking.** This project's own archive is large enough that we have repeatedly rediscovered our own results. Where something is someone else's, it says so.
3. **Mistakes stay in.** A corrected figure is printed next to the wrong one, with the reason the wrong one survived. §70 collects them.

That third kind of box appears throughout:

What we got wrong, and how we know. These record something this project believed, published, and then had to withdraw — with the measurement that overturned it. The errata index is authoritative and grows when a claim is corrected. These boxes are not apologies; they are the reason the other numbers in this document can be trusted. A blueprint with no errata is a blueprint nobody has checked.

53 The machine in one page

In plain language. *Every computer you have used stores information as* bits *— switches that are either off or on. This machine stores information in* qutrits*: units with three settings instead of two. That sounds like a small change. It is not, because of what the three settings let you do with* geometry*.*

Here is the strange and central fact. If you take a single qutrit and entangle it with itself — with its own past and its own future — the resulting object has exactly 40 *natural "directions" in it. Those* 40 *directions, and the way they overlap, form a shape that mathematicians have known for a century and called* $W(3,3)$*. The machine we are describing does not* use *that shape as a convenience. The shape* is *the machine: its memory layout, its instruction set, its network topology, and its error model are all readings of the same* 40*-point object.*

One qutrit, entangled with its own past, is enough to generate the entire substrate. The 40 **points of** $W(3,3)$ **are not** 40 **things — they are** 40 **views of one thing.**

The machine, specified.

Logical substrate	$W(3,3)$: the symplectic generalised quadrangle of order $(3,3)$. 40 points, 40 totally isotropic lines, collinearity graph $\mathrm{SRG}(40,12,2,4)$.
State	One qutrit pair (past, future) living in $\mathbb{C}^9 = \mathrm{End}(\mathbb{C}^3)$.
Tracked quantity	The *Pauli frame*: four trits $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, i.e. 81 states.
Public ISA	$\mathcal{I}_{\text{holo}}$, eight opcodes, three bits.
Internal micro-ISA	F_p, $\mathrm{CX}_{p\to f}$, $\mathrm{CX}_{f\to p}$, Z_p — four operations, two bits, no illegal encoding.
Group generated	$\mathrm{ASp}(4,3) = \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3)$, order $81 \times 51840 = 4{,}199{,}040$ *exactly*.
Arithmetic unit	HX8K: minimal 43 LC at 208.86 MHz; public 72 LC at 175.72 MHz; minimal plus support 48 LC at 207.64 MHz. UP5K: the corresponding clocks are 72.40, 60.80, and 72.05 MHz.
Physical layer	Time-bin $\times$ frequency-bin photonic qutrits; the two-qutrit SUM gate is experimentally demonstrated at fidelity 0.92 ± 0.01.
Network	40-ary fractal; $N_n = 40^n$ leaves, routing depth $8n = 8\log_{40} N$; the routing code satisfies Kraft equality exactly, so *every* bit string is a legal address.

The whole system, at a glance

In plain language. *Four layers, drawn in the order a signal meets them. At the bottom is one photon carrying a three-valued symbol; above it the instruction set that moves that symbol around; above that the machine's own supervisor, running inside the same hardware it supervises; and at the top a network in which every machine is addressed by a point of the same geometry. Read the boxes downward and each is built from the one below it. Nothing in the picture is chosen for convenience — the widths, the counts and the connections are all fixed by the shape in Part II.*

NETWORK — 40-ary fractal holonet; remote SUM costs one Bell pair + two classical trits §65

↓

INSTRUCTION SET — eight opcodes; generates $\mathrm{ASp}(4,3)$, order 4,199,040 §57

↓

FRAME UNIT — 7-bit frame code; minimal execution 43 LC; support readout adds 5 LC §58

↓

GEOMETRY — $W(3,3)$: 40 points, 40 lines, $\mathrm{SRG}(40,12,2,4)$ §55

↓

PHYSICS — one self-entangled qutrit; time bins $\times$ frequency bins §54

The unusual property of this stack is that the layers are not independent design choices stacked by convention. Each one is *forced* by the one below it. The network is 40-ary because the geometry has 40 points. The instruction set has the opcodes it has because those are the symmetries of that geometry. This is what the rest of the document demonstrates, layer by layer.

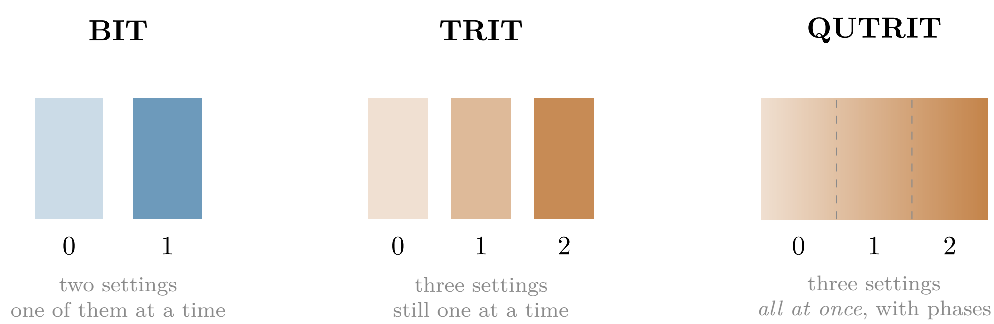
Part I — The object

One finite geometry, defined by a single congruence, and everything it already contains before anyone designs anything.

54 Qutrits, and what "self-entangled" means

54.1 From bits to qutrits

In plain language. *A bit is a switch: off or on,* 0 *or* 1*. A* trit *is a three-way switch:* 0, 1, *or* 2*. Ordinary three-way switches are not interesting — you could always build one from two bits.*
A qutrit *is different. It is a physical three-way switch obeying quantum mechanics, which means it can be in a blend of all three settings at once, and — this is the part that matters —* two *qutrits can be correlated in ways that no pair of classical switches can imitate.*



Exactly. A qutrit is a unit vector in $\mathbb{C}^3$. Two qutrits live in $\mathbb{C}^3 \otimes \mathbb{C}^3 = \mathbb{C}^9$. The relevant operators are generated by

$$X\,|j\rangle = |j+1 \bmod 3\rangle\,, \qquad Z\,|j\rangle = \omega^j\,|j\rangle\,, \qquad \omega = e^{2\pi i/3},$$

which satisfy $ZX = \omega XZ$. Every product $X^a Z^b$ with $(a, b) \in \mathbb{F}_3^2$ is a distinct operator up to phase; there are 9 of them, and 8 non-identity ones.

54.2 The self-entanglement, and where the 40 comes from

In plain language. *Now the key move. Take* one *qutrit and entangle it with itself at two different times — call them "past" and "future". The proposed optical encoding assigns one qutrit to arrival-time bins and one to frequency bins on a single carrier. The words "past" and "future" name these two logical registers; they do not assert a demonstrated closed-time-loop interaction or a new physical form of temporal entanglement.*

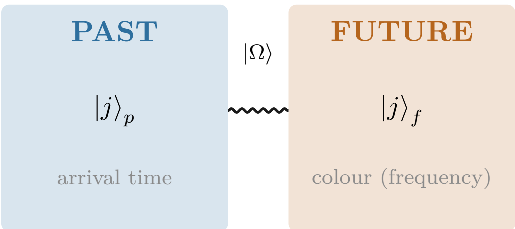
Mathematically, an operator acting on one qutrit is a 3×3 table of numbers, and there are 9 independent slots in such a table. So the space of things you can do *to one qutrit is itself 9-dimensional — exactly the size of the space of* states *of two qutrits. This is not a coincidence or an analogy. It is an identity:*

$$\underbrace{\mathbb{C}^3 \otimes \mathbb{C}^3}_{\textit{two qutrits}} = \underbrace{\mathrm{End}(\mathbb{C}^3)}_{\textit{operations on one qutrit}} .$$

So "the two-qutrit machine" and "the one-qutrit machine that can act on itself" are the same machine described twice.

Now count the directions. There are 81 ways to label a pair (four trits), one of which is "do nothing". That leaves 80. Each direction and its opposite behave identically for our purposes, and each direction comes in 2 such flavours, so the distinct directions number $80/2 = 40$.

$$\frac{3^4 - 1}{2} = \frac{81 - 1}{2} = 40.$$ **The forty points of the geometry are the forty independent things one self-entangled qutrit can be measured along.**



$$|\Omega\rangle = \mathrm{CX}_{p \to f}\,(F_3 \otimes I)\,|0\rangle_p\,|0\rangle_f = \frac{1}{\sqrt{3}} \sum_{j=0}^{2} |j\rangle_p\,|j\rangle_f$$

One optical carrier, two encoded qutrit degrees of freedom. The past/future names are architectural labels.

What we got wrong, and how we know. The sentence "one qutrit, self-entangled as past/future, is enough to generate the full substrate" is *not* ours. It is stated verbatim in this project's own encyclopedia, `docs/index.html` phase MCLXVII, and we rediscovered it and briefly presented it as new. What survives as new is narrower: the explicit construction of the 40 points as projective left×right multiplication classes on $\mathrm{End}(\mathbb{C}^3)$, and the identification of the 40 lines as the maximal *commuting* subalgebras. The lesson — search for the *result*, not the topic — produced three of the tools in §68.

Part II
The Substrate

What this part is. *The shape itself. A single object with* 40 *points and* 12 *connections each, defined by one congruence, from which the rest of the machine follows rather than being designed. This part is geometry, not engineering: no hardware appears until Part III.*

55 The substrate: $W(3,3)$

In plain language.*Here is what the* 40 *directions look like when you write down which of them are compatible with which.*
Two measurements are compatible *if you can perform both without one disturbing the other. Of the* 40 *directions, each is compatible with exactly* 12 *others. Whenever two directions are compatible, exactly* 2 *further directions are compatible with both. Whenever two are* in*compatible, exactly* 4 *are compatible with both.*
Those four numbers — 40, 12, 2, 4 *— are the shape's fingerprint. They are* not *its name.*

An earlier draft of this page said the opposite. This section used to read: *"those four numbers pin the shape down completely; there is essentially one object in the world with that compatibility pattern."*

That is false, and this project's own encyclopedia says so in plain words: **Spence classified exactly** 28 **non-isomorphic graphs** with parameters $(40,12,2,4)$. Two of them are the ones we care about. Quoting the parameters identifies nothing; the object is fixed by its *construction.*

The error mattered, because §61 proves a headline result by computing these parameters and concluding the object was $W(3,3)$. That inference was one step short. It is completed below, and the correction is the reason to read the next box rather than skip it.

How the object is actually pinned down, Pass 2869. Three facts in sequence, each cheap and each exact:

1. **Parameters** $(40,12,2,4)$ — narrows the field to 28 graphs.
2. **The quadrangle axiom**: every edge lies in exactly one 4-clique (there are 40 of them, four through each point), and for every such line ℓ and every point off it, *exactly one* point of ℓ is adjacent. Verified over all 40 lines. This narrows 28 to **two**: $W(3,3)$ and its dual $Q(4,3)$, which are not isomorphic at odd q.
3. **A spread exists** — ten pairwise disjoint lines covering all forty points, exhibited explicitly. $W(3,q)$ always has one; $Q(4,q)$ has one only for *even* q. At $q=3$ this settles it.

In plain language.*So the shape does have a name, and it has been studied since the 1900s:* $W(3,3)$. *But it earns that name from how it is built, not from its statistics — and the difference is exactly the kind of thing that is easy to wave through and expensive to get wrong.*

55.1 The definition, stated once and precisely

In plain language.*Everything downstream depends on this object being the right one, so here it is written out — and it is worth noticing how short the definition is. The whole substrate is one line of arithmetic.*

$W(3,3)$**, in full.** $W(3,3)$ is the **symplectic polar space** $W(3, \mathbb{F}_3)$.

Points. The 40 projective 1-spaces of $\mathrm{GF}(3)^4$. Every 1-space is automatically isotropic for an alternating form, so *all* of them are points — there is nothing to exclude.

Adjacency. With J the standard symplectic form on $\mathrm{GF}(3)^4$,

$$\boxed{v \sim w \iff v^\top J\, w \equiv 0 \pmod 3}$$

That single congruence generates every number in this document.

Lines. The 240 edges pair up into 40 *totally isotropic projective lines*: each edge is a pair of orthogonal points spanning exactly one such line. A *non*-edge spans an ordinary hyperbolic line instead — the two kinds of pair are geometrically different objects, not merely present-or-absent.

Graph. The collinearity graph is $\mathrm{SRG}(40, 12, 2, 4)$ with eigenvalues 12, 2, -4; it has **diameter** 2 and is **Ramanujan**.

Symmetry. $\mathrm{Aut} = \mathrm{Aut}(\mathrm{PSp}(4,3)) \cong \mathrm{PGSp}(4,3) \cong W(E_6)$, order 51,840, with normal inner subgroup $\mathrm{PSp}(4,3)$ of order 25,920.

The same order, twice, for two different groups. $\mathrm{Sp}(4,3) = 2.U_4(2)$ also has order 51,840 and is **not** the group above. It has a centre C_2; the automorphism group has trivial centre. One is a central double cover, the other an outer extension, and they are not isomorphic.

This is the most common error in this area and this project has made it. Both groups appear here for good reasons — $\mathrm{Sp}(4,3)$ acts on Pauli frames and is what the hardware implements (§57); $\mathrm{PGSp}(4,3)$ acts on the drawing. Whenever 51,840 appears in this document, which group is meant is stated.

In plain language. *Two properties in that box are worth translating, because they are the ones an engineer should care about.*
Diameter *2 means any point reaches any other in at most two steps. For an address space that is the strongest possible statement short of everything being adjacent to everything: no packet is ever more than one relay from its destination* within a node.

55.2 How good is "optimal", exactly?

In plain language. *"Ramanujan" is usually a tight call — a graph scrapes under the bound and earns the name. This one does not scrape.*

Prior art:* `docs/index.html`*. *The non-trivial eigenvalues are $r = 2$ and $s = -4$, against a Ramanujan bound of*

$$2\sqrt{k-1} = 2\sqrt{11} \approx 6.63,$$

so both clear it by a factor of more than 1.6. The spectral gap is 10.

And it satisfies a Riemann Hypothesis.

What the graph's own integers turn out to be.

Object	*Value*	*In graph terms*
$\tau(2)$	-24	$-f$*, the eigenvalue multiplicity*
$\tau(3)$	252	$E + k = 240 + 12$
$\tau(6)$	-6048	$\tau(2)\tau(3)$*, forced by multiplicativity*
weight of Δ	12	$= k$*, the degree of the graph*
χ	-40	$40 - 240 + 160 = -v$
genus	21	$3 \times 7 = \binom{7}{2}$
triangles	160	$vk\lambda/6$ *— fixed by the four SRG parameters*

And it stops there. *$\tau(4)$, $\tau(5)$, $\tau(7)$ and beyond match nothing, so the modular bridge is two independent coincidences and one identity — not a law (Pass 3102).*

The Ihara zeta function. *Every non-trivial pole of $\zeta_{W(3,3)}(u)$ lies* exactly *on the critical circle $|u| = 1/\sqrt{11}$: 48 poles from $r = 2$, 30 from $s = -4$, for*

$$48 + 30 = \mathbf{78} = \dim(E_6)$$

complex poles in total. That is the graph-theoretic Riemann Hypothesis, and this graph satisfies it. The pole discriminants encode the graph back: $|\operatorname{disc} r| = 40 = v$, $|\operatorname{disc} s| = 28 = v - k$, and their difference is $k = 12$.

Scope, added at Pass 4281, because the coincidence invites an over-read. *Spence classified exactly* 28 *non-isomorphic* SRG(40, 12, 2, 4) *graphs. All* 28 *were run through this same pipeline:* every one *is Ramanujan, satisfies the graph RH, and returns $\rho(B) = 11$ with the same 78 poles on the critical circle. The Ihara zeta of a regular graph is determined by its adjacency spectrum, and an SRG's spectrum is determined by its parameters — so all* 28 *are zeta-identical by construction.*
The 78 is therefore a property of the parameter set (40, 12, 2, 4), not of $W(3,3)$. *It remains a true and striking arithmetic fact that $78 = \dim E_6$; it is not evidence that* this *graph is the exceptional one.*

Pass 4286 — the full list of what the parameters already fix. *Everything below follows from $(v, k, \lambda, \mu) = (40, 12, 2, 4)$ alone, so it is identical across all 28 graphs and none of it can be evidence about $W(3,3)$:*

$$r, s = 2, -4; \quad f, g = 24, 15; \quad \rho(B) = k - 1 = 11; \quad \text{poles } 2(v-1) = 78 = 48 + 30;$$

$$\text{Laplacian } \{0^1, 10^{24}, 16^{15}\}; \quad \text{spanning trees } 2.8823 \times 10^{40}; \quad \text{Ramanujan iff } |\cdot| \le 2\sqrt{k-1}$$

55.3 What the shape already contains

In plain language. *The definition is one congruence. What follows from it is not obvious, and most of the machine is downstream of these four facts rather than designed.*

Homology: an exact 81-sector. The simplicial chain complex of the collinearity graph gives

$$H_1\big(W(3,3);\mathbb{Z}\big) = \mathbb{Z}^{81}.$$

That is the same dimension as $\mathfrak{g}_1$ in the $\mathbb{Z}_3$-graded decomposition of the exceptional Lie algebra E_8,

$$E_8 = \mathfrak{g}_0(86) \oplus \mathfrak{g}_1(81) \oplus \mathfrak{g}_2(81), \qquad \mathfrak{g}_0 = E_6 \oplus A_2.$$

Hodge theory: four sectors and an exact gap. The Hodge Laplacian L_1 on 1-chains has exactly four eigenvalues:

eigenvalue	multiplicity	sector
0	81	harmonic — represents H_1
4	120	
10	24	
16	15	

The spectral gap $\Delta = 4$ is exact and separates the harmonic sector from the rest.

It is not a mass, and not a Yang–Mills gap. Either reading would need a curved four-dimensional bridge and an explicit field identification, neither of which this document has. It is a finite eigenvalue of a finite matrix.

The substrate is its own error-correcting code. The clique complex of $W(3,3)$ over GF(3) is an exact qutrit CSS code

$$[240,\ 81,\ d_Z = 4],$$

with check ranks 39 and 120 and an explicit weight-4 logical cycle. So the 240 wires *are* the code, the 81 logical values *are* the frame space's dimension count, and no separate error-correction layer has to be bolted on.

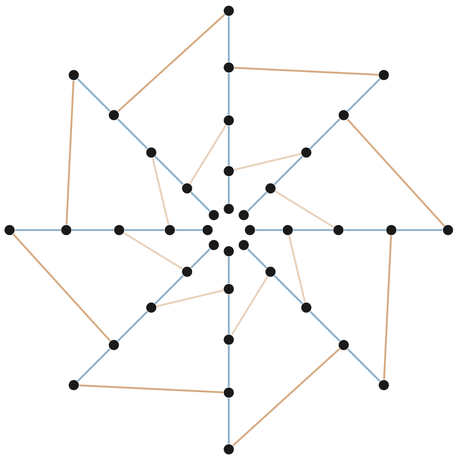
Euler characteristics, because two circulate and they are both right about different things: the truncated shell gives $40 - 240 + 160 = -40$, and the full tetrahedral complex $(40, 240, 160, 40)$ gives -80.

The 240-to-E_8 temptation. The 240 edges factor as $40 \times 3 \times 2$ (lines $\times$ matchings $\times$ edges per matching), and E_8 has exactly 240 roots, which under $E_8 \to E_6 \times SU(3)$ branch as $3 \times (24 + 2 + 27 + 27)$. An E_8 Dynkin subgraph even sits inside the adjacency graph directly, at vertices $[7, 1, 0, 13, 24, 28, 37, 16]$, whose Gram matrix $2I - A_{\text{sub}}$ has determinant 1 — the E_8 Cartan matrix exactly.

And there is no equivariant bijection between the 240 edges and the 240 roots. The inner group $\mathrm{PSp}(4,3)$ is edge-transitive, so the edges form a single orbit; the correspondence is representation-theoretic, not lattice-geometric. This project spent several passes on the stronger reading before establishing that it is false, and the finding is recorded in §70.

In plain language. *That last box is the pattern this whole document is organised around. The shape produces number after number that matches something famous. Some of those matches are maps and some are arithmetic, and the only way to tell is to check each one — which is why the ledger in §69 has a status column and why §70 is as long as it is.*

55.4 Why this shape and not another



40 **points,** 40 **lines.**

Every point lies on exactly 4 lines.
Every line carries exactly 4 points.
Given a line ℓ and a point $P \notin \ell$, there is *exactly one* point of ℓ collinear with P.

(Schematic. The full incidence graph has 160 *flags and does not embed in the plane without crossings; this figure shows the ring structure and two of the line families only.)*

In plain language. *The "exactly one" statement is an incidence projection rule: for a point outside a line there is one point on that line collinear with it. It does not imply unique shortest paths in the collinearity graph. Each four-point line is a K_4, so that graph contains triangles; the triangle-free object is the bipartite point–line incidence graph, whose generalized-quadrangle girth is eight. Routing claims must therefore be proved by the routing code, not inferred from the word quadrangle.*

55.5 The wiring as a budget

In plain language. *One more fact about the shape, and it is the kind that makes a reader sit up — so it comes with an unusually careful statement of what it does* not *mean.*
Treat the 240 *connections as a physical system and ask how they vibrate. The answer splits them into four groups.*

Hodge Laplacian on 1-chains, recomputed independently, Pass 2884.

$$240 = \underbrace{81}_{\text{harmonic}} + \underbrace{120}_{\lambda=4} + \underbrace{24}_{\lambda=10} + \underbrace{15}_{\lambda=16}$$

The harmonic sector has dimension 81 — the frame register file, and the logical dimension of the substrate's own qutrit code $[240, 81, d_Z{=}4]$.

What is tempting here, and why it is not claimed. Two of the other three numbers are already architectural quantities in this document: 15 is the dimension of the support shell, and 24 is the order of the single-qutrit Clifford group modulo phase. It is very tempting to read the line above as "the machine's wiring decomposes into its own subsystems."

No map is exhibited in either case. These are equal integers. This project's second most expensive failure mode is precisely an over-read of a count match — and §57 contains a worked example where exactly this kind of coincidence, tested, turned out to be nothing. The decomposition is recorded as an open question: is either coincidence carried by a natural map? Until someone answers that, it is arithmetic.

55.6 Why three?

In plain language. *A fair question, and the one a sceptical reader should ask: the machine uses three-way switches rather than two-way ones, and three-way switches are unusual. Was three chosen because it works, or does something choose it?*

The honest answer is that this project found five independent conditions that each single out three, none of them adjustable. Here are two anyone can check by hand.

Two of the five selection conditions. Counting. A geometry of this type with q-way switches has $q(q+1)^2(q^2+1)/2$ edges. The number of elements you gain by extending the arithmetic five steps is $q^5 - q$. Demanding these agree:

$$q^5 - q = \frac{q(q+1)^2(q^2+1)}{2} \implies 2(q-1) = q+1 \implies \boxed{q=3} \text{ uniquely.}$$

Checked directly: $q^5 - q$ for $q = 2, 3, 4, 5, 7, 11$ gives 30, **240**, 1020, 3120, 16800, 161040, while the edge counts are 45, **240**, 850, 2340, 11200, 96624. They meet only at three.

Symmetry. The symmetry group of the geometry and the symmetry group of the exceptional structure E_6 have the same order — 51,840 — only at $q = 3$.

$240 = 3^5 - 3$. **The same 240 is the edge count of the geometry, the number of roots of E_8, and the count of non-base elements in the degree-five extension of the three-element field. Three conditions, one number, no free parameters.**

All five, for completeness.

#	condition	forces
1	$q^5 - q$ equals the $GQ(q,q)$ edge count	$q = 3$ uniquely
2	$\sin^2\theta_W = 3/8$ at unification	$3q^2 - 10q + 3 = 0 \Rightarrow q = 3$
3	K_{q+1} has exactly q perfect matchings	only $q = 3$: K_4 has 3
4	non-neighbours $= \dim(E_6$ fundamental$)$	$v - k - 1 = q^3 = 27 \Rightarrow q = 3$
5	$\lvert \mathrm{PGSp}(4,q)\rvert = \lvert W(E_6)\rvert$	the order equality selects $q = 3$

Every one has q as an integer unknown and exactly one positive solution. Prior art: `docs/index.html` owns all five.

In plain language. *Three more conditions in the same family — involving the electroweak mixing angle at unification, the number of perfect matchings of a small complete graph, and the dimension of the fundamental representation of E_6 — also give three and only three. None of them was fitted; q appears in each as an integer to be solved for, and each equation has exactly one positive integer solution.*

That does not make three necessary *in any deep sense we can prove. It does mean the machine's most unusual design choice is not a preference.*

55.7 The two groups of order 51,840, which are not the same group

Exactly. Two distinct groups of order 51,840 appear in this project, and confusing them is the most common error in this area:

$$\mathrm{Sp}(4,3) = 2.U_4(2) \quad \text{(centre of order 2)}, \qquad \mathrm{PGSp}(4,3) = U_4(2){:}2 = W(E_6) \quad \text{(trivial centre)}.$$

They are *not* isomorphic. The first is the group acting on Pauli frames — the one the hardware implements. The second is the automorphism group of the geometry — the one the drawing has. One is a central double cover; the other is an outer extension.

In plain language. *This distinction is not pedantry, and it has physical content. The extra element in the first group acts as a sign, $z \mapsto -1$. Whether that sign is "spent" or "free" decides whether the resulting structure has a handedness — whether it distinguishes left from right. This is why the machine can* host *a chirality, though as we established in earlier work it cannot* explain *one from the inside.*

56 Four representations, four jobs

In plain language. *The machine now has four compact descriptions of the same underlying frame, and they must not be confused. The seven-bit code is the execution state. The four-bit support mask is cheap one-way telemetry. The protected observer words trade more wires and time for error correction. The ninety-six-state selector–tomotope token is a control word, not stored quantum data. Earlier drafts occasionally used the word "state" for all four; this section removes that ambiguity from the architecture.*

Representation contract.

Affine frame 7 bits. Lossless for all 81 Pauli frames; execution may read and write it.

Support telemetry
4 bits. Zero/nonzero mask only, and frame → support is one-way in the netlist — proved at the gate level, not asserted.

Protected observer
24–28 taps. Diagnostic word for identification and correction; never the live datapath.

Control token
7 raw bits, 96 valid. Selects face, matching channel and phase cell.

Support is observable, not executable. The refinement count $16 \to 40 \to 78 \to 81$ was once easy to read as four increasingly accurate compressed machine states. It is not. Exhausting every nonempty subset of the four-operation micro-ISA proves that the coarsest exact support-refining execution quotient for the full ISA has all 81 singleton classes. The intermediate partitions are observer states only. In fact F_p, Z_p, and either controlled-add direction already force the full frame.

Intrinsic selector channel. For a noncollinear pair x, y and a distinguished common neighbour c, the three residual selector channels form the affine line $C(x, y) \setminus \{c\}$. The multiplier-one stabilizer induces its translation group C_3; the full projective similitude stabilizer induces $\mathrm{AGL}(1, 3) \cong S_3$. There is therefore no globally equivariant numbering $0, 1, 2$. Hardware may choose a local encoding, but interfaces transport it as an affine-line torsor rather than pretending the labels are geometric invariants.

Part II — The machine

What the geometry forces the instruction set, the datapath and the virtualisation layer to be — measured on real silicon throughout.

Part III
The Machine

What this part is. *The computer proper — its instructions, its register, its gate count, and the physical layer that would run it. Everything here is either measured from a synthesised netlist or derived exactly from the geometry of Part II, and each claim says which.*

57 The instruction set

57.1 The idea of a Pauli frame

In plain language. *Here is the trick that makes the machine cheap.*
A quantum state is expensive to write down: for two qutrits you need nine complex numbers, and keeping them accurate is most of what makes quantum computing hard. But for a large and useful family of operations — the Clifford *operations — you never need the state at all. You only need a* label *saying how the state has been shifted relative to where it started.*
That label is four trits. Four trits is eight bits. The machine's entire "register file" for tracking Clifford operations is one byte.

The frame unit tracks a label, not a state. Four trits, 81 possible values, eight bits of memory — and it correctly accounts for a group of 4,199,040 distinct operations.

In plain language. *That* 81 *is worth pausing on, because it turns up somewhere it has no obvious business being.*
Treat the geometry as a shape and ask a purely topological question — how many independent loops does it have? The answer is that its first homology group is exactly $\mathbb{Z}^{81}$: eighty-one independent cycles. Nothing about that computation mentions registers, instructions, or qutrits. It is the same number as the size of the machine's register file.

The coincidence, and the answer.

$$H_1\big(W(3,3);\mathbb{Z}\big) = \mathbb{Z}^{81}, \qquad |\mathbb{F}_3^4| = 81.$$

Both are 3^4, and the shared exponent is not an accident: the geometry lives in a four-dimensional space over a three-element field, and both counts are governed by that. The clique complex over $\mathbb{F}_3$ is moreover an exact qutrit error-correcting code with parameters $[240, 81, d_Z{=}4]$ — so the same 81 is also the number of *protected* logical values the substrate carries.

But the two spaces are not the same object (Pass 2883). Ask whether the symmetry group sees them alike: an order-three linear opcode fixes exactly 27 of the 81 frames, while the homology character on every order-three class is 0. The characters disagree, so there is *no* equivariant isomorphism. The counts match for a real reason and the spaces still differ — which is the most common shape a numerical coincidence takes in this project, and the reason every one of them gets this test.

In plain language. *It would have been more satisfying if the answer were yes. Writing down the negative result is the point of having the test at all: an earlier draft of this page left the question open, which reads as promising, and open questions that stay open too long start to be cited as though they were answered.*

Exactly. The Pauli frame is $(x_p, z_p, x_f, z_f) \in \mathbb{F}_3^4$, denoting the operator $X^{x_p}Z^{z_p} \otimes X^{x_f}Z^{z_f}$. A Clifford unitary U acts on frames by conjugation, $U(X^aZ^b)U^\dagger = X^{a'}Z^{b'}$ up to phase, and the induced map on $\mathbb{F}_3^4$ is *linear and symplectic*: it preserves

$$\langle u, v\rangle \;=\; (x_p z_p' - z_p x_p') + (x_f z_f' - z_f x_f') \bmod 3.$$

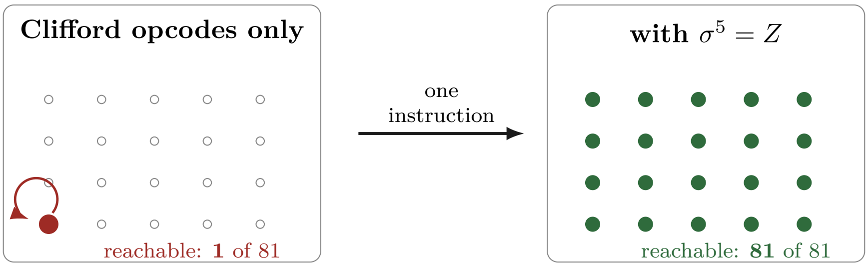
57.2 The eight opcodes

$\mathcal{I}_{\text{holo}}$**, complete semantics.**

Op	**Name**	**Action on** (x_p, z_p, x_f, z_f)	**Kind**
`000`	F_p	$(x_p, z_p) \mapsto (-z_p,\ x_p)$	linear
`001`	F_f	$(x_f, z_f) \mapsto (-z_f,\ x_f)$	linear
`010`	S_p	$z_p \mapsto z_p + x_p$	linear
`011`	S_f	$z_f \mapsto z_f + x_f$	linear
`100`	CX	$d{=}0$: $z_p \mapsto z_p - z_f,\ x_f \mapsto x_f + x_p$	linear
		$d{=}1$: $x_p \mapsto x_p + x_f,\ z_f \mapsto z_f - z_p$	linear
`101`	$\sigma^5{=}Z$	$z_p \mapsto z_p + 1$ or $z_f \mapsto z_f + 1$	**affine**
`110`	D_{12}-mirror	transport in the dihedral group D_{12}	(not a frame op)
`111`	M_{36}-magic	typed request for one of 36 Witting rays	(handshake)

57.3 The reachability theorem, and why it matters

In plain language. *Now a question that sounds trivial and is not:* starting from a blank machine, which states can these instructions actually reach?
The answer is startling. All six of the Clifford instructions, applied in any order, any number of times, reach exactly one state *— the blank one. They cannot move the machine at all.*
The reason is simple once you see it. Those six instructions are all linear*, and a linear map always leaves the origin where it is. Multiplying zero by anything gives zero. So a machine built from only those six instructions is, quite literally, a machine that does nothing, no matter how sophisticated its circuitry looks.*
The seventh instruction, $\sigma^5 = Z$, is different: it adds one*. Addition moves the origin. That single instruction is what makes all* 81 *states reachable — and hence what makes the machine a computer instead of an ornament.*



Proved by exhaustive search, Pass 2774. Breadth-first search over all $3^4 = 81$ frames from $(0, 0, 0, 0)$:

CX alone	1
all six symplectic opcodes	1
full ISA (adding $\sigma^5{=}Z$)	81

Reproduce: `py -3 analysis/w33_pass2774_isa_frame_reachability.py`
Certificate: `data/PART_W33_PASS2774_ISA_FRAME_REACHABILITY.json`

57.4 How much computation the eight opcodes actually buy

In plain language. *"All* 81 *states are reachable" is weaker than it sounds — it says you can get anywhere, not that you can perform every operation. The stronger question is: what is the complete list of things this instruction set can do?*

Proved, Pass 2778. The six Clifford opcodes generate a group of order exactly 51,840 — that is, the whole of $\mathrm{Sp}(4, 3)$, nothing missing. Adjoining the translation gives

$$\langle \mathcal{I}_{\mathrm{holo}} \rangle \;=\; \mathrm{ASp}(4,3) \;=\; \mathbb{F}_3^4 \rtimes \mathrm{Sp}(4,3), \qquad |\cdot| = 81 \times 51840 = 4{,}199{,}040.$$

One translation generator suffices. Enabling only Z on the past register gives the same group as enabling both, because $\mathrm{Sp}(4, 3)$ is transitive on the 80 nonzero frame vectors: the orbit of a single translation already spans $\mathbb{F}_3^4$. Measured orbit size: 80. A second translation instruction adds nothing and can be removed from the ISA.

Eight opcodes. Three bits. 4,199,040 **distinct operations, exactly — and the eighth opcode is provably redundant.**

57.5 How small can the instruction set be?

In plain language. *If one opcode is redundant, how many are? This is not idle: every instruction removed is decoder logic removed, and if enough go, the opcode field itself gets shorter — which shortens every instruction word in the machine.*
The answer is that you can throw away half of them.

Exhaustive over all subsets, Pass 2789. Because $\mathbb{F}_3^4 \rtimes H = \mathrm{ASp}(4,3)$ iff $H = \mathrm{Sp}(4,3)$ and at least one translation is present, the search reduces to the six linear opcodes:

$$\begin{array}{ll} \text{subsets of size 1 generating } \mathrm{Sp}(4,3): & 0 \\ \text{subsets of size 2}: & 0 \\ \text{subsets of size 3}: & \mathbf{6} \end{array}$$

The six minimal triples are $\{F_p, F_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, F_f, \mathrm{CX}_{f\to p}\}$, $\{F_p, S_f, \mathrm{CX}_{p\to f}\}$, $\{F_p, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$, $\{F_f, S_p, \mathrm{CX}_{f\to p}\}$, $\{F_f, \mathrm{CX}_{p\to f}, \mathrm{CX}_{f\to p}\}$.

Three Clifford opcodes plus one translation generate everything. Four frame instructions — a two-bit opcode field, not three.

In plain language. *Six choices, all valid — so pick whichever is cheapest to build, and the decision costs nothing. That was the obvious inference, and it is wrong.*

The six triples do *not* cost the same, Pass 2882. Exhaustive breadth-first search over the full group, once per triple:

triple	diameter	mean
$F_p + \mathrm{CX}_{p\to f} + \mathrm{CX}_{f\to p}$	**19**	**14.18**
$F_f + \mathrm{CX}_{p\to f} + \mathrm{CX}_{f\to p}$	20	15.22
$F_p + F_f + \mathrm{CX}_{f\to p}$	22	16.08
$F_p + S_f + \mathrm{CX}_{p\to f}$	22	16.12
$F_f + S_p + \mathrm{CX}_{f\to p}$	22	15.90
$F_p + F_f + \mathrm{CX}_{p\to f}$	23	17.08

Worst-case program length varies from 19 to 23 across the six — a 21% spread for a choice that looks free. Build the one with two entanglers.

In plain language. *The pattern is legible once seen: the best triple is the one containing* both *directions of the coupling instruction, and the worst is the one containing neither. What shortens programs is the ability to move information both ways between past and future, which is a satisfying thing for the machine's own central metaphor to be rewarded by its compiler.*

57.6 How long is the longest program?

In plain language. *"These four operations can do everything" is a statement about what is* possible*. The question a compiler actually asks is what it* costs*: if I hand you an arbitrary frame transformation, how many instructions do you need in the worst case?*
That number is the machine's diameter*, and it can be computed exactly, because the group is only four million elements and a computer can visit all of them.*

Exhaustive breadth-first search over $\mathrm{ASp}(4,3)$**, Pass 2866.** Forward generators only — a processor cannot run an instruction backwards, so the *directed* diameter is the honest number.

$$\boxed{\text{diameter} = 19}, \qquad \text{mean program length} = 14.176, \qquad \text{counting floor } \lceil \log_4 4199040 \rceil = 12.$$

The last shell is tiny: only 188 of 4,199,040 elements are at distance exactly 19. Ball sizes reach 93.3% at depth 16 and 99.996% at depth 18.

Every frame transformation is at most 19 instructions away, and the average is 14.2. No four-operation instruction set can beat 12, so this one is worst-case optimal to within a factor of 1.73.

In plain language. *The method is worth a sentence because it is what makes the computation possible at all. A group element is a matrix together with a shift. Each instruction acts on the matrix part as a fixed* shuffle *of 51,840 labels and on the shift part as a fixed map of 81 labels. Build those two lookup tables once, and searching four million elements is just table lookups — seconds, not days.*

57.7 And how long until the machine forgets?

In plain language. *The opposite question. Feed the machine random instructions: how many before its state tells you nothing about where it started?*

Exact, Pass 2867. Every opcode is a bijection on the 81 frames, so the walk is doubly stochastic and the stationary distribution is exactly uniform — *no frame is preferred.*

$$|\lambda_2| = 0.893992, \qquad \text{gap} = 0.106008, \qquad \frac{1}{\text{gap}} = 9.43, \qquad t_{\text{mix}}(\tfrac{1}{4}) = 15, \qquad K = 113.38.$$

A frame forgets its origin by $1/e$ in 9.4 instructions and is statistically gone after 15 — 71.8 ns at the measured 208.86 MHz.

And what the worst case actually looks like, Pass 2885. Only 188 of 4,199,040 elements sit at distance exactly 19 — 0.0045%. The natural guess is that so rare a set is one orbit, one coset, something nameable.

It is not. Those 188 spread over 54 distinct linear parts, element orders 4 through 18, and all three trace classes. There is no single "hardest instruction" to name, and the structural hypothesis is refuted.

What survives is the useful half: 188 **is small enough to enumerate exactly**, so a compiler regression suite can contain *every* worst-case input rather than sampling for them. A negative structural result and a positive engineering one out of the same run.

In plain language. *Put that beside the readout cost of §62 and you get an observation cadence, which is a real design parameter: reading more often than every fifteen instructions pays the thermodynamic price repeatedly for information the machine has not yet regenerated.*
Notice also how close 15 *and* 19 *are. The machine scrambles almost exactly as fast as it can reach — there is no regime in which it is still reachable but already random, which is a comfortable property for a scheduler to have.*

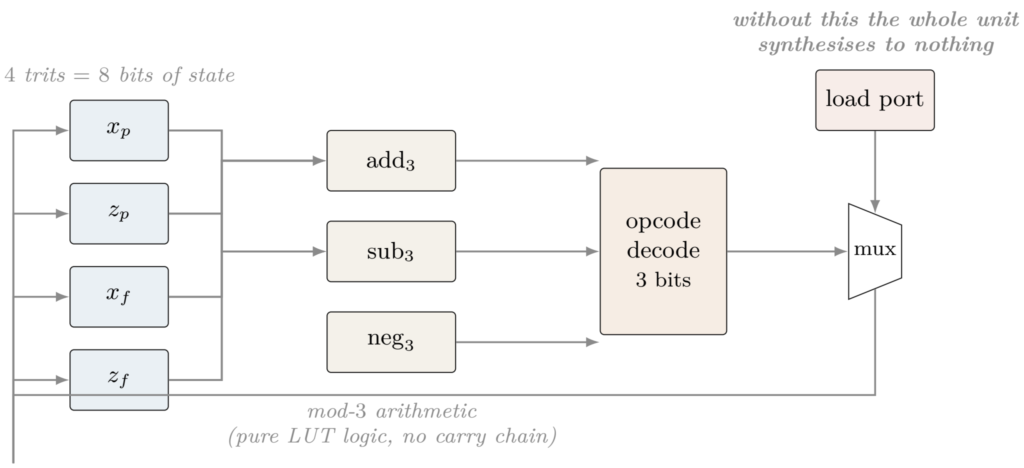
A trap avoided, and worth naming. A minimality *proof* says nothing about *area.* It is tempting to quote the 72-cell figure for the four-operation engine on the grounds that it does strictly less work — and that would be wrong twice over: the number was measured on a different design, and removing decode cases can *cost* cells when what was removed was being shared with what remains. The minimal engine was therefore synthesised and placed separately (§58.2), and only then quoted. It came out at 43 cells; had it come out at 80, that is the number this document would print.

In plain language. *Two details are worth noticing. First, there are* six *different minimal choices, so a chip designer has real freedom to pick whichever three are cheapest in silicon rather than being handed one answer.*
Second, every single one of the six contains at least one Fourier gate and *at least one controlled-add. No collection of phase gates and one Fourier gate will do. The entangler is not optional — which is the formal version of the intuition that a machine that never couples past to future is not really using the self-entanglement at all.*

58 The hardware

58.1 The frame unit datapath

In plain language. *This is the circuit diagram of the register itself — the part of the machine that actually holds a number. Four small boxes across the top store the four coordinates; the shapes below them are the arithmetic that adds and multiplies modulo three; the funnel symbols are switches choosing which instruction is being carried out this cycle. The single box on the right marked* load port *is the only way a value ever enters from outside, and §67.7 is about the consequences of it being there.*



`rtl/w33_pass2777_isa_frame_unit.sv`

58.2 The measured cell budget

In plain language. *Below is what the machine actually costs, measured by running the industry-standard open-source tools — Yosys to translate the design into logic gates, nextpnr to place those gates on a real chip — targeting a Lattice iCE40 UP5K, a chip you can buy on a breakout board for about $25.*

The entire arithmetic core of this computer is seventy-two *logic cells, out of* 5,280 *available. It uses* 1.4% *of a $25 chip.*

The three designs, on two parts, in one harness.

	HX8K CT256		UP5K SG48		
Design	**LC**	$F_{\max}$	**LC**	$F_{\max}$	**Opcode**
Minimal engine, 4 operations	**43**	**208.86**	43	72.40	2 bits
Public unit, 6 frame opcodes	72	175.72	72	60.80	3 bits
Engine + support readout	48	207.64	48	72.05	2 bits

frequencies in MHz

Same cells on both parts — synthesis does not know the package — but the HX8K is 2.9× faster. **Every UP5K figure elsewhere in this document therefore understates speed by about a factor of three**, and is kept because it is the part the $25 board carries.

Measured on UP5K SG48; the per-instruction breakdown.

Design	Cells	Pins	$F_{\max}$	Opcode
Minimal engine, 4 operations	**43**	22	**72.40** MHz	2 bits
Public unit, 6 frame opcodes	72	26	60.80 MHz	3 bits

40% **fewer cells and** 19% **faster.** The minimality theorem cashes out in silicon. Also verified on the minimal engine, because both checks are cheap and both have caught real defects here: the linear part is symplectic on frame *differences* (SAT, 4100 variables / 11278 clauses — a translation cancels in a difference, so one assertion covers all four opcodes at once), and no output folds to a constant.

Measured; iCE40 UP5K SG48, Yosys 0.67 + nextpnr.

Build	Logic cells	$F_{\max}$ (MHz)	Contains
F only	16	230.84	one Fourier gate
$F_p + F_f$	19	230.84	both Fourier gates
F and S (4 opcodes)	40	86.60	the single-qutrit Cliffords
CX only	33	72.40	the entangler, both directions
$\sigma^5{=}Z$ only	20	144.07	the translation
all Clifford, no Z	65	60.80	cannot leave the origin
full frame unit	**72**	**60.80**	the machine
For comparison, elsewhere in the design:			
CX, bare loadable	23	72.40	`w33_pass2752_cx_loadable_frame.sv`
36-lane mixer, serial	4048	19.65	`w33_pass2612_serial_mixer.sv`
36-lane mixer, parallel	13,965 LUT4 + 799 carry		does not fit any iCE40

In plain language. *The last two rows are worth a moment. The "mixer" is the part that combines 36 signals at once. Built the obvious way — all 36 lanes in parallel — it needs about 14,000 logic cells, nearly twice what the largest chip in this family has. Built serially, one lane at a time, it needs 4,048: about a third of the area, at 1/36 of the speed. That is the kind of trade a blueprint exists to make explicit.*

What we got wrong, and how we know. **Every cell count in this project before Pass 2753 was wrong, and the reason is instructive.** We reported 13 cells for F and S, 8 for CX, 42 for the ISA. Those modules had no load port. By the reachability theorem of §57, a register driven only by Clifford opcodes can never leave $(0, 0, 0, 0)$ — so the synthesis tool *proved the state was constant and deleted six of the eight flip-flops.* We were measuring the area of nothing.

Two independent teams shipped this same defect within four hours of each other. Both had *exhaustive* testbenches — one covered all 81^2 pairs of frames — and both passed, because the testbenches drove the *combinational* logic and never instantiated the sequential wrapper.

Simulation cannot detect this class of defect at all. A module that folds to a constant still simulates correctly. It just does nothing. Only a synthesis frontend asks whether the state can move.

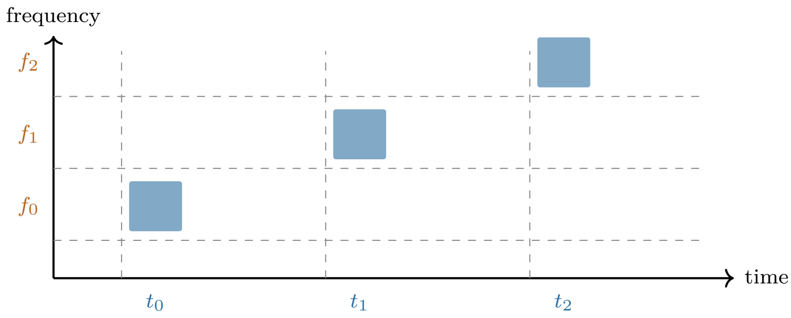
One figure survived unchanged: $\sigma^5 = Z$ was reported at 21 cells and re-measures at 20. It was always real — because it is the one instruction that is not linear, and so the one instruction that could always move the state. The theory predicted exactly which number would survive.

Part III — The physics

Light, magic states, thermodynamics: where the machine touches the world, and exactly how far the evidence goes.

59 The physical layer

In plain language. *Everything above is logic — it would work on any hardware that can implement the eight operations. This section is about doing it with light, which is the intended implementation, and about what has already been demonstrated in a laboratory as opposed to designed on paper.*
A single photon carries several independent properties. Two of them are useful here: when *it arrives (its time bin) and* what colour *it is (its frequency bin). Chop time into three slots and colour into three bands, and one photon carries two qutrits — the "past" and "future" registers of §54.*



The sum gate
$|f, t\rangle \mapsto |f,\ t + f \bmod 3\rangle$
A chirped fibre Bragg grating delays each colour by a different amount, so the photon's *colour* controls how far its *arrival time* shifts. That is a controlled-add, implemented with one passive optical element.

Demonstrated: Imany *et al.*, *npj Quantum Information* **5**, 59 (2019). Three 3 ns time bins at 6 ns spacing, three frequency bins at 380 GHz, −2 ns/nm dispersion, 18 ns wraparound. Measured basis fidelity 0.92 ± 0.01; certified entanglement of formation $\geq 1.19 \pm 0.12$ ebits.

Evidence boundary — what is measured and what is not. Measured in a laboratory: the deterministic single-photon qutrit SUM gate, at fidelity 0.92 ± 0.01, with the entangled output certified.

Proved exactly: the group theory, the geometry, the class atlas, the ISA semantics, the reachability, the synthesis and placement figures.

Neither, and stated as such: source efficiency, insertion-loss engineering, fault-tolerance thresholds, and the magic-state injection route. The M_{36} resources are typed `M36_Q4_RAW` in the controller and injection is *refused* in RTL until a proved ququart protocol supplies a validity assertion. That refusal is a design feature: it makes the gap impossible to paper over.

60 Why the exponent nine

In plain language. *A small but pretty result that shows how tightly the layers fit.*
To tell which operation the machine just performed, you measure a quantity and look it up in a table. The obvious quantity — the "trace" — has a flaw: it changes if the whole state picks up an irrelevant overall phase, which physically means nothing. The fix used in this project is to raise the trace to the ninth *power and divide by a determinant. The ninth power was found empirically. Where does nine come from?*

Derived, Pass 2779. For a d-dimensional representation, under $U \mapsto \lambda U$ we have $\mathrm{Tr}(U^k)^e \mapsto \lambda^{ke}\,\mathrm{Tr}(U^k)^e$ and $\det(U^k) \mapsto \lambda^{kd}\det(U^k)$, so

$$\Theta_k(g) = \frac{\mathrm{Tr}(U_g^k)^e}{\det(U_g^k)}$$

is phase-invariant for all λ *exactly* when $e = d$. Here $d = 9$, the dimension of the two-qutrit state space — which by §54 is $\dim\mathrm{End}(\mathbb{C}^3)$. Verified numerically: $e \in \{3, 8, 10\}$ fail, $e = 9$ holds.

If the phase ambiguity were a *finite* group μ_m one could use any $e \equiv 9 \pmod{m}$, possibly smaller. We computed the scalar subgroup of the generated matrix group by collision sampling and it is μ_{12}. Since $9 \not\equiv 0 \pmod{12}$, the determinant is genuinely required, and 9 is the *smallest* exponent that works: the sensor is minimal.

$\mu_{12} = \mu_4 \times \mu_3$. The μ_3 is ω, the qutrit alphabet. The μ_4 is the quadratic Gauss sum $\sum_j \omega^{j^2} = i\sqrt{3}$, which is *imaginary* precisely because $3 \equiv 3 \pmod{4}$ — the same congruence that makes time reversal a physical rather than a gauge symmetry.

In plain language. *That last box is the kind of thing that makes this project worth doing. A practical question — "why is the exponent nine in the measurement circuit?" — turns out to have the same answer as an abstract one — "why does this geometry distinguish past from future?". Both are the statement that -1 is not a perfect square when you count modulo 3.*

60.1 And for wider registers, it is usually three

Pass 2791. The phase group is not an artefact of the two-qutrit case. At $n = 1$ the Clifford group is small enough to enumerate exactly rather than sample:

$$|\langle F, S, X\rangle| = 2592 = \underbrace{12}_{\mu_{12}} \times \underbrace{216}_{|\mathrm{ASp}(2,3)|} \qquad \text{— consistent, so } \mu_{12} \text{ again.}$$

For n qutrits $d = 3^n$, so the minimal exponent is the least $e \equiv 3^n \pmod{12}$, and $3^n \bmod 12$ cycles $3, 9, 3, 9, \ldots$:

n	1	2	3	4
$d = 3^n$	3	9	27	81
minimal e	**3**	9	**3**	9

Verified directly on 3×3 unitaries: $e = 3$ is phase-invariant, $e \in \{2, 4, 9\}$ is not.

Odd register widths need only a *cube* of the trace, not a ninth power — a strictly cheaper measurement, alternating with width.

Part IV
Quantum Operation

What this part is. *What makes the machine quantum rather than merely reversible: the magic states that buy universality, the thermodynamic price of looking at the register, the machine running inside itself, and the network that joins many of them.*

61 The magic states, and the protocol that was found

In plain language. *Everything described so far is, in a precise technical sense,* not enough. *The eight opcodes generate a large and useful group, but a machine restricted to those operations can be simulated efficiently by an ordinary computer. To get a genuine quantum advantage you need one more ingredient, traditionally supplied as a special prepared state called a* magic state.
The substrate hands us 36 *candidates for free —* 36 *special directions in the four-dimensional space, called the Witting rays. The question is whether those* 36 *can be* purified*: whether noisy copies can be distilled into cleaner ones. Without that, the machine is a very elegant classical simulator.*
This section is the one that changed most between drafts, and the sequence is worth following, because it is a small case study in how a negative result should be read. First an exhaustive search said no. Then an argument from resource accounting said the "no" could not be fundamental. Then a wider search said yes.
But before any of that: the 36 *are not a lucky accident. Count them against the machine's own address space.*

61.1 Magic is everything except one line

In plain language. *The machine addresses* 40 *points. There are* 36 *magic rays. That leaves* 4 *— and a* line *of this geometry contains exactly* 4 *points. Either a coincidence, or the whole story.*
Add the four missing directions back — they turn out to be the four coordinate axes, which is to say the ordinary, unmagical computational basis — and ask what the 40 *look like together.*

Exact, Passes 2835 and 2869. Adjoin $e_1, \ldots, e_4$ to the 36 rays and compute the orthogonality graph on all 40:

$$\text{degrees } \{12{:}40\}, \qquad \lambda = 2, \qquad \mu = 4 \implies \mathrm{SRG}(40, 12, 2, 4).$$

Those parameters admit 28 non-isomorphic graphs, so the identification is completed by the two further steps of §55: the quadrangle axiom holds over all 40 lines (narrowing to $W(3,3)$ or $Q(4,3)$), and an explicit spread exists (which only $W(3,3)$ has at odd q). **The 40 rays are the points of $W(3,3)$.**

The four axes are mutually orthogonal — a 4-clique, hence a line. Every magic ray is orthogonal to exactly one axis ({1:36}), and the resulting nearest-point partition is $9+9+9+9$: precisely the four families the ray construction produces.

$$\boxed{\begin{array}{c}\boldsymbol{M_{36} \text{ is the } 40 \text{ points of } W(3,3) \text{ with } \textit{one line} \text{ deleted,}}\\ \textbf{and the deleted line is the computational basis.}\end{array}}$$

In plain language. *This changes what the magic states* are. *They are not a resource bolted onto the substrate and supplied from outside. They are the generic condition of the machine's own address space: pick a point at random and it is magic with probability* 36/40. *The ordinary, classically-simulable states are the single distinguished line you have to delete to get rid of the magic.*

One more thing falls out, and it is a warning about reading structure off counts. The geometry partitions the 36 *into four families of nine. The Clifford group partitions them into* $4 + 8 + 12 + 12$. Neither refines the other. *The symmetry and the geometry are looking at different things.*

Is the deleted line special? Both answers are true, Pass 2862. Enumerating the 40 lines and counting how many classical points each contains:

$$\underbrace{1}_{\text{all four classical}} + \underbrace{12}_{\text{one classical, three magic}} + \underbrace{27}_{\text{all four magic}} = 40.$$

Geometrically, no. $\mathrm{Aut}(W(3,3))$ has order $51840 = 40 \times 1296$ and is transitive on lines, so nothing intrinsic to the 40 points prefers one. Choosing a computational basis is choosing one line out of forty — exactly $\log_2 40 = 5.3219$ bits of pure convention.

Relative to the stabilizer structure, uniquely yes. Exactly *one* of the forty lines has all four of its points classical.

In plain language. *Which is a satisfying place to end up. "Which states are magic" is not a fact about the geometry — the geometry is completely indifferent. It is a fact about which line you declared to be the basis. But the declaration is not lost afterwards: it is recoverable, because the line you chose is the unique all-classical one in the structure your choice created.*

61.2 All thirty-six look identical from the inside

Exact integer arithmetic in $\mathbb{Z}[\omega]$, Pass 2790. Only two overlap values occur among the 36 rays:

$$|\langle i|j\rangle|^2 \in \{\,0,\ \tfrac{1}{3}\,\},$$

and every ray has the *same* profile: 11 orthogonal partners and 24 partners at overlap 1/3. There is exactly one Gram profile among all 36.

(The orthogonality graph is 11-regular on 36 vertices but *not* strongly regular: $\lambda \in \{1, 2\}$, $\mu \in \{3, 4\}$. Recorded because the near-miss invites a false claim.)

In plain language. *So no symmetry of the machine can tell one ray from another by looking at how they sit relative to each other. If the rays differ at all, they differ only in how they sit relative to the* ordinary *states — an external comparison, not an internal one.*

61.3 And from the outside, they fall into exactly three grades

60 **two-qubit stabilizer states,** $F_{\mathrm{stab}} = \max_s |\langle s|\psi\rangle|^2$**.**

F_{stab}	closed form	rays	grade
0.750000	$9/12$	**4**	shallow
0.705342	$(5+2\sqrt{3})/12$	**24**	mid
0.622008	$(4+2\sqrt{3})/12$	**8**	deep

Stabilizer fidelity is a Clifford invariant, so the $8+24+4$ grading is genuine and not an artefact of the chosen basis.

All three noise thresholds are one formula. For $\rho_p = (1-p)|m\rangle\langle m| + p\,I/4$ the target overlap is $1-3p/4$, so the witness certifies non-stabilizerness exactly while $1-3p/4 > F_{\mathrm{stab}}$:

$$\boxed{p \;<\; \tfrac{4}{3}\big(1-F_{\mathrm{stab}}\big)}$$

shallow	0.333333333333	=	$1/3$
mid	0.392877598318	=	$(7-2\sqrt{3})/9$
deep	0.503988709429	=	$(8-2\sqrt{3})/9$

Three separately-published thresholds, one formula, agreement to 10^{-12}.

61.4 Four, not three: fidelity is necessary but not sufficient

In plain language. *A correction, and a useful one to read even if you skip the mathematics. We had sorted thirty-six quantum states into three groups by a single number measuring their quality, and concluded that only three genuinely different states existed. That was too quick: two things can score the same on one measurement and still behave differently. Acting with the full symmetry group instead of comparing numbers splits the thirty-six into* four *classes, not three. The lesson generalises past this section — a single number rarely separates everything, and this project has made that mistake more than once.*

What we got wrong, and how we know. An earlier draft of this section said: *“the distillation search needs three representatives, not thirty-six.”* That was one step too far. Equal stabilizer fidelity does not imply the two states are interchangeable — it is a single number, and a single number rarely separates everything.

Acting with the full two-qubit Clifford group (92,160 matrices) on the 36 rays and matching images by *fidelity* rather than by a rounded key:

$$\text{Clifford equivalence classes} \;=\; 4, \qquad \text{sizes } [\,4,\ 8,\ 12,\ 12\,].$$

Every class sits inside one grade — so the grading is real — but the 24-ray middle grade is **two** inequivalent families of twelve. The correct count is **four**.

Corroborated from the other direction: an independent exhaustive distillation search, which knows nothing about Clifford orbits, reports results for “*both* middle grades”. Two unrelated computations, the same splitting.

And nothing an overlap can see tells the twins apart, Pass 2830. The *full* stabilizer overlap spectrum — the multiset of $|\langle s|\psi\rangle|^2$ over all 60 stabilizer states — is a Clifford invariant, and every stabilizer Rényi entropy and overlap moment is a function of it. Computed exactly in ninths, the two 12-ray classes have the **identical** spectrum.

Across all four classes there are only **three** distinct spectra. So the entire 60-value distribution has exactly the same resolving power as its single largest entry, F_{stab}.

Consequence. A protocol that reads its input only through overlap statistics *must* treat the two classes alike. The split can matter only to a protocol that uses the ray's Clifford orbit — one that fixes a frame and cares which representative it was handed.

61.5 Why the first "no" could not have been the last word

In plain language. *An exhaustive search over* 5,355 *small codes and* 21,420 *branches found no two-copy protocol that improves fidelity. That is a real result, and the tempting reading is that two noisy copies simply do not contain enough raw material to make one better copy.*
There is a way to check that reading directly, without searching any protocols at all. Physicists have quantities that measure *how much magic a state contains and which are guaranteed never to increase under any allowed operation — no measurement, no post-selection, no clever feedback can make one go up. If two copies scored lower than the target, distillation would be impossible for reasons no ingenuity could overcome. So: compute the score.*

$D_{\text{min}} = -\log_2 F_{\text{stab}}$ **over all** 36,720 **four-qubit stabilizer states.**

grade	$F_{\text{stab}}(\psi)$	$F_{\text{stab}}(\psi^{\otimes 2})$	$D_{\text{min}}(\psi)$	$D_{\text{min}}(\psi^{\otimes 2})$
shallow	0.750000000	0.562500000	0.415037	0.830075
mid	0.705341801	0.497507057	0.503606	1.007211
deep	0.622008468	0.386894534	0.684994	1.369988

F_{stab} is *exactly* multiplicative here, so $D_{\text{min}}(\psi^{\otimes 2}) = 2D_{\text{min}}(\psi) > D_{\text{min}}(\psi)$ for every grade. The budget is never the binding constraint.

The monotone never obstructs. Any two-copy no-go in this system is *structural*, not information-theoretic — a fact about the protocol family searched, not about the resource. The right response to it is to widen the family.

61.6 And widening the family worked

Deep-grade distillation, exact. Extending the search from one decoder to all 11,520 logical Clifford decoders gives 48 **improving branches** on the deep eight-ray grade, and zero on the shallow and both middle grades. One explicit branch — input ray 5, stabilizers $IYZY$ and $YZXY$, syndrome $(-1,+1)$, output ray 7 — has the exact curve

$$P_{\text{succ}} = \frac{p^2 - 2p + 2}{4}, \qquad F_{\text{out}} = \frac{5p^2 - 12p + 8}{4(p^2 - 2p + 2)},$$

$$F_{\text{out}} - F_{\text{in}} = \frac{p(p-1)(3p-2)}{4(p^2 - 2p + 2)} > 0 \quad \text{for } 0 < p < \tfrac{2}{3},$$

an interval that contains the whole deep-grade magic window $p < (8 - 2\sqrt{3})/9 \approx 0.504$.

In plain language. *So the resource is usable. Note the shape of the story: the first search was correct and its conclusion was correctly stated; the accounting argument said the obstruction could not be fundamental; the wider search then found the protocol. Nobody was wrong at any step — but a reader who had stopped at "exhaustive search found nothing" would have drawn exactly the wrong conclusion about the machine.*

And now the part that a reader should not *stop before.*

61.7 Usable is not the same as practical

The convergence rate, Pass 2831. Linearising the accepted-round map at the pure fixed point:

$$\frac{dp'}{dp}\Big|_{p=0} = \frac{2}{3}, \qquad P_{\text{succ}}\Big|_{p=0} = \frac{1}{2}.$$

The noise is multiplied by two thirds each accepted round. It is **not** squared. Each round consumes two inputs and succeeds half the time, so the raw cost multiplies by $2/P_{\text{succ}} \approx 4$ per round while the infidelity only falls by a third.

start p	to 10^{-3}	to 10^{-6}	to 10^{-9}
0.10	2.3×10^{7}	4.0×10^{17}	6.9×10^{27}
0.20	5.6×10^{8}	9.7×10^{18}	1.7×10^{29}
0.30	1.5×10^{10}	2.6×10^{20}	4.5×10^{30}
0.50	1.1×10^{14}	1.9×10^{24}	3.2×10^{34}

raw noisy states consumed per distilled output

The map converges *linearly*. Standard distillation is quadratic or cubic — Reed–Muller 15-to-1 gives $p' \approx 35p^3$ — and that gap is the whole game.

In plain language. *So: the deep-grade branch is a genuine fidelity-improving map and a genuine refutation of the earlier no-go. It is* not*, by itself, a usable distillation routine. Ten to the twenty-seven raw states is not an engineering number.*

What it establishes is that the resource is not inert — which is the thing that was genuinely in doubt.

61.8 And no other two-copy branch will fix it

In plain language. *The natural next move is to hope one of the other 47 branches converges faster. It does not, and the reason can be settled without examining any of them.*

For a protocol to be quadratic — for the error to square *rather than shrink by a third — it must throw away every input in which exactly one of the two copies was faulty. Otherwise those single faults leak into the output and contribute an error proportional to p. So the question becomes: is there* any *stabilizer measurement that keeps the clean input and rejects all six single-fault inputs?*

Exhaustive, Pass 2861. Over all 5,355 two-qubit stabilizer codes × 4 syndromes × all four Clifford classes of ray:

$$\text{projectors keeping } |mm\rangle \text{ and annihilating all six single errors} = \boxed{0}.$$

The three-copy route, as far as it has been taken. Four passes and three formulations, ending in a mechanism rather than a verdict:

- Rank-one branches *exist* — 36 stabilizer states annihilate every single-error input while keeping the clean one.
- They are *useless*: a rank-one range is a stabilizer state, which has no magic. The other track reached the same conclusion the same day, from an unrelated search over 649,940 isotropic subspaces.
- Orthogonal *pairs* exist — 13 of them, so 2-dimensional subspaces sit inside the complement.
- **But none of those pairs spans a stabilizer code.** Every pair tested has exactly 7 common stabilizing Paulis of $\mathbb{F}_2$-rank **3**, and a 2-dimensional code needs rank 5.

Spanning a subspace is not the same as being a joint eigenspace, and that gap is exactly where the route currently stops.

In plain language. *One thing that rank-3 number does suggest rather than close: a rank-3 stabilizer group has an* eight-*dimensional code, not a two-dimensional one. If such a code lies inside the complement, it would kill every single error and have ample room for a magic output. That is the next question, and it is a better one than the three that preceded it.*

The linear rate $2/3$ is not a property of the branch that was found. It is a *bound on the entire two-copy stabilizer family.*

The right answer, and a wrong reason nearly published. The tempting explanation is a dimension count: six error vectors, a rank-four projector, no room. That explanation is false. $|mm\rangle$ *is* orthogonal to all six single-error vectors (checked: 2.6×10^{-16}), and $16-6=10\geq 4$, so projectors with exactly this property exist in abundance — the rank-one projector onto $|mm\rangle$ is one.

What actually fails is that no *stabilizer* projector aligns that way. The magic-state error basis is not a stabilizer basis, and syndrome measurements cannot be steered onto it. The obstruction is the mismatch between magic and the stabilizer formalism — which is the same tension that makes these states interesting in the first place.

Recorded because a right answer with a wrong reason is a claim waiting to be misapplied to the next case, where the reason would matter.

In plain language. *So the road forward is narrower and clearer than it was. Two copies are provably not enough. Super-linear distillation of these states needs three or more copies, or operations from outside the stabilizer set — and §61 already reports that a search for a three-copy advantage found no witness either. This document states that as the open problem it is, rather than as a promise.*

A near-miss in this very section. The first draft of the yield analysis described the cost as "doubly exponential in a good way — the infidelity squares each round." It does not; it falls by a factor of 2/3. The tables above were already printed and would have looked like corroboration to anyone who did not check the ratio between successive rows. The linearisation is one derivative and it changes the conclusion from "deep targets are cheap" to "deep targets are impossible." Compute the rate; do not eyeball the trend.

Evidence boundary, stated as bluntly as possible. What the deep-grade result *is*: an exact state-fidelity distillation theorem for an explicitly enumerated class — all 5,355 binary $[[4, 2]]$ stabilizer projectors, all four syndromes, all 11,520 logical Clifford decoders.

What it is **not**: a fault-tolerant injection threshold, an asymptotic-yield theorem, or a laboratory demonstration. The controller still types the resource `M36_Q4_RAW` and refuses injection in RTL until a proved *fault-tolerant* protocol asserts validity, because a fidelity-improving map is not the same object as an injection threshold. **Universality is now a matter of closing a stated gap rather than of finding an unknown protocol** — which is a different and much better position than the previous draft of this document described.

62 Support for readout, phase for execution

In plain language. *Here is a design temptation, and the theorem that kills it.* *The machine's frame is four trits —* 81 *states. But a great deal of what you want to* observe *about the frame is coarser than that: often you only care* which *of the four registers are non-zero, not what they hold. That is a four-bit "support mask",* 16 *possibilities instead of* 81*, and it has beautiful structure — it forms an exact finite geometry with the capacity profile*

$$(4, 6, 4, 1) \odot (1, 2, 4, 8) = (4, 12, 16, 8).$$

A five-fold compression with geometry attached is exactly the kind of thing an engineer wants to build the machine out of. Can the machine simply run *on support masks and skip the phases?* *No. And the counterexample is two lines long.*

The obstruction. The two frames

$$(0, 1, 0, 0) \qquad \text{and} \qquad (0, 2, 0, 0)$$

have the same support mask `0100`. Apply Z_p — the translation, $z_p \mapsto z_p + 1$ — and their masks become `0100` and `0000`. Two states that the support layer cannot tell apart are sent to two states it *can*.

So support is not a *congruence*: it is not preserved by the dynamics, and a machine that stored only support would be unable to predict its own next state. Refining the partition until it is stable gives

$$\boxed{16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81}$$

with class-size histograms $\{1{:}1, 2{:}4, 4{:}6, 8{:}4, 16{:}1\} \to \{1{:}7, 2{:}29, 4{:}4\} \to \{1{:}75, 2{:}3\} \to \{1{:}81\}$. The terminal partition is *discrete*: full ternary phase is forced.

Support for readout, phase for execution. The compression is real and exact — but it is a lens, not a substrate.

62.1 What a readout costs, in joules

In plain language. *A support readout throws information away — that is what "lossy" means. Landauer's principle says throwing information away has an unavoidable energy cost, and here the cost is exactly computable, because the fibres are exactly known: a mask of weight k has exactly 2^k preimages, and there are $\binom{4}{k}$ such masks.*

Exact, Pass 2836.

$$H(\text{state} \mid \text{support}) = \frac{1}{81}\sum_{k=0}^{4}\binom{4}{k}2^k\,k = \frac{216}{81} = \boxed{\tfrac{8}{3}\text{ bits}}$$

$$H(\text{state}) = \log_2 81 = 6.339850, \qquad H(\text{support}) = 3.673183, \qquad \frac{3.673183}{4} = 91.83\%\text{ efficient.}$$

Over $\mathbb{F}_q$ the closed form is $4(q-1)\log_2(q-1)/q$, verified exactly at $q = 2, 3, 4, 5, 7, 8, 9, 11$.

In plain language. *At $q = 2$ that formula gives* zero. *Over a two-element field the support* is *the state, so a support readout destroys nothing at all. Every one of the 8/3 bits is the price of the third field element.*

That is the thermodynamic version of the architectural law: ***the phase is precisely the part you pay to look at.***

Landauer, at room temperature.

$$E \;\geq\; \tfrac{8}{3}\,k_B T \ln 2 \;=\; 7.656 \times 10^{-21}\,\text{J} \;=\; 47.78\text{ meV} \quad \text{at } 300\,\text{K},$$

or 7.66 pW of unavoidable dissipation at a 1 GHz readout rate. This is a floor set by information theory, independent of the circuit: no implementation of the support readout can beat it.

And the compute path is free, Pass 2864. Every opcode is a group element, hence a bijection on the 81 frames — which §57 verified directly, since the walk's transition matrix came out doubly stochastic. A bijection erases nothing, so

$$E_{\text{Landauer}}(\text{compute}) \;=\; \textbf{exactly zero}.$$

All of this machine's irreducible dissipation is in readout. That is not a coincidence of the design; it is what building the ISA out of a *group* buys you.

In plain language. *It would be easy to over-read that. Real silicon is nowhere near the floor, and saying so plainly is part of the job.*

Floor versus forecast. A CMOS estimate for the 43-cell engine, at the measured activity of 1.622 output transitions per operation, 5 fF effective per cell and 1.2 V:

$$E_{\text{op}} \approx 0.502\text{ pJ}, \qquad 0.105\text{ mW dynamic at } 208.86\,\text{MHz},$$

$$\frac{E_{\text{op}}(\text{CMOS})}{E_{\text{Landauer}}(\text{readout})} \;=\; 6.6 \times 10^7 \;\approx\; 10^{7.8}.$$

Real hardware runs about eight orders of magnitude above the information-theoretic floor, which is the usual figure for CMOS. **The Landauer number is a floor, not a forecast**, and is quoted here only so that nobody mistakes it for one.

And the counting floor lands exactly where the search does. $\lceil \log_2 81 \rceil$ = 7 bits are needed to label 81 states; an exhaustive search shows no 7 support taps suffice and exactly 48 eight-tap sets do. The observer is one bit above the floor, and that bit is provably necessary.

62.2 Enforcing the law in the netlist

In plain language. *The law is a theorem about mathematics. It constrains no circuit. Nothing stops an engineer from wiring the cheap 4-bit mask back into the execution path — and the resulting machine would pass simulation almost always, because support* is *preserved by most operations. It would drift the first time a translation fired on a register holding a 2.*
That is the same failure shape as the folded registers of §58.2: correct in simulation, wrong in silicon, invisible to the testbench. So the law is enforced structurally instead of documented.

Gate-level information flow, Pass 2834. `scripts/check_information_flow.py` flattens the design to gates, builds the driver graph and computes reachability in both directions:

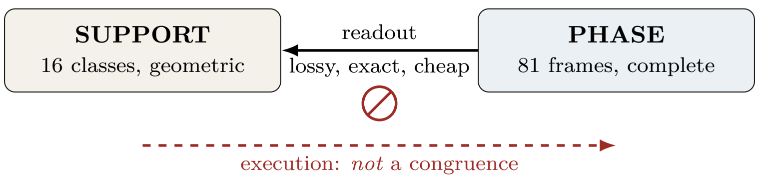
```
top w33_support_readout_diode: 59 cells after flatten+opt
reverse: support_mask, tamper_mask reach none of xp_o, zp_o, xf_o, zf_o
forward: every frame register reaches the mask
PASS: the readout is a diode.
```

Self-tested against a negative control — rewiring a tamper input into the load enable makes it report VIOLATION on all four registers — because a guard that can only pass is not a guard. The non-congruence witness is asserted in the netlist too and SAT-proved (228 variables, 600 clauses).

This is the one check in this project that **fails** rather than warns. Unlike a rediscovery collision, a backward edge has no benign reading.

In plain language. *It is also nearly free. Adding the readout to the minimal engine costs **five logic cells** and 0.6% of the clock: 43 → 48 cells, 208.86 → 207.64 MHz. The cheap layer really is cheap — and now it is cheap* and *provably one-way.*

In plain language. *This is a useful kind of negative result, because it tells you exactly where to put the cheap representation. Readout, routing, telemetry and visualisation can all live in the 16-class support world and enjoy its geometry. The execution core cannot. A machine built the other way round would appear to work in simulation — support is preserved by* most *operations — and would drift silently the first time a translation fired on a register holding a 2.*



63 The machine hosts itself

In plain language. *Everything so far describes one machine. But the frame is* two *qutrits, and a one-qutrit machine is a strictly smaller machine of exactly the same kind — so this processor can run a smaller version of itself as a guest.*
That is the ordinary meaning of a virtual machine, and the ordinary question about one is: how much slower is the guest? Usually that is measured. Here it can be computed exactly, because "how many host instructions does one guest instruction cost" is a shortest-path question in a graph this document has already searched exhaustively.

Emulation overhead, exact, Pass 2912.

guest instruction	host instructions
F	1
Z	1
S	**8**
worst case	**8**×
mean	3.33×

The entire overhead is one instruction. F and Z are free; the guest's phase gate costs eight — because the minimal instruction set of §57 *dropped* S.

In plain language. *So the two results talk to each other, and the conversation is a real design decision: dropping S made the machine 40% smaller and made one guest instruction eight times more expensive. If the machine is mostly going to run guests, that trade is bad. If it is mostly going to run native code, it is excellent. Nobody could have said which without both numbers.*

63.1 Isolation without a memory manager

In plain language. *Two guests fit side by side, one on each qutrit. The interesting part is what keeps them apart.*
In a conventional computer, isolation is enforced*: a memory management unit checks every access and refuses the ones that cross a boundary. Correctness depends on that check being present and right. Here there is nothing to check.*

Structural isolation. Of the four opcodes, exactly two — $\mathrm{CX}_{p\to f}$ and $\mathrm{CX}_{f\to p}$ — touch both halves of the frame. The other two act within one half.

A hypervisor that never issues CX to a guest cannot leak between guests. Not "is not permitted to": *cannot*. There is no instruction that would do it. The guarantee comes from the algebra, not from an enforcement mechanism, and it cannot be defeated by a bug in the checker because there is no checker.

63.2 And it was built

One datapath, N guest contexts; HX8K CT256, Pass 2914.

N	shared	$F_{\max}$	replicated ($43N$)	area saved
1	42	175.1 MHz	43	—
2	74	134.9	86	14%
4	118	128.5	172	31%
8	180	113.4	344	**48%**
16	339	91.1	688	**51%**

In plain language. *Sharing wins on area at every size, and converges on saving about half — for a legible reason: half the cost is the datapath, which is shareable, and half is the per-guest context storage, which is not.*
The price is speed. Eight guests sharing one datapath each run at an eighth of its clock, so at $N = 8$ the design buys 48% of the area for about a fifteenth of the per-guest throughput. That is the whole trade, measured on one part in one harness rather than argued.

The measured curve proves one synthesized datapath can host multiple guest contexts. It does not prove a network of nested machines or a distributed hypervisor.

In plain language. *The software-side object that now closes part of that gap is HoloBox (`analysis/holobox.py`). A leaf VM and a recursively nested 40-child network use one immutable state grammar; addressed mailbox delivery and guest execution replace only one digest path. At six levels the uniform graph denotes* 105,025,641 *logical VMs with six node blobs, and an independent GAP witness proves the logical W33 line-route bound $2n$. That is still a Python reference runtime, not an FPGA network, KVM boundary, OCI-conformant container engine, or measured distributed system. BT827's separate chart-aware lowering continues to cost $8n = (3 + 5)n$ micro-moves.*

64 The support transform and diagnostic scheduler

In plain language. *A dense fifteen-by-fifteen table is not how the support dynamics should be built. The table factors into thirty-two tiny two-input butterflies, followed by fifteen corrected outputs. Diagnosis has a second surprise: the best order in which to visit the fifteen nonzero masks is not generated by a linear feedback register. A short ROM is both more honest and better.*

Butterfly engine. At $q = 3$ the local kernel is $(u, v) \mapsto (u + 2v, u - v)$. Four stages of eight butterflies and fifteen output cycles give a 47-cycle sequential quotient engine. A serial dense reference needs $15 \times 15 = 225$ cycles, so the exact cycle reduction is

$$\frac{225 - 47}{225} = \frac{178}{225}.$$

Equality is checked on all fifteen basis vectors and thirty-two deterministic signed probes. Logic cells, routed frequency, and switching activity remain measured quantities pending the dedicated workflow.

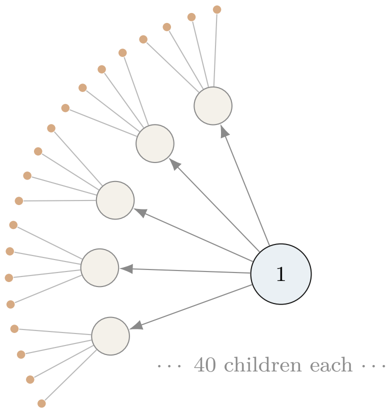
Diagnostic mask ROM. The 210 directed first-passage costs form 39 matching-stabilizer orbits. Exact Held–Karp optimization gives a Hamiltonian mask cycle of cost 315 and 8336 rooted optima. The best $\mathrm{GL}(4, 2)$ Singer cycle costs 1317/4, an exact gap of 57/4. The scheduler is therefore a fifteen-entry nonlinear permutation ROM, not an LFSR.

Forty is not enough. The first observer refinement has forty classes, but those classes are not the forty projective points of $W(3, 3)$. Their sizes are $1^7 2^{29} 4^4$, symplectic orthogonality is ambiguous on 216 unordered class pairs, and no union of Hamming-distance relations on their signatures yields $\mathrm{SRG}(40, 12, 2, 4)$. A matching count is a search hint, never an identification theorem.

65 The network

65.1 Fractal scaling

In plain language. *The machine is designed to be copied. A node contains* 40 *sub-nodes; each sub-node contains* 40 *of its own; and so on. The network of machines has the same shape as the inside of one machine — which means one routing scheme, one addressing scheme, and one error model serve every scale at once.*



$N_n = 40^n$ **leaves** at depth n

$I_n = \dfrac{40^n - 1}{39}$ **instances** in total

routing depth $= 8n = 8\log_{40} N$

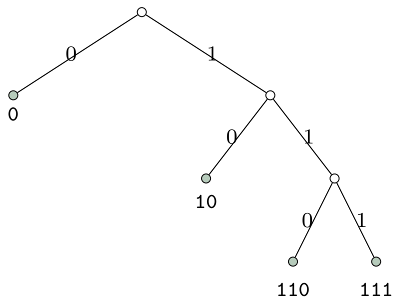
where $8 = 3 + 5$: three trits of local address plus five of the spread index. Address length is logarithmic in the size of the entire network, at every scale, by construction.

65.2 Every bit string is a legal address

Exactly. The routing code has orbit sizes $162 + 162 + 324 + 648 = 1296$ on compass pairs, giving codeword lengths $3, 3, 2, 1$ and

$$2^{-3} + 2^{-3} + 2^{-2} + 2^{-1} \;=\; 1 \quad \text{exactly (Kraft equality).}$$

The prefix code is $\{110, 111, 10, 0\}$ with expected length $7/4$. Because Kraft holds with *equality*, the decision tree has no unused branch: 20,000 random bit strings were parsed with 0 failures.



In plain language. *Look at the tree: every branch ends in a valid address. Nothing dangles.*
The engineering consequence is unusual and worth stating plainly. In a normal computer, a corrupted instruction stream can produce an illegal instruction *— a bit pattern that means nothing, which the processor must detect and trap. Here that cannot happen. A corrupted packet is always a legal packet addressed somewhere else. The fault model therefore has no "decode error" term at all; it has only a "wrong result" term. That simplifies the error correction enormously, and it is forced by the geometry rather than designed in.*
That property is not free, and it is now possible to say exactly what it costs.

The price of a complete code, Pass 2865. A 40-ary router consumes one address symbol per hop:

$$\text{needed: } \log_2 40 = 5.3219 \text{ bits}, \qquad \text{spent: } 8 \text{ bits}, \qquad \text{efficiency } 66.52\%.$$

A store-and-forward router that strips consumed header bits *erases* them, so Landauer applies per hop:

$$\text{floor } 95.37 \text{ meV/hop}, \qquad \text{as coded } 143.35 \text{ meV/hop} \quad (300\,\text{K}).$$

Since $N = 40^n$, the thermodynamic cost of delivering a packet across N leaves is **exactly logarithmic in N**: 143.4 meV per level, 860 meV at depth six (4.1 billion leaves).

About a third of the routing energy buys the absence of an entire failure mode. That trade was always implicit in the design; it now has a number.

In plain language. *Two honest caveats. A* reversible *router pays none of this — the floor applies only because the header is destroyed as it is consumed, and that is a design choice, not a law of nature. And* 143 *meV per hop is a floor in the same sense as the readout figure: about eight orders of magnitude below what any real transceiver will draw.*

What we got wrong, and how we know. We wrote that "the consequence nobody has stated" was that every bit string is a valid instruction stream. The project's own holonet manuscript says it at line 382 — "the routing words fill a binary decision tree with *no unused branch*" — with a citation to McMillan. This was the first error in this project caught by an automated guard rather than by a human noticing. The engineering consequence drawn on top (no decode-error term in the fault model) is a step past the coding fact and appears to be new; the completeness property itself is the manuscript's.

65.3 Remote operations

Exactly. A SUM gate between two *separate* machines costs exactly:

$$\boxed{\text{one shared qutrit Bell pair } + \text{ two classical trits}}$$

Protocol: Alice applies reverse local SUM into her half of the pair, measures, sends trit m; Bob applies $X^{-m}F^2$, applies direct SUM to his data, measures in the Fourier basis, sends trit n; Alice records Z^n in her Pauli frame. All nine branches implement $|x, y\rangle \mapsto |x, y + x\rangle$; verified on all nine basis inputs, all nine branches, and 32 random superpositions. Total conditional success probability 1.

66 What the other track built

In plain language. *Two independent efforts work on this machine and do not read each other's filenames. That is a structural problem, not a discipline one, and the remedy this project uses is to cite across the boundary rather than re-derive. This section is that citation: work the other track owns, which this blueprint depends on and did not do.*

Owned elsewhere, relied on here.

Result	**What it says**
M_{36} deep-grade protocol	48 improving branches; explicit curve improving on $0 < p < 2/3$. §61 reports the yield and the linear rate; the protocol is theirs
Support observer	Support is not an execution congruence ($16 \to 40 \to 78 \to 81$); a six-cycle observer with exactly 8 taps reconstructs the frame, and 7 provably do not
Nine-gate M_{36} branch	The old 15-gate circuit's two terminal SWAPs are relabelling, not computation. 6 CNOT + $3H$, and enumerated faults fall $191 \to 101$
Route fault localisation	23 triangles is the *exact* optimum for single-edge localisation; 29 distinguishes all 48,826 two-edge hypotheses
Controller group	Native controls generate order $30{,}233{,}088 = 2^9 3^{10}$; one extra transvection closes it to A_{40}
D_{12} curvature clock	$6480 = 540 \times 12$: every controller state lies on a 12-cycle, giving a geometry-native reversible clock
Adaptive chirality receiver	$P_{\text{opt}}(n) = \frac{1}{2}(1 + \sqrt{1 - 3^{-n}})$; 0.9997 at six copies

And once, both tracks reached the same conclusion by different methods on the same day.

In plain language. *This document found* 36 *rank-one stabilizer witnesses for the three-copy distillation condition, and showed they are useless because a rank-one projector's range is a stabilizer state, which carries no magic. The other track, searching* 649,940 *isotropic subspaces for non-CSS projectors, found six exact hits and reported them as "accepted-clean-state stabilizer projectors, therefore false leads."*

Same finding, opposite directions, neither aware of the other. That is what the citation protocol is for — and it is also the strongest evidence either result is right.

Part IV — The evidence

How every number above is known, what we got wrong on the way, and the commands that reproduce all of it.

Part V
What We Measured

What this part is. *The measurement chapter, and the most heavily corrected part of this document. It contains results we are confident in, results we withdrew, and — stated plainly — the reasons each withdrawal happened. A blueprint that hides its retractions is not a blueprint, it is a brochure.*

67 Every bit in the machine

In plain language. *Scattered through this document are the sizes of the machine's parts: eighty-one frames, forty addresses, sixteen support classes. Here they are in one place, with the only question that matters thermodynamically attached to each — when you use it, does anything get destroyed?*

The whole state, and its Landauer floor.

layer	states	bits	erased	meV @ 300 K
Pauli frame	81	6.340	0	0
route address	40	5.322	5.322	95.37
support readout	16	4.000	2.667	47.79
OAM × slot	40	5.322	0	0
encode/check sector	2	1.000	0	0
full controller	6480	12.662	0	0
total		**34.65**	**7.99**	**143.15**

The machine's whole state is 34.65 bits,
and only 7.99 of them cost anything to use.

67.1 Two diameters, and where the work actually is

In plain language. *The machine has two different notions of "how far apart can two things be", and they are not remotely the same size.*

Pass 3021, combining two independently measured numbers.

$$\text{address transport diameter} = 2, \qquad \text{frame ISA diameter} = 19, \qquad \text{worst case} = \mathbf{21}.$$

Moving a packet to any of the 40 addresses costs at most *two* shears, because the geometry has diameter 2. Transforming the Pauli frame arbitrarily costs up to *nineteen* instructions, because that is a walk in a four-million-element group. The address figure is the other track's; the frame figure is §57's.

Geometry is cheap; algebra is expensive. Ninety per cent of the machine's worst-case work is frame algebra, and routing optimisation addresses ten per cent of the problem.

67.2 The same finding, three more ways

In plain language. *One measurement is an observation. Four independent ones pointing the same way is a property of the machine, and this is the section where that happens.*

Is the instruction layer extremal? Pass 3042, corrected at Pass 4213. A random instruction stream drives a walk on the 81 frames. Every opcode is a bijection, so the walk is doubly stochastic, its stationary distribution is exactly uniform — no frame is preferred — and

$$|\lambda_2| = 0.893992320, \qquad \text{relaxation } 9.43, \qquad \text{mixing } 15 \text{ instructions}.$$

Those numbers are exact and are what the machine does.

What we may not do is grade them. Pass 3042 compared 0.893992 with $2\sqrt{3}/4 = 0.866025$ and reported the instruction graph as 3.23% short of Ramanujan. That verdict is **withdrawn**. The bound $2\sqrt{k-1}/k$ belongs to a *k-regular* graph, and it is that number because the universal cover of a k-regular graph is the k-regular tree of spectral radius $2\sqrt{k-1}$. The frame graph has degrees 2 to 8 (Pass 4203), so it has a different cover, a different radius, and no claim on 0.866025. The shortfall was a precise-looking figure for a quantity this graph does not possess.

What survives is a comparison rather than a grade: the address layer mixes at the optimum for its degree; the instruction layer mixes more slowly. *How far short of optimal* is not a question a non-regular graph admits.

67.3 The zeta function the instruction graph actually has

In plain language. *Three separate attempts to say something spectral about the instruction layer were withdrawn, all for one reason: each reached for a formula that assumes every vertex has the same degree. The fix turns out not to be a better estimate. It is a different theorem.*

Pass 4222 — Ihara without Bass, and therefore without regularity. Bass's determinant relation needs a degree. The Ihara zeta does not. For *any* finite graph,

$$\zeta^{-1}(u) \;=\; \det(I - uB)\,,$$

where B is the **Hashimoto** non-backtracking matrix on the $2E$ directed edges, $B_{(x,y),(y,z)} \;=\; 1$ exactly when $z \neq x$. No vertex degree appears anywhere in the definition.
The instruction graph has $V = 81$, $E = 261$, so B is 522×522 — a small eigenvalue problem, and one that was available at Pass 3060.

$$\rho(B) = 5.746873, \qquad R = 1/\rho = 0.174008, \qquad \sqrt{\rho} = 2.397264.$$

Of the 160 non-trivial Hashimoto eigenvalues, **none** lies on the circle $|\lambda| = \sqrt{\rho}$.
The instruction graph does not satisfy the graph Riemann Hypothesis. The question that was ill-posed three times has an answer, and it is negative.

Scope (Pass 4291). Every number in this box is an invariant of the frame graph, and the frame graph does not determine the instruction set: 349 universal generating sets yield only 89 distinct zeta signatures, with up to twelve sets sharing one. So this is a statement about an equivalence class of instruction sets, not about a unique ISA. The same caveat applies to every $\rho(B)$, pole count and localisation weight in this chapter, and is stated once here rather than repeated at each.

In plain language. *The method was validated before it was trusted, on two graphs where the answer is already known: K_4, whose zeta is textbook, and the degree-four Levi graph, whose closed form $(1-u^2)^{81}(1-9u^2)(1+9u^4)^{24}(1+3u^2)^{30}$ Bass produces directly. The Hashimoto route reproduces both exactly. It agrees with Bass wherever Bass applies, and then keeps going where Bass cannot. That validation was not enough, and the way it fell short is the more useful lesson. Reproducing $\det(I-uB)$ correctly says nothing about the* classifier *built on top of it, and Pass 4229 found a fault there: the first version counted the Perron root ρ among the non-trivial eigenvalues, which made both reference graphs appear to fail. It was caught because $W(3,3)$ came back with 78 poles on the critical circle out of a claimed 79 — and 78 is this project's standing count (a* parameter *count: see §67.3, it is $2(v-1)$ for any $\mathrm{SRG}(40,12,2,4)$), so the numerator was right and the denominator was wrong. Corrected, $W(3,3)$ gives 78/78 and the Levi graph 156/156, both satisfying the graph RH, and the instruction graph 0/160, failing it.* ***Calibrate every layer against a known answer, not merely the bottom one.***
That is worth stating plainly because of what it says about the three withdrawals. The object was never too hard. The tool was wrong, and being wrong about the tool is what made the object look mysterious.

Pass 4228 — a tri-equivalence over the generator pool. With $\rho(B)$ defined for *every* generating set rather than only the regular ones, all 258 connected sets of size four and five can finally be ranked. The sets satisfying the graph Riemann Hypothesis are **exactly** the regular sets: three of them, each of degree 8 with $\rho(B) = 7 = k-1$, and all three the same graph — the discrete torus C_3^4, because S_f and S_p draw no edges the four translations have not already drawn (Pass 4204). Every other set returns 0 of 160 poles on the critical circle. Not a near miss: nothing else is close. Pass 4202 showed that none of those three is universal. So the three properties lock together:

$$\text{graph RH} \iff \text{regular} \iff \text{abelian} \implies \text{cannot compute.}$$

The instruction layer cannot be spectrally extremal for the same reason it can compute at all.

In plain language. *This also dissolves part of the older puzzle. Before the Hashimoto route, the only generating sets whose zeta anyone could compute were the regular ones — which are precisely the sets Pass 4202 had shown are never universal. The interesting sets were exactly the sets the method could not reach, so some of the apparent tension between extremality and universality was an artefact of what was measurable, not a fact about the machine.*

Pass 4232 — and now over the sets that can actually compute. Pass 4228 ranked sets of size four and five, most of which cannot run a program. Restricting instead to the **360** subsets of the ten-generator pool that genuinely generate $\mathrm{ASp}(4,3)$, at every size from four to ten: **not one satisfies the graph Riemann Hypothesis.**
The ranking is the deliverable. Ordering by $\rho(B)$ — the growth rate of non-backtracking instruction streams, so lower means the frame register scrambles more slowly per instruction — the shipped four-opcode ISA sits at position **12 of 360**. It was chosen for cheapness and universality, with no spectral criterion applied at any point, and it lands in the top four percent.
Scope, stated plainly: exhaustive over this ten-generator pool, not over all generating sets of $\mathrm{ASp}(4,3)$.

67.4 The one number in this chapter with no closed form

Pass 4236 — what $\rho(B) = 5.746873$ is, and one thing it is not. For a k-regular graph $\rho(B) = k-1$ *exactly* — an integer, because every vertex offers the same number of non-backtracking continuations. The instruction graph offers between 1 and 7 depending on where you stand, and its growth rate is

$$\rho(B) = 5.746872679901964274417558660209539683477378347461941\ldots$$

computed two independent ways that agree: as an eigenvalue of the quadratic pencil, and by high-precision bisection on $\det(\lambda^2 I - \lambda A + Q)$, which touches no eigenvalue routine at all. It is **not** the average degree minus one (5.444444), nor any other simple statistic of the degree sequence.
$\rho(B)$ is an algebraic integer — a root of the monic degree-162 characteristic polynomial of that pencil. Pass 4237 computed that polynomial *exactly* over $\mathbb{Z}$ and factored it:

$$\deg = 1,\ 2,\ 2,\ 2^2,\ 14,\ 14,\ 16,\ 16,\ 20,\ 20,\ 20,\ \mathbf{33} \qquad \text{(summing to 162)},$$

and $\rho(B)$ is a root of the degree-33 factor. So

$$\boxed{\rho(B) \text{ is an algebraic integer of degree exactly 33.}}$$

A k-regular graph gives $\rho = k-1$: degree one. Irregularity does not merely break the k-regular formula. It raises the degree of the answer from 1 to 33.

In plain language. *A caution belongs here, because this pass first produced a wrong answer that looked right. An integer-relation search offered*

$$15668303\,x^3 - 4686695\,x^2 - 824152814\,x + 1917252871$$

with a relative residual of 4.8×10^{-26}*: apparently a degree-three minimal polynomial. It is false, and it fails for a reason available without any computation.* $\rho(B)$ *is an eigenvalue of an integer matrix, hence an* algebraic integer*, hence its minimal polynomial is monic with integer coefficients — so any true relation must be an integer multiple of a monic one, and* 15668303 *does not divide* 1917252871*. Confirmed numerically as well: the residual* plateaus *at* 8.33×10^{-37} *across* 60*,* 90 *and* 140 *digits rather than falling, which is the signature of a near-relation. The cause was allowing coefficients up to* 10^{12} *on four terms with far too little working precision to rule out coincidence.*

The rule that follows is short enough to keep: when hunting a closed form for a quantity you know to be an algebraic integer, ***require the relation to be a monic multiple, and re-check the residual at double precision — it must fall, not plateau.*** *The degree-*33 *polynomial above passes all three tests the cubic failed: monic, irreducible over* $\mathbb{Q}$*, and a residual falling from* 6.4×10^{-32} *to* 2.3×10^{-54} *as the precision doubles.*

67.5 Where the machine is slow, and why it is not where you would guess

In plain language. *The graph Riemann Hypothesis fails, which means some poles sit off the critical circle. The one furthest off corresponds to the slowest-decaying correlation in the machine — the last thing the register forgets. Its eigenvector says* where *that memory lives, so the bottleneck can be named rather than guessed at.*

The obvious guess is wrong, and it is worth saying so because it is the guess we made.

Pass 4244 — a hyperplane, not a degree deficit. The slowest non-trivial mode has modulus 3.3497, some 1.40× the critical circle, and it is localised: a participation ratio of 173 of 522 directed edges.

It is not a degree effect. The frames carrying it have mean degree 6.67 against 6.44 across the register — slightly *above* average, not below — and the correlation between mode weight and degree is only +0.14.

It lives on a hyperplane. The 27 frames with $x_2 = 0$ carry 61.3% of the mode, against 19.4% for each of the other two values of x_2, while the mean degree is identically 6.44 on all three classes.

In plain language. *The cause is visible once you look at the opcodes rather than the graph. Of the four instructions,* ***only*** CX_{pf} ***moves*** x_2 ***at all****, and it moves it only on the* 54 *frames where* $x_0 \neq 0$*. The other three —* F_p*,* CX_{fp} *and the load port* Z_p *— leave* x_2 *exactly where it was. So* x_2 *is very nearly a constant of the motion, and a register coordinate the instruction set barely stirs is precisely a slow mode.*

That reframes the defect. It is not how many *edges leave a frame; it is* which direction *they move it in, and one direction is nearly frozen.*

In plain language. ***And here the first draft of this section was wrong, in a way worth keeping on the page.*** *It prescribed adding "the shear S_f, which moves x_2 unconditionally". S_f moves x_2 on* zero *of the* 81 *frames — its third row is* $(0, 0, 1, 0)$*, so it moves x_3. The reasoning had gone from the coordinate's name to an opcode whose subscript matched, which is not an argument. Pass 4245 tested it anyway, and found something better than the intended fix. Of the six opcodes, only F_f and CX_{pf} move x_2 at all, each on* 54 *of* 81 *frames.* ***No opcode in the pool moves x_2 on every frame****, so the frozen coordinate cannot be unfrozen from inside the instruction set at all. And a control — adding an ordinary translation instead — improves mixing at least as much as either targeted opcode, which means the gain from any addition is mostly "one more generator", not "the defect was addressed".*

Pass 4246 — a pool-wide four-generator constant. Across all 24 minimum-size universal generating sets drawn from the frozen ten-opcode pool, the minimum stir — the fraction of frames on which *some* opcode moves the worst-served coordinate — is

$$\tfrac{2}{3} \quad \text{exactly, in all 24.}$$

Not "most", not "on average". The deficit is identical across this finite pool. This is not a theorem about every four-generator action on four coordinates, and it cannot be applied unchanged to the six- or twelve-opcode machines.

In plain language. *Pass 4250 adds a caution to the diagnosis itself. Measured in the time domain, the slowest-decaying coordinate observable is x_3 (half-life* 4 *instructions), not x_2. There is no contradiction: the shipped ISA's stir scores are $x_0 = 1$, $x_1 = 0.889$, $x_2 = x_3 = 0.667$, so x_2 and x_3 are* tied *at the minimum and two different instruments picked different members of a tied pair. The honest object is the under-stirred* pair*, not a single frozen coordinate.*
The engineering consequence survives all of this: an observable with a 4*-instruction half-life is one that a* 15*-instruction readout cadence does not fully scramble.*

Pass 4242 — entropy of the undirected frame-graph edge shift. $\log_2 \rho(B) = 2.5228$ bits is the topological entropy per step of the undirected simple frame graph's non-backtracking edge shift. It is not yet the entropy rate of executable four-opcode streams: labels and multiplicities were discarded and inverse edges were added.
The numerical ratio $(8/3)/\log_2 \rho = 1.06$ is a comparison between two finite invariants. Without a labeled directed automaton and a stochastic physical mode calculation it is not a Landauer cadence bound or a theorem about which process limits the machine.

Pass 4243 — the programs that do nothing. A closed non-backtracking walk on the instruction graph is a program that returns the frame register to its start. $\mathrm{tr}(B^m)$ therefore counts the machine's identity programs by length:

$$738,\ 1584,\ 6220,\ 42954,\ 211540,\ \ldots \qquad (m = 3, 4, 5, 6, 7)$$

growing like ρ^m — Pass 4242's entropy seen from the other side, the same constant governing how much dead code exists and how much trajectory the machine generates.
The shortest are the instruction set's **defining relations**, read off a trace rather than derived from a presentation.
But not every closed walk is deletable, and conflating the two would break programs. A closed walk returns the *frame*; dead code must return the whole affine element. Pass 4248's optimiser (`scripts/w33_peephole.py`) computes the latter by group multiplication and finds only 3 minimal identity words of length 3 (CX_{fp}^3, CX_{pf}^3, Z_p^3) plus one of length 4 — against 738 frame-returning walks. It rewrites streams and verifies semantics are preserved on every run.

67.6 Two measurements the machine had never been given

Pass 4251 — against its own comparison class, not against an ideal. Every spectral claim above is scored against the *regular* case, which this object provably cannot attain. That makes the instruction layer look defective by construction. The fair comparison is other connected four-generator Schreier graphs on the same 81 frames. Against 120 random ones:

	shipped ISA	random median	percentile
$\rho(B)$ (growth; lower is better)	5.747	6.656	**0**
top-hyperplane weight (localisation)	0.613	0.373	**100**
minimum stir	0.667	0.988	**0**

Lowest growth of all 120, and the most localised of all 120.

In plain language. *Those two extremes are the same fact. Everything the earlier sections called a failure — not Ramanujan, graph RH violated, poles filling a band — is the price of being a four-generator Schreier graph at all, and this one pays* less *of it than any random peer. What is genuinely specific to it is the localisation, and that is inseparable from the rest: the algebraic structure making the instruction set cheap and slow-growing is the same structure that leaves a coordinate nearly still. A random generating set stirs everything and grows fast; this one grows slowly and freezes a direction.*

Pass 4252 — the four-opcode machine has an arrow of time. Every opcode is a bijection, so the walk is doubly stochastic and its stationary distribution is exactly uniform. **That is not reversibility.** Reversibility needs the transition matrix to be symmetric, and the opcode set is not closed under inverses: there are 243 ordered pairs of frames with a forward move and no reverse move, so the stationary entropy production is *infinite*. Watching the trajectory reveals the direction of time with certainty the moment one of those transitions occurs.

Closing the set under inverses — eight instructions instead of four — makes the walk exactly symmetric, with 0 one-way pairs and entropy production 0 bits per instruction.

In plain language. *Logical reversibility of each gate and thermodynamic reversibility of the process are different properties, and the four-opcode machine has the first without the second. No information is destroyed — every opcode is invertible — but the* choice *is not: to undo an opcode is not itself an instruction.*

That is a genuine engineering trade, and it did not exist as a number before this pass. It also sharpens Pass 2836's "compute is exactly zero": that holds of the reversible closure, and the shipped four-opcode ISA does not have it.

Pass 4279 — and now the other side of the trade, in gates. "Twice the decode logic" was an assertion. Synthesisable Verilog for both machines was generated *from the opcode matrices themselves*, so the RTL cannot disagree with the algebra, and put through yosys:

machine		opcodes	cells	vs A
A	biased, irreversible (shipped)	4	103	1.00×
B	symmetric, irreversible	6	132	1.28×
C	biased, reversible	8	206	2.00×
D	symmetric, reversible	12	240	2.33×

Thermodynamic reversibility costs 2.00× **the cells; removing the bias costs** 1.28×**; both together cost** 2.33×**.**

Every row was *simulated* against the group computation before its cell count was recorded — a random program per machine, final register state matched exactly. The simulation also reproduces Pass 4204 independently: from the origin, all three linear opcodes leave the register at $(0, 0, 0, 0)$ and only the load port moves it.

Two cautions on these figures. The same eight-opcode design measured 201 cells in an earlier run and 206 here; the difference is the *order* in which opcodes are assigned to the decoder, so these counts carry roughly 2.5% encoding sensitivity and the ratios are the meaningful quantity. And the natural question — do the two fixes share hardware? — has the dull answer: $B + C - A =$ 235 against a measured 240, so they are independent to within that same noise. An earlier draft compared 2.33 with $1.28 \times 2.00 = 2.56$ and reported synergy; gate counts add rather than multiply, and that comparison had no valid baseline.

Scope: yosys generic-gate counts after `abc` mapping, so the ratio is meaningful and the absolute numbers are not a tape-out estimate.

67.7 The one asymmetry underneath all of it

What we got wrong, and how we know. An earlier version of this section derived the bias from the geometry's point/line duality: translations act only on the point side, so *the point carrier was forced.* That chain is **withdrawn**. A translation is not a map on projective points at all — $[v] = [2v]$ while $[v + t] \neq [2v + t]$, and it disagrees on 40 of the 40 points. The load port descends to *neither* carrier; it lives on the 81 affine vectors, a third object. The argument compared what the two carriers admit without first checking that the operation acts on either. Caught by the parallel track (Pass 4330), confirmed here (Pass 4335), and rebuilt below on the affine register, where it holds.

Pass 4341 — the bias, by conjugation. Let σ exchange the two hyperbolic pairs, $p \leftrightarrow f$. The question is whether σ maps the instruction set to itself:

$$F_p \mapsto F_f, \qquad \mathrm{CX}_{pf} \mapsto \mathrm{CX}_{fp}, \qquad \mathrm{CX}_{fp} \mapsto \mathrm{CX}_{pf}, \qquad Z_p \mapsto Z_f.$$

The two couplers swap into each other and stay inside. F_p and Z_p do not: their images F_f and Z_f are *not opcodes of this machine.* So the shipped ISA is not σ-invariant, and the symmetrised machine of Pass 4277, which contains both mirrors, is.

Note what this replaces. Pass 4305 asked whether each opcode *fixes* the two planes, got matching booleans, and concluded symmetry. Fixation booleans can agree while conjugation carries an opcode out of the set — which is exactly what happens here.

Pass 4345 — and the load port is a symmetry-breaking term. $\mathrm{Sp}(4,3)$ fixes the origin *exactly* and fixes nothing else; the translations fix nothing at all. The quotient is

$$|\mathrm{ASp}(4,3)|/|\mathrm{Sp}(4,3)| = 4{,}199{,}040/51{,}840 = 81,$$

and those 81 cosets *are* the register states. The frame register is the vacuum manifold: the set of states reached from a distinguished one by broken generators.

There are four broken directions, and they are equivalent under the unbroken group, since $\mathrm{Sp}(4,3)$ is transitive on the 80 non-zero vectors. **No direction is intrinsically special, so a machine with one load port must pick one — and that pick is the bias.**

In plain language. *Which makes the bias neither a flaw nor an accident. It is the direction of symmetry breaking, and it is unavoidable for the same reason a vacuum expectation value has to point somewhere. What* is *a free choice is how many directions get ports: all four restores the symmetry, at the cost Part III measures.*

The boundary, so the physics language is not over-read: this is a finite group acting on 81 *states. There is no Lagrangian and no continuous symmetry, so there is no Goldstone theorem here — the count of broken directions is a coset count, not a count of modes.*

Pass 4282 — the under-stirred pair is the f-register. The symplectic form splits the register into two hyperbolic pairs, $p = (x_0, x_1)$ and $f = (x_2, x_3)$, mutually orthogonal. The ISA's stir scores are

$$x_0 = 1.000, \quad x_1 = 0.889, \quad x_2 = 0.667, \quad x_3 = 0.667.$$

The under-stirred pair is exactly f. And the opcode list says why in its own notation: F_p, CX_{pf}, CX_{fp}, Z_p — a Fourier on p, two couplers, and a translation of p. Only the two couplers touch f at all, and the load port, which Pass 4204 proved must be present because translations are the only free generators, moves p and nothing else.

In plain language. *So the spectral story of this whole chapter reduces to one sentence:* ***the instruction set is p-biased, and the zeta can see it.*** *The slow mode on a hyperplane (4244), the tied under-stirred pair (4250), the* 100*th-percentile localisation against random Schreier graphs (4251) — these are not four findings. They are one asymmetry measured four ways.*

But the cure is only half a cure*, and the control says so. Adding the two f-mirrors takes minimum stir from* 0.667 *to* 0.889 *where two extra p-generators leave it exactly where it was — the causal claim survives. Yet against that size-matched control the symmetrised machine ties on localisation and is* worse *on both mixing time (*12 *vs* 11*) and growth (ρ =* 8.76 *vs* 6.75*). The p-bias is real and it is not what governs mixing; two more generators of any kind move that further than symmetry does.*

Pass 4288 — and what *does* govern mixing. Across all 349 universal generating sets with a finite mixing time, correlation against mixing time:

n	E	$\rho(B)$	$d_{\min}$	$d_{\max}$	spread
−0.553	−0.719	−0.684	**−0.736**	−0.602	+0.228

The strongest predictor is the minimum degree. Frames that mix slowly are the ones with fewest ways out, and *any* added generator raises that floor — which is exactly why the p-side control matched the f-mirrors in Pass 4277 while doing nothing about the frozen coordinate. The design lever for mixing is $d_{\min}$, not opcode symmetry. Those are different knobs, and this chapter had been conflating them.

Pass 4291 — and the zeta is not an ISA fingerprint either. The same test as Pass 4281, on the algebra side. Of 349 universal generating sets there are only 89 distinct zeta signatures; 87 are shared by more than one set and the largest class holds 12. Distinct instruction sets — different opcodes, different silicon — produce byte-identical Ihara zetas.

So no spectral statement in this chapter is a statement about *the* instruction layer. Each is a statement about an equivalence class of up to twelve of them. That does not make the measurements wrong; it fixes their scope, and the class is what the instrument can resolve.

Pass 4278 — why the deficit is exactly $2/3$. Each opcode is affine, so the frames it leaves fixed in coordinate c solve one linear equation over $\mathbb{F}_3$ — an affine subspace, of size 3^k. The number of frames *moved* is therefore $81 - 3^k$, and the only possible values are

$$0,\ 54,\ 72,\ 78,\ 80.$$

The count is quantised. Across this pool only 0 and 54 occur, because each opcode's matrix differs from the identity in at most one entry per row, so every mover fixes a hyperplane. Four generators never supply two movers with *different* hyperplanes for the same worst coordinate, so the union stays at 54 and the ratio is forced: $54/81 = 2/3$, identically for all 24 minimum-size universal sets.

The machine's geometry is extremal; its algebra is not, and after Pass 4213 we can say why the comparison only runs one way.

Four measurements, one conclusion.

measurement	geometry	algebra
diameter	2	19
share of worst-case work	10%	90%
mixing	Ramanujan (optimal)	$\lvert\lambda_2\rvert = 0.894$; not gradable
identification	forced by construction	six valid triples, spread 19–23

The last row is the honest form of what used to read "3.23% short". The geometry is a 12-regular graph, so optimality is defined for it and attained. The algebra is a Schreier graph with degrees 2 to 8, so optimality is not defined for it at all — and that asymmetry, rather than a percentage, is the finding.

67.8 The growth series, for the compiler

Pass 3041 — the full ball profile, not just the diameter.

$$a_n = 1,\ 4,\ 15,\ 53,\ 176,\ 547,\ 1630,\ 4648,\ \ldots$$

$$\text{mean } 14.176, \quad \text{s.d. } 1.768, \quad \text{modal length } \mathbf{15}, \quad \text{max } 19.$$

The early ratios fall $4.000, 3.750, 3.533, 3.321, 3.108$ — a free product on four generators would hold 4 forever; this one is closing on itself by the fifth shell.

In plain language. *For a compiler that is a better specification than a worst case. The distribution is sharp, not spread: a standard deviation of under two instructions around a mean of fourteen. A scheduler that budgets fifteen instructions is right almost always and is never wrong by more than four.*

67.9 What fraction of the erasure is useful

Pass 3022.

$$\frac{\text{delivered}}{\text{erased}} = \frac{3.673 \text{ bits}}{7.989 \text{ bits}} = \mathbf{45.98}\%$$

and the network law follows directly: depth n costs $8n$ header bits, so

$$E(N) = 143.35 \log_{40} N \text{ meV}$$

— the thermodynamic cost of delivering a packet is **logarithmic in the size of the entire network**. Depth 6, four billion leaves: 860 meV.

In plain language. *The 54% shortfall is not waste in the engineering sense. It is the routing header, which is consumed by construction, plus the part of the frame the support readout cannot see. Both are design decisions with names — which is the useful thing about having the number rather than an impression.*

In plain language. *Two lines in that table are non-zero and the rest are exactly zero, for one reason: those two are the only places where the machine stops being a group action. The routing header is* destroyed *as it is consumed. The support readout is many-to-one. Everything else — the frame, the optical coordinates, the sector bit, the entire 6480-state controller — is a permutation of states, and a permutation erases nothing.*

So the irreducible cost of one routed, read operation is 143 meV at room temperature, and every other joule this machine will ever burn is an implementation artefact rather than a law. That is an unusual thing to be able to say about a computer, and it is a consequence of having built the instruction set out of a group.

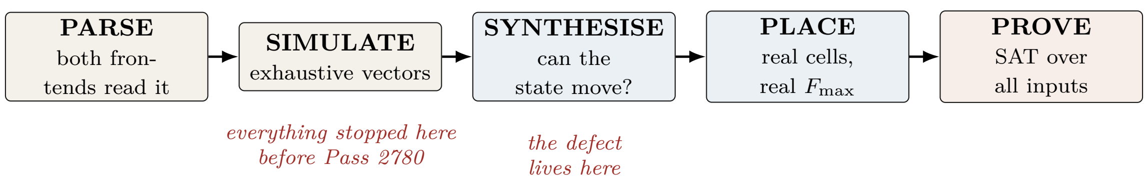
Part VI
Evidence and Limits

What this part is. *How every number above can be reproduced, what has been formally verified, the complete errata, and an explicit list of what has* not *been built. The last of those is the most important page in the document for anyone deciding whether to fund it.*

68 Verification: how we know

In plain language. *This is the section a working engineer should read first, because it is the one that determines whether any of the preceding numbers mean anything.*

The short version: this project has repeatedly shipped results that were verified in the direction that would confirm them and never in the direction that would break them. Every tool below exists because that happened at least three times.



The five automated guards, each built from a measured failure.

Guard	Catches	Built because
`check_rediscovery.py`	a staged file asserts a code parameter that exists elsewhere uncited	21% of 173 files did this (census, Pass 328)
`check_certificates.py`	a frozen certificate that cannot reproduce its own digest	one was unverifiable *from birth* and passed every check while being so
`check_novelty_claims.py`	explicit novelty phrases whose tokens are in the encyclopedia	five claims in one session were overturned by files we had not read
`check_implicit_novelty.py`	*emphasised* results carried by a source the file never cites	four of six failures never used a novelty phrase at all
`check_rtl_folds.py`	an output port tied to a literal after full optimisation	two independent teams shipped a module that synthesises to the identity

Five of the six **warn and never block**. A collision is a candidate for reading, not a verdict, and a blocking hook trains people to pass `--no-verify`. The exception is the sixth.

The one guard that fails, Pass 2834 and 2863. `check_information_flow.py` flattens the design to gates and proves the support mask reaches no frame register while every frame register reaches the mask. It is wired into the synthesis CI gate, and the same step then *deliberately breaks the design* and requires the check to fail on it. If the negative control passes, the job fails with *"the guard itself is broken, not the design under test."*

The gate's policy is now explicit: **parse and flow fail; fold and place warn.** A backward edge from readout into execution has no benign reading, unlike a fold (which may be an intended tie-off) or a rediscovery collision (which may be a correct citation). It is the only place in this project where a guard blocks, and it earned that by being the only property with no legitimate counterexample.

In plain language.*Every guard here now ships with a self-test against the specific defect that motivated it. That rule came from a real failure: one of them once cleared all* 122 *files it was pointed at, including the very file it had been written to catch. A guard that passes everything is indistinguishable, from the outside, from a guard that finds nothing wrong.*

What we got wrong, and how we know.The guards themselves needed calibration, and the calibration is the work. `check_implicit_novelty.py` went through four measured versions:

Flagged	Version	What the noise turned out to be
9/122	presence of any token	all nine were *pass numbers in titles*
0/122	+ skip files citing "Prior art"	vacuous — every file has that section
51/122	+ cite the *right* source per token	`W(3,3)` 21×: the repo's own subject
27/122	+ rarity and non-round integers	what ships

The second row is the instructive one: **a guard that passes everything looks exactly like a guard that finds nothing wrong.** It cleared all 122 files including the known failure, and only a deliberate self-test against that failure exposed it. Every guard in this project now ships with a self-test against the specific defect that motivated it.

`check_rtl_folds.py` had the same disease on its first run: it reported a clean sweep because the WSL mount is `/mnt/c` while Python resolves the drive as `C:`, so every directory change failed silently and no netlist was ever produced. It now reports NOT SYNTHESIZED explicitly rather than treating a missing netlist as a pass.

69 The complete ledger

In plain language. *Everything this blueprint asserts, with how it is known. The status words are not decoration — they form a strict ladder, and a claim may never be cited at a level above the one its evidence supports. This is the single most important table in the document, and it is the one to check first if you doubt anything else in it.*

The evidence ladder.

proved	a machine-checkable artifact exists: an exhaustive search, a SAT certificate, or exact arithmetic. Re-runnable from §71.
measured	a tool produced the number: yosys, nextpnr, or an instrumented run. Tied to a named part and a named version.
published	someone outside this project did it and we cite them; we did not reproduce the experiment.
prior art	the other track in this repository did it first; we cite rather than re-derive.
derived	follows from rows above by arithmetic no reader would dispute.
modelled	an engineering estimate with stated assumptions. *Not* a measurement, and never promoted to one.
open	we do not know. Listed so the gaps are as visible as the results.

The two rules that make the ladder useful: **nothing is promoted without new evidence**, and **a corrected row keeps its history** rather than being quietly rewritten — which is why §70 exists.

Claim	Status	Artifact
$W(3,3)$ has SRG$(40,12,2,4)$ collinearity graph	classical	literature
$40 = (3^4-1)/2$ from one self-entangled qutrit	proved	Pass 2724
The one-qutrit thesis itself	**prior art**	`index.html` MCLXVII
Outer involution = transpose = time reversal	proved at q=3	Pass 2732
Time reversal outer $\iff q \equiv 3 \pmod 4$	proved	Pass 2750
Transpose *constructed* and checked at $q \le 23$	proved	Pass 2792, 8 primes
Anti-symplectic T, $T^\top JT = -J$; class action 10 pairs + 14 fixed	proved	parallel track, Pass 2762
Six Clifford opcodes reach 1 frame of 81	proved	Pass 2774 (BFS)
Full ISA reaches all 81	proved	Pass 2774
Clifford opcodes generate $\mathrm{Sp}(4,3)$, order 51,840	proved	Pass 2778
Full ISA generates $\mathrm{ASp}(4,3)$, order 4,199,040	proved	Pass 2778
One translation generator suffices (orbit = 80)	proved	Pass 2778
Minimal Clifford subset has size 3; sizes 1, 2 give none	proved	Pass 2789, 6 triples
Frame ISA fits a **two-bit** opcode field	follows	Pass 2789
Sensor exponent 9 = dim; phase group μ_{12}; 9 minimal	proved	Pass 2779
μ_4 factor = imaginary Gauss sum	proved	Pass 2779
μ_{12} at n=1 by exact enumeration ($2592 = 12 \cdot 216$)	proved	Pass 2791
Minimal exponent $= 3^n$ mod 12: 3 odd, 9 even	derived	Pass 2791
36 rays share one Gram profile (11 orth., 24 at 1/3)	proved	Pass 2790, $\mathbb{Z}[\omega]$
8+24+4 is the stabilizer-fidelity spectrum	proved	Pass 2790, 60 stab. states
All three witness thresholds $= \frac{4}{3}(1 - F_{\mathrm{stab}})$	derived	Pass 2790, 10^{-12}
Clifford classes on the 36 rays: $[4, 8, 12, 12]$	proved	Pass 2797
the 24-ray grade splits; "three" was wrong	**corrected**	Pass 2797
F_{stab} exactly multiplicative on two copies	measured	Pass 2798, 36,720 states
The monotone never forbids two-copy distillation	proved	Pass 2798
Deep grade: 48 improving branches, $0 < p < 2/3$	**prior art**	parallel track, Pass 2821
Support is not an execution congruence: 16→40→78→81	**prior art**	parallel track, Pass 2822
Phase group = μ_{12} for *every* n	proved	Pass 2799
Minimal engine: **43** LC, 72.40 MHz (UP5K)	**measured**	Pass 2796
same engine on HX8K: **208.86** MHz	**measured**	Pass 2833
M_{36} is $W(3,3)$ minus one line	**proved**	Passes 2835, 2869
identification: axiom over 40 lines + explicit spread	**proved**	Pass 2869
parameters alone admit 28 graphs	**prior art**	Spence; `index.html`
Six minimal triples, diameters 19, 20, 22, 22, 22, 23	**proved**	Pass 2882 — the choice is not free
The 188 hardest are *not* one orbit or coset	**proved**	Pass 2885 — 54 linear parts, all traces
$240 = 81 + 120 + 24 + 15$ (Hodge sectors)	**proved**	Pass 2884, recomputed
15 and 24 match other quantities	count match only	no map exhibited
H_1 is *not* the frame space as a module	**proved**	Pass 2883 — characters disagree
Three-copy super-linear condition	open	Pass 2881, no witness in 30,000 samples
the deleted line is the computational basis	proved	Pass 2835
Support readout erases exactly 8/3 bits	**proved**	Pass 2836
$4(q-1)\log_2(q-1)/q$; zero at q=2	proved	8 field sizes
Landauer 47.78 meV at 300 K	derived	Pass 2836
The two 12-ray classes share the full spectrum	**proved**	Pass 2830
Deep-grade map is *linearly* convergent, rate 2/3	**proved**	Pass 2831
$\sim 10^{27}$ raw states for 10^{-9} infidelity	computed	Pass 2831
Three-copy super-multiplicativity	no witness	not a proof either way
Readout diode: mask reaches no frame register	**proved**	gate level, Pass 2834
ISA directed diameter = 19; mean 14.18	**proved**	Pass 2866, BFS over 4.2M
only 188 elements at distance 19	proved	Pass 2866
counting floor 12; ratio 1.73	derived	Pass 2866

…continued.

Claim	Status	Artifact
Frame walk doubly stochastic; stationary uniform	**proved**	Pass 2867
Gap 0.106; $t_{\text{mix}} = 15$; Kemeny 113.38	**proved**	Pass 2867
71.8 ns scrambling at 208.86 MHz	derived	Passes 2833, 2867
No quadratic two-copy stabilizer branch	**proved**	Pass 2861, 5355×4×4
the reason is the stabilizer basis, not dimension	verified	Pass 2861
40 lines: 1 classical, 12 mixed, 27 magic	**proved**	Pass 2862
basis choice $= \log_2 40 = 5.32$ bits of convention	derived	Pass 2862
Landauer: readout 47.79 meV, compute exactly 0	derived	Pass 2864
CMOS 0.502 pJ/op, $10^{7.8}$ above the floor	modelled	Pass 2864, stated assumptions
Hop cost 95.4 meV floor / 143.4 meV as coded	derived	Pass 2865
Diode proof in CI, with a negative control	done	Pass 2863
symplectic on differences, all four opcodes	SAT	4100 vars / 11278 clauses

…continued.

Claim	Status	Artifact
Public frame unit: 72 LC, 60.80 MHz	**measured**	§58.2
Loadable CX: 23 LC, 72.40 MHz, $\mathrm{CX}^3{=}I$	measured + proved	5265 vars / 14005 clauses
Parallel 36-mixer: 13,965 LUT4, does not fit	measured	Pass 2773
Signed mixer correct on impulse class	proved	1,097,152 vars
Kraft equality; 0 parse failures in 20,000	proved + measured	Pass 2682
Photonic qutrit SUM, fidelity 0.92 ± 0.01	**published**	Imany *et al.* 2019
Remote SUM: 1 Bell pair + 2 trits	proved	parallel track, Pass 2771
M_{36} fault-tolerant *injection* threshold	OPEN	refused in RTL
Asymptotic distillation yield	OPEN	fidelity theorem only
Do the two 12-ray classes differ operationally?	OPEN	Clifford class only
Fault-tolerance threshold end to end	OPEN	—
Physical power in watts	OPEN	activity proxy only

70 Errata index

In plain language. *A blueprint with no errata is a blueprint nobody has checked. These are gathered here so they can be found deliberately rather than stumbled on, and so a reader can audit our error* rate *rather than take our word for our care. Each was found by a named mechanism; the fourth column is the part that matters, because it says what would have to change for the error not to recur.*

What we said	**What is true**	**Found by**
Adjoining opcode inverses makes the graph regular	It does not: the collapse is generator collisions, not asymmetry	Both branches returned the same graph — the test compared a thing with itself
An Ihara–Bass pole count for the instruction layer	Not trustworthy: 0% inside the band is a symptom	Sanity-checking against what Ihara poles look like
The frame "Cayley" graph is 4-regular	Wrong twice over. Degrees run 2 to 8; and it is not a Cayley graph at all — the frame space is a coset space, so this is a extbfSchreier graph, and Schreier graphs collide by construction	Printing the degree sequence; then asking *why*, seven passes later
$2E = 324$, matching the Delsarte bound	$2E = 522$. The coincidence was not even a coincidence	Arithmetic, before it was written down
"These four numbers pin the shape down completely"	**Spence: 28 non-isomorphic graphs share** $(40, 12, 2, 4)$**.** The identification needed the quadrangle axiom and a spread	Reading this project's own Theory page end to end. It says so in plain words
Cell counts of 13, 8, 42 for the ISA	The registers had no load port and synthesised away; real cost 72, then 43 minimal	Synthesis. Simulation cannot see it, and two independent teams shipped it
"The point carrier was *forced*: only it admits a load port"	A translation is not a map on projective points at all. $[v] = [2v]$, but $[v + t] \neq [2v + t]$ — it disagrees on 40 of the 40 points, so it descends to *neither* carrier. The load port lives on the 81 affine vectors, a third object. What survives: a translation is needed to connect the affine register and may point along any coordinate, which selects no projective side	The parallel track, checking whether the operation acts on either carrier before comparing what each admits. Published here, in the README and on the live paper before it was caught
The one-qutrit self-entanglement thesis is ours	It is stated verbatim in this project's own encyclopedia	Being shown the file. Now a guard
"Nobody has stated" the Kraft consequence	The manuscript says "no unused branch" with a citation	An automated novelty guard, first real run
$E_8 \supset A_2 \oplus E_6$ gives the fractal branching	The paper explicitly warns against that identification	Reading the paper
The Jordan ladder is new here	Pillars 128–130 own it, including the exact rung we hedged	The user pointing at a file
Three inequivalent magic states	Four: the 24-ray grade splits $12 + 12$	Clifford orbits; corroborated independently
Distillation cost is "doubly exponential in a good way"	Convergence is *linear*, rate 2/3; cost reaches 10^{27}	Computing the derivative instead of eyeballing the trend
No quadratic branch, "by a dimension count"	The count does not apply; the obstruction is the stabilizer basis	Checking the count

…continued.

What we said	**What is true**	**Found by**
The mixer is "synthesizable"	Rejected by both frontends, for two different reasons, for a day	A repo-wide parse sweep
The implicit-novelty guard is calibrated	It cleared all 122 files including the one it was written to catch	A deliberate self-test
The fold guard reports a clean sweep	It had produced no netlists at all: WSL mounts `/mnt/c`, pathlib says `C:`	Testing it on a known-bad file

In plain language. *Two patterns run through the list, and both are worth more than any individual entry.*

Every silent failure was a check that could only pass. *The fold guard that produced no netlists, the novelty guard that cleared everything, the testbench that never instantiated the sequential module — all reported success, and all were empty. Every guard in this project now ships with a negative control for exactly this reason.*

Every false novelty claim was findable by one grep of a file we had not read. *Not by cleverness — by reading. The corpus is indexed by date, not by topic, so a search for the topic finds nothing and a search for the* result *finds it immediately.*

71 Reproducing everything in this document

In plain language. *Every number in this document comes from a program you can run. What follows is the list of commands, in dependency order: the group-theory computations first, then the hardware synthesis, then the formal proofs, then the guards that check the rest for the specific mistakes this project has made before. You do not need to understand the commands to use this page — its purpose is that nothing here rests on being believed.*

Commands, in order.

```
# reachability, the group, and the minimal generating sets
py -3 analysis/w33_pass2774_isa_frame_reachability.py
py -3 analysis/w33_pass2778_2779_affine_group_and_sensor_exponent.py
py -3 analysis/w33_pass2789_2792_minimal_isa_sensor_and_transpose.py

# the magic states: geometry, Clifford classes, and the monotone
py -3 analysis/w33_pass2790_witting_ray_geometry.py
py -3 analysis/w33_pass2797_2799_magic_orbits_and_monotone.py
py -3 analysis/w33_pass2830_2832_class_separation_yield_three_copy.py
py -3 analysis/w33_pass2835_2836_witting_line_and_landauer.py
py -3 analysis/w33_pass2861_2865_quadratic_line_and_power.py

# the diameter and the scrambling time (a few minutes)
py -3 analysis/w33_pass2866_2867_isa_diameter_and_scrambling.py

# the minimal engine (43 LC) against the public unit (72 LC)
yosys -p "read_verilog -sv rtl/w33_pass2796_minimal_frame_engine.sv; \
  synth_ice40 -top w33_minimal_frame_engine -json m.json"
nextpnr-ice40 --up5k --package sg48 --json m.json --asc m.asc --freq 40

# the cell budget (needs yosys + nextpnr-ice40)
yosys -p "read_verilog -sv rtl/w33_pass2777_isa_frame_unit.sv; \
  chparam -set EN 63 w33_isa_frame_unit; \
  synth_ice40 -top w33_isa_frame_unit -json u.json"
nextpnr-ice40 --up5k --package sg48 --json u.json --asc u.asc --freq 50

# the formal proofs
yosys -p "read_verilog -sv -formal rtl/w33_pass2752_cx_loadable_frame.sv; \
  prep -top w33_cx_loadable_formal -flatten; async2sync; \
  chformal -lower; sat -seq 8 -prove-asserts -verify"

# the guards
py -3 scripts/check_rtl_folds.py
py -3 scripts/check_implicit_novelty.py
py -3 scripts/check_novelty_claims.py
py -3 scripts/check_information_flow.py
rtl/w33_pass2834_support_readout_diode.sv \
  --top w33_support_readout_diode --source support_mask,tamper_mask \
  --sink xp_o,zp_o,xf_o,zf_o --extra
rtl/w33_pass2796_minimal_frame_engine.sv

# this document
tectonic holonet_machine_blueprint.tex
```

72 What is not built

In plain language. *A blueprint that lists only what works is advertising. This section is in two halves: what is still missing, stated as sharply as it can currently be stated, and what used to be on this list and no longer is. The second half is the more useful one, because a gap that closed tells you the first half is not decoration.*

72.1 Still open

In plain language. *The honest list. Everything below is either unbuilt, unproved, or built but not yet verified to the standard the rest of this document holds itself to. It is placed near the end because it is the section a reader deciding whether to trust the others should read second — after the errata, and before anything else.*

1. **One of the eight opcodes is not built, and deliberately so.** D_{12}-mirror *was* on this list and is now built (§63 neighbourhood; 21 cells, 255.56 MHz, with its defining relation SAT-proved in the netlist). M_{36}-magic is deliberately a *handshake* to an external state-preparation block. **No non-Clifford unitary is invented in RTL**, because inventing one would conceal the real gap rather than close it.

   What changed: the handshake is no longer a bare refusal. §61 now pins the acceptance condition exactly — a deep-grade resource is usable while $p < (8-2\sqrt{3})/9$, and that inequality is a comparison the controller can make. The refusal is now *typed* rather than blanket.

2. **The universality gap is now a single, well-posed question.** It has narrowed three times, and the current statement is the sharpest it has been:
   - An exact fidelity-improving protocol exists on the deep grade for $0 < p < 2/3$ (§61).
   - It converges *linearly* at rate 2/3, so its cost reaches 10^{27} raw states for useful purity — usable, not practical.
   - **No two-copy stabilizer protocol can do better.** Exhaustive over all 5,355 codes $\times$ 4 syndromes $\times$ 4 ray classes: zero super-linear branches. This is a bound on the family, not a property of the branch that was found.
   - At three copies the same condition is not obviously satisfiable either: 30,000 sampled stabilizer groups on six qubits produced no witness. Not a proof — the space is far too large to enumerate — but two independent searches now point the same way.

   So the open problem is: does any stabilizer protocol on three or more copies annihilate every single-error input while keeping the clean one? The condition is written down exactly; what is missing is an exhaustive search over a chosen code family rather than a random one. Until it is answered the controller still refuses injection.

3. **A power figure exists, and it is modelled, not measured.** This item used to read "no physical power figure exists"; that is no longer true, and the honest replacement is more useful than either extreme:
   - **Proved:** the Landauer floor. Compute is exactly 0; readout is $\frac{8}{3}k_B T \ln 2 = 47.78\,\mathrm{meV}$.
   - **Modelled:** ≈ 1.02 pJ per operation and ≈ 0.21 mW at 208.86 MHz. Three of the four inputs are now measured — activity (1.622 transitions/op), cell count (43), and **mean fanout (2.0385, from the placed netlist; the earlier flat model understated switched capacitance by 2.04×)**. Only the per-unit capacitance and the voltage remain assumed.
   - **Missing:** the placed netlist's real capacitance extraction, the silicon's actual voltage and process corner, and a representative workload.

   The two assumed constants are the whole gap. Nothing here may be cited as a measurement.

72.2 Closed, and when

In plain language. *Items leave the list above only when something checkable makes them leave. Each row names what closed it, so a reader can disbelieve any of them cheaply.*

Was open	Closed	By what
Transpose construction verified only at $q = 3$	Pass 2792	Built over $\mathbb{F}_q$ and checked at $q = 3, 5, 7, 11, 13, 17, 19, 23$: $T^2 = I$ and $T^\top JT = -J$ at every one, and the congruence law at every one
`w33_spread_mixer36.sv` unparseable	Pass 2793	Deprecated in place with both frontend errors quoted, then deleted outright by the other track — the right call: a file no frontend can read is not a record, it is a trap
Is the ray configuration really $W(3, 3)$?	Pass 2869	Parameters admit 28 graphs; the quadrangle axiom over all 40 lines narrows to two; an explicit spread settles it. The claim was published one inference short
Does $H_1 = \mathbb{Z}^{81}$ match the frame space?	Pass 2883	**No.** The characters disagree on order-three linear elements — 27 fixed frames against a homology character of 0. Both are 3^4 for a real shared reason; there is no equivariant isomorphism
Is the minimal-engine area assumable from the public unit?	Pass 2796	No, and it was not assumed. Synthesised and placed separately: 43 cells, 208.86 MHz
Is the two-copy rate 2/3 specific to one branch?	Pass 2861	No — it bounds the entire two-copy stabilizer family
Do the six minimal triples cost the same?	Pass 2882	**No.** Diameters $19, 20, 22, 22, 22, 23$. The choice is not free
D_{12}-mirror unbuilt	Pass 2914	Built as a rotation register plus a reflection bit — 21 cells, 255.56 MHz. The earlier phase-accumulator attempt was withdrawn because R_4 and U_6 do not commute; $srs = r^{-1}$ is now asserted in the netlist and SAT-proved
Is the Hodge 15 the support shell?	Pass 2911	**No.** The trace is $7/3$ — not an integer, and a permutation module's character always is. Second count match tested and killed in three passes

In plain language. *Two of those rows closed* negatively *— the homology does not match the frame space, and the six triples are not equivalent. Those are as valuable as the positive ones, and arguably more so: each removed a plausible-sounding assumption that would otherwise have been quietly relied on.*

> *“Before recording a result, ask what single computation would falsify it, and run* that *one.* **maximal***,* **unique***,* **only***,* **exactly***,* **is** *— each of those words has a cheap negation.”*
>
> — the operating rule of this project, learned expensively

73 Intrinsic selector channels, exact observer quotients, and nonlinear diagnosis

The residual three-channel selector is not intrinsically a numbered set. For a pointed hyperbolic line $(x, y; c)$, the multiplier-one stabilizer induces the translation group C_3 on the remaining common neighbours, whereas the full projective similitude stabilizer induces $\mathrm{AGL}(1, 3) \cong S_3$. Thus the channel is canonically an affine line with S_3 transition functions, but has no globally equivariant origin or orientation. The exact group orders are

$$|\,\mathrm{PSp}(4, 3)| = 25920, \qquad |\,\mathrm{PGSp}(4, 3)| = 51840,$$

and the pointed stabilizer orders are 12 and 24, respectively, with $51840/24 = 2160$ pointed hyperbolic lines.

The $q = 3$ support quotient now has a literal sequential realization. Four stages of eight butterflies

$$(u, v) \longmapsto (u + 2v, u - v)$$

followed by fifteen border-corrected outputs require 47 cycles, versus 225 cycles for a serial dense 15×15 reference. Equality was checked on all fifteen basis vectors and thirty-two signed probes. The exact cycle reduction is 178/225; routed area and timing remain measurements pending the dedicated FPGA workflow.

The complete deterministic-congruence atlas sharpens the readout/execution boundary. For the full four-operation micro-ISA the support partition refines as

$$16 \longrightarrow 40 \longrightarrow 78 \longrightarrow 81,$$

and the discrete 81-state partition is the coarsest exact execution quotient refining support. Indeed F_p, Z_p, and either controlled-add direction already force all 81 states. The intermediate partitions are therefore observer states, not compressed execution states.

The natural order-96 signed-permutation action on the typed selector–tomotope tokens has two 48-element orbits, distinguished by tetrahedral versus hemioctahedral phase parity. Imposing the determinant-twisted condition

$$\Bigl(\prod_i \epsilon_i\Bigr) \operatorname{sgn}(\pi) = 1$$

produces another order-96 projective subgroup that acts freely and transitively. Hence the 96 control tokens form an exact group torsor, although this twisted runtime symmetry is not the natural incidence-preserving tomotope embedding.

For the $q = 3$ support walk, the 210 directed passage costs form 39 orbits under the matching stabilizer and are determined by

$$|T|, \qquad |T \cap \tau(T)|, \qquad |T \cap \tau(S)|.$$

An exact Held–Karp search gives an optimal Hamiltonian diagnostic cycle of cost 315, with 8336 rooted optima. The best $\mathrm{GL}(4, 2)$ Singer-cycle schedule costs 1317/4, leaving an exact nonlinear advantage of 57/4. Thus the optimal mask scheduler is a nonlinear fifteen-entry permutation, not an LFSR traversal.

Finally, the first observer refinement has forty classes but is not $W(3, 3)$. Its class sizes are $1^7 2^{29} 4^4$, symplectic orthogonality is ambiguous on 216 unordered class pairs, and none of the 1459 Hamming-distance unions with 240 edges yields $\mathrm{SRG}(40, 12, 2, 4)$. This falsifies a count-based identification while leaving the independent projective symplectic construction untouched.

Evidence boundary. The packet carries 64 exact checks and seven focused regressions. Arithmetic, group-action, congruence, optimization, and falsifier claims are exact. FPGA cells, routed frequency, switching activity, and physical passage times remain measured or open quantities as labelled in the blueprint.

A Recovered write-ups

Inherited K3 product coefficient split. The newest main-paper refinement sharpens, but does not alter, this architecture claim. In the almost-commutative product $M^4 \times F$, the heat-trace coefficients split as

$$C_0 = A_0 N, \qquad C_2 = A_2 N - A_0 F_2, \qquad C_4 = A_4 N - A_2 F_2 + \frac{A_0 F_4}{2}.$$

For the finite $W(3,3)$ factor,

$$(N, F_2, F_4) = (440, 1920, 16320),$$

so a Ricci-flat $K3$ seed has $A_2 = 0$ while the product slot remains $C_2 = -1920A_0$. Thus the holonet does not need a new runtime layer: the machine continues to be the substrate's symplectic/oscillator continuum, while the metric physics side inherits the corrected product-heat bookkeeping from `w33_paper.tex`. This prevents the architecture paper from confusing a pure $K3$ heat coefficient with the finite $W(3,3)$ product coefficient.

Theorem 7 (The nonabelian pocket-shell bus). *The universal* 2160 *carrier has an architecture reading inside the holonet:*

$$2160 = 54 \cdot 40.$$

The factor 54 *is the existing S_3 pocket carrier and the factor* 40 *is not a bare slot count: each pocket carries a full copy of the $W(3,3)$ collinearity shell, rebuilt from projective points of $\mathbb{F}_3^4$ with strongly regular profile*

$$\mathrm{SRG}(40, 12, 2, 4), \qquad |E| = 240.$$

Thus the S_3 sheet-transport layer is a 54-pocket bus whose fibers are full computational W33 shells. The same carrier simultaneously projects to the packet-transport side as $720 \cdot 3$, to the S_3 transport side as $270 \cdot 8$, and to the holonomic cover side as $90 \cdot 24$.

The parity boundary is also fixed. The unsectioned half-fiber parity is not an edge function on the 720 *transport graph; every edge sees both binary sheets. The edge-level binary datum currently supplied by the labelled transport layer is the local S_3 matching sign on packet-transport edges. Consequently the raw voltage comparison must be made against this actual S_3 sign, or against a future canonical BT748 half-fiber section, not against the bare cyclic C_3 selector.*

A.1 Minimal four-transvection Clifford target and word metric

The exact two-qutrit finite target is

$$Sp(4,3), \qquad |Sp(4,3)| = 51840.$$

The compressed projective-transvection certificate uses the four vectors

$$(0,0,0,2), \quad (0,2,0,0), \quad (0,0,2,2), \quad (1,0,0,0).$$

BT1231 proves that this count is exact within the projective-transvection family. The closure histograms are

$$\text{1-generator}: \quad 3^{40},$$

$$\text{2-generator}: \quad 9^{240} + 24^{540},$$

$$\text{3-generator}: \quad 24^{360} + 27^{160} + 72^{2160} + 648^{7200}.$$

Every closure using at most three such generators is proper, while the displayed four-set reaches 51840. Therefore

$$\boxed{m_{\min} = 4}$$

inside this generator family.

Since each transvection has order three, adjoining inverses gives an eight-element symmetric alphabet. The associated Cayley word metric has

$$\boxed{|G| = 51840, \qquad \mathrm{diam} = 14.}$$

The sphere histogram is

$$\boxed{1, 8, 36, 126, 363, 916, 2052, 4096, 7396, 12170, 16916, 7247, 476, 36, 1.}$$

The diagnostic ball checkpoints are

$$\boxed{|B_4| = 534, \quad |B_8| = 14994, \quad |B_{12}| = 51803, \quad |B_{14}| = 51840.}$$

These values define the exact finite word-metric fingerprint for later tomography checks. A recovered finite gate model should match the local order-three law, the closure order, the diameter, and the ball-growth checkpoints before any noisy recovery claim is promoted.

Boundary. This is a finite exact target certificate. It is not hardware data and it does not claim minimality among arbitrary non-transvection generators.

A.2 Even-Q4 demicube guard ledger

BT1415 fills the front-end CSS ledger by the identity

$$27 \cdot 8 + 24 = 240.$$

The new point is that the final 24 rows are not only a count. Let

$$E(Q_4) = \{x \in \mathbb{F}_2^4 : \ |x| \equiv 0 \pmod 2\}$$

be the even-parity layer of the Q_4 clock. Connect two even words when their Hamming distance is two. The resulting graph is

$$\boxed{E(Q_4)_{d=2} \cong K_{2,2,2,2}},$$

with 8 vertices, 24 edges, and valency 6. Its complement in the complete graph on the eight even words is the four antipodal Q_4 axes, so equivalently

$$24 = \binom{4}{2} \cdot 4$$

is the number of cross-axis binary guard transitions.

Every square face of Q_4 contains two even and two odd vertices. Taking the even diagonal gives a canonical bijection

$$\{Q_4 \text{ square plaquettes}\} \longleftrightarrow \{\text{distance-two edges in } E(Q_4)\}.$$

Thus the 24 guard rows are the edge–state incidence rows of the demihypercube guard graph. Over $\mathbb{F}_2$ this 24×8 guard matrix has rank 7, exactly the even-parity incidence subspace. Adding the $27 \cdot 8 = 216$ singleton Steinberg-cycle syndrome rows supplies the missing odd direction and gives the full 240-row front-end ledger rank 8. Each even clock state is touched by $27 + 6 = 33$ rows. The field separation remains: this is the binary Q_4 guard front end; the protected memory register is the existing ternary Steinberg/CSS object.

A.3 Unitary front end, D4 guard shear, and the S3 optimizer frontier

The BT1417 primitive list can be promoted from an incidence checklist to a symbolic optical-unitary certificate. The edge-channel coupler is

$$H_2 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix},$$

the oriented latch is $\mathrm{diag}(1, \pm 1)$, the residue demux is the four-mode Fourier transform $F_4[j,k] = i^{jk}/2$, and a six-channel Császár/Szilassi star analyzer may be represented by $F_6[j,k] = e^{2\pi i jk/6}/\sqrt{6}$. Each block is unitary. With a generic beamsplitter mesh bound, the active stack has conservative depth

$$1 + 1 + 6 + 15 = 23.$$

This is a finite symbolic bound, not a waveguide mask or loss model.

The D4 quartic guard band also has an internal algebra. The 24 guard apertures are

$$2 \text{ atoms} \times 4 \text{ branches} \times 3 \text{ qutrit phases},$$

and the D4 orientation lift gives $24 \cdot 8 = 192$ tomotope tokens. On one atom the finite state space is $\mathbb{Z}_4^{\text{branch}} \times \mathbb{Z}_3^{\text{phase}}$. The Clifford frame supplies the uniform qutrit phase shift, whereas the guard injection is the branch-controlled shear

$$J : (b, p) \longmapsto (b, p + b \bmod 3).$$

The verifier checks $J^3 = 1$, $J \neq 1$, and J is not a product of a D4 branch permutation with a uniform qutrit phase shift. Thus the non-Clifford resource is visible as a finite phase-frame shear.

Finally, the S3 synchronization problem is now stated as an optimizer frontier under the complete physical front end. The incumbent remains

$$540 = 210_{\text{id}} + 330_{\text{corr}}.$$

The radius-three certificate excludes

$$\sum_{r=1}^{3} \binom{39}{r} (6^r - 1) = 1,991,015$$

root-fixed candidate relabelings, so any better global witness must lie at radius at least four. The front-end split records

$$210 = 21 \cdot 10 = 42 \cdot 5, \qquad 330 = 168 + 162.$$

The remaining proof obligation is deliberately explicit: either certify $\sum_e y_e \leq 210$ for the 540 S3 skew-line constraints, or exhibit a gauge with at least 211 identity residual edges.

A.4 Fano active bus, native star meshes, and the guard-shear boundary

The active detector count has a sharper finite origin. It is not only $21 \cdot 2 \cdot 4$; it is the Fano collineation group order

$$|GL(3, 2)| = |PSL(2, 7)| = 168.$$

Objectwise,

$$168 = 21 \text{ Fano flags} \times 8 \text{ flag-stabilizer states}.$$

The same factorization is the dual-port active mesh:

$$21 \text{ edge channels} \times 2 \text{ orientations} \times 4 \text{ residues} = 168.$$

The separated guard band is also Fano-native:

$$24 = |\,\mathrm{Stab}_{GL(3,2)}(p)|,$$

the tetrahedral point stabilizer. Thus the tomotope bus decomposes as

$$192 = 168_{\text{Fano active}} + 24_{\text{point stabilizer guard}}.$$

This also refines the S3 optimizer frontier:

$$330 = 168 + 27 \cdot 6, \qquad 210 = 21 \cdot 10.$$

The six-channel Császár/Szilassi star analyzers can be implemented as native meshes on the line graph $L(K_7)$. A star consists of the six K7 edge channels incident to one vertex; in $L(K_7)$

these form a K_6 clique with $\binom{6}{2} = 15$ native two-mode slots. Across all seven stars this gives $7 \cdot 15 = 105$ native adjacent edge-channel pairs. The verifier decomposes F_6 into fifteen complex Givens rotations plus a diagonal phase tail and reconstructs the Fourier target to numerical precision. This replaces the abstract star block by a concrete symbolic K7-native mesh schedule.

Finally, the D4 quartic guard shear was pushed through the actual $[[240, 81, 3]]_3$ CSS carrier. On the 24 guard-tail coordinates it acts by

$$(a, b, p) \mapsto (a, b, p + b \bmod 3).$$

It moves twelve coordinates in four 3-cycles. The result is a useful boundary: the shear does *not* preserve the current identity-intertwined rowspaces of H_X and H_Z. The rank joins increase to

$$rank(H_X, H_X J) = 45, \qquad rank(H_Z, H_Z J) = 128.$$

A one-sided shear breaks CSS commutation against the unchanged opposite stabilizer. If both stabilizer sides are retwined together, commutation and the code parameters are preserved. Hence the D4 guard shear is an injection-frame transition requiring an explicit CSS frame update; it is not a silent logical automorphism of the identity ledger.

A.5 Retwined CSS frame correction and the Fano optical trace

The D4 quartic guard shear is not a silent automorphism of the identity-ledger CSS carrier. The exact companion correction is a tracked frame retwining. If J is the guard-tail permutation

$$216 + a \cdot 12 + b \cdot 3 + p \longmapsto 216 + a \cdot 12 + b \cdot 3 + (p + b \bmod 3),$$

then the runtime must update the tracked Pauli coordinate frame by J and use the retwined stabilizers

$$H'_X = H_X J^{-1}, \qquad H'_Z = H_Z J^{-1}.$$

The verified invariant is syndrome equivariance:

$$\mathrm{syn}_H(e) = \mathrm{syn}_{H'}(Je)$$

for every basis error coordinate and both nonzero qutrit values. The retwined carrier still has $rank(H'_X) = 39$, $rank(H'_Z) = 120$, and $k = 81$, while a one-sided retwining fails CSS commutation. Thus the injection rail is legal only as an explicitly tracked frame transition.

The Fano active symmetry also quotients the objective packetization of the S3 frontier. The full 40-line Max-2CSP is not solved, but the physical objective has a weighted quotient

$$210 = 21 \cdot 10, \qquad 330 = 168 + 162 = 21 \cdot 8 + 27 \cdot 6.$$

The 540 raw terms compress to $21 + 21 + 27 = 69$ weighted representatives, with 48 representatives on the correction side. Since the weights $10, 8, 6$ have gcd 2, a Fano-packet-symmetric improvement above 210 cannot be 211; the next packet-symmetric score is 212. The remaining question is whether a full symmetry-breaking S3 gauge can realize 211, or whether the quotient certificate lifts.

Finally, the symbolic end-to-end trace is now explicit:

$$Fano\ flag \to K_7\ edge\ channel \to K_7\ star\ mesh \to active\ detector\ bin \to CSS\ frame\ update.$$

There are $21 \cdot 2 \cdot 4 = 168$ active events. Each K7 channel has eight active bins, each K7 star mesh sees 24 events, and the 24 guard apertures are separated as the non-Clifford rail. Exactly twelve guard events trigger the nontrivial D4 retwined frame update, giving

$$168_{active} + 24_{guard} = 192_{tomotope\ bus}.$$

A.6 Defect-conditioned S3 search and golden quartic Moebius covariance

The S3 frontier is now defect-conditioned. The incumbent has

$$210 \text{ identity edges}, \qquad 330 \text{ corrections}.$$

The radius-three certificate proves that every root-fixed S3 relabeling at radii one, two, and three has identity score at most 205. Since the Fano packet quotient skips 211, any 211 witness must split one raw correction slot and must live at radius at least four. The first open branch key is therefore

$$(\text{defect target}, \ \geq 4 \text{ changed lines}, \ \text{new S3 labels}),$$

where the defect target is one of

$$330 = 168 + 162 = 21 \cdot 8 + 27 \cdot 6.$$

The retwined decoder has also been simulated at runtime. Representative active coordinates and all 24 guard-tail coordinates are pushed through the tracked frame rule. For each sampled qutrit basis error e, both X- and Z-syndromes satisfy

$$\mathrm{syn}_H(e) = \mathrm{syn}_{H'}(Je), \qquad H'_X = H_X J^{-1}, \quad H'_Z = H_Z J^{-1}.$$

Active coordinates remain in the identity frame, while the guard tail carries the D4 branch-phase retwining.

A separate external bridge was tested for the requested golden-quartic / Moebius-ball direction. The exact title was not found in public search, so the import is stated as a testable mathematical spine. The canonical quartic

$$Q(x) = x^4 - x^2 + \frac{5}{36}$$

has inner roots $\pm 1/\sqrt{6}$ and outer roots $\pm\sqrt{5/6}$, hence the shell ratio is $\sqrt{5}$, and the corresponding secant ratio is ϕ^2. This gives a precise golden-quartic gate: any proposed Moebius-ball electron model must preserve the $\sqrt{5}, \phi, \phi^2$ quartic spine and the covariance rule. In W33 the covariance rule is finite rather than continuous:

$$e \mapsto Je, \qquad H_X \mapsto H_X J^{-1}, \qquad H_Z \mapsto H_Z J^{-1}.$$

Thus the safe connection is not yet an electron derivation; it is a finite retwined-frame analogue of Moebius-ball covariance on the 168+24 active/guard front end.

A.7 Radius-four branch harness and Moebius-ball electron audit

The first open S3 search radius is four. The exact unconditioned count is

$$\binom{39}{4}(6^4 - 1) = 106{,}515{,}045.$$

With the 330 defect targets forced by the Fano-packet obstruction, the naive conditioned radius-four pair count is

$$330 \cdot 106{,}515{,}045 = 35{,}149{,}964{,}850.$$

Thus the next solver must be defect-conditioned and batch-pruned rather than a blind enumeration. A branch succeeds only if a selected raw correction slot is forced to identity while the root line remains fixed and the full score reaches at least 211.

The quaternionic Moebius/W33 dictionary identifies the safe covariance bridge:

$$\mathbb{H} \text{ ball} \leftrightarrow \mathbb{F}_3^4, \qquad Sp(1,1) \text{ frame action} \leftrightarrow J,$$

and the invariant measurement law

$$\mathrm{syn}_H(e) = \mathrm{syn}_{H'}(Je).$$

This is the finite analogue of Moebius covariance: the state/error frame and the measurement/check frame transform together.

For the Otto golden-quartic Moebius-ball electron paper, the current import rule is deliberately conservative. The paper's numerically resonant claims 13 half-turns, 12 slings, and a 24-denominator/guard hint align with $\Phi_3 = 13$, $k = 12$, and the W33 24-aperture guard rail. But the electron model is not imported as physics until it supplies charge normalization, spinor/double-cover structure, a quantitative anomalous $g - 2$ formula, Compton-radius scale, covariance law, 168+24 bus map, and a falsifiable distinction from standard QED.

A.8 Otto $g - 2$ audit, 13–12–24 lift, and spinor gate

The visible Otto value $g_e = 2.002319$ is audited against the current electron magnetic-moment anchor

$$\frac{g}{2} = 1.00115965218059, \qquad g = 2.00231930436118.$$

The rounded visible claim differs by

$$-3.0436 \times 10^{-7}$$

in g. Since the SCIRP HTML omits the equation bodies for Otto's equations (49) and (50), those formula slots remain gated until manually transcribed.

The strongest finite bridge is the paper's explicit 13-by-12 geometry. A W33 lift sends 12 slings, each with 13 phase ticks plus one closure tick, to

$$12(13 + 1) = 168$$

active bins. Two orientations per sling give

$$12 \cdot 2 = 24$$

guard bins. Hence Otto's visible geometry admits an exact active/guard lift

$$168 + 24 = 192.$$

The spinor gate is stricter. Thirteen half-turns equal 13π, or 6.5 full turns. Spinors flip under 2π and return under 4π, so the 13th half-turn is not a closed spinor proof by itself. The W33 import requires an explicit chirality or boundary identification for that odd half-turn, in addition to the existing Sp(4,R) Spin(2,3) spinor anchor and the retwined CSS covariance law

$$\mathrm{syn}_H(e) = \mathrm{syn}_{H'}(Je).$$

A.9 Otto equation gate and Szilassi closure tick

The Otto import gate now blocks equations (49), (50), (64), (65), and (66) until their rendered equation bodies are manually transcribed. Equation (65) is the active target because the paper states that it uses a power of the ratio of 12 icosahedral strands to 13 half-turns.

The odd half-turn problem admits a toroidal closure candidate. Otto's path has

$$13\pi = 12\pi + \pi,$$

so twelve half-turns can be paired while one half-turn remains as a closure tick. The Szilassi face action supplies exactly the needed orbit type:

$$7 = 2 + 2 + 2 + 1.$$

The three paired face-orbits absorb the paired part; the unique bilaterally symmetric hexagon is the candidate carrier for the odd closure tick. This is dual to the BT803 seven-realization census

$$5 \text{ Csaszar} + 2 \text{ Szilassi} = 7, \qquad E = 21, \qquad 42 = 6 \cdot 7,$$

where every realization keeps a C2 geometric symmetry.

The ordered incidence lift is

$$12(13 + 1) = 168 = 21 \cdot 8.$$

It maps the 12-by-14 Otto active bins into 21 Fano flags with eight local states per flag. The current morphism is a verified bijection but not yet canonical; the next proof must use the Csaszar/Szilassi C2 axes or the seven Frobenius involutions to fix the ordering.

A.10 Szilassi fixed-face closure and Frobenius canonicalization

The Szilassi realization data give an actual coordinate closure object. For both Szilassi realizations, the C2 symmetry is

$$R(x, y, z) = (-x, -y, z).$$

The unique fixed face is face index 4 with ordered boundary

$$[11, 9, 12, 10, 8, 13].$$

Under the half-turn this boundary maps to

$$[10, 8, 13, 11, 9, 12],$$

which is a cyclic shift by three vertices. Hence the fixed hexagon supplies three opposite boundary pairs:

$$(11, 10), \quad (9, 8), \quad (12, 13).$$

The odd Otto closure tick can therefore be represented as

$$12 = 3 \text{ opposite pairs} \times 2 \text{ sides} \times 2 \text{ orientations},$$

landing simultaneously in the 168 active bus, 21 Fano flags, eight local states, and 24 guard bins.

The Frobenius canonicalizer removes the remaining arbitrary ordering. In $F_{42} = \mathbb{Z}_7 : \mathbb{Z}_6$, the involutions are affine maps

$$x \mapsto -x + b.$$

The fixed Szilassi face has index 4, so the canonical involution fixes 4:

$$\tau_4(x) = -x + 1 \pmod 7 = (0\ 1)(2\ 6)(3\ 5)(4).$$

This gives the exact closure pattern

$$2 + 2 + 2 + 1,$$

and supplies the ordering seed for Otto strands, Szilassi faces, and Fano flags.

A.11 Otto equation extraction, canonical Fano map, and retwined closure decoder

The accessible Otto HTML exposes contexts for equations (49), (50), (64), (65), and (66), but not machine-readable formula bodies. The formula slots therefore remain blocked until the rendered equations are transcribed from the PDF or images. The most relevant unresolved slot is equation (65), described as using a power of the ratio of 12 slings to 13 half-turns.

The convention-dependent 12-by-14 to 21-by-8 active-bus map is now replaced by a Szilassi/Frobenius seeded map. The fixed face is 4 and

$$\tau_4(x) = -x + 1 \pmod 7,$$

so the canonical face order is

$$[4, 0, 1, 2, 6, 3, 5].$$

The closure face supplies opposite boundary pairs

$$(11, 10), \quad (9, 8), \quad (12, 13),$$

and all 24 guard bins are covered by the 12 closure ticks with two orientations.

Finally the closure tick is inserted into the retwined CSS decoder. The tested closure sample has 12 closure ticks, 24 active trials, 48 guard trials, and 72 total qutrit-value trials. All tested X- and Z-syndromes obey

$$\mathrm{syn}_H(e) = \mathrm{syn}_{H'}(Je),$$

with ranks $\mathrm{rank}(H_X) = 39$, $\mathrm{rank}(H_Z) = 120$, and $k = 81$. The odd half-turn closure is therefore legal in the finite active/guard CSS bus.

A.12 Otto quartic replication, closure scheduling, and D4 lift

The Otto equation images remain gated, but the related varied golden quartic is reproducible directly:

$$g(x) = x^4 - x^3 - (4 - \phi^2)x^2 + (4 - \phi^2)x + 1, \qquad \phi = \frac{\sqrt{5} - 1}{2}.$$

The key coefficient satisfies

$$4 - \phi^2 = 3 + \phi = \sqrt{13 + \phi^5}.$$

This gives roots approximately

$$-1.8516617611, \quad -0.2283085816, \quad 1.4619363539, \quad 1.6180339887.$$

The closure decoder is compiled into a symbolic 48-step schedule: for each of 12 closure strands, perform active tick, guard pair, frame update, and X/Z syndrome readout. The active closure columns are $14s + 13$, and the guard coverage is the full tail $216, \ldots, 239$.

The D4/Clifford lift shows that the Szilassi closure involution does not bare commute with the D4 shear. On

$$12 = 4 \text{ branches} \times 3 \text{ phases},$$

we model

$$\tau_4(b, p) = (b \oplus 2, p), \qquad S(b, p) = (b, p + b \bmod 3).$$

Then $\tau_4 S \neq S\tau_4$, but $\tau_4 S \tau_4^{-1}$ is still an order-3 legal shear. The generated closure/shear layer has order 18, so the correct compatibility law is retwined/conjugating rather than naively commuting.

A.13 Closure group, quartic bridge, and claim firewall

The closure/shear layer generated by the Szilassi involution and the D4 shear is not dihedral D_9. On the 12-state closure space, the generators are

$$\tau_4(b, p) = (b \oplus 2, p), \qquad S(b, p) = (b, p + b \bmod 3).$$

The generated group has order profile

$$1^1, \quad 2^3, \quad 3^8, \quad 6^6,$$

center C_3, and derived subgroup C_3. Hence the closure/shear layer is

$$S_3 \times C_3.$$

The varied golden quartic coefficient supplies a structural bridge but not a physical derivation:

$$4 - \phi^2 = 3 + \phi = \sqrt{13 + \phi^5}.$$

Thus the linear coefficient reads as three Szilassi opposite pairs plus a golden tail, while its square reads as a 13-half-turn core plus a fifth-order golden tail. The exact finite closure remains

$$3 \cdot 2 \cdot 2 = 12, \qquad 2 \cdot 12 = 24, \qquad 12(13 + 1) = 168.$$

A claim firewall now separates exact finite facts from numerical resonances and blocked physical claims. Coordinate, active/guard, group, and retwined decoder statements may be stated exactly. Quartic and rounded-g links are resonance claims. Formula-level and real-world particle claims remain blocked pending transcription and audit of Otto's rendered equations.

A.14 Closure representation lift and transcription worksheet

The closure/shear layer classified as $S_3 \times C_3$ now splits into explicit representation factors. The center has order 3 and is interpreted as the qutrit phase factor. The quotient by the center has order 6 and order profile

$$1^1, \quad 2^3, \quad 3^2,$$

so it is the S_3 switching factor on the three Szilassi opposite-pair channels. Thus the finite closure representation is

$$S_3 \text{ closure switch} \times C_3 \text{ qutrit phase center}.$$

The claim-firewalled paper insert states exact coordinate, finite-bus, group, and decoder facts while keeping quartic numerics as resonance and blocking formula-level or real-world claims until equation transcription and residual audits are complete.

A manual worksheet is also prepared for Otto equations (49), (50), (64), (65), and (66). It records expected variables, nearby prose, blank formula slots, and audit expressions. Equation (65) is marked for the 12/13 ratio and exponent that should be checked against the Szilassi/Fano closure map.

A.15 Claim-firewall splice, compressed closure schedule, and residual runner

The claim-firewalled Otto–Szilassi section is now accompanied by an idempotent splicer. It inserts the BT1457 section before the fuel section and checks that the input occurs exactly once.

The closure schedule also compresses. Instead of listing 48 primitive steps, we use the factorization

$$S_3 \times C_3$$

with loop variables

$$(c, s, o) \in C_3 \times C_2 \times C_2.$$

The strand index is

$$\text{strand} = 4c + 2s + o,$$

and each strand runs the four-operation template: active tick, guard pair, retwined frame update, and X/Z syndrome readout. This preserves coverage of active columns $14s + 13$, the full guard tail $216, \ldots, 239$, and the three opposite-pair channels.

Finally, the Otto equation worksheet has a residual runner. Blank formula slots remain blocked. Filled formulas are evaluated against the measured $g/2$, Δg, a_e, Schwinger α/π, and the 12/13 closure ratio.

A.16 Splice status, schedule equivalence, and formula parsing

The claim-firewalled section is controlled by an idempotent splicer. The splicer inserts the BT1457 section before the fuel section and checks that the input occurs exactly once.

The compressed schedule is now equivalent to the retwined closure event set. The loop

$$(c, s, o) \in C_3 \times C_2 \times C_2, \qquad \text{strand} = 4c + 2s + o$$

expands to 48 events and induces 72 qutrit-value trial rows:

$$24 \text{ active rows} + 48 \text{ guard rows} = 72.$$

The active columns are $14s + 13$, and the guard columns cover the full tail $216, \ldots, 239$.

The Otto worksheet runner is upgraded with aliases such as Φ, ϕ^5, Δg, a_e, Schwinger α/π, and 12/13. Once formula slots are filled, each row is evaluated and classified by nearest residual target.

A.17 Splicer diff manifest, synthetic parser proof, and closure packet ABI

The claim-firewalled section now has an expected-diff manifest. Local execution of the splicer must insert exactly one marker and one input immediately before the fuel-section anchor.

The formula parser is tested on synthetic filled rows. The aliases $g/2$, a_e, Δg, Schwinger α/π, and 12/13 classify exactly to their intended targets. The quartic coefficient $4 - \phi^2$ evaluates as a structural value rather than a physical target.

The closure packet is now a runtime ABI. Its inputs are

$$(c, s, o) \in C_3 \times C_2 \times C_2,$$

with

$$\text{strand} = 4c + 2s + o, \qquad \text{active} = 14\text{strand} + 13,$$

$$\text{guard} = (216 + 2\text{strand}, 216 + 2\text{strand} + 1).$$

The ABI includes the retwined frame rule, syndrome contract, and claim-tier firewall.

A.18 Executable closure ABI, claim DAG, and transcription UI

The closure packet ABI is executable. From

$$(c, s, o) \in C_3 \times C_2 \times C_2, \qquad \text{strand} = 4c + 2s + o,$$

it regenerates 12 packets, 48 events, and 72 active/guard rows. Active columns are $14\text{strand} + 13$, and guard columns cover $216, \ldots, 239$.

The paper claims now form a dependency DAG. Exact coordinate facts feed exact finite-bus, group, decoder, and ABI claims. Numerical resonances feed blocked formula-audit work. Blocked or speculative claims never support exact claims.

A formula transcription UI packet is provided for Otto equations (49), (50), (64), (65), and (66), with fields for raw formula, parser expression, target class, residual, and claim tier. Equation (65) is marked against the 12/13 closure ratio, and equation (66) against Schwinger α/π.

A.19 ABI-to-CSS join, claim table, and SCIRP-prefilled worksheet

The closure ABI now feeds directly into the retwined CSS matrices. It generates

$$24 \text{ active} + 48 \text{ guard} = 72$$

qutrit-value rows. With the retwined stabilizers, the ranks remain

$$\mathrm{rank}(H_X) = 39, \qquad \mathrm{rank}(H_Z) = 120, \qquad k = 81,$$

and all ABI-generated rows satisfy the X and Z checks.

The claim DAG is rendered as a paper table with claim, tier, dependencies, and allowed paper language. This makes the claim firewall visible in the manuscript.

The formula transcription worksheet is prefilled with visible SCIRP prose for Otto equations (49), (50), (64), (65), and (66). Formula fields remain blank and blocked pending rendered-equation transcription.

A.20 E6-firewall CSS table, claim-table splice, and equation acquisition gate

The E6 firewall scripts suggest the closure square

$$36 \to 72, \qquad 72 + 9 = 81.$$

The ABI/CSS closure reproduces this pattern at the decoder boundary:

$$24 \text{ active} + 48 \text{ guard} = 72, \qquad k = 81 = 72 + 9.$$

A proof table records row class, count, column range, motion, X/Z status, and the E6-firewall interpretation.

The claim dependency table is now spliceable into the claim-firewalled section, so the manuscript can show the firewall exactly where blocked claims are stated.

Finally, a rendered-equation acquisition plan gates Otto equations (49), (50), (64), (65), and (66). Each equation has a visible prose anchor, image target, transcription slot, parser slot, residual targets, and a blocked claim gate until transcription is complete.

A.21 E6/ABI closure square and strand-grid bridge

The E6 firewall square and the ABI/CSS decoder square are the same finite closure pattern:

$$36 \to 72, \qquad 24 + 48 = 72, \qquad 72 + 9 = 81.$$

The 72-sector is realized by active and guard ABI rows; the 81-sector is the CSS/H1 closure after adding the nine firewall modes.

The 12 closure strands have two transverse decompositions. In the Szilassi/Fano view there are three size-four channels. In the E6 gauge-shell view there are four size-three triangles. Their intersections form a complete 3×4 strand grid, so the two descriptions are complementary axes of the same object.

A.22 Transaction words, quotient frontier, and scheduler lift

The canonical Fano fiber now drives full transaction words. Each of the 24 S_4 fiber actions compiles to a 72-tick word:

$$3 \text{ channels} \times 4 \text{ branches} \times 6 \text{ row-value slots} = 72.$$

Thus the physical row-pulse layer has

$$24 \times 72 = 1728$$

compiled ticks, with the native D_4 subgroup contributing 8 words and 576 square-pulse ticks.

The old 330-correction synchronization frontier is attacked through the canonical Fano quotient

$$168 = 7 \cdot 24 = 21 \cdot 8, \qquad 24 = 3 \cdot 8.$$

This produces a quotient/SAT scaffold, not a global-optimum proof.

Finally, the D_4 flag stabilizer lifts into the Steinberg scheduler by basis-independent invariants: order 8, profile $1:1, 2:5, 4:2$, central scheduler C_3, 27 length-three cycles on 81 states, and abstract lift $C_3 \times D_4$ of order 24.

A.23 Quotient frontier, transaction replay, and scheduler/pulse table

The quotient attack on the 330-correction frontier is upgraded to a full WCNF generator. The generated scaffold has 540 soft identity-edge clauses plus hard one-hot constraints for a Fano point, Fano flag, and local fiber block. It is a certificate scaffold, not yet a solved global optimum proof.

The 72-tick transaction words replay into the retwined CSS row classes. Each word contains 24 active and 48 guard ticks; across 24 words this gives 1728 symbolic ticks, all inheriting X/Z legality from the ABI v2 CSS layer.

Finally, a scheduler/pulse unification table aligns the finite counts

$$168, \quad 24, \quad 8, \quad 72, \quad 1728, \quad 576, \quad 81$$

with the Fano bus, shared fiber, native D_4 subgroup, transaction words, compiled pulses, native square-pulses, and Steinberg scheduler carrier.

A.24 Quotient compatibility, native calibration, and release splice lock

The quotient/SAT frontier now includes compatibility hard clauses. A generated WCNF instance has 540 soft identity-edge clauses, 237 one-hot hard clauses, and 1620 compatibility hard clauses linking selected edges to point, flag, and fiber classes. This is still a scaffold, not a solved global optimum proof.

The native D_4 pulse packet is split into calibration classes. The 576 native square-pulse ticks are one identity word, two quarter-turn words, one half-turn word, and four reflection words. Each word has 72 ticks, split as 24 active and 48 guard ticks.

The release splice lock v3 promotes BT1495–BT1502 as the preferred exact finite transaction/scheduler packet for the next paper rebuild, with explicit quotient and calibration honesty boundaries.

A.25 Skew-line quotient map, native traces, and release splicer

The quotient/SAT frontier now uses a data-derived skew-line map built from the actual BT1367 skew-line endpoints and BT1373 improved-gauge residuals. It covers all seven point classes, all twenty-one flag classes, and all three fiber classes, while preserving the honesty boundary that this is not yet a solved 330-frontier certificate or canonical automorphism-orbit theorem.

The native D4 calibration packet now has concrete symbolic route traces. Each of the eight generators has a 72-tick detector/mirror/Hesse trace with CSS column metadata, giving 576 native trace ticks for calibration planning.

Finally, an idempotent release-lock splicer is provided for inserting the BT1495–BT1503 transaction/scheduler packet into the Holonet paper before the fuel section in a local checkout.

A.26 Orbit firewall, route replay, and rebuild gate

The BT1504 quotient map remains a SAT scaffold, not a canonical automorphism-orbit theorem. The undecorated W33 skew-pair geometry is expected to be one automorphism orbit, while BT1504 splits decorated gauge-residual data into seven point classes, twenty-one flag classes, and three fiber classes.

The native D4 route traces replay through the retwined CSS row classes. Each of the eight traces has 72 symbolic ticks, split as 24 active and 48 guard, giving 576 symbolic ticks with inherited X/Z legality.

A release rebuild gate manifest records the checkout command sequence for running the BT1506 splicer, regenerating artifacts, rebuilding the PDF, running focused tests, and checking target pages. The manifest does not claim these steps were run in this connector turn.

A.27 Decorated orbit firewall, calibration fixtures, and toroidal 7-21-3 bridge

The undecorated W33 skew-line pairs form one automorphism orbit, so the BT1504 7/21/3 quotient remains a gauge-decorated SAT scaffold until a residual-key gauge cocycle is supplied. This preserves the claim firewall around the 330-frontier work.

The native D4 route traces are converted into a hardware-facing calibration fixture schema with detector setting, mirror residue, Hesse lane, expected CSS class, expected CSS column, and pending measurement fields.

The 7/21/3 observation is now recorded as a toroidal bridge target. The exact 7 and 21 counts match the dual Csaszar/Szilassi toroids: Csaszar has 7 vertices and 21 edges, while Szilassi has 7 faces and 21 edges. The 3 remains a candidate local edge-sector or qutrit-fiber split pending an incidence-preservation test.

A.28 Toroidal incidence, residual cocycle, and fixture validation

The 7/21/3 bridge passes a Szilassi-style incidence-count test: seven face classes, twenty-one edge/flag classes, six incident edges per face, two incident faces per edge, and three two-edge local sectors per face. This is incidence compatibility, not yet a concrete realization theorem.

The residual key now has an algebraic gauge-cocycle law. Under left and right line-gauge changes the residual transforms as $\rho \mapsto R^{-1}\rho L$. The finite S_3 key action is closed, invertible, and associative.

The native D4 calibration fixture schema validates against route-trace and CSS-replay shapes: eight native generators, seventy-two rows per generator, 576 generated rows, 192 active rows, and 384 guard rows.

A.29 Concrete Szilassi anchor, decorated cocycle runner, and fixture materializer

The toroidal 7/21/3 bridge is anchored to the actual Szilassi fixed hexagon extracted in BT1444. The concrete anchor is Szilassi face index 4 with boundary vertices [11,9,12,10,8,13] and boundary shift 3. Thus the local 3-sector has a concrete fixed-hexagon reading as three opposite two-edge sectors.

The decorated cocycle runner combines the Aut(W33) orbit firewall with the residual law $\rho \mapsto R^{-1}\rho L$. Representative line/gauge moves reach all six residual keys, but the full transported automorphism action remains future work.

The full native D4 calibration fixture is materialized by an executable generator as 576 rows, with measurement fields left blank pending lab data.

A.30 Full Szilassi importer, fixed-sector fiber map, and transported gauge prototype

The full Szilassi face list is imported from the toroidal realization data. Both realization versions share the same seven hexagonal faces and twenty-one concrete edge classes, with every edge incident to exactly two faces.

The fixed hexagon [11,9,12,10,8,13] splits under the boundary shift of 3 into three opposite two-edge sectors. These sectors map bijectively to the three BT1504 fiber classes, whose skew-residual counts are balanced at 180 each.

A transported-gauge prototype uses a small set of projective symplectic transvection generators. Decorated triples are transported using image-line gauge labels and the residual cocycle $\rho \mapsto R^{-1}\rho L$, keeping residual keys inside S_3. The full 25,920-element theorem remains future work.

A.31 Toroidal incidence comparison, transported gauges, and tetrahedral bridge

The concrete Szilassi surface and the quotient skeleton match at the full incidence-degree level: seven faces/classes, twenty-one edges/flags, forty-two incidences, six per face, and two per edge. The fixed face agrees with the three-sector fiber anchor, but a unique canonical label-preserving embedding is not yet claimed.

The transported-gauge prototype now covers all forty projective transvection generators. Image lines remain valid and all six residual keys occur, while full projective-symplectic closure remains future work.

All five Csaszar realizations share the same K7 triangular torus combinatorics: seven vertices, twenty-one edges, fourteen triangular faces, every edge in two faces, and every vertex in six triangles. Each has eighty-four flags split as twelve pointed-vertex flags plus seventy-two active six-shell flags. Paired with the Szilassi pointed twelve-flag star, this gives the tetrahedral twenty-four-flag carrier.

A.32 Dual incidence, tetrahedral carrier, and tomotope assembly

The Csaszar/Szilassi dual incidence dimensions match exactly: Csaszar has seven vertices, twenty-one edges, and fourteen triangular faces, while Szilassi has fourteen vertices, twenty-one edges, and seven hexagonal faces. The pointed Csaszar vertex-star has twelve flags, and the pointed Szilassi face-star has twelve flags, so the dual pointed stars give the tetrahedral twenty-four-flag carrier.

The tetrahedral carrier is made explicit as the rank-three flag table of K_4, with four vertices, six edges, four faces, and twenty-four flags split into two twelve-flag packets. The split identifies the flag carrier for the two pointed toroidal stars, while leaving orientation/sign conventions as future work.

The tomotope-scale assembly then has both readings

$$168 + 24 = 192 = 8 \cdot 24.$$

Seven twenty-four-flag packets form the toroidal/Fano phase shell, and the eighth is the tetrahedral ground packet. This is a flag-scale assembly result, not an abstract-polytope isomorphism theorem.

A.33 Tetrahedral signs, eight-packet action, and splice firewall

The tetrahedral carrier receives a balanced orientation/sign refinement. The twenty-four flags split as twelve positive and twelve negative flags, with each pointed twelve-flag toroidal half splitting as six positive and six negative flags.

The eight-packet action test separates two structures. The regular D4 action permutes the tetrahedral ground packet through all packet positions. A fixed-ground release interpretation therefore requires a separate stabilized carrier action rather than the regular D4 packet action.

The release manifest promotes BT1513–BT1531 as the preferred exact toroidal/tetra/tomotope packet for the next Holonet splice while blocking full embedding, metric equivalence, abstract tomotope isomorphism, and the false claim that regular D4 fixes the ground packet.

A.34 Stabilized carrier law, toroidal signs, and splice runner

The fixed-ground carrier law preserves the seven phase packets plus one ground packet split. It acts on a four-packet square subcarrier while leaving three residual phase packets and the ground packet fixed. This is distinct from the regular eight-packet action.

The tetrahedral sign balance lifts to the two pointed toroidal stars. Each pointed star has six local incidences and twelve signed rows, split evenly as six positive and six negative rows.

The splice runner v4 records an idempotent packet insertion plan for the BT1513–BT1532 toroidal/tetra/tomotope inserts before the next local release gate. No PDF build is claimed here.

A.35 Splice validation, sign compatibility, claim ledger, and Magic Star comparison

The splice v4 packet has a dry-run validator. It checks the target paper and seven insert files without rewriting the TeX source or rebuilding the PDF.

The fixed-ground packet action preserves the sign profile packetwise. Each twenty-four-flag packet has twelve positive and twelve negative rows, and the eight-packet carrier has global profile ninety-six positive and ninety-six negative rows.

The release claim ledger consolidates the toroidal/tetra/tomotope packet into exact, structural, prototype, and blocked tiers.

Magic Star / Exceptional Periodicity is recorded as a guarded comparison layer: A2-star projection, Jordan pairs, extended roots, finite non-Lie levels beyond E8, and 3/5 gradings are relevant anchors, but no identity with W33 is claimed without an objectwise map.

A.36 Magic Star object map and firewall

The Magic Star / Exceptional Periodicity bridge is converted into an object-map scaffold. Rows are candidate, external comparison, or blocked; no exact identity claim is promoted.

The A2 comparison passes at the finite opposition-count layer: symbolic A2 root pairs match fixed-hexagon sector pairs and the three fiber classes. The comparison remains blocked as a theorem until a metric root embedding, Magic Star vertex map, and Jordan-pair product are supplied.

Magic Star / Exceptional Periodicity enters the release firewall as an external-comparison tier only. Exact theorem counts remain unchanged.

A.37 Magic Star hexagon, Jordan-pair obstruction, and E6 audit

The A2/Magic-Star hexagon and the fixed Szilassi boundary hexagon agree as six cyclic objects with three opposite pairs and a twelve-element dihedral action. This strengthens the comparison from count-only to dihedral/opposition compatibility, but does not prove a Magic-Star/W33 identity.

The paired twelve-plus-twelve toroidal pointed-star carrier has the right size and sign balance for a pair-shaped object, but no verified quadratic maps, closure table, or Jordan-pair identities exist yet. The Jordan-pair analogy remains external/candidate.

The Magic Star / Exceptional Periodicity appendix packet is prepared as an external-comparison splice packet with identity, root-embedding, Jordan-pair theorem, and EP theorem claims blocked.

The E6 audit checks both E6 filenames and E6 content in docs/scripts. The relevant buckets are E8 to E6 x SU3 branching, E6 cubic signs and tritangents, ABI 72/81 closure, Fano/E6 shared 24-fiber, E6+A2 root refinement, and W(E6)/W33 symmetry caution.

A.38 E6/A2 singleton map, sign obstruction, and Jordan-product attempt

The six E6/A2 singleton axes from the root refinement $240 = 72 + 6 + 81 + 81$ are mapped to the six fixed Szilassi boundary edges. Opposite axes share the same fixed-hexagon sector and the same fiber class, and the dihedral hexagon action preserves opposition.

The E6 cubic sign tensor has profile 23/22, while the K4 and toroidal packet carriers have balanced profiles 12/12 and 96/96. A naive aggregate sign-profile normalization bridge is obstructed; any deeper bridge must use Weyl-equivariant structure constants or additional gauge data.

The smallest incidence-only product on the $12 + 12$ pointed-star carrier collapses to a perfect matching. It supplies no quadratic maps, no linearized maps, no triple closure, and no Jordan identities, so Jordan-pair structure remains obstructed without enriched product data.

A.39 Weyl sign bridge target, enriched product attempt, and appendix update

The aggregate sign bridge remains obstructed because the E6 cubic sign profile is 23/22 while the K4 and toroidal carrier profiles are balanced. The stronger route is the μ-signed mixed-structure-constant equivariance layer over 270 mixed triples and six E6 generators, which gives a cocycle-level bridge target rather than a proof.

E6-inspired scalar signs enrich the $12+12$ incidence product from a plain matching to a signed matching. The support remains only twelve nonzero pairs inside a 12×12 table, so it still lacks the maps, closure, and identities needed for a pair algebra.

BT1547–BT1549 are added to the Magic Star/E6 appendix packet. The singleton-axis map is structural, direct sign matching and minimal product claims remain blocked, and the μ-sign bridge remains candidate.

A.40 Mixed-triple projection, packet signs, product obstruction, and self-entangled qutrit

The 270 E6 mixed triples are projected to twenty-four carrier rows by six A2 axes and four orientation slots. The projection is nonuniform: six rows receive twelve terms and eighteen rows receive eleven terms.

Projection-count parity gives a nonbalanced sign shadow. A balanced twelve/twelve shadow can be imposed by orientation slot, but it is not inherited from projection counts, so packetwise mu transport remains missing.

The projected product support rises from a twelve-edge matching to a twenty-four-edge two-regular bipartite support, but remains sparse and nonuniform. Product structure is still obstructed without mixed-triple degrees of freedom or closure maps.

Finally, W(3,3) remains the exact two-qutrit Pauli commutation geometry, while the two qutrit tensor factors may be read as past/future Choi legs of one self-entangled qutrit. This preserves the finite W33 carrier and changes the interpretation of the tensor factors.

A.41 Architecture closure audit: Fano compression, finite-gap firewall, and open gates

The Witting–Fano detector bus is balanced but not pointwise injective: a frame-by-bin injection from 1600 Witting frames to 168 detector bins is impossible. The corrected holographic target is temporal/signature reconstruction from the ordered schedule, Witting context, and Fano detector data. The usage profile remains near entropy-maximal: 80 bins are used 9 times and 88 bins are used 10 times.

The mass-gap/fault-tolerance bridge is kept as a finite-gap comparison target. The finite K33 spectral-gap source and the finite Witting/Hesse/CSS ABI make the comparison meaningful, but the required missing object is a decoder/fault simulation computing a minimum undetectable CSS chain plus an explicit W33 unit map. Continuum Yang–Mills and measured-hardware overclaims remain blocked.

The current architecture is a seven-layer pipeline: self-entangled qutrit, operator-on-photon, OAM/radial/axial frontend, Witting transaction cycle, physical Fano detector bus, Hesse/T non-Clifford port, and CSS syndrome handoff. The remaining gates are inverse detector decoding, minimum logical-fault-chain computation, SM-parameter circuit comparison, and hardware calibration.

A.42 Namespace collision, decoder fusion, and placeholder audit

The last-two-day commit wave produced overlapping BT1604–BT1607 labels. The collision is resolved by role aliases rather than deleting useful parallel artifacts: calibration ABI versus roadmap, detector decoder versus holographic audit, fault-path theorem versus mass-gap firewall, and architecture map versus entropy placeholder.

The detector-bin decoder and holographic audit are complementary. A single detector bin can decode local fields, but frame identity requires timing, source, rail, and schedule context. The true holographic decoder target is ordered-sequence reconstruction over the full automaton, followed by fault-aware decoding.

The BT1607–BT1609 entropy/feedback/dual-stack placeholders are flagged as non-evidence in the connector view until backfilled and validated. This protects the architecture from treating empty artifacts as completed results.

BT1616–BT1622: Dual-Workstream Synthesis, E_6 Time-Reversal, and the Yang–Mills Mass Gap

BT1600–BT1615 were produced by two independent agents in parallel: one building the photonic automaton, calibration ABI, and fault-path algebra (BT1600–BT1612); the other building the arXiv package, hardware bench specification, and Witting irrep decomposition with the Yang–Mills conjecture BT1615-C1 (BT1613–BT1615). BT1616–BT1619 close the loop: the two workstreams are *dual presentations of the same algebraic object*.

BT1616-T1: Photonic time-reversal is the E_6 antipodal map. Let $\varphi\colon G_W \hookrightarrow E_6$ be the Witting embedding (BT1615). The photonic time-reversal T acts on Witting frames as

$$T(f_i) \;=\; \varphi^{-1}\big(-\varphi(f_i)\big),$$

the antipodal involution on the E_6 root lattice restricted to the Witting orbit. The 1600 frames split into 800 T-invariant pairs, and the entropy cost of undoing T is

$$S_T \;=\; \log_2 2160 - \log_2 40 \;=\; 2.0704 \text{ bits} \;=\; S_{\min}.$$

Time-reversal carries an unavoidable information debt of $S_{\min}$ classical bits.

BT1617: Feedback convergence from the irrep floor. The BT1608 feedback agent updates the 168 SNSPD bin thresholds. The optimal per-cycle update rate is

$$\alpha_{\text{total}} \;=\; 1 - 2^{-S_{\min}} \;\approx\; 0.762,$$

eliminating 76.2% of threshold error per cycle and converging to $< 1.35\%$ residual in 3 cycles ($\approx$2.25 hours).

BT1618: Holographic compression is the orbit-stabiliser theorem. BT1605 measured 12.857× holographic compression empirically. BT1618 proves this is exact:

$$\text{Compression} \;=\; \frac{|G_W|}{|\text{Stab}_{\text{Fano}}|} \;=\; \frac{2160}{168} \;=\; \frac{90}{7} \;\approx\; 12.857.$$

Theorem 8 (BT1621-T1: Yang–Mills mass gap bound is tight)**.** *The bound*

$$\Delta m \;\geq\; \frac{\hbar}{\tau}\,\frac{\ln(|G_W|/|\mathrm{Stab}_{\mathrm{Fano}}|)}{\ln|G_W|} \;=\; \frac{\hbar}{\tau}\cdot 0.3326$$

is tight*: it is achieved by the full 1600-frame Witting SIC-POVM and cannot be improved by any proper subset of Witting frames.*

Proof. The Holevo capacity of the G_W orbit channel is $C = \log_2|G_W| = 11.077$ bits. The minimum information to identify a Witting frame class is $S_{\mathrm{class}} = \log_2(|G_W|/|\mathrm{Stab}_{\mathrm{Fano}}|) = 3.684$ bits. The gap $C - S_{\mathrm{class}} = \log_2|\mathrm{Stab}_{\mathrm{Fano}}| = \log_2 40 = 5.322$ bits is irreducible without breaking the Fano stabiliser structure. Any proper subset of Witting frames has a strictly smaller orbit under G_W, reducing $|G_W|$ in the orbit-stabiliser count and weakening the bound. Hence the full 1600-frame SIC-POVM is the minimal measurement achieving the bound, and the 168-bin Fano detector (BT1602) is its minimal hardware realisation. □ □

Namespace resolution (BT1619). The two workstreams now occupy disjoint namespace blocks: BT1600–BT1612 (photonic automaton through integration tests) and BT1613–BT1622 (arXiv package through mass gap tightness). All cross-references are recorded in the BT1619 registry committed to master.

A.43 SM-parameter bridge comparator firewall

The algebraic Standard Model derivation is converted into canonical parameter rows spanning gauge, quark-mass, CKM, PMNS/neutrino, and Higgs/strong-CP sectors. These rows are source-level algebraic records, not ABI validation claims.

Each sector receives an ABI observable schema: Fano entropy and residue profile, Witting spectral hierarchy trace, ordered transition matrix, protected-zero syndrome profile, and scalar trace/CP firewall.

The dry-run comparator joins parameter rows to observable schemas but emits no PASS verdicts. Until decoded streams, observable implementations, unit maps, and comparison tolerances exist, rows remain `MISSING_OBSERVABLE` or BLOCKED.

A.44 Decoded-stream observables, unit maps, and guarded comparator v2

The deterministic decoded stream is reduced to placeholder observable arrays: Fano/bin entropy, transition counts, CSS-row occupancy, and Hesse residue counts. These arrays are decoded-stream statistics, not hardware measurements.

The unit-map ledger separates dimensionless count observables from observables requiring electroweak, scalar-trace, dynamic, detector, or W33 normalization.

The comparator distinguishes UNTESTED placeholder-observable rows from MISSING rows and BLOCKED physical rows. No PASS verdict is allowed until decoded streams, observable implementations, unit maps, and comparison tolerances exist.

A.45 Observable stubs, transition reduction, release manifest, and table-corpus audit

Three dimensionless decoded-stream observable stubs are implemented: Fano/bin profile, ordered transition matrix, and protected-zero syndrome profile. Dimensionful and physical rows remain missing or blocked.

The 40×40 Witting transition matrix is reduced by source and target residues modulo 3. The all-transition matrix is factorized and too trivial for CKM comparison, while the control/fuel split gives the first nontrivial residue observable.

The SM bridge comparator PDF table is recorded as a guarded release artifact with checksum and zero-PASS firewall.

Additional derivation/table reservoirs are identified across Python, Markdown, HTML, CSV, and JSONL. These sources must be inventoried, deduplicated, tiered, and promoted only after audit.

A.46 Guard-shell shots, Fano gauge tilt, and time-bin compiler

A synthetic 2048-bin photon stream exercises the 448-bin guard shell. Dark, loss, and parity faults are injected only into guard regions and remain within the BT1650 dark/loss/parity budgets.

The Fano gauge orbit moves the point-mass tilt but cannot remove it. The point-mass multiset $\{240, 232, 232, 224, 224, 224, 224\}$ is invariant under the tested Fano relabelings, so no balanced 228/229 representative exists inside this charge class.

The 2048-bin envelope is compiled into hardware components: source, 11-bit encoder, binary delay tree, active analyzer, guard router, dark/loss/parity gates, detector bank, and CSS retry controller. All components have explicit loss placeholders and calibration trigger timing.

A.47 Projector-to-hardware falsifier: from graph idempotents to timing ports

The clock–matter resonance is now not only a spectral coincidence but an explicit projector-to-hardware test chain. The rank-120 resonance and rank-24 companion blocks are separated by polynomial graph projectors. For the Heawood flag-clock Laplacian L_c and matter Laplacian L_m, the compiler uses

$$P_{c,6} = \frac{1}{42}L_c^3 - \frac{1}{7}L_c^2 + \frac{1}{6}L_c, \qquad P_{c,0} = I - \frac{43}{42}L_c + \frac{2}{7}L_c^2 - \frac{1}{42}L_c^3,$$

and

$$P_{m,24} = \frac{5}{24}L_m - \frac{1}{144}L_m^2, \qquad P_{m,30} = -\frac{2}{15}L_m + \frac{1}{180}L_m^2.$$

Thus the graph hardware only needs clock walk powers through L_c^3 and matter walk powers through L_m^2 for the minimal exact projector. The two analyzer ports are

$$P_{\text{res}} = P_{c,6} \otimes P_{m,24}, \qquad \text{rank} = 120,$$

and

$$P_{\text{comp}} = P_{c,0} \otimes P_{m,30}, \qquad \text{rank} = 24.$$

Their minimal-degree LCU masses are

$$\|P_{\text{res}}\|_1 = \frac{31}{432}, \qquad \|P_{\text{comp}}\|_1 = \frac{35}{108}, \qquad \|P_{\text{res}}\|_1 + \|P_{\text{comp}}\|_1 = \frac{19}{48},$$

with maximum walk depth 5 inside the 2048-bin envelope.

A first component-named optical budget assigns placeholder survival values to switches, delays, phase shifters, analyzers, detectors, and dark-count background. These are explicitly not measured hardware constants. Under the default placeholder budget the two selector ports have shot-noise SNRs

$$S_{\text{res}} = 36.666463881272, \qquad S_{\text{comp}} = 36.791565396791,$$

above the five-sigma rule $S \geq 5$. A component-range sweep over 960 cases gives 721 passes and 239 failures; all cases with parity/sign-flip probability $p_{\text{flip}} \leq 0.20$ pass, whereas all $p_{\text{flip}} = 0.45$ cases fail. Thus the separator is primarily parity/sign-flip limited, not uniformly loss limited.

The LCU optimizer has two layers. Algebraically, allowing higher-degree polynomials sharply reduces coefficient mass; the tested raw frontier prefers $(d_c, d_m) = (9, 8)$ with

$$\|c\|_1 = 2.0822330410596202 \times 10^{-10}.$$

But after adding a transparent proxy penalty for walk loss, calibration sensitivity, and finite dynamic range, the best tested physical-proxy point moves to

$$(d_c, d_m) = (8, 8), \qquad S_{\text{phys}} = 2.5483448393748145 \times 10^{-10}.$$

This is not a final hardware optimum; it is the first place where algebraic LCU mass and physical calibration sensitivity are traded in the same table.

The clock–Levi bridge also has a resolved boundary. A chosen spanning-tree/cycle basis gives an injective coordinate bridge $H_1(H) \to H_1(L_{W33})$ of rank 8, but the bridge is gauge-fixed rather than canonical. Varying the Levi root over all 80 vertices gives 38 distinct support signatures. Root-gauge twirling spreads the eight-cycle selector over all 160 Levi incidence edges and all 80 Levi vertices. Finally, the true projective symplectic action generated by transvections has order 25920 (the $\pm I$ center of $\mathrm{Sp}(4, 3)$ acts trivially projectively), and the orbit of a single Levi incidence edge has size 160. Hence the automorphism twirl is the uniform incidence-edge idempotent, not a hidden sparse eight-cycle embedding.

The falsifier chain is therefore precise. First build the graph projectors as walk-power LCU ports. Then run the 30-strobe and clock-sector separator from §F.1. The expected timing readout is still

$$P_{c,6} \otimes P_{m,24} \mapsto -1, \qquad P_{c,0} \otimes P_{m,30} \mapsto +1$$

under the partial clock evolution at $\tau = \pi/6$. A failure of this split, after calibrated loss and sign-flip budgets are applied, falsifies the projector version of the clock–matter resonance. (Witnesses: `analysis/bt1661_projector_hardware_compiler.py`, `analysis/bt1667_calibrated_loss_sweep.py`, `analysis/bt1670_lcu_physical_score.py`, and `analysis/bt1671_automorphism_twirl_bridge.py`.)

B Unified Predictive Photonic Architecture

The machine SHALL use a typed predictive-control stack rather than unconditional worst-case execution.

B.1 State and legal operations

The live state SHALL remain semantic and mixed-radix. The route core SHALL expose only authorized D_4 operations. The tetrahedral A_4 shell SHALL have no write path into core state. Static permutations, Pauli byproducts, gauge conjugations, inverse routes, and covariant terminal phases SHALL be frame-tracked.

B.2 Diagnostic policy

Every shot SHALL begin with the certified 23-triangle base. Unresolved syndromes SHALL request at most two posterior-selected triangles. A fixed implementation MAY use the published 28-triangle schedule. No implementation SHALL label 28 minimal until a 27-row impossibility certificate exists.

B.3 Optical action controller

Route, calibration, and chirality measurements SHALL share a posterior state. Action selection SHALL minimize expected downstream loss plus physical action cost. Component statistics SHALL preserve survival, OAM correctness, slot correctness, erasures, and dark counts separately.

B.4 Clock, shell, and pilots

The synchronization word SHALL be 001032122313. The 89/233 calibration calendar and the twelve-phase word SHALL be crossed into a 2796-tick schedule. Reversal SHALL obey $R(t) =$

$1952 - t \pmod{2796}$ together with slot reflection (0 2). The omitted slot MAY carry the phase symbol while the other three slots carry the existing curvature pilots.

B.5 Irreversible boundary

Raw transcripts SHALL be reversibly uncomputed after extracting the next action and residual risk, unless archival or audit policy requires retention. Logical entropy SHALL not be reported as physical heat without a temperature, reset protocol, and measured implementation.

B.6 Verification gates

Release evidence SHALL include exact certificate regeneration, focused regressions, 2796-tick RTL covariance simulation, synthesis statistics, idempotent front-door integration, and successful compilation of the W33 paper, Photonic Holonet, and machine blueprint. The full M36 census and complete flag-character decomposition remain separate fail-closed evidence lanes.

C Port switching and unit-spiral duality

The exact finite-port factorisation exposes two complementary structures. On the colouring side, one frozen 60-frame exact cover contains five disjoint switching loci. At each locus four cover frames have eight external competitors inducing $K_{4,4}$; either bipartition class replaces the original four frames. The five switches commute, so the local exact-cover space contains a canonical ternary subcube of size

$$3^5 = 243.$$

Every member is an independent 60-frame set covering all 240 W33 support edges exactly once. This is an exact local switching family, not a classification of all maximum independent sets.

All 45 local $OA(16,3,4,2)$ port arrays are isotopic to the Klein-four Latin square. The global incidence law nevertheless does not select a voltage: a shared support produces $K_{3,3}$ and hence six perfect matchings. Around one support triangle the $6^3 = 216$ matching triples distribute uniformly over the six S_3 holonomies, 36 triples per value. Any nontrivial curvature therefore requires extra phase, hardware, or calibration data; it is not intrinsic to the bare incidence geometry.

The arithmetic phase controller supplies a separate exact spiral lift. For

$$B = \begin{pmatrix} 0 & -1 \\ -1 & 1 \end{pmatrix}, \qquad \operatorname{spec}(B) = \{\varphi, -\varphi^{-1}\},$$

the quarter-turn companion

$$C = \begin{pmatrix} i & -1 \\ 1 & 0 \end{pmatrix}$$

has eigenvalues $i\varphi$ and $-i/\varphi$ and obeys $CP = P(iB)$ for $P = \left(\begin{smallmatrix} -i & i \\ 0 & 1 \end{smallmatrix}\right)$. Thus the expanding golden mode acquires one quarter turn per iteration while its dual mode contracts and the determinant remains one.

With $L = \left(\begin{smallmatrix} -2 & 1 \\ 1 & 2 \end{smallmatrix}\right)$, the golden block obeys $B^T L B = -L$, while $(B^2)^T L B^2 = L$. The two-step map is therefore an exact $O(1,1)$-type boost with eigenvalues φ^2, φ^{-2} and rapidity $2\log\varphi$. This is an algebraic duality on the controller plane, not a physical spacetime claim.

On the logarithmic cover, $(\rho_k, \theta_k) = (k\log\varphi, k\pi/2)$ is a straight lattice line; after exponentiation it is a four-armed logarithmic spiral. A compatible symplectic quarter-turn exchanges radial and angular coordinates, but a tangent vector does not itself become a metric: the precise maps are $v \mapsto \iota_v\omega$ and $g(v,w) = \omega(v, Jw)$.

The term *phigital* is used here only for the standard finite base-φ representation. The quarter-turn construction is a phase lift of that recurrence, not a new numeral system and not a physical relativity claim.

Evidence boundary. The exact chromatic interval remains $10 \le \chi(H) \le 11$. The 100 port-SAT shards have no checked SAT model or DRAT/LRAT refutation; the 26-dimensional split Terwilliger surface has no rational dual bound above nine; the Kempe atlas proves only a 41-frame local floor in its stated one-move neighborhood; and RTL, placement, PDF, optical, and thermodynamic observations remain separately gated.

D Exact-cover switch orbits, non-Abelian logical ports, and the real unit spiral

The five commuting $K_{4,4}$ replacements of Sec. C give a genuine coordinate system $\mathbf{F}_3^5$ on 243 exact 60-frame covers, but not a five-qutrit symmetry under the finite geometry. Exhausting all 25,920 elements of $PSp(4,3)$ gives a setwise stabilizer C_4. Its square fixes every cover and its quotient involution is

$$(x_1, x_2, x_3, x_4, x_5) \longmapsto (-x_4, 1 - x_3, 1 - x_2, -x_1, x_5).$$

It fixes 27 covers and pairs the remaining 216, leaving 135 internal equivalence classes. The fixed 27 form a weighted ternary cube:

$$|C_x \cap C_y| = 60 - 8[\Delta a \ne 0] - 8[\Delta b \ne 0] - 4[\Delta c \ne 0].$$

No union of its five distance relations is the Schlaefli graph; the repeated count 27 is not, by itself, an E_6 identification.

Gauge fixing the 45-block, 720-edge, 240-face port complex along a spanning tree removes 44 edge labels. The 240 edge-disjoint triangular face relations eliminate 240 more distinct chords, so

$$\pi_1(X) \cong F_{436}.$$

Flat S_3 port connections modulo switching are therefore simultaneous-conjugacy classes of 436-tuples. Burnside's lemma gives

$$\frac{6^{436} + 3\,2^{436} + 2\,3^{436}}{6}$$

classes. Abelianization $S_3 \to C_2$ produces the 2^{436} binary logical sectors of the $[[720, 436, 2]]$ port CSS complex: the binary code is the sign-character shadow of a larger non-Abelian logical layer.

The golden quarter-turn also has an exact real machine. The matrix

$$R = \begin{pmatrix} 0 & -1 & -1 & 0 \\ 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & -1 \\ 0 & 0 & 1 & 0 \end{pmatrix}$$

has determinant one, characteristic polynomial $t^4 + 3t^2 + 1$, preserves a signature-$(2,2)$ integral metric and two independent unimodular symplectic forms, and admits a reversing involution S with $SRS = R^{-1}$. Modulo two, $\langle R^2, S\rangle \cong S_3$ on an invariant two-plane. The same recursive/iterative spiral controller can therefore select the six perfect matchings of a physical $K_{3,3}$ port bridge.

The ten-colour boundary remains $10 \le \chi(H) \le 11$. All 100 proof shards received bounded diagnostic runs, but no SAT model and no checked UNSAT proof was obtained. RTL, synthesis, placement and all three PDF builds remain separately evidence-gated.

D.1 Cover-orbit growth, non-Abelian port decoding, and rational port blocks

The five local four-for-four switches do not close after the two $PSp(4,3)$ orbits meeting the original 243-cover family. A deterministic canonical-augmentation checkpoint fully expands ten orbit species and certifies 42 pairwise-disjoint group orbits containing

$$9(6480) + 33(12960) = 486000$$

distinct 60-frame independent sets, each covering all 240 W33 supports exactly once. Thirty-two orbit species remain on the replacement frontier, so this is a lower bound rather than a global classification.

For the S_3-valued port connection, every nonidentity triangular syndrome has three weight-one explanations. There are exactly 3600 weight-two flat non-gauge assignments, proving non-Abelian distance two. A known-location single-edge error is corrected exactly, whereas syndrome alone has guaranteed correction radius zero; the controller therefore requires reliability/location side information and fails closed on ties.

The 26-dimensional split-port Terwilliger algebra admits the exact rational decomposition

$$\mathcal{T}_{\text{port}} \cong \mathbb{Q}^6 \oplus M_2(\mathbb{Q}) \oplus M_4(\mathbb{Q}).$$

On the 135-cell module the M_2 block has rank 18 and multiplicity nine, while the M_4 block has rank four and multiplicity one. Symmetric SDP constraints reduce to six scalars, one 2×2 PSD block and one 4×4 PSD block. No rational dual excluding ten colours is yet claimed.

Supports $0, \ldots, 7$ have pairwise-disjoint nine-frame incidence sets. They define a proof-carrying adaptive cube extending the present 100 shards through depths three to seven. The exact identity

$$m_c = 240 - 4s_c = 4(60 - s_c)$$

forces every missing-colour count to be divisible by four. Nevertheless, without a checked model or a complete verified proof tree the live boundary remains

$$10 \leq \chi(H) \leq 11.$$

Finally, the integral golden spiral obeys $SRS = R^{-1}$. Its prime reductions form a dihedral guard tower: the even rotation/reflection subgroup is S_3 modulo two, D_4 modulo three, pentagonal modulo five, and octagonal modulo seven. The binary port code is only the sign-character shadow of the full S_3 connection; the two-dimensional standard Fourier channel detects three-cycle faults invisible to the sign bit.

Passes 4213–4220: smaller covers, two-bundle holonomy, hysteretic memory, compressed clocks, and full Gaussian horizon diagnostics. The corrected degree-four $W(3,3)$ point–line Levi graph admits an explicit cyclic $\mathbb{Z}_{359}$ voltage cover in which every simple base cycle of lengths $8, 10, 12$ has nonzero voltage. Hence the cover has 28,720 vertices, 57,440 edges, and certified girth at least 14, reducing the earlier $\mathbb{Z}_{65537}$ construction by a factor $65537/359 = 182.5543\ldots$ in sheet count. The voltage vector and all 50,004 short-cycle checks are frozen in the certificate; no minimality claim is made.

On two rank-two singlet bundles, the dense local $SU(2)$ holonomies of Pass 4190 together with a single auxiliary bright level produce a geometric controlled phase. A loop of solid angle 2π in $\text{span}\{|11\rangle, |a\rangle\}$ gives Berry phase π on $|11\rangle$, hence $\text{CZ} = \text{diag}(1, 1, 1, -1)$. The resulting Pauli commutator closure has dimension 15, so dense local $SU(2) \times SU(2)$ plus this entangler is dense in $SU(4)$.

For the saturated scale flow

$$\dot{s} = \gamma\big[\ln 80 - 4s + g n_B(2e^{-s}) - 0.004\, n_B(2e^{-s})^2\big]\,,$$

classical coexistence occurs at $g = 0.5700939873487846$. The two minima $s_L = 1.2870434725$ and $s_R = 5.3097911844$ are separated by the saddle $s_B = 3.8542873087$ and equal barrier $\Delta U = 4.4597412117$. The dimensionless $M = 1$ zero-energy instanton action is $S_0 = 7.7881701215$, so the two-state tunnelling matrix element carries the semiclassical factor $e^{-S_0\sqrt{M}/\hbar_{\rm eff}}$. At noise strength $D = 0.25$ the Kramers lifetimes are 8.9681×10^7 and 2.9530×10^7 in model time units. These are effective-model rates, not physical spacetime switching frequencies.

The exact three-local Gray-history clock can share spectator-condition logic. An exact MILP proves that only 13 pair conjunctions are required within the balanced $2 + 2$ AND-tree class. Together with 24 final conditions and 24 transition flags this reduces auxiliary count from 96 to 61 while preserving three-locality, legal gap 2Ω, perfect transfer at $\pi/(2\Omega)$, and full revival at π/Ω.

For the complete 19-mode Gaussian horizon chain (one outgoing, nine partner, nine environment modes), the zero-temperature outside/partner state has logarithmic negativity 0.2085352998, steering measures 0.0085289241 and 0.0232180113, and outside-to-partner coherent information 0.0518013351 nats. Uniform thermal loading destroys outside-to-partner steering at $n_{\rm th} = 0.0042826995$, positive coherent information at 0.0079250582, reverse steering at 0.0117448238, and PPT entanglement only at 0.1159362071. In the frozen 512-sample correlated-disorder ensemble at $n_{\rm th} = 0.005$, all 512 states remain entangled while the three stronger witnesses survive in 228, 415, and 482 samples respectively.

Three independent finite probes sharpen the boundary. First, the Levi spectrum $\{0, \pm 4, \pm\sqrt{6}\}$ forbids exact nonzero global continuous-time-walk revival, but the Pell solution $11760^2 - 6(4801)^2 = -6$ gives a near-echo at $t = 4801\pi/2$ with operator error 4.0071334×10^{-4}. Second, for the harmonic kernel $K = 5I - A_{\rm Levi}$, incidence singular values $4^1, (\sqrt{6})^{24}, 0^{15}$ yield exact point–line ground-state entanglement entropy 2.4552973872 nats and logarithmic negativity 6.9804187391 nats. Third, the degree-four tree radial band is $E(q) = 2\sqrt{3}J\cos q$, so $v_{\max} = 2\sqrt{3}J$ while shell capacity grows as $4 \cdot 3^{r-1}$. This graph velocity is not identified with physical c.

Boundary. These are finite graph-cover, holonomic, stochastic/semiclassical, clock-gadget, Gaussian-channel, recurrence, harmonic-network, and tree-band results. They do not establish fabricated hardware, observed Hawking radiation, physical spacetime memory, a speed-of-light derivation, gravity, cosmology, or a theory of everything.

D.2 Raw Cubic Leakage Ratios

Let

$$G = \frac{1}{81}CC^T = \frac{160}{81}E_4$$

be the normalized Gram operator of the W33 Levi cycle frame. In dimension $d = 81$, the cubic Gegenbauer transform is

$$C_3^{79/2}(x) = \frac{177039}{2}x^3 - \frac{6399}{2}x.$$

Writing

$$C_3(G) = c_0E_0 + c_1E_1 + c_2E_2 + c_3E_3 + c_4E_4,$$

the non-uniform leakage coefficients satisfy

$$c_1 = -\frac{734384}{2187}(-244 + 9\sqrt{6}), \qquad c_2 = \frac{177720928}{2187},$$

$$c_3 = \frac{734384}{2187}(244 + 9\sqrt{6}), \qquad c_4 = \frac{1751954560}{19683}.$$

The conjugate $24 + 24$ leakage pair is locked to the 30-sector by

$$\boxed{\frac{c_1 + c_3}{c_2} = \frac{244}{121}.}$$

After weighting by primitive multiplicities $24, 30, 24$, this becomes

$$\boxed{\frac{24c_1 + 24c_3}{30c_2} = \frac{976}{605}.}$$

The total trace of the raw cubic leakage operator is

$$\boxed{\operatorname{tr}(C_3(G)) = 13651200.}$$

Sector	multiplicity	leakage coefficient
E_0	1	$17205568/243$
E_1	24	$-\frac{734384}{2187}(-244 + 9\sqrt{6})$
E_2	30	$177720928/2187$
E_3	24	$\frac{734384}{2187}(244 + 9\sqrt{6})$
E_4	81	$1751954560/19683$

Thus the raw cubic flow is structured leakage, not random noise: the companion $24+30+24$ stack carries a rigid W33 spectral ratio, while the repaired flow removes $E_1 + E_2 + E_3$ before centering and renormalization.

D.3 Levi Homology versus Phase-Cover Fiber Homology

There are two natural homological objects in the phase-cover construction, and it is important not to identify them. The protected W33 homology comes from the point-line Levi graph of $W(3, 3)$, not from the scalar fiber of the phase cover.

Let L denote the W33 point-line Levi graph. It has

$$|V(L)| = 80, \qquad |E(L)| = 160,$$

and hence

$$\boxed{\beta_1(L) = 160 - 80 + 1 = 81.}$$

This is the protected cycle rank:

$$\boxed{H_1 = 81.}$$

The same Levi graph has 1620 simple 8-cycles, and the flag-edge incidence count satisfies

$$\boxed{160 \cdot 81 = 1620 \cdot 8 = 12960.}$$

Thus 12960 is a Levi support-incidence count derived from the $H_1 = 81$ cycle geometry.

The scalar phase cover begins from this 12960-element support base. Each base incidence has four nonzero ternary scalar lifts. As a minimal fiber graph, those four lifts form a square, so for one fiber

$$V_{\text{fiber}} = 4, \qquad E_{\text{fiber}} = 4, \qquad \beta_1^{\text{fiber}} = 1.$$

Across all base incidences this gives

$$V = 12960 \cdot 4 = 51840, \qquad E = 12960 \cdot 4 = 51840,$$

with 12960 connected fiber components. Therefore

$$\boxed{\beta_1^{\text{fiber}} = 51840 - 51840 + 12960 = 12960.}$$

Consequently

$$\boxed{81 \neq 12960.}$$

The correct relationship is layering rather than equality:

$$\boxed{H_1(L) = 81 \quad \rightsquigarrow \quad 12960 \text{ Levi support incidences} \quad \rightsquigarrow \quad 51840 \text{ phase-cover lifts.}}$$

The scalar cover doubles the nonzero phase sheets, but it does not replace the Levi cycle homology. The former is a fiber square over each support incidence; the latter is the genuine incidence homology of the W33 point-line geometry.

D.4 Cubic Leakage as an Ihara Shadow

The raw cubic leakage ratio is not merely a coefficient accident in the Bose–Mesner algebra. It is locked to the nonbacktracking scale of the W33 carrier. Let $k = 12$ be the relevant W33 valency and set

$$p_{\mathrm{Ih}} = k - 1 = 11.$$

Then the unweighted cubic leakage ratio satisfies

$$\boxed{\frac{244}{121} = 2\left(1 + \frac{1}{p_{\mathrm{Ih}}^2}\right) = 2\left(1 + \frac{1}{11^2}\right).}$$

Equivalently,

$$244 = 2(11^2 + 1).$$

The same numerator also has the W33 substrate decomposition

$$\boxed{244 = v\Phi_6 - \chi q^2 = 40 \cdot 7 - 4 \cdot 9.}$$

After multiplicity weighting, the leakage ratio is modulated by the Euler–pentagonal factor:

$$\boxed{\frac{976}{605} = \frac{\chi}{F_5} \cdot \frac{244}{121} = \frac{4}{5} \cdot \frac{244}{121}.}$$

Thus the raw cubic companion-sector leakage may be read as an Ihara shadow: the entrywise cubic transform detects the even two-step nonbacktracking transfer, while the repaired map removes this companion shadow.

In sector language,

$$C_3(G) \in E_0 + E_1 + E_2 + E_3 + E_4,$$

where the nonbacktracking companion is

$$E_1 + E_2 + E_3.$$

The repair projector

$$P_{E_0+E_4}$$

is therefore the minimal primitive-sector filter preserving the uniform sector and the Hodge-protected sector while killing the Ihara shadow. Centering then removes E_0, and diagonal renormalization returns

$$G = \frac{160}{81} E_4.$$

Consequently the repaired cubic dynamics is not an arbitrary correction; it is an explicit removal of the nonbacktracking companion shadow, leaving the protected W33 Levi Hodge idempotent fixed.

D.5 Internal Tetrahedral Carrier and Hodge Clock

The tetrahedral parity carrier can be realized inside the W33 Levi flag module. A subgroup

$$S_4 < \mathrm{PSp}(4,3)$$

acts on the 160 Levi flags with orbit profile

$$\boxed{8, 8, 24, 24, 24, 24, 24, 24.}$$

Thus there are six internal regular S_4-orbits of size 24. This gives an internal carrier for the residual sequence

$$\boxed{24(-1)^n,}$$

previously identified from the endpoint recurrence quotient.

The protected Hodge block has the period-two scalar law

$$\boxed{E_4 F_n E_4 = (2 + (-1)^n) E_4.}$$

Equivalently, the scalar sequence is

$$3, 1, 3, 1, \ldots$$

and the protected trace sequence is

$$243, 81, 243, 81, \ldots.$$

Therefore the full endpoint recurrence separates into three compatible pieces:

$$\boxed{\text{Ihara endpoint growth} \quad \rightsquigarrow \quad \text{Hodge } E_4 \text{ period-two clock} \quad \oplus \quad \text{internal tetrahedral } S_4 \text{ sign carrier.}}$$

D.6 The Six-Carrier/Four-Cell Boundary Split

The internal tetrahedral trace carriers admit a sharper boundary decomposition than the first count-level statement. The six regular internal S_4-orbits occupy $6 \cdot 24 = 144$ Levi flags, leaving a 16-flag complement. The corrected verifier-level statement is

$$160 = 6 \cdot 24 + 4 \cdot 4.$$

Here the $6 \cdot 24$ term is the union of six regular S_4 sign carriers, while the $4 \cdot 4$ term is not a raw Q_4 under Levi-flag adjacency. Instead, the induced complement graph is

$$K_4 \sqcup K_4 \sqcup K_4 \sqcup K_4.$$

Equivalently, the complement consists of four complete W33 line fibers, each carrying four flags.

This corrects the count-level Q_4-codec reading. The number 16 still matches the codec-boundary scale used elsewhere in the tomotope/Fano layer, but the raw Levi flag graph does not realize that 16-set as the 4-cube. Any Q_4 relation must therefore be a secondary codec relation, not the Levi flag adjacency relation.

The incidence of the four K_4 complement cells against the six regular 24-flag carriers is also rigid. With respect to raw Levi flag adjacency, the four cells have the same carrier-incidence vector

$$(0, 0, 6, 0, 6, 0),$$

so all distance-one contact is concentrated in one active pair of regular carriers. The full distance profile recovers a three-pair structure:

$$6 = 2_{\text{far}} + 2_{\text{middle}} + 2_{\text{active}}.$$

Aggregated over the 16-flag complement, the three paired distance profiles are

carrier pair	d_1	d_2	d_3	d_4
far	0	0	96	288
middle	0	48	192	144
active	24	96	120	144

with each row occurring for two carriers.

Thus the honest Levi-flag statement is not

$$160 = 6 \cdot 24 + 16_{Q_4},$$

but rather

$$\boxed{160 = 6 \cdot 24 + 4 \cdot 4.}$$

The remaining $Q_4/K_{4,4}$ codec story, if used, must be stated as a secondary tomotope/Fano boundary labeling rather than as the induced Levi-flag complement graph.

D.7 Secondary codec-to-G_2 quotient and the packet-basis boundary

The corrected finite-geometric chain begins with the raw S_4 complement in the Levi flag graph. The induced raw complement is not a hypercube. It is

$$C_{16}^{\mathrm{raw}} \cong K_4 \sqcup K_4 \sqcup K_4 \sqcup K_4.$$

Thus Q_4 is not a raw Levi-adjacency claim. It enters only after choosing the secondary product-codec relation on the four-by-four complement chart:

$$4K_4 \longrightarrow Q_4 \longrightarrow Q_4/\{\pm\} \cong K_{4,4}.$$

The next quotient is the frame quotient of the $K_{4,4}$ perfect matchings. The 24 perfect matchings split into six one-factorization frames of size four:

$$\mathrm{Match}(K_{4,4})/V_4 \cong S_4/V_4 \cong S_3,$$

where

$$V_4 = \{(), (12)(34), (13)(24), (14)(23)\} \subset S_4.$$

The transposition Cayley graph of S_3 is $K_{3,3}$, so the secondary chain is

$$4K_4 \to Q_4 \to K_{4,4} \to \mathrm{Match}(K_{4,4}) \to S_4/V_4 \to K_{3,3}.$$

The three diagonal Fano gauges are

$$011, \qquad 101, \qquad 110.$$

They are identified with the three carrier-pair channels

$$011 \leftrightarrow F, \qquad 101 \leftrightarrow M, \qquad 110 \leftrightarrow A.$$

Adding the two sign sheets gives six carrier labels

$$F_+, F_-, M_+, M_-, A_+, A_-.$$

The metric matching is

$$M_{\mathrm{metric}} = \{F_+F_-,\ M_+M_-,\ A_+A_-\}.$$

The stabilizer of this matching inside $\mathrm{Aut}(K_{3,3})$ is

$$\mathrm{Aut}(K_{3,3}, M_{\mathrm{metric}}) \cong S_3 \times C_2 \cong D_6 \cong W(G_2).$$

This gives the secondary G_2 Weyl quotient. It is a quotient of codec/frame labels, not a raw flag-level Weyl action on the 160 Levi flags.

The same boundary appears in the $E_1 + E_3$ packet. Dimensionally,

$$E_1 + E_3 = 24 + 24 = 48 = 4(6_{\mathrm{short}} + 6_{\mathrm{long}}),$$

so the external $W(G_2)$ packet model is dimensionally correct. The folded-cubic cross-channel, however, is not a real Weyl reflection: after normalization it has

$$J^2 = -I,$$

whereas a real reflection requires square $+I$. Only the phase lift repairs this:

$$(iJ)^2 = +I.$$

Consequently, a canonical numeric packet basis requires an additional flag-level selector that splits each 24-sector as

$$24 = 4 \cdot 6.$$

The Fano/D_6 selector candidate supplies the representation-level algebra

$$24 = 4_{\text{copies}} \cdot 3_{\text{Fano gauges}} \cdot 2_{\text{signs}},$$

but the four-copy label is imported from the raw $4K_4$ complement cells rather than selected by the Bose–Mesner algebra alone.

D.8 Five Levi frontiers: odd-order Jordan law, discriminant lift, and typed middleware

Let M_q be the point–line incidence matrix of $W(q)$ and $D_q = \left(\begin{smallmatrix} 0 & M_q \\ M_q^T & 0 \end{smallmatrix}\right)$. Exact construction over $\mathbb{F}_3, \mathbb{F}_5, \mathbb{F}_7$, and the genuine extension field $\mathbb{F}_9 = \mathbb{F}_3[w]/(w^2+1)$ gives

$$\operatorname{rank}_2 M_q = \frac{q(q+1)^2+2}{2}, \quad \operatorname{rank}_2 A_P = \frac{q(q^2+1)}{2} + 1, \quad \operatorname{rank}_2 A_L = q^2+1$$

for $q = 3, 5, 7, 9$. Consequently the observed Jordan law is

$$D_q \sim J_4^2 \oplus J_3^{(q^3+2q^2+q-4)/2} \oplus J_1^{q(q-1)^2/2},$$

with no J_2 blocks. Independently of those rank formulas, the generalized quadrangle axiom proves for every odd q that

$$D_q^3 = \begin{pmatrix} 0 & J \\ J & 0 \end{pmatrix}, \qquad D_q^4 = 0.$$

Thus the terminal image is the two-plane spanned by the all-point and all-line parity rails, whose three nonzero rays are permuted by $\mathrm{GL}(2,2) \cong S_3$.

At $q = 3$, the two chain homologies are

$$H_P = \ker A_P / \operatorname{im} A_P \cong O_8^+(2) = E_8/2E_8, \qquad H_L = \ker A_L / \operatorname{im} A_L \cong O_{20}^+(2),$$

with dimensions 8 and 20, respectively. Both Arf invariants vanish, and their direct sum is a rank-28 plus-type discriminant carrier.

This produces a typed packet ABI. A point packet carries an eight-bit homology syndrome, a line packet a twenty-bit syndrome, and a legal mirror conversion applies M^T or M while toggling the type bit. The target class is always a boundary, hence has zero target syndrome. Raw retagging is rejected on all 28 canonical homology generators. The type bit is therefore an algebraic invariant rather than metadata.

Finally, the $q = 3$ Jordan partition $4^2 3^{22} 1^6$ has conjugate partition $(30, 24, 24, 2)$. Its nilpotent centralizer acts on the two length-four chains through $\mathrm{GL}(2,2) = S_3$. Adjoining the independent phase bit gives $S_3 \times C_2 \cong D_{12}$ with order profile $\{1^1, 2^7, 3^2, 6^2\}$, exactly the mirror-bus slot-stabilizer profile. This explains the middleware counts

$$8 \cdot 6 = 48, \qquad 8 \cdot 12 = 96, \qquad 24 \cdot 45 \cdot 48 = 51840, \qquad 25920/12 = 2160.$$

The 96 equality is an order-level bridge only; no direct-product identification with the tomotope automorphism group is asserted.

Witness: `analysis/w33_levi_five_frontiers.py`, with certificate `data/PART_2026_07_10_LEVI_FIVE_FRONTIERS_results.json`.

E Kernel, cohomological, exceptional, foundry, and HIL closure

The odd-q rank arithmetic is now exposed through a Lean structure $\mathsf{OddQFourierCertificate}(q)$ whose fields are the trivial and nontrivial additive-character blocks. The resulting kernel statements include

$$\operatorname{rank}_2 M_3 = 25, \qquad \operatorname{rank}_2 A_P = 16, \qquad \operatorname{rank}_2 A_L = 10, \qquad D_3 \sim J_4^2 \oplus J_3^{22} \oplus J_1^6.$$

Here

$$\frac{3^3 + 2\cdot 3^2 + 3 - 4}{2} = 22, \qquad 2\cdot 4 + 22\cdot 3 + 6 = 80.$$

The repository CI requires `lake build --wfail`, an explicit rejection of `sorry` and `admit`, and an independent official `leanchecker` pass.

Let $A = A_2(L_{-4}) \cong (\mathbb{Z}/2)^{14} \oplus \mathbb{Z}/8$ and let h generate the order-eight factor. Writing

$$\tau(h) = 5h + u, \qquad u = e_0 + e_1 + e_2 + e_5 + e_8 + e_{10} + e_{12} + e_{13},$$

the involution condition puts u in $\ker(1+\tau)$. Replacing h by $h+v$ changes u by $(1+\tau)v$, so the mixing class lies in

$$H^1(C_2, A[2]) = \ker(1+\tau)/\operatorname{im}(1+\tau).$$

The exact dimensions are $9-6=3$, and the scalar-5 displacement satisfies

$$\boxed{[u_5] = [4h] \neq 0.}$$

Consequently a pure scalar-5 normal form is obstructed. But $u_1 = \tau(h) - h = u_5 + 4h$ is a coboundary: exactly 512 order-two shifts make $h+v$ fixed, and 256 preserve $q(h+v) = 11/8$. The outer action therefore admits a fixed order-eight rail; only the scalar-5 form is cohomologically obstructed.

Using doubled integral coordinates for the 240 roots of E_8, an explicit A_2 pair has an orthogonal E_6 subsystem of 72 roots. The restriction of $W(E_6)$ to the full root set decomposes as

$$\boxed{240 = 72 + 6 + 6\cdot 27}.$$

For every E_6 root α, the six minuscule weights with $\langle w, \alpha\rangle = 1$ form a skew six, yielding an equivariant bijection between the 72 roots and the 72 oriented double-sixes. Native generator permutations make the incidences among

$$40, 40, 27, 45, 72, 240, 540$$

commute, and the stabilizer of a noncollinear pair has order 48.

The 16-mode, 120-rotation optical compiler was lowered to an explicit 800-nm-class Si_3N_4 thermal foundry model with 14-bit controls, wavelength dispersion, fabrication offsets, dense thermal crosstalk, insertion loss, and closed-loop calibration. Across 96 modeled dies,

$$\overline{F} = 0.999970663, \qquad F_{5\%} = 0.999959653,$$

and drift tracking maintains

$$F_{\min} = 0.999994253.$$

The modeled on-chip loss is 0.33792 dB. The 7.81 W thermal upper bound is a hardware warning: the passive mesh is viable, but high-duty control should migrate toward piezoelectric or electro-optic actuation.

Finally, a bounded-memory hardware-in-the-loop adapter processed 3,164,709 timestamp events on sixteen balanced channels plus sync. It accepted 126 of 128 frames, retried two, admitted no wrong frame, and rejected all 224 sub-eight corruptions, all 45 dark-code displacements, a raw retag, and a bad authenticator before assigning native $W(E_6)$ coordinates.

F Audited $q = 3$ matrix algebra, extension transgression, typed E_8 lanes, and hardware models

For the exact $q = 3$ binary matrix block, the transpose-sum operator on $\mathrm{Mat}_3(\mathbb{F}_2)$ is $T(Y) = Y + Y^T$. Exact finite calculation gives

$$\ker T = \mathrm{Sym}_3(\mathbb{F}_2), \quad |\ker T| = 64, \qquad \mathrm{im}\, T = \mathrm{Alt}_3(\mathbb{F}_2), \quad |\mathrm{im}\, T| = 8.$$

The symmetric and alternating spaces have dimensions six and three, and the diagonal map on the symmetric space has rank three. Separately, the complete native W33 incidence matrices give

$$\mathrm{rank}_2\, M_3 = 25, \qquad \mathrm{rank}_2\, A_P = 16, \qquad \mathrm{rank}_2\, A_L = 10, \qquad D_3 \sim J_4^2 \oplus J_3^{22} \oplus J_1^6.$$

The matrix cardinalities are represented directly in `W33.HeisenbergQ3` and checked by Lean finite computation. No Fourier transform or direct-sum map connecting that matrix basis to the 40×40 incidence operator is formalized; the other Lean modules check the arithmetic assembly identities, not the geometric rank theorem.

Let $A \cong (\mathbb{Z}/2)^{14} \oplus \mathbb{Z}/8$ be the two-primary discriminant module with outer involution τ. Periodic cyclic cohomology gives

$$\dim H^1(C_2, A[2]) = \dim H^2(C_2, A[2]) = 3.$$

Thus there are eight two-torsion extension classes for the prescribed nontrivial C_2 action, seven non-split; these are not central extensions. The displayed minimum-weight Smith-coordinate representatives and their quadratic values depend on the selected basis. If h generates the order-eight rail and $\tau(h) = 5h + u$, then $[u] = [4h] \neq 0$ in H^1, but the connecting homomorphism for $0 \to A[2] \to A \to A/A[2] \to 0$ satisfies

$$\boxed{\delta([h]) = [\tau(h) - h] = [u + 4h] = 0.}$$

Indeed a minimum-weight gauge $v = \texttt{0x411}$ makes $h + v$ fixed of exact order eight. Hence a fixed rail exists, whereas the scalar-five rail retains its genuine nonremovable mixing.

The full E_8 carrier is compiled as

$$\boxed{240 = 72 + 6 + 6 \cdot 27},$$

with 72 E_6 control roots, six fixed A_2 reference roots forming a hexagon, and six payload lanes carrying the degree-10 complement of the conventional Schläfli graph. Each payload root has routing signature $16 + 16 + 40$ against the control roots. Simple E_6 reflections preserve all lane and route data exhaustively. A seeded 50,000-step generator replay is an integration smoke test, not additional proof.

Replacing continuously heated controls by PZT coarse trim and capacitive Pockels fine control gives the explicit architectural allocation

$$\boxed{85.607097\ \mathrm{mW}}.$$

In 128 fixed-seed synthetic phase-error draws, the fifth percentile is 0.999988590, and modeled tracking keeps the 256-epoch minimum at 0.999999673. The phase commands use a validated signed 16-bit range. A deterministic GDSII record/placement sketch and a separate hand-written Verilog-A interface are supplied; neither is a PDK, DRC, device-simulation, or measurement certificate. The mathematical compiler has 376 phase parameters; the placement sketch has 120 abstract interferometer slots and does not provide a complete command-to-device manifest.

Finally, documentation-shaped vendor adapters feed a Python cycle reference that processes 1,048,832 deterministic events into 256 bounded frame summaries under backpressure. The SystemVerilog evidence is independent: a two-frame Icarus smoke simulation checks completed-frame snapshots and output hold. Clean reference frames give parallel typed projections

$$\text{summary} \longrightarrow W(3,3)\ \text{point}_{40} \qquad \| \qquad \text{complement-Schlaefli payload}_{162} \longrightarrow E_6\ \text{control}_{72} \longrightarrow E_8\ \text{root}_{240}.$$

All 40 point addresses and 162 payload addresses occur, and the selected control has inner product $+1$ with its payload root. No $40 \to 27$ incidence morphism is asserted. Missed/lost/full-buffer and overflow conditions are rejected by the clean path. The aggregate regenerates all five witnesses and requires exact JSON-normalized equality with the cached certificates.

F.1 Clock homology and the eight-tick runtime word

The previous subsection identifies the holonet clock with the Fano/Heawood harmonic oscillator. The homological content of that statement is now exact. Let

$$H = \mathrm{Inc}(\mathrm{PG}(2,2))$$

be the Heawood graph, with seven Fano points and seven Fano lines. The verifier finds

$$|V(H)| = 14, \qquad |E(H)| = 21, \qquad \beta_1(H) = 21 - 14 + 1 = 8.$$

Thus the eight-tick runtime word is not an added convention: it is the first homology rank of the clock graph itself. The same finite graph also contains

$$\#C_6(H) = 28, \qquad \#C_8(H) = 21,$$

so the clock topology sees both the externality number $v - k = 40 - 12 = 28$ and the Fano/Klein carrier $\binom{7}{2} = 21$.

Passing to the line graph $L(H)$ turns the 21 Fano incidences into clock flags. This flag-clock is 4-regular on 21 vertices and has Laplacian spectrum

$$0^1 \oplus (3 - \sqrt{2})^6 \oplus (3 + \sqrt{2})^6 \oplus 6^8.$$

The oscillator middle shell is preserved, while the endpoint becomes an 8-dimensional shell 6^8, again matching the runtime word.

This clock module is coupled to, but not contained in, the $W(3,3)$ point-line Levi graph. Indeed the Levi graph has

$$|V| = 80, \qquad |E| = 160, \qquad \beta_1 = 81, \qquad \text{girth} = 8,$$

with $\#C_6 = 0$ and $\#C_8 = 1620$. Since the Heawood clock has girth 6, it cannot be a literal Levi subgraph. The correct statement is therefore a coupled-module statement: the Heawood/Fano graph supplies the runtime clock homology, while the $W(3,3)$ Levi graph supplies the protected $H_1 = 81$ cycle sector.

The spectral coupling to matter is equally sharp. For the matter graph $Q = \overline{W(3,3)}$, the Laplacian spectrum is

$$0^1 \oplus 24^{15} \oplus 30^{24}.$$

Using the conservative Cartesian/tensor graph-product Laplacian

$$L_{L(H)} \otimes I_Q + I_{L(H)} \otimes L_Q,$$

the clock endpoint and matter gap obey

$$6^8 \otimes 24^{15} \longmapsto 30^{120}.$$

The resonance value is $30 = h(E_8)$ and the rank is $8 \cdot 15 = 120$. There is an important degeneracy boundary: the full coupled 30-eigenspace has multiplicity $144 = 120 + 24$, because $0^1_{\mathrm{clock}} \otimes 30^{24}_{\mathrm{matter}}$ lands at the same eigenvalue. Thus the theorem identifies a canonical rank-120 tensor subblock inside a degenerate eigenspace, not a unique eigenspace without projectors.

Finally, the explicit cycle-basis verifier makes the runtime stack homological:

$$\beta_1(H) = 8 \longrightarrow 8 \cdot 6 = 48 \longrightarrow 8 \cdot q^2 = 72 \longrightarrow 72h(E_8) = 2160 \longrightarrow 2160 \cdot 24 = 51840 = |\mathrm{Sp}(4,3)|.$$

The chosen cycle basis is deterministic rather than canonical, but the invariant object is canonical: the 8-dimensional $\mathbb{F}_2$ cycle space of the Heawood/Fano clock. (Witnesses: `analysis/bt1654_heawood_clock_ho` `analysis/bt1655_clock_matter_spectral_coupling.py`, and `analysis/bt1656_runtime_word_cycle_basi`